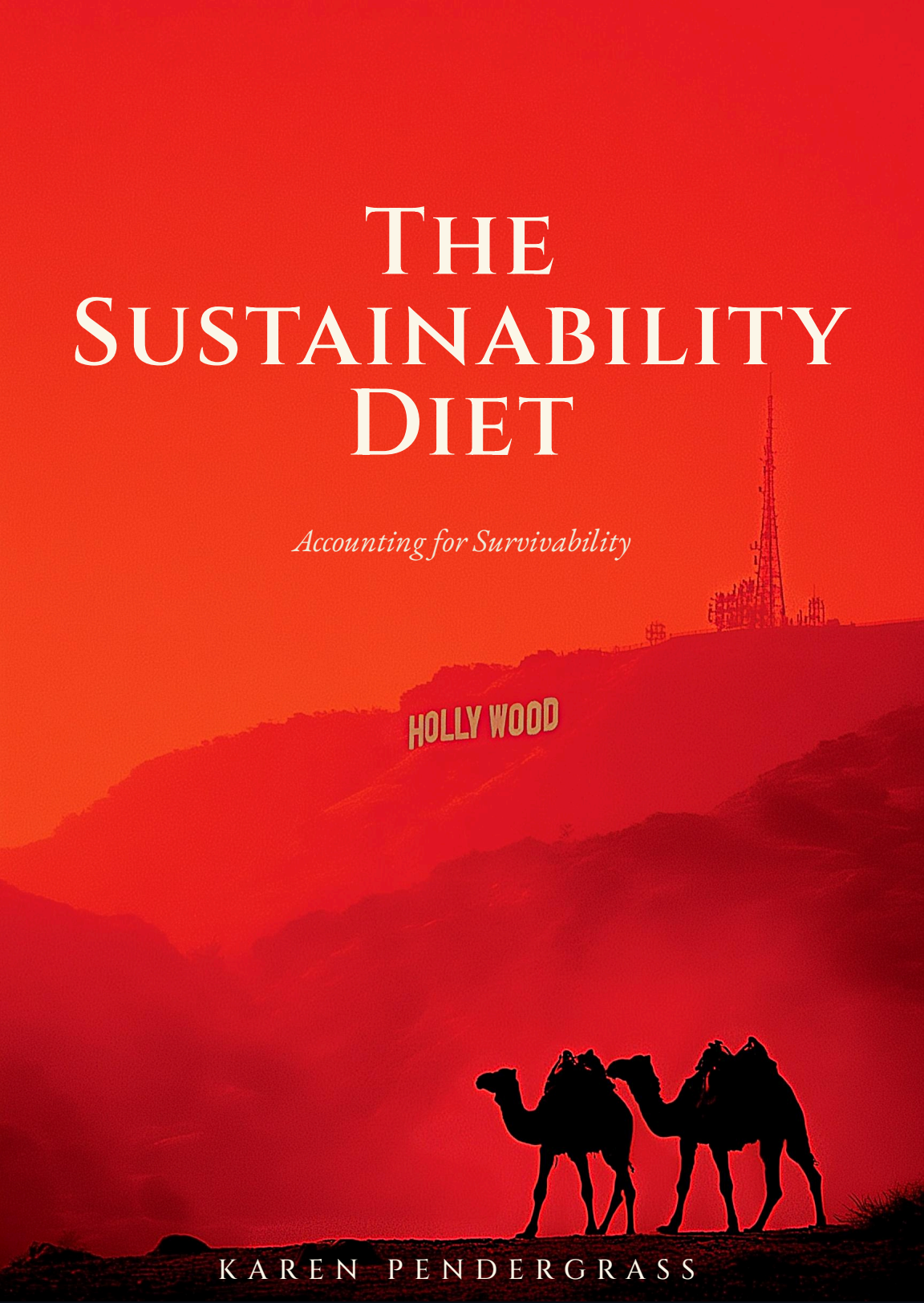
The Sustainability Diet

Accounting for Survivability



KAREN PENDERGRASS

FIRST PROOF EDITION · MMXXVI

The Sustainability Diet

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Accounting for Survivability



KAREN PENDERGRASS

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FIRST PERSONAL PRINT EDITION, 2026.

SET IN EB GARAMOND AND CINZEL.

For Nefling.

What is left of this continent is yours.
What returns to it, with luck, will be ours together.

AUTHOR'S NOTE

This book makes one claim and walks the consequences of it through every food system worth evaluating.

The claim is that the framework most often used to ask whether food production is sustainable — economic viability, social fairness, environmental fit — describes how a system performs under stable conditions. It does not describe how the system performs when the conditions change. That is a different question, and it has been left out of the framework we have been using to score food for forty years. I call the missing axis *survivability*. The rest of the book is the case for taking it seriously.

The argument is timely because the conditions are changing. Climate volatility now reaches a single growing season. Supply-chain shocks reach the grocery shelf within weeks. Soil base, irrigation supply, pollinator population, antibiotic ladder, fossil-fuel input — every variable food production rests on is moving, and most of them are moving in directions the current framework has no language for. A food system can be the very best system the present-day inputs can produce and still fail the survivability test. This book exists to make that test visible, applicable, and unavoidable.

I am writing it from inside the certification work, not outside it. Sixteen years of standards-writing for the Paleo

Foundation taught me how to score food production in operation; the four-axis framework here is what I now believe we should also have been scoring all along.

The book ends in a hypothesis: that one of the food systems that scores best on all four axes is one we lost twelve and a half thousand years ago and could choose to bring back. The chapters arrive there in their own time.

— *K. P.*

A NOTE ON WHERE I AM WRITING FROM

This book could be read as a sixty-thousand-word advertorial for a certification ecosystem I have been part of for sixteen years. Reading it that way is not unfair, and I want to address it before it becomes the loudest thing in the room.

Since 2010, I have been architecting standards for the Paleo Foundation. That work began with Paleo Approved, a farm and ranch certification that was a flop, and later expanded into Certified Paleo, Keto Certified, and most recently Heavy Metal Tested & Certified — which is where the heavy-metals chapter of this book sits with respect to my current professional work.

The Afterword names four sites I run alongside the certification work — wikibiome.com, microbiomemedicine.com, heavymetaltested.com, and heavymetalindex.com — and the heavy-metals chapter cites two of my own preprint papers as part of its evidence base.

At first glance these look like disparate interests: agriculture, diet, microbiome medicine, microbial metallomics, heavy metals. They are not. They are what comes of pulling on the same threads for sixteen years in different directions. They are intricately connected, and the book is the place where some of the connections finally come into view.

I am disclosing this here because the alternative — burying it in the Acknowledgments — would not change the conflict; it would only make finding it harder. The conflict is real. I want to be honest about what it is and what it isn't.

What it isn't: this book is not a sales document for HMTc certification. The certification work is a smaller proposition than the book is. HMTc tests for a specific set of contaminants in specific food categories under specific protocols. The book is an argument for a four-axis evaluative framework that applies to food systems whether or not anyone certifies anything. The framework was not built to backfill the certification work. The certification work was built because the framework, applied to the parts of the food system I happened to be looking at most closely, kept producing the same result: the heavy-metal load is the input nobody is measuring at the input level, and somebody had to start.

The heavy-metal load had consequences that could not be understood without connecting dots across agriculture, diet, the microbiome, human health and disease, and microbial metallomics. The patterns emerged, and AI will only make those patterns louder in coming years. But that's beyond the scope of this book. Maybe in the next one.

What it is: sixteen years of standards-writing taught me, faster than any other route I can imagine, what it actually takes to score a food production system against a stated criterion. You write the criterion. You go look at the production. You discover what the criterion did not anticipate. You revise. You discover what the revised criterion did not anticipate. You revise again. After enough rounds, you start to see the structural shape of what the criterion was always missing, and the missing piece is usually the same piece

across categories. The four-axis framework is what I think the missing piece, in the most general form I have been able to find, actually is. The reason the preprints are cited is not to send readers to my certification work; it is to show that I have been taking the question seriously enough to publish what I have found, in formats anyone can read, before asking anyone to trust the framework.

I am writing this book from inside that work, not outside it. The reader can decide what weight to give claims that come from inside that vantage. My obligation is to make the vantage visible.

— K. P.

THE FRAMEWORK, IN ONE DIAGRAM

The standard sustainability framework asks how a food system performs now. The survivability axis asks how it performs as the conditions change.

ECONOMIC VIABILITY

Can the system pay its way at the farm gate, today?

Margin. Labor cost. Capital outlay. Subsidy structure.

SOCIAL FIT

Does it work for the people who run it and depend on it?

Labor conditions. Access. Affordability. Cultural fit.

ENVIRONMENTAL FIT

Does it match the place it sits in at present?

Soil. Water. Biodiversity. Emissions.

THE STANDARD FRAMEWORK

a snapshot of the system as it is

SURVIVABILITY

does the system keep working as the conditions move?

climate volatility · input depletion · supply shock · time

A FOOD SYSTEM IS WORTH KEEPING ONLY IF IT PASSES ALL FOUR.

A system can score perfectly on all three classical axes and still fail the fourth. Most twentieth-century food systems do. The chapters that follow apply the test.

◆

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PROLOGUE

One Hundred Camels

♦ ♦ ♦

From Daniel H. Janzen & Paul S. Martin, Science (1982)

"Neotropical Anachronisms: The Fruits the Gomphotheres Ate."

A continent's flora carries the signature of the animals it evolved with. When those animals vanish inside a geological instant, the flora is left producing for mouths that no longer exist.

— D. H. Janzen & P. S. Martin, 1982

It is not lost on me that for three million years, millions of camels lumbered through the very place where I first sat down to write this book. A cafe in Santa Monica called 212. The cafe is gone now. The camels have been gone much longer.

The camel was *Camelops hesternus*, "yesterday's camel," the Western camel, native to North America for almost the entire existence of its genus. It stood roughly seven feet at the shoulder—somewhat larger than a modern dromedary—built for the shortgrass and sage of the late Pleistocene. Its bones turn up across the American West, from the La Brea tar pits in southern California to kill sites in the Great Basin to arroyos in the southern Plains. It ate mesquite and prickly pear. It browsed juniper. It moved in herds across a continent where horses, mammoths, ground sloths, American lions, and American cheetahs shared the same country. Then, about thirteen thousand years ago, it was gone.[1]

Most readers know this story, if they know it at all, in reverse. The camels Americans picture are Old World animals: dromedaries and Bactrians, the camels of the Silk Road and the Sahara, introduced to the Americas once, briefly, by the US Army Camel Corps in the 1850s, then failed out of service and absorbed into folklore. The ancestors of those Old World camels evolved on the North American continent and migrated out across the Bering land bridge millions of years ago. They diversified in Eurasia and Africa. They returned to their continent of origin only as imports. The camel's deep home is here. We have forgotten.

The forgetting is not personal. It is structural, and it is the reason I am writing this book.

The framework most of us have inherited for deciding what belongs on this continent begins in 1492. Animals present here before Columbus are "native." Animals introduced after Columbus are "non-native," "invasive," "exotic" — the vocabulary varies with the disposition of the speaker, but the baseline does not. The line is drawn at the date of European contact. What was here at that moment became the reference point against which all subsequent ecological change is measured.

It is an oddly narrow window. Camels had been absent from North America for roughly twelve and a half thousand years by the time Columbus made landfall. Horses had been absent for about the same span. Mammoths, mastodons, ground sloths, American cheetahs, American lions, giant beavers, dire wolves, and most of the continent's megafaunal assemblage were already gone. The continent the Spanish sailed into was not a pristine ecosystem frozen at its baseline—it was a depleted one, long depleted, more than half of its large-bodied fauna already missing.[2] The Columbian reference date captures the moment of European contact. It does not capture anything ecological about the land itself.

Paul S. Martin spent most of his career at the University of Arizona trying to put this observation into wider circulation. Martin was a paleoecologist who had worked on the Late Pleistocene extinctions for decades. By the time he published *Twilight of the Mammoths* in 2005, he had arrived at a position that is still, twenty years later, outside most popular conservation conversation. He proposed that the question "what belongs here" should be answered on a deeper timescale than 1492. He called the conservation paradigm that stops at Columbus "the Columbian Curtain." He proposed dropping it.[3]

On the other side of the Curtain is a different continent. Mammoths in the boreal north. Ground sloths the size of bears browsing the forest edges. American horses on the plains. American cheetahs in the canyons (which is why the pronghorn, the fastest land animal in North America, still runs at speeds no living predator requires; its legs are calibrated to a cat that no longer exists). And camels. *Camelops* across the basin and range. *Hemiauchenia*, a larger, stouter relative, through the woodlands. Smaller llamas along the southwestern edge. A continent arranged around a set of animal relationships that have, almost entirely, gone unreciprocated for a hundred and twenty centuries.

Martin was not sentimental about this. He was a scientist with the fossil record to work with. His argument was that ecological function does not care about the date on which a species disappeared. A prairie that evolved with proboscideans behaves differently in their absence than one that evolved without them. A predator-prey dynamic misses both parties when either one is removed. A fruit tree whose seeds evolved to be eaten by ground sloths is still waiting for the ground sloths. The ecosystem is a set of relationships, and most of the relationships on this continent in the late Pleistocene have been one-sided for a very long time.

If you grew up in the American Midwest or the mid-South, you have almost certainly met an osage orange tree. *Maclura pomifera*. It is common along fence rows and in old hedgerow plantings — it was planted massively in the nineteenth century as a thorny living fence before barbed wire made it obsolete — and its fruit is the size and approximate color of a softball, grainy-skinned, slightly fragrant, faintly sticky with a white latex when cut. To any modern animal of the continent, that fruit is essentially inedible. Horses will nose one and move on. Cattle will not eat them. Squirrels pick through the seeds if they can reach them, which is rarely. The fruit falls, rots in place, and stains the ground. A tree with a bumper crop looks like it is producing for no one.

Because, in a real sense, it is. In 1982, Daniel Janzen and Paul Martin published a paper in *Science* titled "Neotropical Anachronisms: The Fruits the Gomphotheres Ate."[4] The argument was straightforward; once you have read it, you cannot un-see it. Many of the large, sugary, otherwise-puzzling fruits of the Americas (avocados, pawpaws, honey locust pods, osage oranges, a long list of tropical species) carry the structural signatures of having co-evolved with large-bodied dispersers that are no longer there. A gomphothere, a mastodon-relative proboscidean that ranged widely

across North and Central America until the end of the Pleistocene, would have taken an osage orange whole, walked some miles, and deposited the seeds in a dung pile prepared for germination. Without the gomphothere, the tree continues to produce the fruit the tree evolved to produce. The fruit is a message to a mouth that no longer exists.

Janzen and Martin called these trees ecological anachronisms. The phrase stayed with me.

TRY THIS

The Osage Orange

Next time you see an osage orange fruit on the ground, pick it up. Feel how heavy it is. Look at it for tool marks. There will not be any. No mouse has chewed it. No deer has browsed it. No squirrel has cracked a single seed out. The fruit is, for the present fauna, inert.

Now picture the mouth the fruit was made for. A gomphothere stood roughly nine feet at the shoulder and weighed several tons. Its molars were the size of bricks. A fruit that looks large to you would have looked like a snack to it.

If you want a small exercise: count the osage oranges under one tree in late autumn. Note how many show any sign of being opened, by any animal, over the course of the winter. Most years, on most trees, the answer is zero. The tree produced a hundred-pound crop for a mouth that is not coming.

The data are not earth-shaking. But they are yours, and they are specific to a tree you can walk to.

An anachronism, strictly, is something out of its time—a character in the wrong century, a gesture from the wrong era. The osage orange is out of its time because its partner is out of its time. The tree is stranded in a present that has forgotten what it is for. Multiply this by every species with a large-fruit architecture, every grassland assemblage that evolved under the hooves and digestive tracts of animals weighing tons, every predator community whose prey has been gone for longer than there has been writing, and you have something like the actual ecological state of this continent. The landscape is not depleted in a vague or poetic way. It is missing specific animals. Their absence has specific consequences. The consequences have

compounded across twelve thousand years of quiet rearrangement that most of us have not been taught to notice.

This is what the Columbian Curtain hides. Drop the baseline back to the end of the Pleistocene and the continent you are standing on is not recovering from European contact; it is recovering from something much older, and most of that recovery has not yet begun.

I should say that I have held various positions on this material over the years. I first read Martin in the late 2000s, and my initial reading left me convinced of a simpler story than I now think the evidence supports. For a long time I believed that the extinctions had been driven primarily by humans arriving across the Bering land bridge (the "overkill hypothesis," which Martin himself had done much to establish in his earlier career) and that the case for rewilding followed from a kind of collective human debt. Later work on rapid climate oscillation during the Younger Dryas, on the paleoclimate record more broadly, and on the difficulty of cleanly dating the extinctions against human arrival, has made me less confident. The question of cause is not settled, and I no longer think the rewilding argument depends on its resolution. Chapter 3 returns to this in the detail it deserves. The prologue does not need to.

What I want to put on the table first, before any framework or argument, is the shape of the absence itself. The animals are not here. The fruit is still being made. The grasslands still stand in the assemblages that evolved around grazers that are gone. The pronghorn still runs from a cheetah. The continent is a library that has lost most of its librarians, and it is somehow still trying to operate.

I would like, somewhere in my life, to see one hundred camels in an osage orange orchard.

This is plainly said because there is no more elegant way to say it. I do not mean it as policy, or as provocation, or as a headline. I mean it as a wish—a specific one, with specific animals, in a specific arrangement. A hundred camels because that is enough for a small, real herd; large enough to be a population and small enough to manage honestly. An osage orange orchard because I would like to know what those trees look like when their fruit has a mouth. Camels are not gomphotheres, but they are closer in body plan, gut size, and browsing habit than anything now available on the continent, and they are already here—feral in parts of the American West, stabled on

hobby ranches elsewhere, just not in the ecological role they once had. I would like to know whether the fruit gets eaten. I would like to know whether the seeds, passed through a large mammalian gut and deposited with fertilizer, germinate at higher rates than the current state of abandonment allows. I would like to know whether the ground around the trees begins, even slightly, to resemble something other than rot.

I am not sure anyone would fund the experiment. I am not sure the permits would be approved, or the ecological risk assessments would come out favorably, or the neighbors would agree. I know what serious critics of Pleistocene rewilding would say about it. I have read their arguments, and the book takes them seriously in the chapters where they belong. The wish is not a program. It is the thing the program would be for.

This book is, for the most part, an analytical argument about how to evaluate food systems. It proposes a framework, applies it, and reaches conclusions I think are defensible. But the analytical case is not what drives it. What drives it is this: the orchard, the camels, the fruit with a mouth. The sense that we have been living for twelve thousand years on a continent whose ecological vocabulary has been spoken at half-volume, and that there is no reason to treat that half-silence as a permanent condition.

Paul Martin spent his career trying to make the silence legible. I am writing this book, more than anything else, to return his argument to a conversation that has mostly moved on without it.

The framework comes later. The camels come first.

CHAPTER ONE

The Meat Counter

♦ ♦ ♦

In September of 2009 I was standing at the meat counter of a major natural-foods retailer in Overland Park, Kansas, with my mother, scanning the selection and trying to make sense of what I was looking at.

A note on naming. I am keeping the retailer anonymous throughout this account. The events are exactly as they happened. Naming the chain in print invites a category of legal dispute the book does not need, and the chain is recognizable enough from the geography and the era that anyone who wants to know which one will know. The choice to keep it anonymous here is to keep this book out of court.

Overland Park is a suburb of Kansas City, the kind of sprawling Johnson County development where the grocery stores are large and clean and lit like operating theaters. The store was relatively new to the area. It occupied the spot in the local retail ecology that a retailer of that scale always occupies: the store where the prices are higher and the assumption, embedded in the brand, is that the higher prices are buying you something the other stores cannot guarantee. I was there because I believed that assumption. I had reason to.

I had been sick for the better part of a year. By September I was on the other side of the worst of it, eating carefully, paying attention to what I put in my body in a way I had never done before. The wrong food could set me back weeks. The right food (clean protein, known sourcing, no hidden grains) was not a preference but a medical necessity, and I was standing in the meat section of a store I trusted because the store's entire brand proposition was that it had already done the vetting for me. The grass-fed beef was what I had come for. The label said grass-fed. The price reflected

grass-fed. The display case was arranged with the quiet confidence of a store that believed it was selling you exactly what you thought you were buying.

I picked up a package, turned it over, and asked the person behind the counter a question that would not have occurred to me six months earlier. "Is this a hundred percent grass-fed?"

He said yes.

I asked again — politely, but with the specificity that comes from having spent months learning that the wrong food could put me back in a condition I did not want to revisit. "A hundred percent? Grass the whole way through?"

He paused. "It's grass-fed," he said, "but it's grain-finished for the last three months."

My mother and I looked at each other. She said what I was thinking before I could say it: "If they're feeding them grain to fatten them up, how can you call it grass-fed?"

The man behind the counter was not being evasive. He was, as far as I could tell, repeating what he had been told. The beef was marketed as grass-fed. It had been raised on grass for most of its life. It had then been transitioned to grain for the final ninety days before slaughter (a practice called grain finishing, common in the beef industry, which puts weight on the animal quickly and produces the marbling American consumers have been trained to prefer). The man behind the counter was not the person who had decided to call this product grass-fed. He was the person standing between the decision and the customer, doing his job, and his job did not include explaining the regulatory architecture that made the label possible.

I did not buy the beef. I stood there for another minute, looking at the case — the chicken, the turkey, the pork, the lamb, all of it arranged under labels I had been reading as promises — and something shifted. Not dramatically. Not with any sense of revelation. Quietly, the way a crack appears in a wall you have been looking at for a long time without seeing it. The label said one thing. The practice was another. And if this label, on this product, in this store (the store whose entire value proposition rested on the reliability of its sourcing claims), could mean something other than what it appeared to mean, then what else in the case was I reading wrong?

The gap between the label and the practice was not an accident or an error. It was structural. I did not yet have that word for it. I only had the feeling, which was the

feeling of having trusted a system that had not earned the trust, without knowing that the trust was unearned until I asked one specific question on one specific afternoon.

FIELD NOTE

What Actually Happened at the Meat Counter

The version of the scene above is the cleaned one. The actual scene was louder. This is the dirty version.

I had been sick for over a year. By the end of 2009, at twenty-five, I had been diagnosed with insulin-resistant PCOS, PMDD, schizophrenia, grade-three anemia, cirrhosis, a kidney mass, GERD, peripheral neuropathy, optic neuritis, bacterial vaginosis, menopause, ascites with chronic edema, endometriosis, fibromyalgia, and finally, celiac disease. I had gained fifty pounds in three months. The anemia was severe enough that the physician treating me had already said I needed someone else to drive me around. Despite following the red-meat diet he suggested, I was still deteriorating.

That day at the store, I had already been crying. I had been grocery shopping for over half an hour, and the only thing I had added to the basket was apples. They were the only thing I could eat at the time without adverse effects, and I avoided grain-finishing because it had previously caused digestive issues. That is why I was so intent on getting grass-finished beef. I needed it. This was a dire situation. I was emotionally exhausted.

The man behind the counter had just told me that what I had been eating was grain-finished, which meant it had been making me sicker for months, which meant I had been paying premium prices to be poisoned more politely. There were expletives. There were a lot of them. Ten out of ten completely unhinged.

Security was called. My mother had to drag me out, and she did, and I have never loved her more. There were expletives all the way to the car. Then more crying.

I later found out that the incident traveled. It went up the chain to corporate headquarters, and there was an emergency meeting about what employees could and could not say about the grass-fed claim. For a while I was famous at that Overland Park store, in a way I did not want to be famous and have no reason to be quiet about now. That store is gone now, too.

I am not in the slightest bit embarrassed of 2009 Karen either. She was sick, she had finally figured out that the system she was paying premium prices to trust was not telling her the truth, and she said so, loudly, in the kind of place where loudly is not how you are supposed to say it.

FIELD NOTE

What I Did Next

I should say what I did next, because it is part of the same thing.

The version of me who got dragged out of that store went home and started writing standards. Actual certification standards, the kind producers have to meet to put a label on a package. Paleo Approved came first, in 2010. It was an animal standard, and a strict one. It was also a flop. Ranchers did not yet understand the importance of it, and the Paleo community itself was still embryonic and not creating the demand. I was too early.

Paleo as a popular movement did not really take off until 2013 or 2014, and by then we had abandoned ship to move toward where the demand actually was, which was in consumer packaged goods.

What followed, under the Paleo Foundation, was Certified Paleo. Although I initially felt like a sellout for abandoning farm and ranch certifications, the most consequential ecological work I have done in my life is bound by non-disclosure agreements I signed willingly and intend to honor.

In other words, the thing I am proudest of in that entire arc happened under the Certified Paleo label, not Paleo Approved.

I am telling you this now because the Field Note above ends with me losing it at a meat counter. The version of me writing this book has a clearer view of the gap between what is on a label and what is in the package, in part because I have spent sixteen years writing the standards for the labels — and because there are things I wish I had done better.

◆ WHAT "GRASS-FED" MEANT ◆

At the time I was standing in that store, the term "grass-fed" on beef packaging in the United States was governed by a voluntary marketing claim standard administered by the USDA's Agricultural Marketing Service. The standard, finalized in 2007, defined

grass-fed as meaning that the animal's diet had been derived solely from forage (grass, legumes, hay, silage) with the exception of milk consumed prior to weaning.[1] Under this definition, grain finishing would have disqualified the beef from a grass-fed claim. The standard existed. On paper, it was clear.

In practice, the standard was voluntary, and its enforcement relied on a verification process that most producers did not undergo. The USDA did not inspect every operation making a grass-fed claim. It certified those that applied for certification and met the documentation requirements; those that did not apply were subject to the same labeling regulations that governed all meat products, which is to say, subject to review only if a complaint was filed or an audit was triggered. The gap between the standard's existence and its application was wide enough to drive a grain truck through. A retailer could sell beef labeled "grass-fed" that had been grain-finished, and unless someone with standing complained to the right agency, the label stood.[2]

The standard was later withdrawn entirely. In January 2016, the USDA's Agricultural Marketing Service revoked its grass-fed marketing claim standard, citing concerns about the verification process and the scope of the standard's applicability.[3] After the withdrawal, the term "grass-fed" on beef packaging became, in regulatory terms, essentially unregulated at the federal level: any producer could use it, under any definition they chose, provided they did not make a claim that was demonstrably false under the Federal Trade Commission's general deceptive-practices authority. The practical effect was that "grass-fed" became a term whose meaning was determined by whoever was printing the label.

I did not know any of this in 2009. I was a consumer standing in a store, reading a label, and trusting the label to mean what I thought it meant. What I thought it meant was that the animal had eaten grass. What the label actually certified — to the extent it certified anything — was something narrower, more hedged, and more dependent on the specific practices of a specific producer than the two words on the package could convey. The gap between my assumption and the reality was not a gap I had been invited to notice. It was a gap the label was designed to bridge without my having to look down.

◆ THE PATTERN ◆

I might have treated the meat-counter moment as an isolated problem — a single misleading label on a single product in a single store — if the same pattern had not kept appearing.

Consider "free-range," as applied to poultry. The term appears on egg cartons and chicken packaging across the country, and its presence reliably commands a premium. Consumers pay more for free-range eggs than for conventional eggs, often significantly more, on the reasonable assumption that the word describes a meaningfully different life for the bird that produced them.

The USDA's Food Safety and Inspection Service defines free-range, for the purpose of poultry labeling, as meaning that the birds have been "allowed access to the outside."[4] The definition does not specify the size of the outdoor area. It does not specify the duration of access. It does not specify the quality of the outdoor space: whether it is grass or concrete or bare dirt. It does not specify whether the birds actually go outside, only that a door exists through which they could, if they chose to, if they could reach it, if they knew it was there. A producer can raise ten thousand birds in a single enclosed structure, open a small door at one end of the structure for some portion of the birds' lives, and label the resulting product free-range. The birds nearest the door may occasionally step through it. The birds at the far end of the structure — hundreds of feet from the opening, in a flock so dense that individual movement through the crowd is physically constrained — may never see daylight. The label does not distinguish between these birds. It certifies the existence of a door.

Most consumers, asked what "free-range" means, will describe something that looks like a farm: birds on grass, moving freely, doing the things birds do when they have room and sunlight and ground to scratch. The mental image is not irrational. It is the image the language invites. "Free-range" is a compound of two words that each carry spatial meaning: freedom and range, openness and distance. The phrase produces a picture. The regulatory definition behind the phrase produces a door.

I came to understand this only gradually, in the months after the meat counter, as I began reading labeling regulations with the attention I had previously reserved for ingredient lists. The gap between the picture and the door is not a failure of consumer education. It is not a case of consumers being too lazy or too credulous to read the fine print. There is, in most cases, no fine print to read. The label says "free-

range." The regulation defines it. The consumer encounters only the label. The regulation sits in the *Federal Register*, available to anyone who knows where to look, written in the language regulations are written in, useful to anyone with the time and the training to parse it. The system is not hiding the definition. It is relying on the distance between the definition and the consumer to do the work that active deception would do if the system were less sophisticated.

"Pasture-raised" is a variation on the same theme. The term has no federal regulatory definition at all. It appears on packaging because producers choose to put it there, and its meaning is whatever the producer's own practices happen to be. Some operations using the "pasture-raised" label run birds on genuinely open pasture in mobile shelters, moved daily, with real access to grass and insects and sunlight. Others use it more loosely. The consumer cannot tell the difference from the package. Third-party certifications (Certified Humane, Animal Welfare Approved, and others) have stepped into the regulatory gap with their own standards for what "pasture-raised" means, and some of those standards are rigorous. But the label itself, the two words on the carton, certifies nothing in particular. It is a description without a definition, a promise without a guarantor.

Or consider "organic," which is the most comprehensively regulated of the major food labels and still produces a version of the same gap. The USDA's National Organic Program, established under the Organic Foods Production Act of 1990 and administered through a body of regulations codified at 7 CFR Part 205, defines organic production as a system that integrates cultural, biological, and mechanical practices to foster cycling of resources, promote ecological balance, and conserve biodiversity.[5] The standards are real, detailed, and enforced through a third-party certification process that is substantially more rigorous than anything governing "grass-fed" or "free-range." An organic label on a food product means something. It means the producer has been inspected, has met specific input and practice requirements, and has documentation to support the claim.

What the organic label does not mean, and has never meant, is what many consumers take it to mean. Organic certification does not certify that the food was produced locally. It does not certify that the animals were raised on pasture. It does not certify that the soil is being built rather than depleted, only that certain prohibited inputs (most synthetic fertilizers, most synthetic pesticides, most antibiotics, genetically engineered organisms) are absent from the production

process. A large-scale organic operation can look, structurally, a great deal like a conventional industrial operation: confined animal housing, monoculture crop rotations, long-distance commodity distribution, significant fossil-fuel inputs for field work and transportation. The organic label certifies what is not being used. It does not certify how the system as a whole is functioning.

This is not a criticism of organic certification. The organic standard was designed to do what it does, and what it does is useful: it removes a specific set of chemical inputs from the production process and provides a verification mechanism for that removal. The problem is not that the standard fails at its own task. The problem is that the label, as encountered by a consumer in a grocery aisle, communicates something substantially broader than the standard certifies. The word "organic" carries associations (with naturalness, with ecological care, with a certain quality of attention to the land) that the certification process was not designed to guarantee and does not guarantee. The consumer's assumption is reasonable. The label's scope is narrower than the assumption. The gap is the same gap I encountered at the meat counter, wearing different clothes.

CASE

Three Labels, Three Definitions, One Gap

The three labels this chapter has walked through carry three different relationships to their definitions. "Grass-fed," under the 2007 USDA marketing-claim standard, had a precise definition that was voluntary and effectively unenforced; the standard was withdrawn entirely in 2016, after which any producer could use the term under any definition. "Free-range," under FSIS poultry labeling, requires only that birds have "access to the outside," which in practice is a door. "Organic," under the 1990 Organic Foods Production Act and its 2002 implementing regulations, carries the most rigorous definition of the three, verifiably prohibiting specific synthetic inputs through third-party certification.

Three different labels, three different definitional rigors, three different gaps between what the consumer assumes and what the label certifies. The gap is not uniform across labels. It is structural across the labeling system.

◆ THE ARCHITECTURE OF THE GAP ◆

Standing at that meat counter that September, I did not have the vocabulary for what I was noticing. I had a feeling, which was the feeling of having been misled without anyone having lied to me, and I had a question, which was how a system this large and this well-funded could consistently produce labels that communicated less than consumers believed they were being told.

The answer, which I arrived at slowly over the years that followed, is that the labels are not failing. They are succeeding at a task different from the one consumers assume they are performing.

A food label in the United States is, in regulatory terms, a marketing claim. It exists at the intersection of the producer's desire to differentiate a product, the retailer's desire to command a price premium, and the regulatory apparatus's desire to prevent outright fraud without unduly burdening commerce. The label's job, within this intersection, is to communicate *enough* to support the marketing claim while remaining narrow enough to survive regulatory scrutiny. "Grass-fed" communicates pastoral images. "Free-range" communicates open space. "Organic" communicates naturalness. Each term does this reliably, for the majority of consumers, and each term is defensible under the regulatory definition that governs it. The system works. It works for the producer, who can differentiate a product. It works for the retailer, who can charge a premium. It works for the regulator, who can point to a definition. The only party for whom it does not quite work is the consumer who assumed the label was telling them something about the *system* their food came from, rather than about a single certified attribute of that system.

This is, I think, the structural observation the meat counter moment eventually led me to. The food labeling system is not broken. It is designed to communicate less than consumers want to know. Each label certifies one thing — one attribute, one absence, one practice verified at one point in the production chain. Consumers want to know many things — how the animal lived, what it ate, how the land was managed, what inputs were used, what the operation's relationship to its surrounding ecology looks like, what would happen to the system if one of its dependencies were removed. No single label, even a good one, can answer all of those questions. The labels were not designed to answer all of those questions. They were

designed to answer one question each, narrowly, defensibly, in language that happens to evoke a broader picture than the narrow answer supports.

The gap between the narrow answer and the broad picture is not a conspiracy. It is an architecture. It is the predictable result of a system in which labeling is designed to facilitate market differentiation rather than to provide comprehensive transparency about production practices. Producers are not villains for using the labels as designed. Retailers are not villains for stocking labeled products. Regulators are not villains for writing narrow definitions. Consumers are not fools for reading the labels the way the labels invite themselves to be read. Everyone in the system is doing what the system was designed to let them do. The system itself is the problem, and the problem is not that it fails but that it succeeds at something less than what most of us need it to do.

The architecture has a further consequence worth naming, because it bears on everything the rest of this book discusses. A label that certifies a single attribute can, by its presence, create the impression that the system behind the product has been evaluated comprehensively. A package of chicken bearing an organic label and a free-range label and a "no antibiotics ever" label looks, to the consumer scanning the case, like a product that has been thoroughly vetted. Three labels. Three assurances. Surely the system has been looked at from multiple angles. In fact, what has happened is that three narrow attributes have each been verified independently, with no requirement that anyone has ever looked at the operation as a system: as a whole set of practices, dependencies, inputs, outputs, and vulnerabilities functioning together. The labels are point measurements. The consumer needs a map. The multiplication of labels can actually make the map harder to read, because it produces the feeling of comprehensiveness without the substance of it.

◆ WHAT I DID NOT KNOW YET ◆

I want to be careful, here, about what I am not claiming. I am not claiming that I understood any of this at the meat counter. I understood almost none of it. I understood that the label had misled me, and I understood that I was angry, and I understood that the anger was not usefully directed at the person behind the counter, who was doing his job and had answered my question honestly when I

pressed him on it. Beyond that, I was a sick person in a grocery store who had just learned that the food she trusted was not what she thought it was.

What happened next took years, not minutes. I went home. I started reading. I read about labeling standards. I read about certification processes. I read about USDA definitions and the regulatory comment process and the difference between a mandatory standard and a voluntary marketing claim. I read about organic certification's scope and limits. I read about the word "natural," which has no regulatory definition at all on most food products and functions as pure marketing language attached to nothing.[6] I read, eventually, about the deeper questions underneath the labeling problem: about food systems, about sustainability, about what it means to ask whether the way we produce food can continue to work under conditions different from the ones we are used to.

The reading took me to Paul Martin. It took me to Allan Savory and Joel Salatin and the regenerative agriculture literature. It took me to the paleontological record, to the Columbian Curtain, to the near-time extinctions, to the question of what this continent looked like before most of its large-bodied fauna disappeared. It took me to questions I could not have anticipated from a meat counter in suburban Kansas: questions about timescale, about ecological baselines, about what it means for a food system to survive under conditions different from the ones it was designed for.

It took me, eventually, to the framework this book proposes (a framework I did not have in 2009, did not begin to develop until several years later, and did not fully articulate until the middle of the last decade). The path between the meat counter and the framework was not a straight line. It included a number of positions I have since revised, a number of arguments I made with more confidence than the evidence warranted, and a number of assumptions I held longer than I should have. The authorial revisions that have shaped this book (about dietary universality, about the causes of the Pleistocene extinctions, about the tone in which disagreement should be conducted) are part of the trajectory. The meat counter was not the moment I found the answer. It was the moment the question became unavoidable. The distance between the question and whatever answer this book offers has been covered on foot, over sixteen years, with wrong turns.

FIELD NOTE

What I Wrote Down

I have my journals from 2009. I did not start writing seriously until later, but there is a page from that fall where I wrote, in capital letters, *labels are lies*. That was the first thing I wrote down.

It was wrong in the specific sense. The labels on the meat I was looking at were not legally false. But it captured what I had felt at the counter, which was the feeling of having been told something technically true and operationally misleading. I underlined it. I did not yet know that this was a decades-old problem in food policy, or that the regulatory architecture had been engineered to produce exactly that feeling without producing a prosecutable deception.

I was twenty-five. I thought I had discovered something. What I had discovered was a question a lot of other people had already asked. But it took sixteen years to get from the question to the framework, and I do not regret the underlined sentence. It was the beginning of where I am now.

I am noting this because the trajectory matters to the book's honesty. The temptation, in a chapter like this one, is to present the early moment of recognition as though it contained the later understanding, as though I stood at the meat counter and saw, fully formed, the structural inadequacy of the labeling system and the deeper questions it pointed toward. That is not what happened. What happened was small and specific: a label that did not mean what I thought it meant, a feeling of having been misled, and a question that did not resolve itself for years. The understanding came later, incrementally, through reading and experience and — as the revisions that have shaped this book will show — through being wrong about a number of things along the way. The meat counter was the beginning of a question, not the arrival at an answer.

◆ WHAT LABELS CANNOT TELL US ◆

The labeling problem is, in the end, a problem about information: about what the consumer is given and what the consumer would need. The gap between the two is not a gap that better labels can fully close, because the questions a thoughtful

consumer wants answered about a food system are not questions that fit on a package.

A label can tell you that an animal was fed grass. It cannot tell you whether the grassland the animal grazed on is building soil or losing it (whether the land is gaining organic matter year over year or slowly losing the biological capital that makes it productive). It cannot tell you whether the operation depends on purchased inputs from industrial commodity markets for parts of its production cycle, or whether it is genuinely self-sustaining on its own acreage. It cannot tell you whether the operation could survive the loss of a single input (a subsidy, a commodity feed source, a supply chain) or whether it would collapse. It cannot tell you what the land looked like before the operation was there, or what it will look like after. It cannot tell you whether the system the food comes from is one that could continue to function under conditions meaningfully different from today's, or whether it is riding on a set of assumptions about stable inputs, stable climate, and stable policy that may or may not hold.

These are not unreasonable questions. They are, I would argue, the questions that matter most: not just for the consumer deciding what to buy on a Tuesday evening, but for the society deciding what kinds of food production to support, to subsidize, to build infrastructure around, to invest in for the next century. The questions are about systems, not about attributes. They are about how the whole thing works, not about whether one variable meets one standard. And the labeling system, by design, answers attribute questions. It was built for attributes. It does attributes well. It does not do systems at all.

There is a reason the labeling system was built this way, and the reason is not sinister. Attributes are verifiable. You can inspect an operation and determine whether the birds have access to the outdoors, whether synthetic pesticides were applied, whether antibiotics were administered. These are binary or near-binary questions, and a regulatory system can answer binary questions through documentation and inspection. Systems-level questions — is this operation ecologically sustainable, is it resilient to disruption, could it continue to function under changed conditions — are not binary. They require frameworks of evaluation that the regulatory apparatus was not designed to perform and that the labeling system was never intended to communicate. Asking a food label to convey systems-

level information is asking it to do something it was not built to do. The question is not why the labels fail at this. The question is what else we have.

This is where the meat counter moment, small as it was, connects to the larger argument the book is about to make. The label's inadequacy is not a labeling problem in the narrow sense. It is not a problem that can be solved by writing better definitions for "grass-fed" or tightening the enforcement of "free-range" or adding more labels to packages that are already covered in them. It is a problem about the kind of information we have access to when we make decisions about food, and the kind of information we would need. The gap between the two is where most of the interesting questions in food-systems thinking live. It is the gap the rest of this book is written to address.

I do not have a better label to offer. I want to be clear about that, because this chapter could easily be read as a setup for a pitch — here is what is wrong with existing labels, and here is the label that fixes it. That is not where this book goes. The problem is not that the right label has not been invented. The problem is that labels, as a category of communication, cannot carry the weight of what consumers need to know about the systems their food comes from. The solution, if there is one, is not a better label but a better framework: a way of evaluating food systems that addresses the questions labels were never designed to ask.

The prologue planted an image: camels in an osage orange orchard, a continent missing its animals, a longing for restoration. This chapter has planted a different kind of image: a woman at a meat counter, reading a label, discovering that the label is not enough. Both images are starting points. The prologue's starting point is emotional — the feeling that something is missing from the landscape. This chapter's starting point is practical — the feeling that something is missing from the information. The two feelings converge on the same set of questions. If the labels cannot tell us what we need to know about our food systems, what can? If the dominant framework for evaluating those systems is measuring the wrong things, or measuring the right things too narrowly, what would a more adequate framework look like? And if such a framework exists, what does it reveal about the systems we rely on — and about the ones we have not yet tried?

The next chapter begins the work of answering those questions, by going back much further than a meat counter in Kansas, to the moment an entire continent's ecological vocabulary was quietly, catastrophically reduced.

I am going to give the answer this book ends in away here, because I would rather you read for the argument than for the suspense. One of the food systems that scores best on all four axes — including the one I will argue most twentieth-century work has been missing — is one we lost twelve and a half thousand years ago, and could choose to bring back. The chapters between here and there are the case for taking that claim seriously. The reader who finds it ridiculous now is the reader I am writing for; the reader who finds it ridiculous at the end is the reader I have failed to convince. I am willing to be told I have failed. I am not willing to write the book without making the case.

CHAPTER TWO

The Columbian Curtain

♦ ♦ ♦

From Daniel Pauly,
Trends in Ecology & Evolution (1995)

Each generation of fisheries scientists tended to take the species composition and abundance of the oceans at the start of their own careers as the reference condition against which subsequent change was measured.

The cumulative effect, across several generations, was that the "natural" state of the oceans slid progressively further from anything the oceans had been before industrial fishing began.

— paraphrase of Pauly, "Anecdotes and the shifting baseline syndrome of fisheries"

The story most Americans carry about the horse is this: horses came from Spain. The Spanish brought them in the early 1500s, and before that there were no horses on the continent. This is, as far as it goes, true. It is also among the most misleading true things in the country's ecological imagination, because the sentence after it almost always gets dropped.

The sentence after it is that horses evolved here.

The genus *Equus* originated in North America roughly four million years ago and diversified on this continent for most of its evolutionary history.[1] Horses walked out across the Bering land bridge, populated Eurasia and Africa, and continued to evolve there, eventually producing the animals that would be domesticated on the Eurasian

steppe some six thousand years ago. Meanwhile, the horses that had stayed in their continent of origin went on being American horses until the end of the Pleistocene, when all of them disappeared in the same wave of extinctions that took the camels and the mammoths.[2]

DEFINITION

Equus and the deep equid record

The genus *Equus* — the lineage that includes every living horse, donkey, and zebra — is the crown of a much longer equid story. Before *Equus* the continent already had a succession of horse-family genera: *Dinohippus*, its likely direct ancestor, already essentially modern in body plan and single-toed; before that, the three-toed *Pliohippus*; before that, the grazing and browsing radiation of *Merychippus* and its contemporaries; and before those, the small, multi-toed browsers of the early Miocene. The equid family has been running on this continent for close to twenty million years. *Equus* is the last four million of that twenty. When paleontologists say horses evolved in North America, they are not saying one species showed up here, stayed a while, and left. They are saying the entire deep history of the family played out on this continent, and the Spanish ships of the sixteenth century were returning the crown of a tree whose roots had been American the whole time.

CASE

The American horses by name

The late-Pleistocene equid fauna of North America went by several specific names, and the specifics matter if only to show that this was not one horse but many. *Equus scotti* was large, stout-bodied, and widely distributed across the southern and central portions of the continent; its body mass places it comfortably within the range of modern domestic horses. *Equus conversidens* was smaller and more gracile, centered on what is now Mexico and the American Southwest. *Equus lambei*, the Yukon horse, persisted in the Alaskan and Yukon refugia; the Thistle Creek genome that confirmed the deep relationship between late-Pleistocene American horses and modern *Equus caballus* came from this population. Other named forms — *Equus francisci*, *Equus complicatus*, *Equus occidentalis* — fill out the record with varying degrees of preservation. The point of the list is not taxonomic tidiness. American horses were a diversified fauna

occupying a range of habitats and body sizes, and the wave of terminal-Pleistocene extinctions took the whole set.

DEFINITION

Camelops and Its Kin

Camelidae is the family. Its living members include the dromedary and the Bactrian of Asia and Africa, *Camelus dromedarius* and *Camelus bactrianus*, and four South American species: the llama, the alpaca, the vicuña, and the guanaco. Llamas are camels.

The American Pleistocene fauna held two named camelid genera that are no longer with us. *Camelops hesternus*, the dominant North American camel of the late Pleistocene, ranged from the Yukon to central Mexico and is the species the rewilding proposal is most directly about. *Hemiauchenia macrocephala*, the large-headed llama, occupied the same continental range and is preserved at La Brea, alongside *Camelops*, in the asphalt fossil record of southern California. Both were extinct by the terminal Pleistocene pulse around twelve thousand years ago. Recent paleogenomic work places *Camelops* as sister to the Old World dromedary and Bactrian lineage, not to the South American camelids morphology had previously suggested.

What is known about *Camelops* from the La Brea record and from skeletal material elsewhere on the continent: roughly seven feet at the shoulder, a foot taller than a modern dromedary, longer-necked, with leg proportions consistent with a sustained running speed in the neighborhood of forty miles per hour. Tooth wear and microwear analysis identifies it as a browser, feeding on shoots, leaves, and the woody growth its long neck would have placed within reach.

Whether *Camelops* carried the fatty hump or humps of its living Old World relatives is unresolved. Soft tissue does not preserve in the fossil record, and the vertebral column shows no anchoring features that would force the answer either way.

A pedantic but important note. *Camelops* itself is extinct. The rewilding hypothesis this book ends in does not propose its resurrection. What it proposes is the introduction of modern camelids — dromedaries, Bactrians, or, depending on the landscape, llamas, alpacas, vicuñas, or guanacos — into a continent that is the family's ancestral home. Heintzman 2015 and adjacent paleogenomic work establishes that *Camelops* sat within the Camelini tribe alongside the modern Old World camels, not within the South American camelid lineage. The animal that

walked North America for most of the existence of the camelid family is gone. Its closest living relatives are the dromedary and the Bactrian. The proposal is to put a related camelid back into the niche, not to put *Camelops* back. The book's argument requires the niche to be real and the family to belong here. It does not require resurrection biology.

What is known, however, with certainty, is that *Camelops hesternus* is as American as apple pie.

Genetic work in the last two decades has tightened the story. In 2013 a team led by Ludovic Orlando reported a complete nuclear genome recovered from a horse bone frozen in permafrost at Thistle Creek in the Canadian Yukon. The bone is dated to the middle Pleistocene, around seven hundred thousand years old. The genome clustered within the diversity of modern *Equus*, confirming that the domestic horse and the Late Pleistocene American horse are close members of a single long-running lineage — the same animal, in a deep sense, with its genetic lights never quite going out.[3] Earlier mitochondrial work by Jaco Weinstock and colleagues had reached a similar conclusion from Alaskan and Yukon Pleistocene material: North American late-Pleistocene horses fell within, not outside, the modern *Equus caballus* / *Equus ferus* group.[4] The phylogeny does not particularly care which ships crossed which ocean. A horse on the plains in 1600 is the same kind of animal as a horse on the plains in 15,000 BP. The gap between the two dates is the question.

The same story holds, with different details, for *Camelops*. In 2015, Peter Heintzman and colleagues recovered paleogenomic data — mitochondrial genomes and partial nuclear genomes — from three western-camel specimens excavated from interglacial deposits along Hunker Creek in the Yukon, dated by paleoecological context to roughly seventy-five to one hundred thirty thousand years ago. The phylogenetic placement was unequivocal in both data sets: *Camelops* is sister to the Old World *Camelus* lineage, the dromedary and the Bactrian, and not to the South American camelids of *Lama* and *Vicugna* as morphology had previously suggested. Divergence between the *Camelops* and *Camelus* lineages dates to the Late Miocene, around seven and a half to six and a half million years ago.[16] What this means for the rewilding argument is direct. The dromedary and the Bactrian are not merely large camelids of convenience; they are the closest living relatives of the animal the American landscape lost.

The gap is also not, by any reasonable ecological measure, a long one. Twelve and a half thousand years sounds like a great deal in the scale of a human life. It is not very much in the scale over which grassland plant communities, ungulate digestive physiology, or soil microbial assemblages evolve. The grasses of the central plains spent tens of millions of years evolving in the presence of large-bodied grazers of the horse's general architecture. Their defenses, their growth habits, their response to compaction and disturbance are calibrated to those animals and to their relatives. Thirteen thousand years of relative quiet on that front is not enough time for those calibrations to disappear. When the Spanish horse arrives on the American plains, it does not enter an ecosystem naive to its body plan. It enters one that has been waiting.

That gap is what the standard story hides. Told as "the Spanish brought horses," the sentence puts the entire evolutionary record of the genus outside the frame. Told as "horses returned to their continent of origin with the Spanish, after an absence of roughly thirteen thousand years," the sentence includes the record, and the ecological meaning of the animals' arrival changes. They are not an introduction from elsewhere. They are a homecoming that took the scenic route.

This is the conceptual move the chapter is about. Nothing in the fossil record changes whether we call the horse native or non-native. What changes is our choice of when to start counting.

◆ THE BASELINE IS A CHOICE ◆

The dominant framework of American conservation treats the year 1492 as the effective ecological baseline for the continent. Animals present here when Columbus made landfall are categorized as native. Animals not present are categorized as non-native, invasive, exotic, or feral, depending on the disposition of the speaker and the legal category in which the animal has been slotted. The vocabulary varies; the baseline does not. Under this framework, the horse that returns with the Spanish is non-native. The burro that escapes into the Grand Canyon watershed in the nineteenth century is feral. The cattle that replace bison across the western grasslands are livestock rather than wildlife, even where they live in effectively free-ranging herds. Each classification depends not on the animal's relationship to the land but on

whether its species was on the continent on a particular October morning in the late fifteenth century.

Paul S. Martin, who spent most of his career as a paleoecologist at the University of Arizona, named this framing the Columbian Curtain. The prologue has introduced it. I want us to take the name more seriously here than the prologue had room to. A curtain is the right image. It is not a hard wall. It is not a border or a boundary in any physical sense. It is a visual instrument — a piece of fabric hung at a particular point in time, with everything behind it obscured from view. The observer standing in front of the curtain treats whatever is in front of it as the whole of the room. Whatever is behind it is out of sight and, therefore, out of the conversation.

Martin's argument was not that the curtain is wrong in the sense that 1492 is a fake date. European contact is a real event, and it does mark one of the largest biotic exchanges in the planet's recent history. His argument was that the curtain is arbitrary. There is nothing about the ecological state of North America in 1492 that makes it the correct reference point against which to measure restoration. The continent Columbus encountered had been losing large-bodied fauna for twelve and a half thousand years by the time he arrived.[5] More than forty species of large mammals were already absent from North America before a single Spanish sail appeared on the horizon. Calling that assemblage "the native condition" is a bit like calling a library "complete" on the strength of its remaining shelves, after most of its collection has been carted off.

Granted, every historical science has to choose a reference point somewhere. Paleontology works in sliding windows and does not pretend they have natural edges. The choice of 1492 as the baseline for "native" was not a deliberate scientific decision so much as a cultural inheritance. The framing came downstream from the history we were telling ourselves about the continent. The arrival of Europeans was the episode after which record-keeping, in the form that came to count, began. The animals present at that moment became the reference animals, and the ones that had been here earlier dropped out of the frame along with everything else the record did not carry forward. The baseline is a cultural artifact dressed as an ecological one.

Notice what the baseline does once it is fixed. A horse that shows up in the sixteenth century is, under this frame, an introduction. A horse that was here in the fourteenth millennium BP is not an ancestor to which the arriving animal is

returning; it is an unrelated animal in an unrelated world. A burro roaming the Colorado Plateau is not occupying an ecological niche left behind by extinct American equids; it is a non-native species degrading a native landscape. The entire vocabulary of invasion biology, as applied to these animals, assumes the 1492 baseline. Change the baseline and you do not change whether the animal is doing what it is doing. You change what you call it.

The Bureau of Land Management currently manages roughly eighty-three thousand free-roaming horses and burros across ten western states, at levels its own appropriate-management-level estimates describe as well above the range at which the agency believes the rangeland can be sustainably grazed.[6] The resulting program involves gathers, long-term holding facilities, fertility-control trials, and a political conflict that has continued for decades without much resolution. I am not here to resolve the management dispute. I want only to point out that the argument, as currently constructed, rests on a baseline choice. A horse on BLM land is treated as a non-native animal whose population must be reduced to protect a native range. Push the baseline back eleven thousand years and the same horse is a partially returned native whose relationship with the range is something the range has not experienced in a long time. The animals are the same. The analytical frame is different. The management implications are different. And the frame is almost never examined.

This is the first thing the Columbian Curtain hides: that it is a frame at all, rather than a fact.

◆ THE VOCABULARY OF INVASION ◆

The baseline does not stand alone. It is propped up by a vocabulary that carries the baseline's assumptions into every sentence in which the vocabulary appears. Once the vocabulary is in circulation, one can argue within it for decades without ever noticing that the argument has already been shaped by what the words assume.

"Invasive" is the clearest case. The term is doing real work in ecology and conservation biology — there are species introductions that cause measurable ecological harm, and a name for the category is useful. But the term is defined, operationally, against the 1492 baseline. A species is invasive, under the standard definitions deployed by state and federal agencies, if it has been introduced to an ecosystem in which it was not previously present and is causing or likely to cause

harm.[12] "Previously" here, in nearly every operative framework, means "at the time of European contact." The definition is internally coherent, and it reflects a genuine ecological concern about the rate at which human commerce has moved species across biogeographic barriers in the last five centuries. What it cannot see, by construction, is any species whose "introduction" is in fact a reintroduction at a deeper timescale.

"Non-native" carries the same assumption in a softer form. "Exotic" adds a faint aesthetic flavor, the suggestion that the animal is elsewhere's animal visiting. "Feral" contains an additional implication about the animal's relationship to human management, and typically marks an animal as out of place. All of these words are useful within their proper scope. All of them encode the 1492 baseline as a hidden premise.

The vocabulary produces one further effect worth flagging. A well-adjusted ecological assemblage (one that, under a 1492 baseline, is categorized as intact) can be assumed to be self-sustaining unless shown otherwise. A species classified as invasive can be assumed to be harmful unless shown otherwise. The burden of proof is asymmetric, and the asymmetry follows from the baseline. Reverse the baseline and the burden shifts. A species whose lineage was present on the continent for most of the last four million years does not become categorically benign by virtue of its lineage — there can be real ecological reasons to manage specific populations at specific densities — but the question of its presence stops being foreclosed by vocabulary. It becomes a question to argue.

This is, I think, the quieter cost of the Columbian Curtain. The baseline is not sitting in conservation textbooks as a debated proposition. It is sitting in the verbs and adjectives used to describe every animal on the continent, and most practitioners are using it without noticing.

◆ NEAR TIME ◆

A paleoecologist looking at the continent in 1492 does not see a native baseline. They see a system very recently stripped of most of its large-mammal biomass, in the middle of a long readjustment that has not yet stabilized. The frame in which this is visible is what Ross MacPhee, in the 1999 edited volume that gave the idea its current name, calls *near time* — the last fifty thousand years or so of Earth history, the

window during which most of the world's large-mammal extinctions have occurred and the present biota has taken its current shape.[7]

Near time is not a sentimental concept. It is the timescale on which the actual cast of the current ecological play was assembled, because most of the changes that produced the present continent happened within it. Plate tectonics and long-term climate shifts operate on vastly deeper scales. The specific animals and plants in any given ecosystem today, and the gaps in those ecosystems where animals used to be, settled into their current arrangement inside the near-time window. If we want to know what a continent was before humans arranged it into the present, near time is the window to look through.

MacPhee's framing also sets the stakes. The volume assembles case studies from every inhabited continent and major island group, and the synthesis across cases is unambiguous: the world has lost an extraordinary proportion of its large-bodied fauna within this window. Island losses are in some cases total. Continental losses are proportional but massive. Martin's own summary of the North American figure — more than forty large-mammal species extinct within near time, and thirty-five or so of them gone in the terminal Pleistocene pulse around thirteen thousand years ago — sits within this larger pattern rather than outside it. South America lost more than fifty genera. Australia lost nearly all of its megafauna tens of thousands of years earlier. The near-time extinctions are the single largest event shaping the modern ecological assemblages we inherit.[8]

The comparison to what was there is sobering in a way statistics alone cannot convey. Using the late-Pleistocene fossil record, paleontologists estimate that the mean body mass of large mammals in North America during the period immediately before the terminal extinctions was roughly comparable to the modern East African savanna.[9] The Great Plains supported proboscidean biomass. The southwestern basins supported camelid and equid biomass. The eastern woodlands supported an assemblage of ground sloths, peccaries, and tapirs whose nearest living analogues are neotropical. The continent Europeans encountered was a structurally different place from the continent that had existed twelve thousand years earlier, and the difference was not an improvement or a maturation or a natural development. It was a subtraction.

Near time is the window in which that subtraction happened. To take the 1492 baseline as "native" is to take the post-subtraction assemblage as the reference. It is to treat the end result of a mass extinction as the default state. The near-time frame does something different. It asks what the continent was like across the whole window in which the present system came together, and it lets the observer see the losses as losses rather than as the background against which all subsequent change is measured.

This is not a rewilding argument yet. It is a framing argument. The near-time window is the analytically honest timescale for thinking about what any given continent is now, because it is the window in which what it is now was assembled. Using any shorter window — five hundred years, two hundred years, the span of European land-use records — is not wrong, exactly, but it is structurally unable to see the shape of the assemblage's losses. A shorter window can tell us how a given forest has changed since the timber surveys of the 1870s. It cannot tell us what the forest is missing on a scale that matters to how the forest functions.

The near-time argument has a close cousin in fisheries science, and the parallel is worth drawing because it arrived at a similar insight from a different direction. In 1995 the fisheries biologist Daniel Pauly published a short essay in *Trends in Ecology & Evolution* naming what he called "shifting baseline syndrome."[13] Pauly had noticed that each generation of fisheries scientists tended to take the species composition and abundance of the oceans at the start of their own careers as the reference condition against which subsequent change was measured. A scientist trained in the 1970s considered the 1970s ocean normal and was alarmed by the changes since then. A scientist trained in the 1990s considered the 1990s ocean normal and did not find the reduced abundance alarming, since it was what they had always known. Each generation ratcheted the reference condition down. The cumulative effect, across several generations, was that the "natural" state of the oceans slid progressively further from anything the oceans had been before industrial fishing began.

Pauly's essay was a disciplinary turning point within fisheries biology. It did what this chapter is trying to do for terrestrial ecology: it named the assumption hiding in the reference condition. The syndrome Pauly described is not quite identical to the one the Columbian Curtain produces — Pauly's diagnosis involves a rolling baseline that shifts with each observer, while the Columbian baseline is fixed at a particular date — but the underlying structure is the same. Both are cases where the reference condition is a recent artifact, shaped by the observer's position in time, mistaken for a

deep property of the system. Both can be corrected only by stepping back to a longer frame.

In the fisheries case, the longer frame looks to historical records — ship logs, cannery data, colonial-era fish markets, archaeological fish-bone middens — to reconstruct what abundance was before the baseline slipped. In the terrestrial case, the longer frame is paleontological: the fossil record, the dung caves, the bone beds, the radiocarbon chronologies. The data are different. The corrective move is the same. Find the frame the observer has inherited, locate the assumption it is making about where to start counting, and then look further back.

I find the parallel reassuring in the sense that it was arrived at independently by people working on an unrelated problem. The 1492 baseline on land and the rolling baseline at sea are not identical pathologies, but the cognitive mechanism behind both is the same, and the correction in both cases is the same.

◆ WHAT A BASELINE IS FOR ◆

A baseline does three kinds of work in conservation. It tells us what to count as native. It tells us what to count as damage. And it tells us what to count as restoration. Change the baseline and all three categories rearrange.

Consider the first. Under the 1492 baseline, the native vertebrates of Arizona are the ones present when Coronado's expedition passed through in the 1540s — a set that includes pronghorn, mule deer, bighorn sheep, elk, a handful of smaller ungulates, and the surviving predators. Under a near-time baseline, the native vertebrates of Arizona include those animals *and* the camelids, equids, and proboscideans whose fossils are distributed across the state's late-Pleistocene sediments, some of which are well enough preserved to appear in local natural-history museums within an hour's drive of many of the state's largest school districts. The animals are the same animals either way. Whether they count as native to the state is entirely a function of which window the question is being asked across.

Consider the second. Damage, in an ecological assessment, is usually measured against a baseline — the reference condition the system has diverged from. Under the 1492 baseline, damage to the American shortgrass prairie is the conversion of grassland to cropland, the loss of bison, the encroachment of mesquite, the

overgrazing of exhausted cattle regimes. All of those are real and serious. Under a near-time baseline, damage to the same prairie includes all of those *and* the absence of the proboscidean biomass that likely contributed to the system's historical openness, the absence of the American cheetah that shaped the pronghorn's running speed, the absence of the ground sloths that may have played a role in seed dispersal for some of the legumes still standing there now. The losses the Columbian Curtain hides do not cease to be losses because the curtain is hiding them. They simply cease to be counted.

Consider the third. Restoration, under the 1492 baseline, means returning a degraded landscape to something resembling its state at the time of European contact. It is the frame against which Park Service management plans, Forest Service stewardship documents, and most published habitat-restoration studies are written. Under a near-time baseline, restoration can still include all of the 1492 targets, but the target list expands: it includes restoring, where possible, the large-bodied animals and ecological relationships that were present before the terminal Pleistocene losses. The two frames do not necessarily conflict in practice on short time horizons — returning bison to the plains scores well under both frames, and reforesting cleared eastern hardwoods scores well under both — but they begin to diverge on the margin. Reintroducing American burros to public land scores as damage under the 1492 frame and as partial restoration under the near-time frame. Considering camels as proxies for the extinct American camelids of the Southwest is, under the 1492 frame, not even a conversation worth having; under the near-time frame, it is a hypothesis worth testing.

The point is not that one frame is correct and the other is mistaken. The point is that both frames are choices, with different consequences, and the choice is not intellectually neutral. Conservation biology has, on the whole, chosen the 1492 baseline without examining the choice closely. The result is that a great many restoration proposals that would be coherent under a near-time baseline are ruled out before they can be evaluated. They are non-starters because of the frame, not because of the evidence.

I want to be precise about what I am claiming. I am not claiming that any restoration proposal that invokes the Pleistocene baseline is automatically a good one. I am not claiming that species introductions should be attempted casually. I am not claiming that conservation biologists who work under the 1492 baseline are

foolish, or that the 1492 baseline produces worse outcomes than alternative frames in every case. Many of the people who have shaped the 1492 baseline into its current form are serious ecologists whose work I have learned from. The dominant paradigm is not dominant by accident.

What I am claiming is that the baseline is a choice, that the choice shapes the outcome, and that serious conservation thinking requires naming the baseline rather than treating it as a given. Once the baseline is named, it can be argued about. Until it is named, the argument is smuggled into every premise.

◆ WHAT IS ON THE OTHER SIDE ◆

The prologue made the point through camels and an osage orange orchard. I will not revisit either here, except to say that they are instances of a pattern, and the pattern is worth naming.

On the other side of the Columbian Curtain, the North American fauna is a different assemblage from the one Americans grew up knowing. The flatlands held proboscideans. The broken country of the Southwest held equids and camelids in numbers that, by modern standards, seem implausible. Eastern hardwoods held ground sloths — at least two genera of them, *Megalonyx* and *Eremotherium*, the smaller of which was roughly the size of an ox and the larger of which approached the size of a modern elephant. The canyons of the West held cats whose closest living relatives are African. The boreal north held mammoths until approximately 10,500 years before present in some refugial populations.[10]

The list, stated all at once, has a faintly unreal quality for most readers. It reads like a natural-history exhibit rather than an inventory. But the animals were not mythical. They left bones in amounts that were once, and in some places still are, routinely exposed by arroyo cutting after heavy rains. They left dung in caves dry enough to preserve it for ten thousand years.[11] They left tracks turned to stone. They left, in some cases, well-characterized genetic material. They were as concrete as any animal on the continent today. The only thing odd about them, from a modern standpoint, is that they are gone.

The recovery record is not speculative. It is grounded in physical evidence that can be held, measured, dated, and in many cases sequenced. Ground sloth coprolites

from dry caves on the Colorado Plateau have yielded preserved plant material detailed enough to reconstruct the animals' browse preferences across thousands of years of occupation.[14] Mammoth material recovered from permafrost has supported nuclear genome sequencing in the same rough ballpark as the Thistle Creek horse. Tar-seep deposits at Rancho La Brea in southern California have produced preserved animal bodies by the tens of thousands, from dire wolves to saber-toothed cats to the American lion *Panthera atrox*, with radiocarbon-dated material spanning roughly fifty thousand years.[15] The late Pleistocene is not prehistory in the sense of being inaccessibly remote. It is prehistory in the sense that it predates writing. Much of what we know about it, we know at a resolution comparable to the resolution at which we know the historical Roman road network — from the ground, from what was left behind, from dating the physical evidence.

I am not going to make the rewilding case here. That case belongs to later chapters — to the chapter on what proxy species might plausibly be considered, to the chapter on the specific restoration work already under way, to the chapter that takes the critics of Pleistocene rewilding seriously and answers the criticisms I think can be answered. Here I only want the reader to hold, briefly, the shape of the other side.

The continent on the other side of the Curtain is not exotic. It is the continent the land itself still remembers. Grasses on the plains evolved under the hooves and digestive tracts of animals whose modern descendants are managed as livestock or warehoused on reservations or gone altogether. Soil communities in the Southwest evolved under trampling patterns that have not been produced in twelve thousand years. Predator-prey dynamics in the high canyons evolved around hunters whose bodies are in museum drawers. The landscape is not recovering from a bad decade. It is not recovering from European contact. It is, on the whole, still in the long thaw after a much larger subtraction.

The Columbian Curtain is the frame that makes this hard to see. Drop the frame and it becomes legible.

◆ NATIVE TO WHEN ◆

I opened this chapter with the horse because the horse is the cleanest case for the argument, and because it is a case the evidence has already essentially made on its own. The horse of the Plains is both a sixteenth-century immigrant and a thirteen-

thousand-year returnee. Under a 1492 baseline, it is the former. Under a near-time baseline, it is the latter. The only question is which window we are using to look.

The question the chapter lands on is not whether horses are native, or whether camels would be native if reintroduced, or whether burros belong in the Grand Canyon. Each of those is a specific question that requires its own specific argument. The question the chapter lands on is the one upstream of all of them, the question the 1492 baseline has silently answered without asking: *native to when?*

Every conservation decision rests on an implicit answer to this question. The answer is rarely stated, and it is often buried under vocabulary that presents the baseline as natural rather than chosen. Once the question is stated in the open, every subsequent decision becomes a live argument rather than a settled one. We can argue about the right window. We can argue about what to do about particular species under particular windows. We can argue about the risks of different proposals. What we cannot do, once the question is in the open, is pretend that conservation has been doing anything other than drawing a time-line through the ecological past and calling one side of it native.

I want to note one thing about how the argument will develop from here, because it matters for the register in which the rest of the book is written. The point of naming the Columbian Curtain is not to discredit the practitioners who have worked within it. A great deal of serious conservation biology has been done under a 1492 frame, and a great deal of that work has produced real results. Wetland restoration in coastal North America, the recovery of the bald eagle and the American alligator, the return of the gray wolf to parts of its historic range, the ongoing expansion of bison onto grassland that has not seen significant bison presence in over a century — all of these operate within the 1492 baseline and all of them represent genuine conservation gains. None of this chapter's argument dismisses any of that. The argument is additive, not subtractive. The near-time frame does not replace the shorter-window frames used in existing conservation work; it makes visible a set of additional considerations that the shorter windows cannot see. The chapters that follow ask what those additional considerations are, what we should do with them, and where they lead.

The baseline is a choice. It has always been a choice. The work of the rest of the book, in large part, is to argue that the choice should be made deliberately rather than

inherited without examination, and that when it is made deliberately, the near-time window is the one most consistent with what we actually know about the ecosystems we inhabit. A baseline chosen deliberately is still a baseline; it will have its own exclusions and its own assumptions, and it will need to be defended with the same honesty required to defend any other analytical frame. The difference is that a deliberate baseline can be argued against, refined, and corrected. The inherited one, by design, resists each of those moves.

But the argument begins here. With the horse returning to its continent of origin after a long absence, and a culture that mostly has not noticed the return. With a baseline dressed as a fact. With a continent whose memory is older than its paperwork. Native to when is the question. Near time is, I think, the defensible answer.

The next chapter asks what was lost in that window, and when, and what we know and do not know about why.

CHAPTER THREE

Near Time

♦ ♦ ♦

From Paul S. Martin,
Twilight of the Mammoths (2005)

"A climatic catastrophe forced extinctions in Australia around 46,000 years ago, and in North and South America 10,000 to 12,000 radiocarbon years ago." If so, "neither event registered in New Zealand."

"I will lay aside impartiality. I think the overkill theorists have the more convincing argument for what happened in America 10,000 years ago."

— P. S. Martin, chs. 7–8

Paul Martin came to the extinction question in the 1950s, at roughly the moment the question first became answerable. He was a young paleoecologist at the Desert Laboratory in Tucson — Arizona's field station for the study of arid-land ecology, perched on Tumamoc Hill above the city — and he had been trained in the generation for which the fossil record was a stratigraphic problem. Animals disappeared from the record at some point in the strata. The strata told you roughly when. What the strata could not tell you, at the resolution then available, was whether one animal disappeared at the same moment as another, or centuries earlier, or centuries later, or whether any of them disappeared at the same moment as anything else: a climatic event, a vegetation shift, the arrival of a new predator. The granularity was not fine enough to see coincidence at the scale where coincidence becomes causation.

Willard Libby changed that. Libby's radiocarbon technique, published in the late 1940s and refined through the 1950s, allowed organic material up to about fifty

thousand years old to be dated at a resolution of centuries rather than millennia.[1] Martin later described what this did for the field in plain terms: it took a question that had been stratigraphic and made it chronological. Instead of asking which layer a bone came from, you could ask what year (within a century or two) the bone had lain down in its layer. For the first time, the timing of specific extinctions could be matched against the timing of specific other events. What the dating began to show, across the 1950s and 1960s, was that the timing of the great North American extinctions sat uncomfortably close to one specific other event, and the coincidence bothered Martin for the next half century.

The event was the arrival of humans on the continent.

◆ WHAT WAS LOST ◆

The chapter has to begin with the inventory, because without the inventory the rest of the argument does not land. The prologue named a few of the animals as emblems. Chapter 2 gestured at the catalog without unfolding it. Here the catalog unfolds.

North America north of Mexico lost at least thirty-five genera of large mammals in the terminal Pleistocene pulse, concentrated in the few centuries around thirteen thousand calendar years before present.[2] That is a genus-level figure, not a species count; counted at the species level the losses are higher. Martin's own tabulation in *Twilight of the Mammoths* ran to more than forty North American species above forty-five kilograms of body mass, or roughly a hundred pounds (the threshold biologists often use to separate large-bodied mammals from medium-sized ones).[3] The loss on the proboscidean front alone involved both of the continent's mammoth species, the Columbian mammoth *Mammuthus columbi* and the woolly mammoth *Mammuthus primigenius*, together with the American mastodon *Mammut americanum*, a separate lineage with a range that extended through the eastern hardwoods. Three proboscideans, all gone in the same window. The closest parallel anywhere on Earth today would be to imagine Africa losing both species of elephant in the course of a human lifetime.

The ground sloths went with them. Martin counted at least four North American genera: *Megalonyx jeffersonii*, the Jefferson's ground sloth, widely distributed through the east; *Nothrotheriops shastensis*, the Shasta ground sloth, familiar from the

dung-dense caves of the Colorado Plateau; *Paramylodon harlani*, Harlan's ground sloth, common enough at La Brea to be an exhibit standard; and *Eremotherium laurillardi*, a massive southern genus that ranged north into the Gulf Coast and occasionally farther. Two of these animals (*Megalonyx* and *Eremotherium*) reached sizes comparable to modern elephants. The others were bear-sized. None of them is present in any modern ecosystem. Their closest living relatives are the two- and three-toed tree sloths of Central and South American rainforests, which weigh a few kilograms and hang in canopies that the extinct North American sloths would have flattened.

The equids went with them. The genus *Equus* had diversified across the continent for the entirety of its evolutionary history, as the previous chapter described. By the terminal Pleistocene pulse several species of native American horse were still in place (*Equus scotti*, *Equus conversidens*, *Equus lambei*, and others whose taxonomy paleontologists continue to revise), and every one of them disappeared in the same window.[4] The camelids went the same way. Martin described three North American genera going out together: *Camelops*, the Western camel of the prologue's concern; *Hemiauchenia*, the larger of the two native llamas; and *Palaeolama*, the smaller. The bovids and cervids took losses too, though not total ones; the woodland musk ox *Bootherium*, the stag moose *Cervalces*, and several pronghorn relatives *Capromeryx*, *Stockoceros*, and *Tetrameryx* all went out in the pulse.

The predator assemblage that had evolved around these herbivores collapsed with them. The American lion *Panthera atrox*, larger than any living cat, disappeared from a range that had extended from Alaska to central Mexico. The saber-toothed cat *Smilodon fatalis* and the scimitar-toothed cat *Homotherium* both went. The American cheetah *Miracinonyx*, whose pursuit pressure had shaped the pronghorn's still-extant running speed, was gone. The dire wolf *Aenocyon dirus* was gone. The short-faced bear *Arctodus simus*, the largest carnivorous land mammal of the North American Pleistocene and in some reconstructions of its ecology the most aggressive scavenger the continent has ever hosted, was gone.[5] Beyond the predators, the bear-sized giant beaver *Castoroides*, the capybara-like *Neochoerus*, Jefferson's extinct peccary lineages *Platygonus* and *Mylohyus* — on and on, and then the list stops.

The list does not stop because the subtraction tapered gently. It stops because the continent had little more above forty-five kilograms of body mass to lose. Counting what remained in the Holocene — bison, elk, moose, bighorn, mountain goat,

pronghorn, the surviving deer, and the surviving carnivores of wolf and grizzly and cougar — the North American large-mammal fauna at the beginning of written history was roughly half of what it had been, by genus, a few thousand years earlier.[6] The same pattern, at varying magnitudes, appeared on every inhabited continent in near time. South America lost more than fifty genera, a greater proportional loss than North America; the glyptodonts, the car-sized armored herbivores *Doedicurus* and *Glyptodon*, the massive ground sloth *Megatherium*, the native ungulates *Macrauchenia* and *Toxodon*, and a range of sabertooths and short-faced bears all exited the continent's fauna in the same terminal Pleistocene window. Australia's losses were earlier, roughly forty-six to fifty thousand years ago by current chronology, and were nearly total for the continent's megafaunal assemblage: the rhinoceros-sized diprotodontids, the marsupial lion *Thylacoleo*, giant kangaroos, and the flightless bird *Genyornis* all disappeared. Eurasia lost the woolly mammoth, the woolly rhinoceros *Coelodonta antiquitatis*, the steppe bison *Bison priscus*, the giant deer *Megaloceros giganteus*, the cave bear *Ursus spelaeus*, and the cave hyena, each on its own regional chronology distributed across the late Pleistocene and into the early Holocene. Madagascar's losses came more recently still, within the last two thousand years: the giant lemurs, the elephant birds, and a dwarf hippopotamus disappeared after Malagasy colonization. Each continent's loss profile differed in timing and in which specific animals were involved. What each continent shared was the shape: disproportionate culling of the largest-bodied terrestrial fauna, with little parallel effect on small mammals, plants, marine species, or insects.[7]

This is the subtraction the previous chapter framed as the reason the 1492 baseline is an inadequate reference point. The chapter has not yet said why it happened. The literature has been trying to say why it happened for the seventy years since Martin first began trying, and the literature has not, on the whole, settled.

◆ THREE HYPOTHESES ◆

Three explanations have done most of the work in the scholarly debate. They are not strictly mutually exclusive, and the current synthesis involves combinations of them, but it is easier to see the shape of the combination if the three are taken one at a time.

The first is overkill: the hypothesis that the extinctions were caused, primarily or substantially, by predation from newly arrived human hunters. Its most forceful articulation is the one Martin himself gave it, refined over four decades and summarized in *Twilight of the Mammoths* in 2005.

The second is climatic forcing: the hypothesis that the late-Quaternary climate oscillations, in particular the abrupt cooling known as the Younger Dryas around 12,800 calendar years before present, pushed species past thresholds their populations could not recover from. This was the dominant view among most vertebrate paleontologists before Martin's formulation entered the field, and it remains a serious position today, though the version that prevails now is more chronologically precise than the one Martin engaged with in the 1960s.

The third is cosmic impact: the hypothesis, developed in the peer-reviewed literature over the last two decades, that the Younger Dryas cooling itself was initiated or amplified by an airburst or impact event near the Laurentide ice sheet at approximately 12,900 calendar years before present. This is the Younger Dryas Impact Hypothesis, and it interacts with both of the other two: if the hypothesis is correct, the climatic mechanism has a proximate trigger it otherwise lacks, and the timing of the extinction pulse with respect to human arrival becomes a triple rather than a double coincidence.

Each of these is worth walking through in its strongest form, because each has been argued in its strongest form by serious researchers whose work deserves that level of presentation.

REVISION

The Overkill Hypothesis

2011: I was confident the Late Pleistocene extinctions were caused by humans arriving across the Bering land bridge. The overkill hypothesis, as laid out in Martin's earlier work, was the version of the story I believed. Human arrival, large mammals gone within a few thousand years, clean narrative, rewilding justified as a kind of debt repayment.

2017: I began reading the paleoclimate work on the Younger Dryas and the challenges to the timing of human arrival. The fossil record did not line up as cleanly with the human signal as the overkill story required. At least some of the extinctions ran ahead of the first firm dates for human presence.

2024: I am no longer confident in overkill as the single cause. I think humans were involved. I think climate was involved. I think the mixture varies by species and by region, and I am not sure it is knowable from the evidence we currently have.

This matters for the book only in that the case for rewilding used to depend, in my thinking, on human responsibility. It no longer does. The case now rests on ecological function. The landscape is operating below capacity because animals that shaped it are missing. Why they are missing is a separate question, and the rewilding case holds either way.

I changed my mind. Because I changed my mind, I considered other ideas and arguments against its premise more seriously. It sharpened the thinking, and narrowed the arguments. The book is better for it.

◆ OVERKILL, AS MARTIN MADE IT ◆

Martin did not invent the overkill hypothesis. Richard Foster Flint at Yale had floated human involvement earlier. Darwin's contemporary Richard Owen had proposed something similar for New Zealand's moas in the nineteenth century. What Martin did (working through the 1960s, 1970s, and 1980s, often with collaborators in simulation work, comparative island ecology, and radiocarbon calibration) was make the case in a form rigorous enough that climate-centric alternatives had to argue against a specific alternative rather than merely assert their own.[8]

The case, as Martin stated it in *Twilight of the Mammoths*, rests on three observations about near-time extinctions that climate-centric explanations struggle to accommodate.

The first is time-transgressiveness. Near-time megafaunal extinctions did not happen at the same moment around the world. Australia's pulse is roughly forty-six thousand years ago. Continental North America's is around twelve to thirteen thousand years ago. New Zealand's moa extinctions occurred less than a thousand years ago. Madagascar's losses date to within the last two thousand years. The Caribbean island sloths persisted into the middle Holocene and disappeared only after human colonization of the islands. What does fit each of these extinction events closely is the arrival of humans on the landmass in question. Climatic shifts of

comparable magnitude, if they are invoked as the primary mechanism, have to be invoked at different times in different places, without a unifying global pattern of climate that matches the extinction pattern. "A climatic catastrophe forced extinctions in Australia around 46,000 years ago, and in North and South America 10,000 to 12,000 radiocarbon years ago," Martin wrote. If so, "neither event registered in New Zealand."[9] Time-transgressiveness is the signature the overkill hypothesis predicts and the climate hypothesis does not.

DEFINITION

Time-transgressive Extinction

A time-transgressive extinction is one that occurs at different times in different places: the same pattern recurring across landmasses, offset from each other by thousands or tens of thousands of years.

The near-time megafaunal losses are the paradigmatic example. Australia's pulse dates to roughly 46,000 years before present. Continental North America's centers on 12,000 to 13,000 years. New Zealand's moa extinctions occur less than 1,000 years ago. Madagascar's sit within the last 2,000. The pattern that unifies these events is not a single global climate shift, which would affect all landmasses simultaneously. It is the arrival of human populations, which occurred at different times on different continents.

Time-transgressiveness is the diagnostic signature by which paleoecology distinguishes human-driven from climate-driven extinction events, and it is the single observation the climate hypothesis has the hardest time accommodating.

The second is selectivity. Near-time losses fell almost exclusively on large-bodied terrestrial mammals. Trees and shrubs show essentially no contemporaneous extinction in the North American fossil record. Small mammals show essentially none. Marine mammals show essentially none. Beetles, birds outside flightless insular lineages, and other less-hunted groups show essentially none. The earlier Quaternary had repeatedly imposed severe climatic switches on the same flora and fauna without similar losses. If climate was the primary mechanism, its selectivity for mammals above forty-five kilograms (the threshold at which human hunting returns become energetically attractive) would be an extraordinary coincidence. Human predation,

by contrast, predicts exactly that size cutoff. Large animals reproduce slowly, track easily, and yield high energetic returns per successful hunt. They are the animals a novel predator would exhaust first.

The third is the uniqueness problem. Climatic changes as dramatic as the Allerød–Younger Dryas switch had occurred repeatedly during the Quaternary. Greenland ice cores record dozens of them. None of the earlier switches, over hundreds of thousands of years, are associated with comparable megafaunal die-offs. For a climatic explanation of the near-time losses to be adequate, the climate shift at the end of the Pleistocene had to be uniquely severe. The ice-core evidence does not obviously support that uniqueness.[10]

Martin paired the three criteria with a quantitative demonstration. In 1975, Martin and the statistician James Mosimann published a simulation showing that a founding human population of a few thousand, advancing at plausible rates through naive megafaunal prey, could in principle extinguish the continent's large-bodied mammal fauna in three to five hundred years. The simulation was sensitive to assumptions (population growth rate, hunting success, prey density, the rate of population spread), but its central finding was robust across a wide band of those assumptions. The claim was not that the simulation proved overkill had happened; it was that a critic could no longer reject overkill on the grounds that a small human population could not plausibly cause continental extinction. A small human population could. That was the point the simulation made, and its significance for the debate was that it shifted the burden of argument. Afterward, the climate camp had to explain why something that was computationally plausible had not, in fact, happened.[8] Martin's later articulation of a "blitzkrieg" model — a rapid wave of human advance through an ecologically unprepared fauna — is the popular form of the same argument, and it remains the most vivid framing of the overkill case.

Martin did not claim the overkill hypothesis was complete. He did claim — openly, and with the pugnacity the climate-change camp sometimes found tiresome — that it was the hypothesis most consistent with the global pattern, and that the climate-change camp had a harder time explaining the pattern than they usually admitted. "I will lay aside impartiality," he wrote in the opening of the chapter of *Twilight* that takes the debate head on. "I think the overkill theorists have the more convincing argument for what happened in America 10,000 years ago."[11]

It is important to hold the integrity of this argument in view. Martin was not a polemicist. He was a paleoecologist who had spent half a century on the evidence and had reached a considered conclusion. The hypothesis carried his name, fairly, because he had done more than any other researcher in his generation to put it on firm empirical footing. Anyone who wishes to retire the overkill hypothesis, as many working in the field now effectively have, owes Martin the courtesy of retiring it for reasons the evidence supplies, not by gesturing vaguely at discomfort with its implications.

◆ CLIMATE, RECONSIDERED ◆

The climate-change hypothesis, as Martin encountered it in the 1950s, was the inherited default position of vertebrate paleontology. Most vertebrate extinctions in the Cenozoic fossil record had been accounted for by climate change, often in the absence of any more specific mechanism. Applying the same default to the near-time losses was, from inside the discipline, the conservative move.

Martin's three-criterion critique — time-transgressiveness, selectivity, uniqueness — landed hard on the default. The default response, in the decades since, has been to refine the climate mechanism rather than abandon it. Current climate-centric accounts do not usually claim that a single global cooling event extinguished the world's megafauna in one stroke. They claim, more modestly, that rapid regional shifts in temperature, precipitation, vegetation structure, and seasonality interacted with the life-history traits of particular species to push specific populations below recovery thresholds in specific places. Under this account, the climate mechanism is local rather than global, continuous rather than catastrophic, and cumulative rather than discrete.[12]

The Younger Dryas cooling (an abrupt return to near-glacial conditions in the northern hemisphere around 12,800 calendar years before present, lasting roughly 1,300 years) does receive particular weight in these accounts for continental North America. The timing of the Younger Dryas onset aligns closely with the terminal Pleistocene extinction pulse in North America. The severity of the climate shift was substantial, involving rapid drops in North Atlantic surface temperatures and significant reorganization of precipitation patterns across the continent. The leading proximate mechanism in the paleoclimate literature is a freshwater pulse from the

catastrophic drainage of glacial Lake Agassiz (an enormous proglacial lake along the retreating margin of the Laurentide ice sheet) whose outflow into the North Atlantic is thought to have disrupted the thermohaline circulation and triggered the cooling. The details of that mechanism, including the precise routing of the meltwater, continue to be actively argued in the literature; what is not argued is that the cooling was abrupt and that its continental consequences were substantial.[13] Whether the shift was sharp enough to drive extinction of animals that had weathered dozens of comparable earlier shifts remains the unresolved point. Ice-core records from Greenland do show the Younger Dryas clearly; they also show earlier oscillations of comparable magnitude. The selectivity problem Martin identified — why this oscillation rather than the last twelve — continues to cut.

What the climate camp has credibly retrieved from the Martin critique is that *regional* climate dynamics matter. The loss of mammoth populations in the Siberian arctic, where some refugial mammoths persisted on Wrangel Island until roughly four thousand years ago and on St. Paul Island until roughly five thousand six hundred years ago, is difficult to explain by human pressure alone; the chronology shows that these populations outlasted terminal Pleistocene conditions by thousands of years, then disappeared under what appear to be largely climatic or habitat-fragmentation stresses.[14] What the climate camp has not credibly retrieved is the claim that a single, unique, catastrophic climatic event explains the synchronous disappearance of an entire large-bodied mammal fauna. That stronger claim runs into Martin's critique as soon as it is made globally.

◆ THE YOUNGER DRYAS IMPACT HYPOTHESIS ◆

A third candidate entered the peer-reviewed literature in 2007. Richard Firestone and colleagues, in a paper in the *Proceedings of the National Academy of Sciences*, proposed that an extraterrestrial impact event at approximately 12,900 calendar years before present (a comet or fragmenting bolide, most likely an airburst over or near the retreating Laurentide ice sheet) had initiated the Younger Dryas cooling and contributed to the megafaunal extinctions. The authors identified a dark sedimentary layer, which they called the "black mat," at archaeological and paleontological sites across North America, dating to the Younger Dryas boundary.

The black mat contained elevated concentrations of magnetic microspherules, carbon spherules, nanodiamonds, iridium, and other markers they interpreted as consistent with an impact signature.[15]

The paper was controversial on publication and has remained so. Critics have challenged the claimed peak concentrations, the interpretation of the spherule and nanodiamond markers, the statistical robustness of the chronology, and the plausibility of the proposed mechanism. Several rigorous critical reviews have argued that the evidence for the impact is insufficient to support the strong causal claim, and that alternative explanations for some of the proposed markers (cosmic dust background, wildfire signatures, biological spherules) have not been adequately ruled out.[16]

Two subsequent papers extended the physical case. In 2013, James Wittke and colleagues reported evidence for deposition of approximately ten million tonnes of impact spherules across roughly fifty million square kilometers on four continents, centered on the Younger Dryas boundary layer.[17] In 2015, James Kennett and colleagues applied Bayesian chronological analysis to the black-mat horizons at twenty-three sites across four continents and reported age ranges consistent with synchronous deposition at 12,835 to 12,735 calendar years before present.[18] The synchrony result does not by itself prove an impact — a rapid globally distributed sedimentary horizon can be produced by processes other than impact — but it is a feature of the record the hypothesis predicts and alternatives have to account for.

The hypothesis remains contested. It is not the consensus position in the paleoclimate or paleontological literature. It is, however, a peer-reviewed hypothesis that has accumulated a non-trivial evidentiary body over the last two decades, and it is the correct place for a serious discussion of catastrophe at the Younger Dryas boundary to take place. Any reader interested in the question of what happened at 12,900 BP should read Firestone 2007, Wittke 2013, and Kennett 2015 in the original, along with the major critiques of each, and form their own judgment.

I want to state clearly what this book's engagement with the hypothesis is and is not. This book engages the Younger Dryas Impact Hypothesis through the peer-reviewed literature only. It does not engage catastrophist popularizations of the hypothesis circulating in trade publishing or documentary media, which have tended to detach specific claims from the evidentiary context in which the peer-reviewed

work has made them. The distinction matters. The peer-reviewed work is a serious scientific hypothesis, contested but defensible; the popular accounts are a separate cultural artifact and do not carry over.

THE STRONGEST CURRENT CASE FOR HUMAN CAUSATION

While the impact hypothesis was accumulating evidence, a distinct body of work was re-examining the human-causation case at global scale with tools Martin did not have. In 2014, Christopher Sandom, Søren Faurby, Brody Sandel, and Jens-Christian Svenning published a paper in *Proceedings of the Royal Society B* that applied spatially explicit global analysis to the late-Quaternary extinctions, testing the relative contributions of climate change and human arrival as predictors of extinction severity. The analysis used extinction data for roughly 178 mammal species across all ice-free continents, paired against reconstructed climate change metrics for the period 132,000 to 1,000 years before present and against hominin colonization chronology for each region.

The result was unambiguous. Extinction severity was strongly and consistently predicted by hominin arrival; it was not consistently predicted by climate change. The authors wrote that their findings "provide empirical support for prehistoric humans being the main driver of the Late Quaternary Megafauna Extinction."[19]

The Sandom paper is, in my reading, the strongest current empirical case for human causation as the primary driver. It is not Martin's argument — Martin had made the comparable point at regional scale with the tools available to him in 2005 — but it is Martin's argument extended with global data and statistical methods that did not exist when Martin was writing. Anyone attached to the overkill hypothesis has a stronger set of references to cite in 2026 than they did in 2006. The hypothesis has aged, but it has also sharpened.

What Sandom and colleagues did not claim is that climate played no role. Their result was about the *relative* contributions. Humans appeared to be the dominant driver; climate appeared to modulate the outcome in specific regions and for specific species. That is a multifactorial finding dressed as a human-cause headline, and it is best read that way.

TOWARD A MULTIFACTORIAL SYNTHESIS

Anthony Stuart, in a 2015 synthesis in *Geological Journal*, drew the continent-by-continent picture the last decade of work had assembled.[20] Stuart's synthesis does not crown a winner. It argues, across Eurasia, North America, South America, and Australia, that the near-time extinction pattern is best explained by the interaction of climate change and human pressure, with the relative weight of each varying by region, by species, and by the phase of the late Quaternary under examination.

In some regions, the climate contribution appears to have been substantial. The mammoths that persisted on Siberian refugia well past the terminal Pleistocene pulse and then disappeared in the middle Holocene were not, as far as the archaeological record shows, subjected to sustained human hunting pressure in a way that would explain their final extinction; habitat fragmentation and climate-driven vegetation change are the more plausible proximate causes. In other regions, the human contribution appears to have been decisive. The New Zealand moa extinctions within a few centuries of Polynesian arrival cannot be accounted for by climate change; there was no substantial climate change to account for. Most of the world's near-time extinctions sit between these cases, with both factors present and the relative contributions unresolved.

Stuart's synthesis is not a retreat into vagueness. It is a recognition that the question "what caused the near-time megafaunal extinctions" is itself a shorthand for a family of more specific questions, and the specific answers differ. What caused the loss of *Mammuthus columbi* in Arizona is not necessarily what caused the loss of *Mammuthus primigenius* on Wrangel Island five thousand years later. The synthesis allows the specific answers to be specific.

Notice what this does to the larger interpretive framework. If the causation is multifactorial, with climate and human pressure interacting differently in different places and for different species, the clean story of either a single human culprit or a single climatic catastrophe is replaced by a more layered account. The layered account is less satisfying as a headline. It is probably closer to what actually happened.

WHERE I NOW STAND

I have held more than one position on this material. I read *Twilight of the Mammoths* in 2012, shortly after finishing the first version of this book, and was persuaded quickly. For a long stretch afterward, into the essays I published in the middle of the 2010s, I wrote as though the overkill hypothesis was effectively settled, and as though the rewilding case carried an ethical weight drawn from collective human responsibility for the extinctions. My 2015 and 2016 writing, which I have since retired, states the human-causation position flatly. I now think that framing was too confident.

What I missed at the time was exactly the material the present chapter has walked through: the peer-reviewed work on the Younger Dryas oscillation and its mechanisms, the Firestone and Wittke and Kennett papers on the impact hypothesis, the Stuart synthesis, and the global statistical work (including Sandom 2014) that has made the human-causation case more defensible but also more bounded than Martin's original formulation was. Read across the whole corpus, I am now not confident that the extinctions had a single dominant cause. I am confident they were heavily human-involved in most regions. I am also confident that climate oscillation, potentially triggered or amplified by the impact event the Firestone hypothesis describes, contributed materially in at least some regions and at least for some species.

This revision is not a repudiation of Martin. Martin's critique of mid-century climate-centric accounts was correct on the terms in which those accounts were being made. The subsequent refinement of the climate side of the debate has partly answered Martin's critique, and the subsequent accumulation of impact-hypothesis evidence has supplied a mechanism Martin did not have available to consider. The overkill hypothesis in its strong 2005 form is not quite where the field now stands. The overkill hypothesis in a weaker, multifactor-inflected form is, I think, the best reading of the current evidence.

I want to be explicit about what changes when one moves from a strong-overkill view to a multifactorial view, because the change is not merely scholarly. It alters the moral inflection of the rewilding case. The strong-overkill framing carries with it, often implicitly, an accounting of debt: humans caused the extinctions, humans ought to consider repair. The multifactorial framing does not carry that accounting. Humans may or may not have caused the extinctions — in any given region, in any

given case, with any given confidence — and the case for restoration stands or falls on something other than collective human guilt. I think the case stands. I think it stands more cleanly when it is not entangled with an attribution question the evidence does not cleanly resolve.

It is worth naming, plainly, that I arrived at the strong-overkill position by reading Martin first and reading the challenges to Martin much later. That is a reasonably common trajectory for readers drawn to the rewilding argument, and I am not the only author to have walked it. The corrective, when it came, came from reading the Quaternary climate literature carefully, from taking the impact hypothesis seriously as a peer-reviewed proposition rather than dismissing it because of what had been made of it downstream, and from the simple act of sitting with Stuart's synthesis long enough to notice that it was not a split-the-difference move but a closer reading of the actual evidence than either pole had offered. A revised position of this kind is not a loss of confidence in the subject. It is a gain of confidence in the shape of what is known and what is not.

◆ WHAT THE REWILDING CASE ACTUALLY DEPENDS ON ◆

The argument the rest of the book is going to make rests on an ecological observation, not a historical one.

The observation is that an ecosystem missing its keystone large-bodied animals functions differently than one in which those animals are present, and functions worse by a number of measurable metrics: seed dispersal, nutrient cycling, vegetation structure, predator-prey regulation, fire regime, soil compaction and turnover. The observation is independent of why the animals are absent. The continent is without mammoths, ground sloths, and native equids for whatever reason. The reason does not change the ecological consequence. A grassland that evolved under proboscidean herbivory is poorer in structure and function without proboscidean herbivory, whether the proboscideans were hunted to extinction, cooled to extinction, or died in some combination of the two. A fruit evolved for a gomphothere is abandoned by its disperser regardless of the cause of the gomphothere's disappearance. A predator-prey calibration between cheetah and pronghorn is one-sided whether the cheetah was speared by a Clovis hunter or starved in a Younger Dryas winter.

This is the chapter's load-bearing point, and it is where the chapter lands. Near time is the window in which the present ecological assemblage was stripped of much of its large-bodied biomass. The stripping was probably multifactorial. Apportioning blame is a worthy scientific question that the literature will continue to refine. The decision of whether to attempt restoration does not depend on the blame question's resolution. It depends on whether we take the observation seriously that the stripped system is still, twelve thousand years later, functioning as a stripped system — incompletely, with unfilled niches, with species waiting on partners that are not coming back on their own.

The continents that lost their megafauna have not recovered the function those animals performed. No continent has. The fossil fruits still fall, the grasslands still miss their grazers, the pronghorn still runs from nothing. Whatever the cause of the missing animals was, the absences are still here, and they are still absences. The next part of this book asks what, knowing that, we should do.

CHAPTER FOUR

The Fourth Axis

♦ ♦ ♦

From Tendall et al.,
Global Food Security (2015)

Food system resilience: "the capacity over time of a food system and its units at multiple levels, to provide sufficient, appropriate and accessible food to all, in the face of various and even unforeseen disturbances."

The concept, they note, remains underspecified in most applied assessment frameworks.

— Tendall, Joerin, Kopainsky, et al.

Picture a poultry operation in the Piedmont of Virginia: five hundred acres, rotationally grazed, solar panels on the barn roof, organic certification on the feed. The birds live outdoors in mobile shelters moved daily across perennial pasture. The soil organic matter has climbed two percentage points in a decade. The operation employs eight people from the surrounding county, pays above the local median, and sells direct to consumers within a hundred-mile radius. A journalist doing a sustainability profile would find everything in order. The economic numbers work at the operator's scale. The social indicators — employment, community retention, wage levels — are positive. The environmental metrics are strong: no synthetic pesticide runoff, measurable soil carbon gains, arthropod and avian biodiversity substantially above the row-crop fields across the road. By any standard assessment of sustainability, this is a model operation. It would score well on every axis the standard framework knows how to measure.

Now remove one thing. Not a catastrophe. Just one input. Suppose the commodity corn supply that feeds the birds becomes unavailable. Not forever. For one season. A drought in the Midwest reduces the corn harvest. Grain futures spike. The regional feed mill that supplies the operation cannot fill its orders at a price the farm can absorb without selling birds at a loss. The operation has no on-farm grain production. Its acreage is in perennial pasture, which is why the soil numbers are so good. The birds cannot eat grass. They are monogastrics. They require concentrated energy and protein, and the concentrated energy and protein comes from somewhere else, and the somewhere else has, for one season, failed to deliver.

The operation does not adapt. It does not pivot to an alternative feed source, because at the volumes required there is no alternative feed source available at a survivable price on the timeline the birds' metabolisms demand. It does not switch to a different species, because its infrastructure, its processing relationships, its customer base, and its brand are built around poultry. It does what an operation in this position does: it reduces the flock, absorbs the financial loss, and waits for the commodity market to restabilize. If the disruption lasts one season, the operation survives diminished. If it lasts two, the operation is in financial distress. If it lasts three, the operation that scored so well on every other measure of sustainability is gone.

This is not a hypothetical constructed for rhetorical convenience. Versions of it have played out across the pastured-poultry sector in every year the corn market has spiked significantly: in 2012, when Midwest drought drove corn futures above eight dollars a bushel for the first time; in 2022, when the disruption of Ukrainian grain exports compounded existing supply-chain stress to push US feed costs to levels that several direct-market poultry operations, in interviews and trade-press accounts, described as existential.[1] The operations that survived those spikes survived them by drawing on reserves, raising prices past what some of their customers would pay, or reducing production. They did not survive them by being sustainable in the sense the word is usually used. They survived them by having enough financial margin to absorb a shock the system had not been designed to withstand.

The gap the example is meant to illustrate is not a gap in the operation. It is a gap in the framework we use to evaluate the operation. The three-axis sustainability model — economic, social, environmental — measured the operation and found it sound. What the model did not measure, because the model was not built to measure

it, is what happens to the operation when one of its underlying conditions changes. The model assessed the system's performance under current conditions. It did not ask what the system requires to keep performing, or what happens when those requirements are not met. That question — the question of what the system is riding on and whether the ride continues — is the question this chapter exists to introduce.

FIELD NOTE

The Zombie Apocalypse Litmus Test

The question this chapter is built around — what does a food system require to keep functioning, and what happens when those requirements stop being met — is older than any framework I have built around it. The first time I asked it was while watching an episode of *The Walking Dead*.

The show was set in a world where the supply chains had collapsed. The episode involved a farm. Hershel — the older farmer character with the beard — had a flock of chickens, and the show had taken the trouble to mention what he was feeding them. Corn and soy. Industrial poultry feed. It dawned on me: farmer Hershel is in trouble. There was no commodity grain in his world anymore. No gasoline-powered transport. His chickens were on a clock the show had not yet noticed.

I went home and started thinking about what a sustainable food system would have to look like if you removed the assumption that the inputs would keep arriving. Most of the systems I had been calling sustainable did not survive that test. They were *fragile*.

I gave the test a name. In the previous version of this manuscript I called it the zombie apocalypse litmus test of sustainability.

I have a different name for it now — survivability — and a more careful (and serious) framework around it. But the question is the same question. I just asked it for the first time on a couch while watching a fictional farmer feed his chickens industrial feed.

Imagine being this invested in a plot hole in a TV show. Yet, here we are.

DEFINITION

Fragile, Robust, Antifragile

Nassim Taleb's 2012 taxonomy names three distinct responses a system can have to volatility.

A *fragile* system performs well under stable conditions and breaks under stress: concentrated supply chains, high-debt financial instruments, monocultures. A *robust* system withstands stress without breaking but does not improve from it: a well-engineered bridge, a conservative portfolio. An *antifragile* system gains from disorder, strengthened rather than merely preserved by exposure to shock: evolutionary populations under selection pressure, decentralized networks that route around failures, certain biological immune systems.

Most industrial food systems are fragile in Taleb's sense, optimized for efficiency under stable conditions and brittle under the conditions we can now see coming. The survivability axis measures this property directly, and it is the dimension most twentieth-century food-system assessment simply omitted.

◆ THE THREE AXES AND THEIR ASSUMPTIONS ◆

The framework that most sustainability assessments operate within was not handed down as a finished schema. It accumulated. Its current shape — the familiar triad of economic viability, social equity, and environmental integrity — crystallized over roughly two decades of international development and environmental policy conversation, from the Brundtland Commission's 1987 report through the 1992 Rio Earth Summit and into the early-2000s proliferation of corporate sustainability reporting, triple-bottom-line accounting, and the institutional architecture that now supports the field.[2]

The Brundtland Commission's formulation is worth returning to, because the formulation contains the assumption. The commission defined sustainable development as "development that meets the needs of the present without compromising the ability of future generations to meet their own needs."[3] The definition is elegant, widely cited, and quietly dependent on a premise it does not examine: that the conditions under which "needs" are being met remain roughly

stable over time. The future generations in the Brundtland formulation are imagined as living under conditions recognizably continuous with the present, different in degree, perhaps, but not in kind. The framework asks whether a given system can keep going. It does not ask what happens if the ground the system stands on shifts.

This is not a criticism of the Brundtland Commission or of the researchers and policymakers who built the three-axis framework on its foundation. The framework was built during a period when the premise of continuity was reasonable, or at least more reasonable than it has since become. The late twentieth century was, by historical standards, a period of unusual stability in the global systems that food production depends on: fossil energy was cheap and reliably available, international commodity trade was expanding under relatively stable geopolitical arrangements, the Green Revolution's yield gains were still compounding, and climate variability, while present, had not yet produced the frequency and severity of disruption that the early twenty-first century has delivered. The framework was built for the world it was built in, and in that world its omission was less visible.

The three axes, as they came to be operationalized in sustainability assessment, each do real work. The economic axis asks whether the system generates sufficient returns to persist as an economic entity: whether the farm stays solvent, the business model holds, the value chain from producer to consumer functions at each link. The social axis asks whether the system's benefits and costs are distributed in ways that are equitable or at least defensible: whether workers are treated fairly, whether communities bear disproportionate externalities, whether access to the system's outputs is broadly available. The environmental axis asks whether the system's ecological footprint is within bounds that the surrounding ecology can absorb: whether soil is being built or depleted, whether water is being contaminated or conserved, whether biodiversity is being maintained or eroded, whether greenhouse gas emissions are being managed.

Each of these axes captures something real about a food system's relationship to the world it operates in. The scholarship behind each axis is extensive, the metrics have been refined over decades, and the institutional infrastructure for measuring them (from life-cycle assessment protocols to social-impact auditing to the environmental management standards codified in the ISO 14000 series) is among the most developed bodies of applied assessment methodology in existence.[4] The framework is not casual. It is not superficial. It is, within its scope, serious and well built.

Consider what the framework produces in practice. Imagine a mid-sized organic dairy cooperative commissioning a sustainability assessment under the Global Reporting Initiative standards, one of the most widely adopted three-axis reporting frameworks in use.[5] A competent assessment of this kind will document performance across all three axes: the cooperative's member farms are economically viable under the prevailing organic price structure, the social metrics show fair labor practices and positive community economic impact, and the environmental indicators — soil organic matter trends, water quality in adjacent watersheds, biodiversity surveys on member farms — are favorable. A reader encountering the report would conclude, reasonably, that the operation is sustainable. What the report does not document (because the framework it is built on does not ask) is that the cooperative's economic viability depends entirely on the organic price premium, which depends in turn on consumer willingness to pay a substantial markup over conventional milk, which depends in turn on household disposable income remaining stable. The organic dairy sector's recurring price-premium compressions (documented in USDA Economic Research Service reporting on organic milk markets across multiple years of the past two decades) have repeatedly produced the pattern in which farms that scored well on three-axis assessments exited production when the premium narrowed.[6] The three-axis assessment finds the system sustainable. The system's dependency on a single economic condition (the premium) is not part of the assessment, because the framework does not measure dependencies. It measures performance.

The scope, however, has a boundary, and the boundary is the one the Brundtland premise draws. All three axes describe a system's performance under a given set of operating conditions. The economic axis asks whether the system is profitable *given current input costs, current market structures, current policy arrangements.* The social axis asks whether the system distributes its effects equitably *given current labor markets, current community structures, current regulatory environments.* The environmental axis asks whether the system's ecological footprint is sustainable *given current climate patterns, current hydrological regimes, current soil conditions.* Each axis takes the operating conditions as given and evaluates the system's behavior within them. None of the three asks what happens when the operating conditions themselves change.

This is not an esoteric distinction. It is the distinction between asking "is this building structurally sound?" and asking "is this building structurally sound in an earthquake?" The first question is answerable with one kind of engineering assessment. The second requires a different kind, because it asks about behavior under conditions the building was not specifically designed for. Both questions matter. The first tells you whether the building will stand on a calm day. The second tells you something about what the building is made of: what it depends on, where its tolerances are, what breaks first. A sustainability framework that answers only the first question tells you about calm days. It does not tell you about earthquakes. And the 2020s have been, for food systems, a decade of earthquakes.

◆ THE DECADE THAT MADE THE GAP VISIBLE ◆

The assumption of stable operating conditions did not break all at once. It frayed across a series of events that, taken individually, could each be absorbed as an anomaly, and taken together, constituted a pattern that the three-axis framework had no native vocabulary to describe.

The COVID-19 pandemic, beginning in early 2020, exposed the fragility of just-in-time supply chains across the US food system in a matter of weeks. Meat processing — concentrated into a small number of very large facilities, each handling volumes that made the facility a single point of failure for its region — shut down or dramatically reduced throughput as infection spread through workforces.[7] The result was not a gradual adjustment. It was a simultaneous surplus of live animals on farms and shortage of meat in retail outlets, a condition that the supply chain's efficiency had made possible by eliminating the redundancy that would have buffered the shock. At the same time, direct-to-consumer vegetable operations that had built their distribution around restaurant sales and farmers' market attendance lost both channels simultaneously and had to rebuild their customer-facing infrastructure on the fly. The pandemic did not discriminate between conventional and alternative food systems. It discriminated between systems with slack and systems without it, and the decades-long optimization of the US food system for efficiency over redundancy meant that most systems, of whatever kind, had very little slack.

The Texas winter storm of February 2021 added a different lesson. Winter Storm Uri dropped temperatures across the Texas grid to levels that the state's energy infrastructure had not been designed for, producing cascading power failures that lasted days in some areas. Agricultural impacts were immediate and severe: an estimated $600 million in crop losses across the Rio Grande Valley citrus industry, widespread livestock losses in operations without backup power for water and heating systems, and disruption to the cold-chain infrastructure on which the state's export-oriented produce industry depends.[8] The event was not a food-system crisis in the classic sense. It was an energy-system crisis that became a food-system crisis, because the food system's dependence on continuous energy had not been tested against a disruption of this magnitude. The three-axis framework, applied to the Texas agricultural sector before the storm, would have found operations scoring well on economic, social, and environmental metrics. The storm did not change any of those scores. It revealed that the scores were contingent on a condition (reliable grid power) that the scoring had taken as given.

The disruption of Ukrainian grain exports following Russia's invasion in February 2022 made the dependency structure of global commodity agriculture visible at a scale that domestic disruptions had not. Ukraine and Russia together accounted for roughly twenty-eight percent of global wheat exports and a significant share of corn and barley before the invasion. The blockade of Black Sea ports and the destruction of Ukrainian agricultural infrastructure removed a substantial fraction of global grain supply from the market in a matter of months.[9] The price effects rippled through every system that purchased commodity grain as an input. In the United States, feed costs for livestock operations (including the pastured-poultry operations whose vulnerability the opening example described) rose sharply. In North Africa and the Middle East, where dependence on Ukrainian and Russian wheat imports was highest, the price spikes contributed to food-security crises that the affected governments had limited capacity to buffer. The event did not cause a global famine. It demonstrated that the global commodity system on which most food production depends is a single interconnected substrate, and that a disruption at one node propagates to every node that draws on the same supply.

Climate volatility has compounded the pattern. The 2012 US drought — still, at the time of writing, among the most severe and extensive single-year droughts in the modern record of the US Corn Belt — reduced the national corn yield by twenty-

seven percent and pushed commodity prices to levels that imposed real financial stress on every operation in the country that purchased corn as a feed input.[10] The drought was not unprecedented in a climatic sense; comparable events appear in the paleoclimate record and in the instrumental record of the twentieth century. What was new was the degree to which the US food system had been optimized around the assumption of a reliable corn harvest. The system had removed the buffers — the stored reserves, the regional crop diversity, the on-farm feed production — that would have absorbed the shock in an earlier era. The three-axis framework, evaluating the system before the drought, would not have flagged this removal as a sustainability concern. The removal improved efficiency. Efficiency, under stable conditions, looks like a strength. Under unstable conditions, it is the mechanism of transmission.

I do not want to make the 2020s sound unique in kind, because disruptions to food systems are not new. What is new — or newly visible — is the frequency and the compounding. Droughts, energy failures, pandemics, and geopolitical disruptions have always occurred. What has changed is that the food system's optimization for efficiency under stable conditions has simultaneously increased its vulnerability to disruption and reduced its capacity to absorb disruption when it occurs. The system has become better at performing under the conditions it was designed for and worse at surviving conditions it was not designed for. The three-axis framework captures the first quality. It does not capture the second.

The compounding is worth pausing on, because it is the feature that separates the current moment from earlier disruption episodes. In 2022, a pastured-poultry operator in the mid-Atlantic faced not one of these disruptions but several simultaneously: feed costs elevated by the Ukrainian grain-export disruption, diesel costs elevated by the same geopolitical event, labor costs elevated by the lingering effects of pandemic-era workforce contraction, and a customer base whose own purchasing power had been reduced by the inflationary environment all of these inputs were feeding. No single disruption was necessarily fatal. The combination was, for some operations, unmanageable. The system's dependency profile was not a single thread that could snap; it was a web of threads all under tension at the same time, and the three-axis framework had no mechanism for registering the difference between a system with one vulnerability and a system with five vulnerabilities compounding simultaneously.

This is the gap the opening example was designed to make tangible, and it is the gap this chapter proposes to name.

THE FOURTH AXIS: SURVIVABILITY

The framework I want to introduce adds a fourth axis to the three the sustainability literature already uses. I call it survivability, because the word captures precisely what the axis measures: not whether a system functions under current conditions, but whether the system survives when conditions change.

The definition is worth stating with some care, because the precision matters for everything the framework is used for in the chapters that follow.

Survivability, as I use the term, asks two linked questions about any food system under evaluation. First: what does this system require to keep functioning? The requirements include inputs (energy, water, feed, fertilizer, seed, labor, capital, infrastructure), market conditions (commodity prices, consumer demand, distribution networks), policy conditions (subsidies, regulations, trade agreements), and ecological conditions (climate stability, soil fertility, hydrological patterns, pollinator populations). Every food system has a dependency profile: a set of things it needs from outside itself in order to continue operating. The first question maps that profile.

Second: what happens when one or more of those requirements is not met? This is the stress question, and it is the question the three-axis framework does not ask. A system whose dependency profile is narrow (one that requires only conditions the system itself produces or that are reliably available independent of any single external supply chain) scores high on survivability. A system whose dependency profile is broad — one that requires continuous diesel, continuous commodity feed, continuous federal subsidies, continuous just-in-time logistics, continuous stable climate — scores low on survivability, regardless of how well it performs on the other three axes when all of those requirements are being met.

The axis is not binary. It is a continuum, and the continuum has a direction. A system that requires fewer external inputs and can tolerate the loss of any one input without collapsing is more survivable than a system that requires many external inputs and collapses when any one of them is removed. A system that degrades

gracefully under stress (that reduces output without failing catastrophically) is more survivable than one that operates at full capacity until a threshold is crossed and then fails entirely. A system whose dependency profile is composed of conditions that are themselves stable and likely to persist is more survivable than one whose dependency profile is composed of conditions that are volatile, contested, or historically contingent.

The concept is not, in itself, novel. Ecologists have been studying the behavior of systems under stress for decades. C. S. Holling's foundational 1973 paper on ecological resilience drew the distinction between stability (a system's tendency to return to an equilibrium state after a small perturbation) and resilience (a system's capacity to absorb disturbance without shifting to a qualitatively different state).[11] Holling's insight was that the two qualities are not the same, and that a system can be stable (returning quickly to equilibrium after small shocks) while being fragile (collapsing entirely when the shock exceeds a threshold). The distinction has been enormously productive in ecology, and versions of it have been imported into food-systems thinking. Tendall and colleagues, in a 2015 review in *Global Food Security*, defined food system resilience as "the capacity over time of a food system and its units at multiple levels, to provide sufficient, appropriate and accessible food to all, in the face of various and even unforeseen disturbances," and noted that the concept remained underspecified in the applied literature, that the food-systems field had adopted the vocabulary of resilience without having fully operationalized it into assessment frameworks.[12]

Nassim Nicholas Taleb's work on antifragility (the property of systems that gain from disorder rather than merely surviving it) has added a further conceptual layer. Taleb's 2012 *Antifragile* argues that the categories of fragile, robust, and antifragile describe a spectrum of system responses to volatility, and that most modern institutions and supply chains are fragile precisely because they have been optimized for efficiency under stable conditions at the expense of robustness under unstable ones.[13] The argument applies to food systems with obvious force. A concentrated industrial supply chain is the paradigmatic fragile system in Taleb's vocabulary: it performs superbly under the conditions it was designed for and breaks catastrophically under conditions it was not.

Survivability, as I use it, draws on these intellectual predecessors without duplicating any of them. It is narrower than Holling's resilience, which is a property

of ecological systems at scales ranging from a pond to a biome. It is narrower than Taleb's antifragility, which is a property of systems and institutions across domains. It is specific to food systems, and it is designed to function as an axis: a dimension of evaluation that can be applied alongside the existing three, producing a four-axis assessment that captures both a system's current performance and its behavior under changed conditions.

The four axes, stated together, are:

Economic viability: does the system generate sufficient returns to persist as an economic entity under current conditions?

Social equity: does the system distribute its benefits and costs in ways that are equitable or defensible under current conditions?

Environmental integrity: does the system operate within ecological bounds that the surrounding environment can sustain under current conditions?

Survivability: what does the system require to keep functioning, and what happens when those requirements are not met?

Notice the structural difference. The first three axes share a temporal orientation: they describe performance under *current* conditions. The fourth axis asks about performance under *changed* conditions. The first three axes tell you how the system is doing. The fourth axis tells you what the system is made of: what it is standing on, how deep the foundations go, and what happens when the ground moves.

A system can score well on three axes and poorly on the fourth. This is, in fact, the most common configuration in the contemporary US food system. A well-managed industrial grain operation can be economically profitable, can provide employment and community economic activity, and, under best-management practices with cover cropping, reduced tillage, and precision nutrient application, can score reasonably on environmental metrics. It can do all of this while being deeply dependent on continuous fossil energy, continuous synthetic fertilizer derived from natural gas, continuous federal commodity-support payments, and continuous favorable climate for the specific cultivars it has planted. The three axes find the system sustainable. The fourth axis finds it piggybacking on a set of conditions that are historically contingent, geopolitically vulnerable, and climatically uncertain. Both findings are accurate. They are measuring different things.

A system can also score moderately on the first three axes and well on the fourth. A household-scale subsistence polyculture, growing diverse crops and raising a few animals on its own feed with minimal purchased inputs, may not score impressively on economic viability (the returns are low by commercial standards), may have limited social reach (it feeds a family, not a community), and may have an environmental footprint that is positive but modest in scale. On survivability, however, it scores high: it requires little from outside itself, it can tolerate the loss of any single input without collapsing, and it can continue to function under a wide range of conditions because it was never optimized for any single set of conditions in the first place. The three-axis framework would rank it below the well-managed industrial operation on most metrics. The four-axis framework reveals that the two systems have fundamentally different relationships to the future.

◆ WHAT THE AXIS DOES NOT DO ◆

I want to be precise about the limits of the framework, because precision about limits is what distinguishes a tool from a sales pitch.

The survivability axis does not measure moral worth. A system that scores poorly on survivability is not evil; it is structurally vulnerable. The industrial grain system that feeds most of the planet scores poorly on survivability for reasons that are structural and historical, not because anyone set out to build a fragile system. The fragility is a side effect of the optimization for efficiency, and the optimization for efficiency was, in its historical context, a rational response to the problems it was designed to solve: feeding a rapidly growing global population at declining real cost. The system succeeded at what it was built to do. Its survivability deficit is a feature of its design, not evidence of malice.

The survivability axis does not predict specific failures. It identifies dependency profiles and maps vulnerability, but it does not tell you which specific disruption will occur or when. A system that depends on continuous diesel, continuous commodity corn, continuous federal subsidies, and continuous stable climate has four distinct vulnerability surfaces. The axis can identify all four. It cannot tell you whether the disruption will come from diesel, corn, subsidies, or climate, or whether it will come from something the analysis did not anticipate. The axis is a diagnostic tool, not a

crystal ball. It tells you where the fault lines are. It does not tell you when the earthquake comes.

The survivability axis does not produce a single numerical score. Attempts to reduce complex systems to single scores are, in my experience, more useful for marketing than for understanding. The axis produces a profile: a description of what the system depends on, how deep each dependency runs, and how the system behaves when each dependency is stressed. The profile can be compared across systems, and the comparison is informative, but the comparison is qualitative and contextual rather than a simple ranking. A system that depends on diesel is vulnerable in a different way than a system that depends on federal subsidies, and collapsing both vulnerabilities into a single number would obscure more than it reveals. The framework is a tool for thinking, not a scoreboard.

The framework is also not personally branded. I developed the core of this argument in a 2016 essay, where I proposed that sustainability assessments should include survivability as a dimension and sketched the evaluative logic that the current formulation extends. The essay named the framework with my own name, which I no longer think was the right choice. The framework is a tool. Tools are described by what they do, not by who made them. What matters is whether the four-axis assessment produces useful distinctions that the three-axis assessment does not. I believe it does, and the next two chapters are the test of that belief.

FIELD NOTE

Testing the Score Against What I Liked

I tested the framework on the systems I liked, and that part is not in the chapter body.

The week I finished the 2016 essay, I scored monocultures, and the result is the one the REVISION sidebar in the next chapter records. Then, deliberately, I turned the framework on the cases I had been writing about with admiration. I scored Allan Savory's holistic-management work. I scored rewilding, in the form the literature was using at the time. I needed to know whether the framework was a piece of analysis or a way of dressing up disdain in quantitative clothing. I needed to know whether it would punish the systems I disliked while flattering the ones I admired. If it was the latter, it was not worth keeping.

It cut both ways. Some of what I admired held up. Some did not. The cases that did not are part of why this book exists in the form it does: the conversation with Savory that Chapter 7 records, the conflicting views about Polyface that close out Chapter 6. The framework had to be willing to find against my favorites for me to trust it. It did. I am still using it.

WHAT THE FRAMEWORK REVEALS

The value of a framework is in the distinctions it makes visible. The three-axis framework makes visible whether a food system is currently functioning well: whether it is profitable, equitable, and ecologically responsible under the conditions it is operating in. These are real and important distinctions. The four-axis framework makes visible an additional distinction: the difference between a system that functions well under current conditions because it is well designed and a system that functions well under current conditions because the conditions happen to be favorable.

This distinction matters because conditions change. They have always changed — climate has oscillated, trade routes have opened and closed, energy regimes have shifted, empires have risen and collapsed and taken their agricultural systems with them. What is different now is not that conditions change but that the food systems we have built are more sensitive to change than most of their predecessors, because they have been more thoroughly optimized for a specific set of conditions than most of their predecessors. The global commodity system is the most efficient food-production apparatus in human history. It is also, by the metric of survivability, among the most dependent on the persistence of its specific operating conditions. The efficiency and the dependency are the same feature, viewed from different angles.

The chapters that follow apply the framework. Chapter 5 takes the two dominant industrial food systems — industrial feedlot animal production and industrial grain monoculture — and runs them through the four-axis assessment in detail, with the survivability axis producing findings that the three-axis assessment alone would not. Chapter 6 has already done the same for regenerative agriculture, and the finding there is instructive in a different way: the system scores substantially better than industrial production on the first three axes and meaningfully better on the fourth, but the survivability score is still constrained by the structural ceiling the chapter

identified: the dependency on industrial commodity feed for monogastric animal production, which is a survivability deficit embedded in the system's relationship to its inputs rather than in its own management practices. The framework does not declare winners and losers. It maps the dependency profiles and shows where the vulnerabilities live. What the reader does with the map is the reader's decision.

The work of Part II of this book, which this chapter opens, is to use the framework and see what it shows. The survivability axis is the analytical addition the book proposes. It is not the book's conclusion. The conclusion lives in Part III, where the framework is brought to bear on a set of food-system possibilities that the current sustainability conversation has mostly overlooked: possibilities that score well on all four axes, including the fourth, because their dependency profiles are structured differently from anything the dominant food system currently offers.

There is one further point worth making before the framework is put to use, because it bears on how the rest of the book's argument is constructed. The survivability axis does not exist to condemn any existing food system. It exists to ask a question that the existing assessment framework does not ask, and to make visible a set of distinctions that the existing framework cannot see. An industrial operation that scores poorly on survivability is not thereby a failure. It may be, under current conditions, the most productive, most affordable, most efficient food-production system available. The survivability axis does not erase those achievements. It adds a dimension to the assessment: a dimension that asks whether the current conditions on which those achievements depend are conditions we can count on persisting. If they are, the three-axis assessment is sufficient and the fourth axis adds nothing of practical consequence. If they are not — if diesel prices are volatile, if commodity supply chains are geopolitically exposed, if climate patterns are shifting in ways that affect crop yields, if the policy environment that subsidizes current production is politically contestable — then the fourth axis is measuring something the first three cannot, and the measurement matters.

REVISION

On Discovering Resilience Thinking

Sometime after I had written the four-axis framework as it appears in this chapter, a reader pointed me at C. S. Holling's 1973 paper on ecosystem resilience, then at Walker and Salt's *Resilience Thinking* (2006), then at the broader literature

developed by Folke, Berkes, Ostrom, and others over fifty years. I had not arrived at the framework through that field. I arrived at it through certification work, and through asking, repeatedly, which of the food systems I had been writing standards for would still be food systems the day after their inputs got harder to count on.

It would have been a better book had I known about the resilience-thinking field when I started. I did not. The framework here was developed independently, and that fact does not shield me from the criticism that I am reinventing wheels with my own labels.

What I will say is this. When independent thinkers, working in different fields, with different prompts, arrive at the same structural observation — that systems described well at one moment in time can be described badly across changes in their conditions — the convergence itself is evidence. Evidence that the observation is not a quirk of one camp's framing. Evidence that something is missing in the framework most of us have been using. The reader should pause at the convergence.

I owe the resilience-thinking field a careful read. I will not pretend to have done that read in time for this book. Where my framework and theirs say the same thing, I want to be clear that they were there first. Where they say different things, I want to be clear about what the difference is, and I do not yet know enough to do that work in this volume. It is the next thing I owe, and I will write it.

The framework is a lens. The next chapters are what the lens is pointed at.

CHAPTER FIVE

Scoring the Industrial System

♦ ♦ ♦

The previous chapter introduced a framework. This chapter puts it to work.

The four-axis assessment was designed to do something the three-axis model cannot: to distinguish between a food system that functions well under current conditions and a food system that functions well under current conditions because those conditions happen to be favorable. The distinction matters only when the framework is applied to specific systems and the application produces specific findings. Abstractions do not feed anyone. The question is what the framework shows when it is pointed at the systems that actually feed most of the people reading this book.

Two systems dominate the American food landscape, and between them they account for the overwhelming majority of calories and protein consumed in the United States. The first is industrial feedlot animal production, the confined animal feeding operation (CAFO), which finishes roughly ninety-nine percent of the meat animals sold in US retail markets on concentrated grain rations in confined facilities.[1] The second is industrial grain monoculture: the corn, soy, and wheat production system that supplies both the feed for the first system and the caloric base for most processed food consumed in the country. They are structurally linked. The feedlot depends on the grain monoculture for its feed inputs. The grain monoculture depends in turn on the feedlot, on the export market, and on the ethanol mandate for its demand. Neither system, in its current form, would exist without the other. Scoring them separately is an analytical convenience; in practice they are two faces of a single industrial substrate.

The chapter takes each system through all four axes in turn. The work is analytical, and I want to be clear about what that means. The findings are structural, not moral. A system that scores poorly on a given axis is not thereby condemned. It is described: its dependency profile mapped, its vulnerabilities identified, its genuine achievements registered alongside its genuine costs. The framework is a diagnostic instrument. It is not a verdict.

REVERSAL

The Strongest Case for the Industrial System

The strongest version of the case for the industrial system goes like this.

Nine billion people, by mid-century. A carrying capacity question that preindustrial agriculture has never met. Norman Borlaug's Green Revolution averted famine at a scale no other twentieth-century intervention can claim. Industrial grain yields per acre are several times what the same land produced a century ago.

The system, whatever its externalities, has fed more people per unit of land and per unit of labor than any system in human history. To propose replacing it with anything less productive is to propose, by arithmetic, that some fraction of those nine billion does not eat.

This is not a straw man. It is the argument the FAO has made in its food-security publications. It is the argument made in peer-reviewed journals by economists at land-grant universities whose work I respect. I had to take it seriously before I could write a book that proposes the system has to change.

My answer is in the chapters that follow. But the argument deserves to be heard at full strength before the rebuttal, because the consequence of being wrong in the other direction is famine, and that is not a small matter.

REVISION

On Monocultures

2012: I treated monocultures as a moral fact. They were destructive. They were unsustainable. There was no version of the case for them I considered worth taking seriously. After all, there was no such thing as a closed-loop agricultural system.

2016: I scored monocultures of animals and grains on the four-axis framework of survivability. I came up with the concept that year, when I still felt I had something to prove, and called it the *Pendergrass Score of Survivability*. They scored abysmally. My disdain for monocultures continued. The name did not survive. The framework did.

2017: I read Matt Ridley's *The Rational Optimist*. Ridley took seriously a case I had been refusing to take seriously: that yields per acre are not incidental, that what is required to feed a population at industrial scale is a real constraint and not a propaganda line, that the people working on technological improvement of agricultural systems are not, mostly, villains. I had been arguing against a caricature. And, most importantly, that over time these systems will inevitably improve. That is the benefit of time × human ingenuity.

2018: Monocultures have serious and structural sustainability problems — the four-axis framework lays them out. But they also have meaningful potential for improvement under cover-cropping, reduced-tillage, perennial-integration, precision-input, and microbial-soil-management regimes. The case I had to take seriously was not "monocultures are fine." It was "the strongest argument for monocultures is not the one I was attacking."

This evolution of thought matters for the chapter that follows because the scoring is structural, not moral. Even though I softened my stance against monocultures, industrial grain monoculture scores poorly for specific reasons that the four-axis framework makes legible. That is a different argument from "monoculture is evil."

The binary position of 2012 is a reflection of where I sat on the Dunning-Kruger curve at the time. And, while holding space for two conflicting ideas at the same time, it is clear that this chapter still owes a great deal to ideas from more than a decade ago that held strong.

◆ INDUSTRIAL FEEDLOT ANIMAL PRODUCTION ◆

Economic Viability

At the point of sale, the industrial feedlot system is an economic success by the metric most consumers encounter: price. Inflation-adjusted retail prices for beef, pork, and chicken in the United States have declined substantially over the last half

century. The USDA's Economic Research Service tracks this in real terms: between 1960 and 2020, the inflation-adjusted retail price of chicken fell by roughly forty percent, and beef and pork prices have remained lower in real terms than they were at mid-century peaks, despite periodic spikes driven by drought, disease, or commodity-market disruption.[2] For a consumer standing in a grocery store, the system delivers protein at a price point that most household budgets can absorb. This is not a trivial achievement. Cheap animal protein has been one of the defining features of the American diet for two generations, and the system that produces it has been remarkably effective at the narrow task of minimizing per-unit cost at the point of purchase.

The point-of-purchase price, however, is not the full economic accounting, and the gap between the two is where the system's economic picture changes. The Pew Commission on Industrial Farm Animal Production, an independent panel convened by the Pew Charitable Trusts and the Johns Hopkins Bloomberg School of Public Health, published its comprehensive assessment in 2008, and its finding on this point was direct: the retail price of industrial animal products does not reflect the external costs the system imposes on public health, environmental quality, and rural economies.[3] The externalities are real, they are large, and they are borne by parties other than the producer or the consumer at the point of sale.

The antibiotic-resistance externality is among the most economically consequential. A 2019 review published by the Centers for Disease Control and Prevention estimated that antibiotic-resistant infections cause more than 2.8 million illnesses and more than 35,000 deaths annually in the United States, with associated healthcare costs and productivity losses running into the tens of billions of dollars.[4] The connection to agricultural antibiotic use is not casual. The FDA reported that in 2021, approximately sixty-six percent of medically important antibiotics sold in the United States were sold for use in food-producing animals, administered not primarily to treat clinical disease but to promote growth and prevent illness in the high-density conditions that confined operations require.[5] The mechanism by which sub-therapeutic antibiotic use in livestock selects for resistant bacteria that subsequently affect human health has been documented across multiple peer-reviewed studies and was identified as a significant public health concern by the World Health Organization as early as 2001.[6] The cost of that resistance is external-ized. It does not appear on the price tag of the chicken breast or the ground beef, but

it appears in the healthcare system, in the insurance premiums, and in the reduced efficacy of antibiotics that were once reliable clinical tools.

Pollution remediation adds a second layer of externalized cost. The EPA estimated in 2003 that animal feeding operations were contributing to water quality impairment in more than twenty thousand miles of rivers and streams and had contaminated groundwater in seventeen states.[7] The cleanup costs for that contamination are borne by municipalities, by taxpayers, and by downstream agricultural and recreational operations rather than by the CAFOs whose waste produced the contamination. Property values in communities adjacent to large-scale confined operations decline measurably — studies in North Carolina, Iowa, and Minnesota have documented decreases ranging from six to eleven percent in residential property values within a three-mile radius of a CAFO, with effects persisting for years after the operation's establishment.[8]

Federal subsidies add a further economic distortion. The commodity support system, administered through the farm bill and its various iterations since the 1970s, has kept the price of corn and soybeans below the full cost of production for most of the last four decades, and those below-cost commodity grains are the primary input for confined animal feeding. Researchers at Tufts University calculated that between 1997 and 2005, the industrial livestock sector saved an estimated $35 billion through below-cost feed made possible by federal commodity support payments.[9] That saving is not an efficiency gain in any meaningful economic sense. It is a transfer from the taxpayer to the industrial system, and it renders the retail price of industrial meat artificially low relative to what the system actually costs to operate.

The economic axis, then, produces a split finding. At the point of sale, the system scores well: it delivers affordable protein to consumers. When the externalities are accounted — healthcare costs from antibiotic resistance, pollution remediation, property value losses, and taxpayer-funded commodity subsidies — the economic picture reverses. The system's apparent efficiency is substantially subsidized by costs it does not pay.

Social Equity

The social axis asks whether the system distributes its benefits and costs equitably or defensibly. For industrial feedlot production, the distribution is sharply asymmetric.

Worker conditions in meat processing have been documented extensively. The Government Accountability Office reported in 2005 that meatpacking workers experienced injury and illness rates substantially above the national average for manufacturing, with the Bureau of Labor Statistics recording injury rates roughly two and a half times the average across all private industry.[10] The COVID-19 pandemic made the workforce conditions newly visible: the CDC documented infection clusters at 115 meat and poultry processing facilities across 19 states in the first two months of the pandemic alone, with facility-level attack rates that reflected the combination of close working proximity, high line speeds, and inadequate protective measures.[11] The workers in these facilities are disproportionately immigrant, disproportionately non-English-speaking, and disproportionately employed under conditions where reporting injuries or unsafe practices carries economic and sometimes legal risk. The social cost of industrial meat processing is concentrated in the workforce that makes the system operate.

Community health impacts extend beyond the processing facilities. Residents of communities adjacent to large-scale confined operations report elevated rates of respiratory symptoms, headache, nausea, and psychological distress in epidemiological studies conducted across North Carolina, Iowa, and other states with high CAFO concentrations.[12] The communities bearing these health effects are disproportionately rural, disproportionately low-income, and disproportionately communities of color, a pattern that has led environmental justice researchers to identify CAFO siting as a significant environmental equity concern.[13]

The displacement of family farmers is a structural feature of the system's economic logic. Vertical integration in the poultry and pork industries has consolidated production into contract-grower arrangements in which the processor owns the animals, the feed formula, and the processing relationship, while the grower owns the capital infrastructure: the houses, the land, the debt. Congressional Research Service analyses have documented that the number of independent hog farms in the United States declined from more than 660,000 in 1980 to fewer than

67,000 by 2012, while total hog production increased over the same period.[14] The consolidation was not the elimination of pork production. It was the elimination of the people who had been doing it, replaced by a smaller number of larger-scale contract operations producing more animals at lower margins under tighter corporate control. The social cost of that consolidation has been documented by rural sociologists for decades, and the industry itself does not seriously contest it. It shows up in rural depopulation, in the loss of small-town economic activity, and in the civic infrastructure that contracts when the people who supported it leave.

The social axis scores net negative: the system's benefits flow primarily to consumers in the form of cheap protein and to the corporate entities that control the processing and distribution chain; its costs are concentrated in workers, in adjacent communities, and in the rural regions whose agricultural economies the system has restructured.

Environmental Integrity

The environmental axis is where the industrial feedlot system's costs are most extensively documented and least disputed.

Waste management is the entry point. The EPA's regulatory definition of a large CAFO, an animal feeding operation confining more than 1,000 animal units at 1,000 pounds each, means that a single large cattle feedlot can produce waste volumes equivalent to a small city. The USDA estimated in 2009 that confined animal operations in the United States produce approximately 500 million tons of manure annually, roughly three times the volume of human sanitary waste produced by the entire US population.[15] That waste is typically stored in anaerobic lagoons or applied to surrounding cropland at rates that frequently exceed the land's absorptive capacity for nitrogen and phosphorus. When the application exceeds the capacity, the surplus runs off into surface water.

The water quality consequences are measurable at continental scale. Nutrient runoff from agricultural operations, of which confined animal waste is a leading component, is the primary driver of the hypoxic zone in the Gulf of Mexico, a seasonal dead zone that covered approximately 6,334 square miles in 2024 according to NOAA survey data, making it one of the largest areas of oxygen-depleted water in the world.[16] The Chesapeake Bay, closer to the concentrated poultry production of

the Delmarva Peninsula, has experienced a parallel pattern of nutrient-driven hypoxia that has degraded its fisheries for decades.

The numbers are real. What they describe is also visible. Louisiana shrimpers working the waters off the Atchafalaya Delta have watched the dead zone's expansion reshape their industry over three decades. In peak hypoxia years — midsummer, after the spring flush of nitrogen from the Mississippi watershed reaches the Gulf — trawlers that once worked nearshore waters are pushed miles further out, burning more diesel to reach the oxygenated water where the shrimp still congregate. The catch does not disappear. It migrates, and the boats follow it, and the economics tighten. In 2023, the dock-side price for Gulf brown shrimp hit its lowest point in over a decade, driven in part by import competition but compounded by the operational costs of fishing around a dead zone the size of Connecticut. The hypoxic zone is not an abstraction measured only by NOAA research vessels. It is a feature of the working landscape that the people who fish the Gulf navigate around every season, and its growth tracks the nutrient load from the agricultural system upstream with a fidelity that the science has been documenting since the early 1990s.

Greenhouse gas emissions from confined animal production are significant in the global accounting. The FAO's widely cited 2013 assessment estimated that global livestock supply chains contribute approximately 14.5 percent of all anthropogenic greenhouse gas emissions, with the confined production segment contributing disproportionately through enteric methane from cattle, nitrous oxide from manure management, and carbon dioxide from feed crop production and transportation.[17] The feedlot system, specifically, contributes on the production side through methane from enteric fermentation and manure storage, and on the input side through the full emissions profile of the commodity grain system that feeds it: the diesel for planting, harvesting, and transporting the corn, the natural gas for synthesizing the nitrogen fertilizer the corn requires, and the emissions from the land-use practices under which the corn is grown.

The system sequesters no carbon. Feedlot pens are bare dirt or concrete. The land on which the animals stand contributes nothing to soil carbon accumulation. The land from which the feed is grown, under annual row-crop cultivation with conventional tillage, is losing soil carbon rather than gaining it, as the next section documents.

The environmental axis scores net negative across every metric the framework measures: water pollution, greenhouse gas emissions, soil degradation via feed production, and zero contribution to carbon sequestration.

Survivability

The survivability axis asks what the system requires to keep functioning and what happens when those requirements are not met. For industrial feedlot production, the dependency profile is broad and deep.

The system requires continuous commodity grain inputs. A single large cattle feedlot finishing 30,000 head per year requires, at roughly six to eight pounds of grain per pound of weight gain over the finishing period, millions of pounds of corn annually.[18] That corn must be available, at a price the feedlot can absorb, delivered on a schedule that matches the metabolic demands of the animals currently on feed. A disruption to the corn supply — drought, export-market competition, ethanol-mandate pressure on available supply — translates directly and immediately into financial stress on the feedlot.

The system requires continuous federal subsidies for its feed inputs. As the economic section documented, below-cost commodity grain has been the structural condition of the feedlot's profitability for decades. Without the commodity-support system, feed prices would reflect the full cost of production, and the feedlot's margins, already thin in most years, would narrow further or invert. The policy structure is the floor the system stands on.

The system requires continuous diesel for transportation. Commodity grain moves from the Corn Belt to feedlots across the country by truck, rail, and barge. Finished animals move from feedlots to a small number of large processing facilities. Processed meat moves from those facilities to distribution centers and retail outlets nationwide. Each link in this chain requires petroleum-derived fuel, and the chain is long. A sustained increase in diesel prices propagates through every link.

The system requires continuous antibiotics for animal health at the stocking densities the system operates at. Confining thousands of animals in close quarters at the stress levels inherent to the feedlot environment creates disease pressure that sub-therapeutic antibiotic administration manages. Remove the antibiotics and the

disease pressure reasserts itself at densities that make clinical treatment of individual animals economically unfeasible.

Remove any one of these requirements — grain supply, federal subsidies, diesel, antibiotics — and the system cannot continue to function at its current scale and cost structure. This is not a prediction of imminent failure. It is a description of the system's dependency profile. The system operates successfully because all four of these conditions are currently being met. Its survivability score reflects what happens when any one of them is not.

The survivability axis scores very low, not because the system is poorly managed, but because the system's design requires continuous external inputs that are historically contingent, geopolitically exposed, and, in the case of climate-dependent grain supply, increasingly volatile.

◆ INDUSTRIAL GRAIN MONOCULTURE ◆

The second system is, in a structural sense, the first system's substrate. Industrial grain monoculture — the large-scale, chemically intensive production of corn, soybeans, and wheat that dominates the American agricultural landscape — supplies the feed for the feedlot system, the caloric base for most processed food, and the raw material for the ethanol industry. It covers roughly 250 million acres of US cropland, and the three crops together account for the majority of planted acreage in the country.[19]

Economic Viability

The economic picture for industrial grain monoculture mirrors the feedlot's in structure: positive at the point of sale, substantially different when externalities are accounted.

For centralized producers — the large-scale operations with enough acreage to capture economies of scale in equipment, input purchasing, and commodity marketing — the system is profitable in most years, though the margins are sensitive to commodity prices, input costs, and the federal support structure. The farm bill's commodity support programs — direct payments, counter-cyclical payments, crop insurance subsidies, and their successive reformulations — have provided a floor

under commodity grain profitability for decades, smoothing the volatility that would otherwise make large-scale monoculture economically unsustainable in low-price years.[20]

For family-scale operators and local economies, the picture is different. The consolidation of US grain production has followed the same trajectory as livestock production: fewer, larger operations producing more output. The number of farms in the United States has declined from approximately 6.8 million in 1935 to roughly 2 million today, while average farm size has more than tripled, and the majority of production value is now concentrated in a small fraction of operations.[21] The economic benefits of the system flow disproportionately to the largest operators and to the input suppliers, processors, and commodity traders that the system depends on. Rural communities in the Corn Belt and the wheat-growing regions of the Plains have experienced the same pattern of depopulation, retail contraction, and civic thinning that the livestock consolidation produced in poultry- and pork-producing regions.

The system delivers cheap calories to urban consumers. The calories come from corn syrup, soybean oil, and refined wheat flour, and their prices have kept food expenditure as a share of household income at historically low levels. This is a real benefit to household budgets. It is also a benefit whose costs appear elsewhere: in the healthcare system, where diet-related chronic disease driven in part by the overconsumption of cheap refined carbohydrates and seed oils imposes costs that the National Institutes of Health has estimated at hundreds of billions of dollars annually; in the environmental costs documented below; and in the rural economic hollowing the consolidation has produced.[22]

Social Equity

The social axis for grain monoculture is structurally similar to the feedlot case: the system's benefits are distributed broadly to consumers in the form of cheap food, and its costs are concentrated in the communities where the production occurs.

Rural communities in intensive monoculture regions bear the environmental health burden: pesticide exposure, nitrate contamination of drinking water, respiratory effects from dust and chemical drift. They bear it without capturing a proportional share of the economic value the production generates. The value is

captured at the processing and distribution stages, which are physically and economically distant from the fields. Farm labor in large-scale grain operations has shifted from a human workforce to a mechanical one; the tractor, the combine, and the chemical application system have replaced the hands, and the people those hands belonged to have, for the most part, moved elsewhere.

The displacement narrative for grain monoculture is quieter than for livestock. There are no CAFO siting controversies, no concentrated processing facilities with high-visibility labor conditions. The underlying structural change is the same. Fewer people are needed to produce more grain, and the communities that were organized around the people are contracting as the people leave.

Environmental Integrity

The environmental costs of industrial grain monoculture are extensive, well documented, and, in some dimensions, cumulative in ways that compound across decades.

Topsoil loss is the foundational concern. David Pimentel, whose career at Cornell established much of the empirical basis for energy-input analysis in agriculture, estimated that US cropland loses topsoil at an average rate of roughly ten times the natural soil-formation rate, with annual erosion rates on conventionally tilled cropland averaging approximately 1.8 billion tons across the country.[23] The Natural Resources Conservation Service's more recent modeling using the Revised Universal Soil Loss Equation places the figure at roughly four to five tons per acre per year on average for conventionally managed cropland, against a natural formation rate of roughly half a ton per acre per year.[24] The loss is quiet. It does not produce dramatic events in most years. It produces a slow, cumulative thinning of the soil horizon that reduces the land's productive capacity over generational timescales. The soil that took centuries to build is being removed in decades.

Biodiversity reduction is a direct consequence of the monoculture model. A cornfield is, by design, one species of plant over a very large area. The organisms that would normally occupy the ecological niches of a diverse plant community — the insects, the birds, the soil microbiota, the mycorrhizal fungi — are either absent or suppressed by the chemical regime required to maintain the monoculture. Tilman and colleagues' landmark 2002 paper in *Nature* documented that intensification of

agricultural production practices was the single largest driver of biodiversity loss in terrestrial ecosystems globally, operating through habitat conversion, chemical inputs, and the simplification of plant communities from diverse assemblages to single-species stands.[25]

Groundwater contamination from nitrogen fertilizer and pesticide leaching is a chronic and geographically extensive problem. Nitrate contamination of drinking water from agricultural sources affects millions of Americans, particularly in rural areas overlying the shallow aquifers of the Corn Belt and the Great Plains. The USGS National Water-Quality Assessment has documented nitrate concentrations exceeding the EPA's maximum contaminant level in a significant fraction of shallow groundwater wells in agricultural areas, with the highest concentrations correlated with the most intensive corn and soybean production.[26]

Mycorrhizal damage from fungicide application is a less visible but ecologically significant consequence. Mycorrhizal fungi, the symbiotic networks that connect plant root systems to soil nutrient pools and to each other, are suppressed by the fungicide applications that modern grain production relies on for disease management. The disruption of mycorrhizal networks degrades the soil's capacity to cycle nutrients, retain water, and support the biological activity that builds soil organic matter over time. The consequence is a system that becomes progressively more dependent on synthetic inputs to replace the biological functions it has disabled, a dependency spiral that the environmental literature has documented across multiple cropping systems.[27]

Nutrient runoff from grain monoculture is the leading cause of non-point source water pollution in the United States. Synthetic fertilizer is applied at rates calibrated to maximize yield in an average year, which means applied in excess of what the crop uses in any below-average year. The nitrogen and phosphorus that the crops do not absorb moves through the soil profile into groundwater and across the surface into streams, rivers, and eventually into the coastal waters where it drives the eutrophication and hypoxia described in the feedlot section. Foley and colleagues, in their 2011 *Nature* paper surveying the environmental footprint of global agriculture, identified nutrient management in intensive grain systems as one of the highest-priority leverage points for reducing agriculture's environmental impact.[28]

Heavy-metal contamination is a quieter dimension of the same inventory, and it is worth slowing down for. The reader does not have to be a toxicologist to follow the argument; the test the framework cares about is whether each metal accumulates faster than the system can clear it. Most of them do.

Phosphate rock carries cadmium as a co-occurring trace element, and the soils that have been receiving phosphate fertilizer for half a century carry a legacy cadmium load that moves into the food chain through plant uptake. Arsenic entered US cropland through approved organoarsenical poultry-feed additives whose drug approvals the FDA withdrew after 2011, with decades of accumulated soil arsenic still in place. Lead persists in the upper soil horizons of much of the Corn Belt as legacy load from the leaded-gasoline era. Nickel cycles into agricultural soils through phosphate fertilizers, manure, and atmospheric deposition. The metals do not move through the system on the same timescales as nitrogen does. They accumulate. The soil that is being thinned by erosion is also, where it remains, slightly more contaminated with each crop.[29]

Nickel deserves a specific note. Nearly every pathogen on the WHO's antimicrobial-resistance priority list — *Helicobacter pylori,* virulent *Escherichia coli, Klebsiella pneumoniae,* and others — uses nickel-dependent virulence factors to establish and maintain infection. Nickel itself has no established essential role in human nutrition. The asymmetry — essential for pathogens, nonessential for hosts — is what makes agricultural nickel accumulation a microbial-system question rather than a contamination question alone, and it is one of the inputs the survivability axis is measuring that the three-axis framework is not.[30]

The connection to the mycorrhizal disruption noted above is not incidental. The soil microbial communities that mediate metal speciation and bioavailability — the communities that determine whether a given soil cadmium load reaches a wheat kernel or stays bound — are the same communities the fungicide regime suppresses. The chapter has space here only to name the connection and to score the axis. The full pathway-by-metal exposition, with the primary literature, is in the Technical Appendix at the back of the book; readers who want to follow the chemistry into the soil microbiology should turn there.[31]

The environmental axis scores net negative: topsoil loss, biodiversity reduction, groundwater contamination, mycorrhizal disruption, heavy-metal accumulation, and nutrient runoff as the nation's leading non-point source water pollutant.

FIELD NOTE

The Rabbit Hole

What started as one question — what did my chronic conditions have in common — has not stopped asking the next question for sixteen years.

The first question opened onto food. What I was eating, and what I had been told I was eating. The second question opened onto agriculture. What was being fed to the animals I was eating, and what that feed was grown in. The third opened onto soil. What the soil was, and what the agricultural systems had done to it. The fourth opened onto microbes. The soil microbiome and the human gut microbiome are not, on close inspection, the distinct and unrelated systems I had assumed they were. They are parts of the same continuum, exchanging metabolites, signals, and loads through every meal.

The fifth opened onto heavy metals. The same agricultural systems that simplified the soil microbial communities had been concentrating cadmium, arsenic, lead, and nickel in those soils for decades. The metals do not just sit there. They are mediated, sequestered, transported, and sometimes mobilized by the soil microbial communities the fungicide regime has been disrupting for the same decades. They reach plants. Plants reach plates. The same gut microbiomes the agricultural regime has been simplifying are also the ones doing the most consequential work on those metals once the metals arrive.

This is where my interest currently sits, and it has pulled me harder than I expected. The certification work the Field Note in Chapter One named — Paleo Approved, then Certified Paleo — is now organized around heavy-metal testing under a label called HMTc. The Afterword names where the rest of that work has gone.

The book in your hands is built on the first four questions. The fifth has its own work, in progress.

Survivability

The survivability axis for grain monoculture reveals a dependency profile that is structurally parallel to the feedlot's.

The system requires continuous chemical inputs. Synthetic nitrogen fertilizer, produced from natural gas through the Haber-Bosch process, is the single largest energy input in grain production. Pimentel and Pimentel's analysis in *Food, Energy, and Society* documented that roughly forty percent of the total energy input in corn production is accounted for by nitrogen fertilizer alone.[32] Without it, corn yields fall to a fraction of their current levels, because the monoculture regime has depleted the biological nitrogen-fixation capacity that diverse systems maintain through legume rotation and mycorrhizal function. The system has substituted a chemical input for a biological process, and the substitution is not reversible on a short timeline.

The system requires continuous fossil fuel for field operations and irrigation. Every pass of the tractor — planting, cultivating, applying chemicals, harvesting — requires diesel. In the western Corn Belt and the Great Plains, where rainfall is insufficient for reliable yields, irrigation from the Ogallala Aquifer and other groundwater sources requires energy for pumping, and the Ogallala itself is being drawn down at rates that exceed recharge in much of its extent, making the water supply a diminishing resource rather than a renewable one.[33]

The Ogallala drawdown is worth pausing on, because it is the survivability axis made physically legible. The aquifer underlies approximately 174,000 square miles across eight states from South Dakota to the Texas Panhandle, and it accumulated its water over millions of years, recharged slowly by precipitation filtering through the High Plains soils above it. In southwestern Kansas — Grant, Stanton, and Haskell counties, the heart of the irrigated corn and feedlot corridor — the saturated thickness of the aquifer has declined by more than sixty percent since large-scale pumping began in the 1950s. The Kansas Geological Survey has documented wells that produced reliably for decades going dry or dropping below the threshold at which irrigation is economically viable, and the projection for much of the southwestern Kansas portion of the aquifer is functional depletion within a generation.[^30a] This is not a speculative scenario. It is a measurement. Farmers in the region can tell you the rate their water table is dropping; many of them track it year to year. The arithmetic is not ambiguous. The system that depends on this water

is drawing down its supply faster than geology can replace it, and when the water is gone, the irrigated agriculture it supports goes with it. The dependency the survivability axis measures is, in this case, a physical quantity with a known depletion rate and a calculable endpoint.

The system requires continuous federal subsidies. Commodity grain production in the United States has operated under federal price support, crop insurance subsidy, and conservation-compliance payment structures for decades. These programs smooth the volatility of commodity markets and keep production profitable in years when market prices fall below the cost of production. Without the subsidy structure, the marginal operations, those farming the least productive land at the highest per-acre cost, would exit production, and the system's total output would contract.

The system requires continuous favorable climate for predictable yields. The cultivars planted across the Corn Belt are selected for performance under a specific range of temperature, moisture, and growing-season conditions. Deviations from the expected range produce yield losses that propagate through the entire commodity system. The 2012 drought reduced the national corn yield by roughly twenty-seven percent in a single season.[34] Climate projections for the central United States indicate increasing frequency and severity of growing-season heat stress and drought stress over the coming decades, which means the climate condition the system was optimized for is itself becoming less reliable.[35]

Remove any one of these — chemical inputs, fossil fuel, federal subsidies, favorable climate — and the system cannot continue to produce at its current scale and cost structure. As with the feedlot system, this is not a prediction of when the disruption occurs. It is a mapping of the dependency profile. The system's survivability score reflects the breadth and depth of its dependencies, each of which is contingent on conditions that are volatile, contested, or trending in directions unfavorable to the system's continuation in its current form.

◆ THE IMPROVEMENT ACKNOWLEDGMENT ◆

I want to register something that the axis-by-axis scoring, taken alone, might obscure.

Industrial grain monoculture is not a fixed technology. It is a set of practices, and the practices are changing. Cover cropping means planting a secondary crop between cash-crop seasons to hold soil, fix nitrogen, and maintain mycorrhizal networks. The practice has expanded substantially in the United States over the last decade, and the evidence for its benefits across multiple dimensions is strong and growing. The USDA's Census of Agriculture reported that cover crop acreage in the US increased from roughly 10 million acres in 2012 to over 18 million acres by 2022, a pace of adoption that suggests the practice is moving from the margins of the system toward something closer to standard practice in progressive operations.[36] Reduced tillage and no-till systems, which leave crop residue on the soil surface rather than turning it under, reduce erosion rates dramatically and improve soil biological activity. Integration of perennial vegetation into row-crop systems — buffer strips, alley cropping, silvopasture adjacent to grain fields — can restore some of the habitat complexity and biological function that pure monoculture eliminates. Precision agriculture, variable-rate application of inputs guided by GPS mapping and soil sensing, can reduce fertilizer and pesticide use per unit of output by applying inputs where the crop needs them rather than uniformly across the field.

These improvements are real, and they are meaningful. Gabe Brown's operation near Bismarck, North Dakota, is the case that cover-crop advocates cite most often, and the case is worth citing because the numbers are specific. Brown transitioned a 5,000-acre grain and livestock operation from conventional tillage and continuous monoculture to a no-till, diverse cover-crop, integrated-grazing system over roughly two decades. His soil organic matter measurements, documented in NRCS technical visits and in Brown's own published accounting,[37] rose from less than two percent to over six percent on fields that had been under the system longest. His input costs dropped substantially: no synthetic fertilizer purchases, no fungicide, no insecticide, herbicide use reduced to a fraction of the county average. His water infiltration rates, a direct measure of soil biological health, increased severalfold, meaning his fields absorb rainfall that would run off a conventionally managed field carrying topsoil and nitrogen with it. The operation is profitable, and the profitability does not depend on commodity support payments in the way his neighbors' operations do, because his input costs are low enough that the margin holds without the subsidy floor.[38]

Brown's case is not typical. It is a best-case demonstration of what the improving trajectory produces when a skilled operator commits to the transition fully. But it establishes that a grain monoculture operated with winter cover cropping, no-till practices, precision nutrient application, and integrated perennial buffer strips would score materially better on the environmental axis than the same acreage operated under conventional practices. It would build soil rather than lose it, support biological activity rather than suppress it, and reduce nutrient runoff rather than contribute to it. The practices exist, they work, and their adoption is expanding.

The book does not present monocultures as categorically destructive. It presents the current dominant industrial implementation, the version of monoculture that operates across most of the 250 million acres of US cropland without these improvements, as scoring poorly on the four-axis framework for specific, structural reasons. The distinction matters because the structural analysis and the technological trajectory are both true at the same time. The dominant implementation has the vulnerabilities the framework identifies. The improving implementation has the potential the evidence supports. Both findings are accurate. They describe different populations of practice on the same landscape.

What the improvement trajectory does not resolve, however, is the survivability question. A cover-cropped, no-till, precision-managed corn operation is still a corn operation. It still requires seed, fertilizer, and diesel — in reduced quantities, but still required — and a climate regime compatible with corn production. It still sells into a commodity market whose price is shaped by global supply and demand and by federal policy. The improvements narrow the dependency profile. They reduce the system's dependence on any single input by making the system more biologically self-sustaining. But they do not eliminate the profile. A system that depends on fewer external inputs is more survivable than one that depends on more, and the improving monoculture is more survivable than the conventional one. It is not, by the framework's measure, as survivable as a system that depends on very few external inputs and can tolerate the loss of any one of them without collapsing. The improvement is real. The structural constraint remains.

◆ WHAT THE SCORING SHOWS ◆

Two systems, eight axis scores, one pattern.

Industrial feedlot animal production and industrial grain monoculture both score positively on the economic axis at the point of sale, and both score negatively when externalized costs are accounted. Both score net negative on the social axis, with benefits flowing to consumers and corporate entities and costs concentrated in workers, adjacent communities, and the rural regions whose economies the systems have restructured. Both score net negative on the environmental axis across essentially every metric. Both score very low on the survivability axis, with dependency profiles that are broad, deep, and contingent on conditions that are volatile.

The pattern is structural rather than incidental. The two systems were not designed by malevolent architects. They evolved through seventy years of economic incentive, agricultural policy, technological development, and consumer demand, each reinforcing the others, producing a food system that is extraordinarily efficient at converting fossil energy and federal dollars into cheap calories under stable conditions. That efficiency is real. It feeds people. The framework does not deny this. What the framework adds is the observation that the efficiency is built on a specific set of conditions, and that the conditions are not guaranteed.

The question the scoring raises is whether an alternative exists that escapes these structural constraints: a food system that scores well on the first three axes without depending on the specific set of external conditions that make the industrial system's performance possible. The next chapter takes up that question, and the answer it finds is instructive in a way I did not expect when I began this work. The leading alternative — regenerative agriculture, holistic management, the pasture-based model — scores substantially better than industrial production on every axis. It also scores better on the survivability axis, which is the axis where the industrial system is most vulnerable. But it does not score as well on survivability as its advocates might hope, and the reason it does not is a structural one that the framework makes visible in a way that three-axis assessment cannot.

The industrial system's survivability deficit is the floor. The question is what can be built above it.

CHAPTER SIX

Pasture, Honestly

♦ ♦ ♦

Joel Salatin's Polyface Farm sits in the Shenandoah Valley of Virginia, on about five hundred and fifty acres of rolling pasture, woodlot, and pond that have been in his family since the early 1960s. The operation is the most publicly described small farm in the United States. A steady stream of agricultural interns, journalists, documentary crews, and Polyface's own mail-order customers have made the farm into the de facto reference site for what has come to be called, interchangeably, regenerative agriculture, holistic management, or pasture-based farming. Salatin has earned that attention. By his own published accounting, the soil organic matter on Polyface rose from roughly one and a half percent when his parents bought the land in the early 1960s to around eight percent by the time he was writing *You Can Farm* at the end of that century, against a continental cropland average closer to one to three percent.[1] That is not a trivial gain. An inch of topsoil takes a century or more to form under conventional agriculture. Polyface has built measurable inches within a human career.

Nothing in this chapter should be read as denying that it is working. The previous chapter applied the four-axis framework, economic, social, environmental, survivability, to the two dominant industrial systems and watched them score poorly on each axis, for reasons that were structural rather than incidental. The obvious next question is whether the leading alternative scores better. It does. By a substantial margin, on the axes where it is easiest to measure, it does. The harder question, which is what this chapter is actually about, is whether scoring better is the same thing as escaping the structure. I have come to think it is not, and I have come to think this in a way that takes Salatin seriously rather than the other way.

WHAT THE WORK ACTUALLY ACHIEVES

The case for regenerative agriculture, as a practice rather than a brand, is strongest when it is presented specifically. I want to take the time to present it specifically before anything else.

Start with soil. The conventional row-crop regime in the United States has been losing topsoil for a century. USDA Natural Resources Conservation Service modeling using the Revised Universal Soil Loss Equation places average annual erosion rates across conventionally managed cropland at roughly four to five tons per acre, against natural soil-formation rates of a half ton or so. The result, on aggregate, is the loss of something on the order of ninety centimeters of topsoil across the major US cropping regions since the start of European farming on the continent.[2] Rotationally grazed pasture, by contrast, does not erode at anything like that rate. Machner and colleagues, reviewing long-term grazing trials across the southern and central United States, reported that well-managed perennial pasture under adaptive multi-paddock grazing can build soil organic carbon at rates on the order of 0.3 to 1 ton of carbon per hectare per year in the upper soil horizon over multi-decadal time frames, in a system from which the ruminants are simultaneously producing meat.[3] The soil carbon and the meat are not trading against each other. They are being produced together.

The animal-welfare comparison is not close. A broiler chicken in a conventional confined house lives thirty-five to forty-two days in a structure whose stocking density is regulated, in the United States, by the industry's own standards rather than by federal law.[4] The bird typically cannot turn around without stepping on another bird and cannot exhibit most of its species-typical behaviors at all. A broiler on pasture lives in daylight on grass, in a covered mobile structure moved once a day, with access to insects, seeds, and whatever small animals it can catch. The life is shorter than wild chickens' lives and is still ended by slaughter, but within the window of the bird's life the quality of that life is, by any reasonable behavioral measure, incomparable. The same comparison holds for laying hens, for pastured pigs, and for cattle, though with different gradients. Confined animal feeding operations have imposed a kind of welfare compression that has no real historical

precedent in Western agriculture, and pasture-based systems do not operate at that compression. On the welfare axis, the difference is not a matter of degree but of kind.

The operation's internal design deserves credit on its own terms as well. Polyface's signature move, and the move most imitated across the pastured-livestock literature, is the integration of species in rotation across the same ground. Cattle graze first. The pasture is then rested for a set interval, during which the cow manure ages and fly larvae develop in the pats. The broilers follow in mobile field shelters, eating the larvae and other insects mobilized by the cattle, adding their own manure to the surface, and continuing to be moved forward. Laying hens, in a separate mobile structure Salatin calls the "eggmobile," circulate through the same ground in a slightly different phase of the rotation, completing the fly-cycle disruption and contributing a different manure-deposition pattern. Pigs are used elsewhere on the operation for targeted tillage and forest-floor disturbance. The design moves biological work: pest control, manure distribution, vegetation management, nutrient cycling, from purchased inputs and mechanical operations into the animals themselves. It is a serious piece of agroecological engineering — and it has earned the attention it has received. Any fair assessment of the operation starts by acknowledging that the integration works and works well within its own perimeter.

The rural economic picture is similar in structure. A typical industrial broiler contract puts the grower in capital debt to a vertically integrated processor, with most of the margin on the bird captured at processing and distribution.[5] The grower absorbs the capital cost, the environmental externalities, and the labor; the processor absorbs most of the profit. A pasture-based operation at Polyface's scale, by contrast, captures significantly more of the value of the bird on the farm itself. Salatin has argued for decades that rural economic renewal runs through the direct-marketed farm rather than through the contract-grower arrangement, and he is right about this as a matter of basic accounting. Where pasture-based operations exist at scale in a region, they support local processing infrastructure, farm-to-consumer distribution networks, agritourism, and a retained agricultural workforce that the contract system has hollowed out. That hollowing has been measurable for decades. The rural sociology literature, some of the longest-running sociological work on American agriculture, has documented across fifty years of comparative studies beginning with Walter Goldschmidt's original comparisons of Arvin and Dinuba in the 1940s that the consolidation of US livestock and commodity production has coincided with

measurable declines in rural population, retail employment, small-town median income, and civic participation across regions where the transition from diversified to contract production has been most complete.[6] The pasture-based operation is working against that curve. The people Polyface employs live in the Shenandoah Valley. The money Polyface earns is spent in the Shenandoah Valley. The operation is not extracting value from its region toward a distant headquarters — which is what the contract model does structurally.

The carbon case for well-managed grazing, finally, is genuine. This is not the same thing as saying it is sufficient. Richard Teague and colleagues argued in a 2016 paper in the *Journal of Soil and Water Conservation* that adaptive multi-paddock grazing, if scaled across existing North American grasslands, could plausibly offset a meaningful fraction of North American agriculture's greenhouse gas footprint through soil carbon accumulation and reduced reliance on synthetic nitrogen fertilizer.[7] The Teague paper is the most forceful peer-reviewed articulation of the climate case for regenerative grazing, and I take it seriously. I will come back to its limits later in this chapter — because the limits are real — but the positive case is not fabricated. Well-managed grazing genuinely pulls carbon into soil — and it does so while producing food — and the effect is not a rounding error.

Put all of that together and you have a system that is, by any fair accounting, substantially better across several important axes than what it is trying to replace. It builds soil. It produces food. It restores something close to normal animal lives. It supports rural economies. It sequesters carbon. None of those achievements is imaginary — and none of them should be held at arm's length in what follows.

◆ APPLYING THE FRAMEWORK ◆

Run the four-axis framework across this version of the operation — and the scoring changes noticeably from the industrial comparison in the previous chapter.

On the environmental axis, pasture-based operations score well. Soil carbon accumulates rather than erodes. Chemical inputs — synthetic fertilizer, herbicide, insecticide, fungicide, antibiotic — are absent or dramatically reduced, meaning the downstream externalities of runoff, resistance, and non-target toxicity are absent or dramatically reduced. Biodiversity on and around the operation is higher: perennial pastures host arthropod, avian, and small-mammal assemblages that row-crop fields

cannot support.[8] Water infiltration rates into rotationally grazed soil are higher than into tilled soil by substantial multiples, with consequent reductions in flood peaks and downstream sedimentation.[9] The environmental axis, in a pasture-based system run competently, moves from the deep negative register of the industrial score into a positive register that is not marginal.

On the social axis, the picture is similarly improved. Worker conditions on a pasture-based operation are not the worker conditions of an industrial slaughterhouse. The operation sits inside the community it feeds rather than at the end of a commodity chain that terminates at a processor two states away. Rural community indicators — employment, retention, median income for agricultural families — track with the presence of diversified direct-marketed agriculture in ways that have been documented for a generation now.[6] The social axis improves.

On the economic axis, the scoring is conditional and depends on scale. At the scale of a Polyface or a comparable operation, several hundred acres, owner-operated, direct-marketed, the economics work: the farm is profitable, the family is supported, the next generation has a ladder onto the operation if one of them wants it. The work is not easy and the margins are not wide, but the operation is economically viable in a way that much of conventional small-scale commodity agriculture is not. At the scale of the continental food supply, the economic axis becomes much harder to score because the system as currently practiced has never operated at that scale. Scaling a direct-marketed pastured-poultry model from one operation in the Shenandoah Valley to, say, the broiler demand of a metropolitan Atlantic seaboard would require land, processing infrastructure, labor supply, and concentrated feed in volumes that the existing direct-market architecture has not produced and the existing commodity architecture cannot produce without compromising the ecological distinctions that make the pasture-based operation worth doing in the first place. This is not a case against the operation at its own scale. It is a case for treating the operation's achievements as operation-scale achievements, which is how Salatin himself has generally framed them. The claim that what Polyface does could be extended to feed the country is a claim that has rarely been made by Polyface itself. It has more often been made by journalists and advocates around the operation, and the operation is not responsible for the claims made on its behalf.

On the survivability axis, pasture-based operations score substantially better than industrial ones, which is the axis the industrial system scored worst on. The

operation does not depend on continuous diesel for its own field work in the same way a row-crop operation does, because most of the field work is being done by the animals themselves. It does not depend on continuous antibiotic administration, because the stocking densities and the rotation pattern handle the disease pressure that makes antibiotics necessary in confined systems. It does not depend on continuous federal subsidies for its own outputs, because those outputs are priced into a direct market rather than a commodity one. The system is more survivable than what it replaced.

But the survivability score is not the maximum possible score, and the reason the score is not maximum is the subject of the rest of this chapter. The operation, as currently practiced, still depends on something the industrial system supplies — that dependency is the structural ceiling.

◆ THE STRUCTURAL CEILING ◆

Regenerative agriculture, in its current operative form in the United States, is structurally tethered to the industrial commodity system it opposes. This is not a statement about any particular operator's motives or honesty — it is a statement about how the economics and the supply chain of the current US agricultural system constrain even the operations explicitly trying to operate outside it.

The ceiling shows up in different places at different operations. For operations that raise ruminants on pasture — grass-finished beef, pastured lamb, grass-fed dairy — the ceiling is lower and the tether is thinner, because ruminants can live most or all of their lives on forage produced on the farm itself and require relatively little from outside the operation. For operations that raise monogastric animals on pasture — broilers, layers, pigs — the ceiling is higher and the tether is thicker, because monogastrics cannot live on forage alone and must be supplemented with concentrated protein and energy that, under current conditions, comes from somewhere else.

Polyface is the worked example because it is the operation whose numbers are most completely in the public record. The numbers are worth walking through carefully.

THE POLYFACE MATH

Polyface's broiler operation, at the scale Michael Pollan described in *The Omnivore's Dilemma*, processes approximately twelve thousand broiler chickens per year in mobile field shelters moved daily across pasture.[10] Salatin's own *Pastured Poultry Profit$* gives the feed conversion ratio for the birds as roughly three pounds of feed for each pound of finished body weight — with an average finished weight in the neighborhood of four pounds per bird.[11] The arithmetic from those numbers is direct. Twelve thousand birds, at four pounds of body weight each, at three pounds of feed per pound of body weight, comes to approximately one hundred and forty-four thousand pounds of feed per year for the broiler flock alone.

The composition of that feed is also in the public record. Polyface has published its layer and broiler ration specifications; independent writing on the farm, including longer profiles and follow-up interviews over the years, has confirmed the proportions. The ration is built around roughly forty-nine to fifty percent corn and roughly thirty to thirty-one percent soy, with the remainder made up of mineral premix, fish meal, kelp, and a few other supplements.[12] Applied to the total feed figure, that means roughly seventy-one thousand five hundred pounds of corn and roughly forty-four thousand five hundred pounds of soybean meal per year, for the broiler flock alone. The layer flock, smaller but year-round, adds to these totals. So do the pigs that spend part of their finishing period on concentrate.

The corn and soy do not come from Polyface's own acres. The farm's perennial pasture system is not arranged for row cropping, and the Shenandoah Valley's climate and soils are not particularly suited to corn on the scale required. The feed comes from off-farm sources — grain sold by neighboring farmers and feed mills in the broader region of the valley and adjacent states. Salatin has written openly about this sourcing, and has emphasized that the farm works to buy from local non-GMO producers rather than from anonymous commodity channels.[13] The sourcing preference is real. The underlying point is that even with the most scrupulous regional sourcing, the corn and soy feeding the Polyface birds are grown under the conditions that most corn and soy in the United States are grown under: annual row crop, tillage or herbicide-based residue management, synthetic nitrogen inputs, petroleum-powered field operations, and — directly or indirectly — the federal commodity support system that keeps the price of corn and soy low enough for a pastured-

poultry operation to purchase them profitably. The industrial substrate is the substrate.

I want to be careful about how this finding is registered, because the framing matters.

It is not a gotcha. Salatin publishes these numbers himself, in his own books, for farmers who want to replicate his system. The feed proportions are printed in *Pastured Poultry Profit$*; the scale is in *The Omnivore's Dilemma* and subsequent profiles. There is no concealment to expose. Anyone who has read the source material with any attention has encountered the math. The math was already public, in that sense, when any of us started having this conversation.

It is also not a personal failure. The operator is working inside the structural conditions of US agriculture as those conditions currently exist. Grass-finishing a broiler in any meaningful volume is not, under present conditions, economically possible at a price point that sustains a family farm. The price of corn and soy has been held down by half a century of agricultural policy; the cost of land, labor, and processing has not. An operator deciding, in 1995 or 2005 or 2025, whether to build a pastured broiler enterprise has faced a narrow set of real choices, and at every point in that window the economically survivable choice has involved purchasing grain from someone else. The alternative — growing the grain on the farm under perennial-compatible practice — is not absent from the regenerative literature, but it has not, to my knowledge, been implemented at any scale that makes broiler operations like Polyface's replaceable.

The point is worth sharpening, because the sharpening matters for everything that comes after. An operation like Polyface is doing the most that is currently doable inside the existing system. The ceiling on what it can do sits not at the gate of the farm but at the edge of the commodity market the farm buys from. Raising the ceiling would require changing the commodity market, which is a different kind of problem than changing how a farm is managed. An individual farmer can change how the farm is managed; an individual farmer cannot change the structure of US corn and soy production. The work Salatin and operators like him have done on the manageable side is the work they could do. The work on the substrate side is the work the rest of the food system has not done.

What the finding is, then, is a structural observation. Even the best version of regenerative agriculture as currently practiced, the operation with the most documented soil gains, the most direct-marketed income, the most visible animal welfare improvements, the most sustained rural economic impact, still has its monogastric-animal enterprises running on carbohydrate and protein inputs produced by the industrial commodity system it opposes. The system it is offering as the alternative is, in its concentrate-dependent component, riding on the substrate it wants to replace.

This is the structural ceiling. It is not the end of the argument — it is the thing the argument has to take seriously before it continues.

◆ THE MCWILLIAMS-SALATIN EXCHANGE ◆

The substrate question became public in a way most non-specialists would remember when James McWilliams published "The Myth of Sustainable Meat" in the *New York Times* in April of 2012.[14] McWilliams, a historian at Texas State, had been arguing for several years that the environmental advantages of pasture-based animal production were overstated. The op-ed compressed the argument. It made several claims, some of which were wrong, one of which was right, and the combination produced a public exchange that illustrated the limits of the conversation as then framed more clearly than either side would likely have wanted.

McWilliams's central environmental claim, that pasture-raised beef produces substantially more methane per unit of beef than feedlot beef because grass-finished animals live longer and eat more lignocellulose, was overstated as applied. The methane-per-unit-beef numbers are genuinely higher on pasture, but the comparison is incomplete without the soil carbon side of the ledger, which can offset a meaningful fraction of the methane difference in well-managed systems. The Teague analysis and the subsequent literature have laid this out in more detail than the op-ed allowed for.[7] The land-use claim, that pasture requires more acres per pound of finished animal than feedlot, was true on its own terms but also incomplete. The relevant question is not acres-per-pound but which kind of land is being used, for what, under what set of effects on downstream ecosystems. A great deal of pastureland is land that cannot productively support row crops.

The claim McWilliams made that landed, and landed in a way neither he nor his critics particularly wanted to admit, was the feed-sourcing claim. He pointed out that pasture-based operations raising monogastric animals still relied on purchased grain produced in the conventional industrial system, and that this reliance made the sustainability advantages of the pasture component narrower than their advocates claimed. On the feed-sourcing point, the math is the math, and it was already public in Salatin's own books.

Salatin's public response came through Polyface Farm's own channels and through subsequent essays and interviews.[15] The response was vigorous and, in places, sincere in a way the op-ed had not been about Salatin's work. Several of his counterpoints were substantive: the methane comparison was being handled more simplistically than the science warranted; the land-use comparison was ignoring land-use distinctions; Polyface's local non-GMO sourcing was real and deserved to be distinguished from anonymous commodity sourcing. One argument in particular deserves engagement on its own terms, because it is the most serious thing said in the exchange and because it gets close to the structural question without quite reaching it.

Salatin argued, in effect, that the movement of grain from grain-growing regions into grazing regions via animal feed is not a pathology of the industrial system. It is an ancient pattern. Historically, lowland grain-growing regions and upland grazing regions have exchanged nutrients through exactly this kind of trade: grain flowed to the uplands as animal feed, and animal products flowed back to the lowlands as food. The pattern long predates petroleum agriculture, commodity subsidies, and the US Department of Agriculture. A pastured-poultry operation buying corn from a farmer fifty miles away is, under this framing, participating in a trade relationship as old as agriculture itself.

There is real history in that framing. The highland-lowland nutrient exchange is genuine, and it is defensible as a way of organizing agricultural landscapes. What the framing does not quite reach, in my reading of Salatin's reply, is the point underneath McWilliams's otherwise flawed argument. The historical highland-lowland exchange was regional and operated at a scale that did not require a continental commodity system. The modern version of the exchange operates at a scale that does. When Polyface sources non-GMO corn from within the Shenandoah region, it is still sourcing from farms whose own inputs, seed, fertilizer, herbicide,

field equipment, transportation to market, run through the continental industrial substrate that corn and soy have been embedded in since roughly the Nixon administration's rewrite of the farm bill.[16] The regional-sourcing move reduces the transportation footprint and the genetic-engineering exposure. It does not detach the feed from the underlying substrate.

Salatin's response, read carefully, is best understood as a defense of the farming model's integrity at the farm-gate scale, and at that scale it is largely correct. It is not a response to the structural observation that the farm, as a node in a larger agricultural system, cannot fully escape that system as currently organized. McWilliams had framed the feed-sourcing point as an accusation, which it did not need to be and which made the response defensive. The point was better made as a structural observation. This is how I am trying to make it now.

What the 2012 exchange illustrated, in retrospect, was the limit of the conversation as then conducted. McWilliams wanted to indict the pasture model. Salatin wanted to defend it. Between the two positions there was not much room for the observation that the pasture model was genuinely doing most of what it claimed to do and was still riding on the industrial substrate to do it. Both of those are true. Neither side's framing in the 2012 exchange made that doubled picture easy to see. The conversation has moved somewhat since then. Versions of the structural observation now appear in places where they did not appear in 2012: in *Civil Eats* coverage, in the peer-reviewed agroecological literature, and in the internal conversations of regenerative-agriculture advocates who have been working on perennial-grain and on-farm feed alternatives precisely because the ceiling is a ceiling they can see from inside the movement.

FIELD NOTE

The Things Worth Doing

The first version of this chapter was easily the best chapter in the book. Everyone said so, from Robb Wolf to the publisher. They also did not want me to publish it.

It was scathing. *The math was scathing.* The chapter was passed on to another individual, someone close to the system the math was taking apart. Later that year, she approached me at a dinner in Austin and said, "If you publish this chapter, it would be the end of Organic farming as we know it."

I leaned in ominously and said, "If little ol' Karen Pendergrass could say something that crushed that system, then it deserves to crumble. And it deserves to be built up on a better foundation."

That scathing chapter still lives in my head.

The same person was a guest speaker at Expo West in Anaheim in 2015. She said something to me there I have never forgotten: "The things worth doing — are worth doing poorly at first." I have repeated it more times than I can count, in conversations she is not in. Those words *softened* me. She was right, and I was wrong.

She has, on other occasions, eye-rolled the whole idea of camels — *will people seriously eat camel?* — and I have eye-rolled back. Who knows. There is, for instance, a real demand for camel milk for autistic children. The question is worth answering nonetheless.

Today I hold two conflicting views about Polyface. Both are true. But one, I believe, is truer.

The things worth doing — are worth doing poorly at first.

◆ GRAZED AND CONFUSED ◆

The climate side of the ledger deserves the same honest treatment. The case for well-managed grazing as a substantial climate mitigation lever is real but bounded. It is bounded in ways the most enthusiastic popular framing has not always absorbed.

Tara Garnett and colleagues at the Food Climate Research Network at the University of Oxford published *Grazed and Confused?* in 2017, which remains the most rigorous systematic assessment of grazing's climate claims.[17] The report reviewed the accumulated peer-reviewed evidence for soil carbon sequestration under improved grazing management and concluded that the sequestration effect is genuine, measurable, and at a global scale able to offset somewhere on the order of twenty to sixty percent of annual ruminant greenhouse gas emissions under optimistic assumptions. The report also concluded that the effect saturates after several decades. A grassland accumulating carbon under improved management does not continue accumulating at the initial rate indefinitely. The upper end of the estimated sequestration potential requires management conditions that are difficult to achieve and maintain at scale.

Teague and colleagues have pushed back on parts of the *Grazed and Confused?* analysis, arguing that the report underestimates the sequestration potential of adaptive multi-paddock grazing specifically and that it aggregates the grazing-management literature in a way that flattens differences between management regimes that are in fact ecologically distinct.[18] Their case is not implausible. The specific management regime matters: a continuous grazing system and an AMP system are not interchangeable from the soil's perspective, and the literature has not always treated them as distinct.

The saturation mechanism in the Garnett analysis deserves a moment of its own, because its implications are easy to miss in popular framings. Soil carbon accumulation under improved grazing is fastest in the early years after management change, when the soil is most below its equilibrium carbon level for the new management regime. As the soil approaches the new equilibrium, a process that takes in temperate grasslands somewhere on the order of twenty to forty years, the annual sequestration rate declines toward zero. The soil is not permanently scrubbing carbon out of the atmosphere the way an unsaturated chemical sink would. It is moving from one equilibrium to a higher equilibrium, and the movement is bounded. The total amount of carbon ultimately stored is meaningful; the rate of annual accumulation has an arc. Any long-run climate accounting for grazing has to incorporate this shape, and the popular framings often do not. This is not a criticism of Polyface or of any particular operator; it is a criticism of the rhetorical compression that has sometimes attended the climate argument for regenerative agriculture in the broader public conversation.

Read against each other, Garnett and Teague produce an honest calibration. Well-managed grazing is genuinely carbon-positive at the operation scale. It is not climate-neutral for ruminant production overall, even in the most optimistic scenarios. It is a meaningful contribution to the climate side of the agricultural ledger, substantially better than poorly managed grazing or feedlot, and not by itself a climate solution for ruminant agriculture. The claim that pastured livestock operations are climate-positive, stated carefully, is about the operation's own carbon balance rather than about the emissions the operation is also producing. Stated carefully, the claim is defensible. Stated as a blanket net-negative framing for ruminant agriculture as a sector, it is not.

None of this undoes the soil, welfare, or rural-economy case. All of those stand on their own. It does mean that the climate claim specifically needs to be stated with care. This care is compatible with the operation being ecologically serious, which it is.

◆ WHAT THE CEILING MEANS ◆

The chapter has, I hope, been doing two things at once. It has been taking regenerative agriculture seriously as the most developed alternative to the industrial system on offer — and it has been looking at what the most developed alternative can and cannot do as currently practiced. Both of those are honest descriptions. The operation does what it does, and it does not do what it does not do, and the honest account requires holding both.

What the structural ceiling means, stated as plainly as I can state it, is this. Regenerative agriculture as currently organized in the United States is significantly better than what it is trying to replace. It is also still riding on the substrate it is trying to replace. Its monogastric-animal enterprises import tens of thousands of pounds of corn and soy per year from the industrial commodity system. Its ruminant enterprises depend less directly on that system but still operate inside the broader infrastructure: the diesel for equipment, the steel for fencing, the mineral supplements, the veterinary care, that the industrial substrate supplies. An operation like Polyface is an achievement inside a system whose boundaries the operation cannot fully escape.

This is not a defeat. It is a limit. Limits can be mapped and worked against. A substantial fraction of the regenerative-agriculture literature over the last decade has been exactly this work: perennial-grain breeding programs trying to make cereal production compatible with perennial root systems,[19] silvopastoral systems integrating timber, browse, and ruminant production on the same acreage, and multi-species rotation schemes that reduce the dependence on external concentrated feed. Every one of these is a real response to the structural ceiling. None of them has, as of this writing, produced the fully untethered alternative the framework I have been using would score highest.

The Land Institute's intermediate wheatgrass program, marketed under the trade name Kernza, is the most commercially advanced of these responses. After decades of breeding work in Salina, Kansas, Kernza now has commercial acreage in the low

thousands of acres across the upper Midwest, a published grain market, and a small but real downstream supply chain in the craft-brewing and specialty-baking sectors. It is not yet a corn replacement at any scale that would matter to a Polyface feed order. It is not even a credible partial replacement at that scale. What it is is existence-proof that a deep-rooted perennial can be bred toward commercial grain production, and that a seed company and a farmer can, between them, make the numbers work at the margin. Silvopasture research in the southeastern United States and the Mediterranean dehesa literature offer similar existence proofs on the animal side: systems in which ruminants, browse, mast, and tree crops coexist on the same acreage under management regimes that resemble pre-industrial agroforestry more than they resemble modern pasture-based operations. The point is not that any of these is ready to replace the current substrate. The point is that the work of designing an alternative substrate is actively under way, even if the finished substrate is not yet here.

The question that closes this chapter is the question Part III of the book exists to take up. If the best version of the currently-available alternative is structurally tethered to what it opposes, even at the scale of a farm as studied and careful as Polyface, what would actually escape the substrate? What would an operation look like whose feed comes from the landscape rather than from the commodity system, whose animals are selected for the landscape rather than for the commodity market, whose relationship to the land the continent sits on runs through the land's deeper history rather than through the last two centuries of row-crop engineering?

I think the answer lives in the direction Paul Martin has been pointing us since the 1970s. The landscape this continent is, considered on the scale the continent has actually existed on, is not a landscape that evolved under row-crop corn and feedlot cattle. It is a landscape that evolved under a suite of large-bodied animals most of which have not been present for twelve thousand years. The animals we use to feed ourselves now are, in the main, animals whose ecological partnership is with somewhere else. The animals whose ecological partnership is with this continent are, in the main, not being used to feed us at all. That mismatch is the hinge of what follows.

Regenerative agriculture has taken the food system as far as it has been able to take it while working within the cast of species it inherited and the commodity substrate that cast depends on. The distance it has covered is real and should not be given back.

What comes next is a different question: whether the cast of species can be reopened, and whether, reopened, it changes what sustainability looks like on this continent. The next chapter starts there.

CHAPTER SEVEN

The Wrong Proxy in the Right Place

♦ ♦ ♦

From Paul S. Martin,
Twilight of the Mammoths (2005)

"From fossils we know that the grasses, forbs, shrubs, and trees of the Americas coevolved with a much greater variety of large herbivores than exists today."

The thorns are not decorative. They are anti-herbivore defenses calibrated to animals the continent has not carried for twelve thousand years.

— P. S. Martin, p. 185

The first time I saw Allan Savory speak in person was at his very first conference. There were maybe twenty-five people there, total. Maybe. It was a small classroom: no podium, no stage, no microphone. A handful of ranchers, a few people who had read the book and wanted to meet the man behind it, and me, in my twenties, with a marked-up copy of Paul Martin's *Twilight of the Mammoths* in my bag. I had read Savory's book twice by that point. I was not there to be convinced. I was there because I had a question.

The room was small enough that you could see everyone's face. Savory spoke with the calm authority of someone who had been making the same argument for decades and had outlasted most of his critics, and even at his age, there was something about him that felt younger than I had expected: a quickness, an energy, a sense that the

ideas were still alive in him rather than calcified into a script. The talk he gave was genuinely good, substantive, grounded in decades of observation. He showed photographs of degraded rangeland in Zimbabwe and the same rangeland after years of planned grazing. He showed soil profiles. He showed vegetation transects. The evidence was real, and he presented it without overselling, which I respected. A lot of people in the regenerative space oversell. Savory did not need to. Only a small number of people can say they were in that room to witness such a great man on the way to becoming who he is today. I photobombed him afterward.

I asked him whether holistic management worked everywhere, or whether it worked better in some landscapes than others.

He paused. Then he said it was not working in the arid Southwest.

The room was small enough that you could hear the shift — the way twenty people adjusted in their chairs when the answer was not the one they expected. Savory did not hedge. He did not qualify it into nothing. He said the results had not tracked in the Southwest the way they had tracked in southern Africa or the Great Plains. He said it with the particular kind of candor that serious people use when they are being honest about something they have not yet resolved — not defensive, not dismissive, just honest.

I spent the rest of the session thinking about it. The talks continued. I took notes on a few of them, but my attention kept circling back to what Savory had said. The arid Southwest. The results had not tracked. I thought I knew why.

The food truck was parked at the edge of the conference grounds, one of those converted Airstream trailers with a sliding window and a short menu of things that could plausibly be called lunch. I got a plate and a bottle of water. Savory had also gotten a plate, and instead of eating outside we both took our food back into the classroom, and that is where the conversation happened. A few others wandered in and listened. I monopolized him during the entire lunch period. The food was good. Neither of us ate it until much later.

I should say what I had with me, because it matters. I had my conference bag, my phone, and a copy of Paul Martin's *Twilight of the Mammoths*. It was not a new copy. I had read it three times by that point, and it looked like it. The spine was cracked, the pages were dog-eared in places I had found worth returning to, and I had marked passages in the margins with the kind of shorthand that means nothing to

anyone else: page numbers cross-referencing other pages, small arrows connecting ideas across chapters, question marks next to claims I had wanted to verify. It was a working copy of a book I had been thinking with for several years. I did not bring it to impress him. I brought it because I thought I might need to show him specific passages, and I did.

The conversation started where the Q&A had ended. I told him I thought I understood why the Southwest had been difficult, and that the answer was not in his framework but in the fossil record beneath the landscape his framework was being applied to. He looked at the book on the table between us. I opened it.

I introduced myself. I told him I ran a certification organization and that I had been working on food-system evaluation for years, which was true and also not the point. The point was the book. I told him I had been reading Paul Martin, the paleoecologist at the University of Arizona who had spent his career on the late Pleistocene extinctions and their ecological consequences. I thought Martin's work explained something about the Southwest that Savory's framework, on its own, could not.

He looked at the book on the table between us. He said he actually did know who Paul Martin was. And something in me shifted — I remember holding my hand out toward the book, a little wide-eyed, thinking: *Why has this work not been integrated with your work yet?* The fossil record and the management framework were sitting in the same intellectual space, pointed at the same landscapes, and no one had connected them. I told him the version I had been turning over in my head for months: that the Pleistocene fossil record preserves, in extraordinary detail, the specific animals that co-evolved with specific landscapes across the American continent, and that the record for the Southwest tells a story that does not involve cattle.

What I walked him through, and what I want to walk the reader through now because the argument is the hinge of everything that follows in this book, is a specific claim about the relationship between animal species and landscape. The claim is this: the fossil record tells us which animals co-evolved with which landscapes, and the American Southwest's fossil record does not say cattle.

Martin had spent decades building the evidentiary case. His chapter on the Grand Canyon, titled "Grand Canyon Suite," is the most concentrated version of it, and it is

the chapter I had marked most heavily. The cave deposits along the Colorado River and its tributaries preserve one of the most detailed records of late Pleistocene fauna in the American West. Rampart Cave, a limestone overhang above the river in western Grand Canyon, contained stratified deposits of Shasta ground sloth dung, five feet of it in places, layered over thousands of years, preserved so well that the plant material in it could be identified to genus and sometimes species.[1] Stanton's Cave, downstream in Marble Canyon, held the bones and dung pellets of Harrington's mountain goat, an extinct species whose remains turned up alongside those of bighorn sheep, small equids, and a scattering of bison. Condor remains appeared at multiple cave sites. The assemblage that emerged from the deposits was specific: ground sloths, mountain goats, equids, camelids, condors. Not one of those animals is a bovid.

I explained this to Savory as plainly as I could. The ground sloths at Rampart Cave had been eating globe mallow, Mormon tea, saltbush, catclaw acacia, yucca, and cactus. Dick Hansen's composition analysis of the dung showed that desert globe mallow leaves and stem fragments made up roughly half the plant parts in the deposit, with Mormon tea pollen accounting for eighteen percent and the remainder distributed among saltbush, acacia, common reed, and various succulents.[2] What Hansen found almost none of was grass. The Shasta ground sloth, a thousand-pound browser that had occupied the Southwest for tens of thousands of years, was not a grazer. Neither were the camelids whose bones turn up across the basin-and-range country. The equids grazed, and so did the mammoths, but the overall assemblage was weighted heavily toward browsing, toward animals whose digestive systems and foraging behavior were calibrated to the woody, thorny, armed vegetation that still dominates the arid Southwest today.

Martin made the connection explicit in his later chapters. The thorny, oil-rich, tannin-laden shrubs of the Southwest — creosote, mesquite, catclaw acacia, the entire armored understory of the Sonoran and Chihuahuan deserts — carry the structural signatures of vegetation that evolved under heavy browsing pressure from large-bodied animals.[3] The thorns are not decorative. They are anti-herbivore defenses, and the herbivores they evolved against are the ones the fossil record preserves: ground sloths, camelids, and to a lesser extent the equids and proboscideans that ranged through the region. Martin put the principle as directly as anyone has: "the grasses, forbs, shrubs, and trees of the Americas coevolved with a

much greater variety of large herbivores than exists today."[4] The vegetation is still shaped by the animals. The animals are no longer there.

FIELD NOTE

The Contract

For the record, when I eat fruits, I eat the seeds. I have ever since it occurred to me that there was a contract that we made with these fruits.

The contract is simple. The plant grows a sweet, fragrant, nutritious package around a small hard seed. Something eats the package, carries the seed some distance from the parent tree, and deposits it in its own fertilizer. The plant invests in the package because the seed gets dispersal and a fertilized start. The animal invests in finding the fruit because the package is calorie-dense, and its appearance is, literally, an advertisement.

Most of the animals the fruit trees of this continent made their contract with are gone. Mastodons, gomphotheres, ground sloths. The mouths that osage orange and honey locust and pawpaw grew their fruit for are not coming. The tree drops its crop every autumn and most of what is down there will rot where it fell.

I am not a mastodon. But I can keep one small piece of the contract. I eat the seeds when I eat the fruit. Dispersal in fertilizer is not guaranteed the way it once was. But there are few things in this world that actually want to be eaten. Fruits want to be eaten. Milk wants to be drunk. Grass wants to be grazed down. It seems to me that if a thing is asking, the polite response is to answer.

Savory listened. He did not interrupt, which I noticed. People who are performing attention interrupt. People who are actually paying attention do not. He asked a question about the camelids: where specifically the fossil evidence placed them, whether the Arizona deposits had been dated reliably. I turned to the pages I had marked and showed him the relevant passages. He read them. He asked about the equids, and I walked him through the Grand Canyon deposits and the broader Western record that Chapter 3 of this book has already covered. He asked about the dating methods, and I explained that the radiocarbon dates on the Rampart Cave deposits were among the most robust in the North American Pleistocene record: tens of thousands of years of stratified organic material, independently dated by

multiple laboratories, cross-referenced against the geological stratigraphy of the cave itself.

He sat back in his chair. His plate was mostly untouched. He knew Martin's work. He said so. But he had not read it from this angle, had not connected the proxy question to the management question. He said something I will paraphrase rather than quote, because I did not record the conversation and I do not trust my memory to preserve his exact words across the years since: he said that in southern Africa, where the framework had been developed, the large herbivores were still present. The wildebeest, the zebra, the buffalo, the elephant — the animals the African grasslands had evolved under were still there, and the management methodology had been designed around their behavior. He had always known that the American situation was different in some respects. He had not thought about it from the direction I was pointing.

This was the moment the conversation pivoted, and it is the moment I want the reader to sit with, because the pivot matters. Savory was not admitting a flaw in his framework. He was recognizing a variable his framework had not needed to specify in the context where it was developed. In southern Africa, the proxy question does not arise, because the original animals are still there. You do not need to ask "which herbivore?" when the wildebeest and the zebra and the buffalo are standing in front of you. The question only becomes visible when the original animals are gone and you are choosing a substitute. And in the Americas, the original animals have been gone for twelve thousand years, and every herbivore currently on the landscape is a substitute, some closer to what the fossil record prescribes and some further away.

What I was suggesting was simple enough to say at a folding table over lunch, and difficult enough that almost no one had said it yet. The arid Southwest was calling for equid species: donkeys, horses, the animals whose lineage actually belonged to that landscape. It made sense to take a look at the fossils to get a better idea of the match with the flora of the area. Not just the animal, but the relationship between the animal and the vegetation: the specific co-evolutionary partnership that the fossil record preserves and the current management regime ignores.

I want to state the broader version of this argument with the same care I stated the lunch version at that table.

Allan Savory's holistic management framework is a genuine intellectual achievement. I do not say this as a courtesy before a critique, because what follows is not a critique. It is a refinement, one that I believe makes the framework significantly stronger.

The core of holistic management, as Savory and Jody Butterfield laid it out in *Holistic Management: A New Framework for Decision Making*, is the proposition that the relationship between large herbivores and grassland is not incidental but constitutive.[5] The grassland needs the animal. Not the animal standing in place, overgrazing a single patch until the roots give out, but the animal moving — concentrated by predation pressure or management pressure into dense herds that graze intensively, trample standing dead material into contact with the soil surface, deposit urine and dung, and then move on, leaving the grazed area to recover for an extended period before the herd returns. The cycle of impact and recovery is what the grassland evolved under, and in its absence the grassland degrades: the standing dead material oxidizes in place rather than being incorporated into the soil, the root systems weaken without the stimulus of defoliation, and the bare ground between grass plants widens until the system tips into desertification.

The framework is powerful, and I do not use that word casually. It explains a great deal about the dynamics of grassland degradation that rest-based conservation management has struggled to address, and the empirical results from southern Africa, from the Great Plains, and from managed operations around the world are real. The ranchers and land managers who have adopted holistic management and seen their landscapes recover are not imagining it. The grass is coming back. The soil biology is rebuilding. The water-infiltration rates are improving. The framework works.

The honest thing to add is that the empirical record on holistic management's specific claims is mixed at best and damning at worst. Briske and colleagues' 2008 systematic review of the rangeland literature found no support for the overstocking-reverses-desertification claim that has been the loudest version of Savory's argument in popular circulation. Hawkins's 2017 follow-up review reached similar conclusions. Replicated trials including those run by the Carbon Drawdown Initiative have found that holistic management does not produce, at the scales it has been tested at, the outcomes Savory's TED talk described.

I do not write that as a takedown. I write it because the chapter you are reading is taking Savory seriously, and taking him seriously means taking the criticism seriously too. The specific claims that have failed peer review are claims about cattle overstocking on degraded rangeland in arid zones reversing desertification. Those claims have not held up. What I want from this chapter is the underlying observation — that the missing-proxy problem is real and that some animal disturbance, on some land, in some configuration, is part of how the system kept itself running before we removed the disturbance. The argument is not that Savory was right about cattle. The argument is that the diagnostic question Savory was asking — what is the missing animal disturbance — is the right question, and the work of figuring out which proxies actually work in which places is still in progress.

I am not on team Savory. I am on the team of figuring things out, and that team needs the people who have run the systematic reviews more than it needs another defense of any single practitioner. The refinement that follows in this chapter is offered in that spirit.

What the framework does not do — and this is the refinement, not the critique — is specify which animal. Holistic management tells you that the landscape needs a large herbivore, managed in a way that mimics the evolutionary pressure the landscape evolved under. It tells you how to manage the animal: the timing, the density, the recovery period, the monitoring. What it does not tell you is which animal belongs in which landscape. It treats the herbivore slot as generic: fill it with whatever large ruminant is available, manage it correctly, and the system responds.

For the Great Plains, this works. The shortgrass prairie evolved under bison, and cattle, which are bovids like bison, occupy a similar enough ecological niche that the substitution holds. The grass species, the root architectures, the soil microbiology, the grazing-and-recovery dynamics are all calibrated to a large-bodied bovid grazer, and cattle are a large-bodied bovid grazer. The proxy is close enough that the framework's management methodology can do the rest of the work. Holistic management on the Great Plains is, in a meaningful sense, restoring a relationship that the landscape recognizes.

The Southwest is different. And the reason it is different is the reason Savory's results had not tracked there, and the reason I had come to the conference with Martin's book.

The Southwest did not evolve under bovids. The fossil record is unambiguous on this point. The dominant large herbivores of the late Pleistocene Southwest were ground sloths, camelids, and equids: browsers and mixed feeders whose digestive systems and foraging behaviors were adapted to the woody, thorny, browse-heavy vegetation of the arid basins and canyons. Bison were present in the Southwest in the late Pleistocene, but they were a minor component of the fauna, not the dominant one. The bones in Stanton's Cave included some bison, but the assemblage was overwhelmingly Harrington's mountain goat and equids.[6] The Rampart Cave deposits were ground sloth from top to bottom. The landscape's primary large-herbivore relationships were with animals that no longer exist, and those relationships were not with grazers.

When you put cattle on that landscape, even well-managed cattle under the best holistic management protocol available, you are asking a bovid grazer to fill an ecological role that was held by browsers and mixed feeders. The grass species of the arid Southwest, to the extent that they exist, did not evolve under the kind of intensive grazing pressure that the Great Plains grasses did. The shortgrass species of the Plains — the blue grama, the buffalo grass, the sideoats grama — co-evolved with bison over millions of years; their growth points sit at or below the soil surface, their root systems respond to defoliation with renewed vigor, their entire architecture is an adaptation to heavy, periodic grazing. The browse species of the Southwest — the acacias, the mesquite, the saltbush, the globe mallow, the yucca and prickly pear — evolved under a fundamentally different kind of pressure. They evolved to be eaten by animals with the body mass and the gut capacity to process woody, thorny, chemically defended plant tissue. Cattle do not browse acacia. Cattle do not eat globe mallow. Cattle do not strip mesquite pods. Cattle graze, and when the grass in the arid Southwest gives out under grazing pressure, the system does not recover the way a Great Plains grassland recovers, because the system was never organized around that kind of pressure in the first place. The browse thickens. The mesquite encroaches. The bare ground between the surviving grasses widens. And the conventional diagnosis — overgrazing, poor management, insufficient rest — misses the deeper cause. The animal is wrong for the landscape.

This is what I told Savory in the classroom over plates neither of us was eating. The holistic management framework provides the management methodology — the how. Martin's fossil record provides the species guide — the which. Together, they

are stronger than either one alone. Separately, the framework has a blind spot in exactly the landscapes where the proxy mismatch is largest.

He did not reject it. He sat with it. He asked me whether anyone had tried running camelids in the Southwest under managed grazing protocols, and I told him the honest answer, which was that almost no one had, because the conversation about proxy selection had barely begun. Josh Donlan and his colleagues had published their provocative essay in *Nature* in 2005 proposing Pleistocene rewilding as a conservation strategy, arguing that the ecological histories of the Americas make the case for restoring large vertebrates, including proxies for extinct species, to landscapes that had co-evolved with them.[7] Donlan's longer treatment in *The American Naturalist* the following year had laid out the agenda more carefully, with specific proposals and specific risk assessments.[8] But the rewilding conversation had remained largely in the conservation-biology lane. Almost no one had connected it to the food-systems conversation. Almost no one had asked the question that seemed, to me, obvious: if the fossil record tells us which animals belong in which landscapes, and if those animals can be managed for food production, then the rewilding agenda and the food-systems agenda are the same agenda.

Savory, to his credit, saw the connection immediately. The man had spent his career arguing that the landscape needs the animal. He did not need to be persuaded of the principle. What the fossil record offered was a specification of the principle that his framework had left unspecified. The landscape needs *this* animal, in *this* place, because *this* is the animal the landscape's vegetation and soil biology evolved under. The proxy is not interchangeable. The proxy is chosen by the fossil record.

The principle extends far beyond the Southwest, and it extends far beyond the conversation I had with Savory in that classroom. Let me widen it.

CASE

Proxies Already in Use

The proxy concept this chapter has been developing — large herbivores selected to match the co-evolutionary record of a landscape — is not theoretical. Versions of it are running, on the ground, in several places. The catch is that in almost every case, the proxy chosen has been the proxy that was *available,* not the proxy the landscape co-evolved with.

California: goats. Cal Fire and various municipal fire-prevention contractors deploy domesticated goat herds across thousands of acres of chaparral each year, driven there to reduce the fuel load that drives California's increasingly catastrophic fire seasons. The goats are effective at brush reduction. They are also Old World animals introduced to a continent whose chaparral co-evolved under browsing pressure from large American camelids, ground sloths, and a now-extinct mountain goat (*Oreamnos harringtoni*), none of which were goats in the Eurasian sense.

Australia: cattle. Crown-land grazing experiments in the savanna woodlands of the Northern Territory have used cattle to reduce native grass fuel loads ahead of fire seasons, with measurable effects on fire intensity. The cattle are doing useful work. They are also doing it in a landscape whose original large grazers were marsupial — *Diprotodon* and the now-extinct megafauna of the Pleistocene Australian interior — and the proxy match between cattle and what was there is even looser than the cattle-for-bison match in North America.

The pattern repeats. The proxy in use is the proxy at hand: cattle and goats are domesticated, available, regulatorily uncomplicated, and economically embedded in the food system. The proxy the fossil record points at — for the American Southwest, large camelids; for southern Africa, the still-living elephant; for the central Asian steppe, the species Pleistocene Park has been working with — is, in most cases, harder to procure, harder to permit, and easier to skip.

Nobody has, to my knowledge, tried camels at any scale in the American Southwest specifically as a fire-and-desertification management proxy. The biology is in their favor: dromedaries do thrive on the thorny browse the chaparral and basin-and-range produce; the historical record places *Camelops* in those landscapes for most of their geological history; the regulatory framework Chapter Thirteen describes does not yet exist for proxies of any kind, but if it did, this would be a candidate. The experiment has, for the southwestern American case, never been run.

If the fossil record provides the species guide, and I believe the evidence from multiple continents and multiple research programs shows that it does, then the question of which animals belong in which landscapes is answerable on a continental scale. The answer is not uniform. It varies by region, by biome, by the specific evolutionary history of the specific landscape in question. And the variation is exactly what makes the principle useful, because the variation tells you something the current food system does not know it needs to know.

The Great Plains, as I have said, want bovids. The shortgrass prairie is a bovid landscape, and the best available proxy for the extinct *Bison antiquus* is its living descendant, *Bison bison*. Bison restoration on the Plains, the work American Prairie is doing in Montana and tribal nations are doing across the northern Plains, is proxy selection that the fossil record endorses. The management methodology that Savory provides is appropriate here, because the proxy is appropriate here. The framework and the fossil record agree.

The Southwest wants something different. The arid basins and canyons of Arizona, New Mexico, and the northern Sonoran and Chihuahuan deserts evolved under camelids, equids, and ground sloths. The ground sloths have no living proxy; no extant species closely replicates a thousand-pound herbivorous xenarthran. But the camelids and equids do. Camels are already in the Americas: feral in parts of the West, stabled on hobby ranches, used in tourism and sport. Dromedaries thrive in arid heat, browse thorny shrubs that cattle will not touch, and produce milk with higher fat and protein content than cow's milk, a fact that matters if the question is not just which animal belongs here but what the animal provides. Horses and donkeys are already in the Americas: feral in vast herds across the basin and range, managed on public and private land, embedded in the cultural and economic fabric of the West. Donkeys in particular are superbly adapted to the arid Southwest: hardy in heat, efficient on sparse forage, and browsers as readily as grazers. The small equids whose bones turned up in Stanton's Cave were closer in size to a donkey than to a modern ranch horse. The species are here. What is missing is the ecological framing that would put them in the role the fossil record identifies as theirs. A camel on Southwest rangeland is not an exotic introduction. It is a proxy for *Camelops*, returning to a landscape its lineage shaped for millions of years before a twelve-thousand-year interruption. A donkey or a horse on that same rangeland is not a feral pest. It is the closest living relative of the Pleistocene equids whose bones lie in the canyon deposits, returning to an ecological role that the vegetation still remembers. The Southwest does not need more cattle. It needs the animals it lost.

The eastern hardwood forests present yet another case, and one that connects directly to the ecological-anachronism argument the Prologue introduced. The Pleistocene fauna of the eastern woodlands included mastodons, the dominant large-bodied browser of the deciduous forests, whose range extended from the Great Lakes to the Gulf Coast, along with woodland-adapted ground sloths, elk, deer, and a suite

of smaller herbivores. The osage orange, the pawpaw, the honey locust, the Kentucky coffee tree. These megafaunal-dispersed fruit trees, which Janzen and Martin identified as ecological anachronisms, are eastern species, and the mouths they evolved for were mastodon and ground-sloth mouths.[9] There is no living proxy for an American mastodon. But the principle still operates: the eastern hardwood food system should be oriented around the animals that the eastern hardwood fossil record identifies, which are browsers and mixed feeders, not open-grassland grazers. Silvopastoral systems in the eastern woodlands — integrating ruminant browse, mast, and tree crops on the same acreage — are closer to the fossil record's prescription than any open-pasture cattle operation, even a well-managed one.

The Gulf Coast and the subtropical Southeast present their own puzzle. The Pleistocene fauna of Florida and the coastal plain included tapirs, capybara, llamas, ground sloths, glyptodonts, and a suite of animals whose nearest living relatives now occupy Central and South America. The fossil presence of vampire bats and tapirs in Arizona and California, animals whose modern ranges are exclusively tropical, tells us that the ecological communities of the southern United States were connected to Neotropical faunas in ways that the current landscape does not reflect. Martin noted that tapirs ranged from the Sonoran lowlands to over six thousand feet on the Colorado Plateau, and that their late-Quaternary disappearance is more reasonably modeled as a retreat to tropical refugia under predation pressure than as a response to climate change.[10] The proxy implications for the subtropical Southeast are different from those for the Southwest or the Plains: the animal assemblage the landscape evolved under was more diverse, more Neotropical in character, and farther from anything the current food system provides.

The boreal and subarctic tell a different story again. The Pleistocene fauna of the far north included woolly mammoths, steppe bison, horses, muskoxen, and caribou. Caribou and muskoxen survive. The mammoth and the steppe bison do not. The proxy question here is whether elephants, African or Asian, could serve as mammoth proxies in the boreal, a question Donlan and his colleagues raised explicitly and that the Pleistocene Park experiment in Siberia has begun to test with horses, bison, muskoxen, and yak.[11] The point is not that elephants are about to be introduced to Alaska. The point is that the principle, the fossil record tells you which animal, applies to every landscape on the continent, and the answers it gives are not the same from one landscape to the next.

The map that emerges from this exercise is nothing like the map the food system currently operates from. The current map is simple: cattle everywhere, with regional adjustments for breed and forage type. The fossil-record map is complex: bovids on the Plains, camelids and equids in the arid West, browsers and mixed feeders in the eastern hardwoods, Neotropical assemblages in the subtropical South, Arctic-adapted species in the far north. The current map treats the continent as a single production surface. The fossil-record map treats it as what it actually is — a mosaic of distinct evolutionary histories, each with its own set of animal relationships, most of which have been unreciprocated for a hundred and twenty centuries.

This is what the current food system misses entirely. The industrial system treats animals as interchangeable production units and landscapes as interchangeable production surfaces. The regenerative-agriculture movement, for all its real achievements, has treated the proxy question as settled — cattle for grasslands, chickens and pigs for everything else — without consulting the record that would tell it whether those proxies are correct. Savory's framework gets the management right and leaves the species unspecified. Martin's fossil record specifies the species and does not provide the management. The synthesis, holistic management guided by paleontological proxy selection, is what neither framework achieves alone.

I am not proposing that the synthesis is easy to implement. I am proposing that it is the correct framework, and that the difficulty of implementation is not an argument against its correctness. The permits for running camels on Southwest rangeland do not currently exist in most jurisdictions. The infrastructure for processing camel or horse meat for the US market does not currently exist at any meaningful scale. The cultural resistance to eating horses in the United States is real and historically deep, and I am not dismissing it. That resistance has its own history, its own logic, and its own emotional weight, and a framework that requires people to change what they eat had better take seriously the reasons they eat what they eat now. The ecological risk assessments for introducing proxy species to landscapes they have not occupied in twelve thousand years are genuinely complex, and the critics of Pleistocene rewilding, Rubenstein and colleagues' careful objections in particular, deserve the engagement they will receive in Chapter 9 of this book.[12] None of these obstacles are trivial. All of them are navigable.

What I am saying is that the question "which animal belongs here" has an answer, and the answer is in the ground beneath our feet. The fossil record is not a metaphor.

It is a dataset. It contains the bones and the dung and the pollen and the diet profiles of the animals that co-evolved with every landscape on this continent, and those profiles tell us — specifically, empirically, verifiably — which species the landscape's vegetation, soil biology, and ecological dynamics evolved under. When we ignore that record and put cattle on a landscape that evolved under camelids, we are making a proxy-selection error. When we compound that error by building an entire food system around the wrong proxy, we are making the error structural.

The chapter before this one ended with a question: what would actually escape the substrate? What would an operation look like whose animals are selected for the landscape rather than for the commodity market?

The answer begins with the fossil record and a worn copy of Martin's book on a folding table between two people who care about the land. Savory spent his career arguing that the landscape needs the animal. Martin spent his career identifying which animal the landscape needs. The food-truck conversation was the moment I understood that the two frameworks are halves of the same argument, and that the argument, stated whole, reorganizes everything the food system thinks it knows about which species belong where.

I think about that conversation often. I think about Savory sitting with the information, not arguing, not deflecting, just absorbing it and asking the right questions. I think about Martin's book on the table between us — marked up, worn out, full of someone else's lifetime of work distilled into a set of observations so straightforward that the only mystery is why they have not yet reorganized the conversation they belong to. I think about the folding table, the food truck, the conference going on inside without us, and two people outside eating lunch and talking about which animals belong in which places on a continent that has been making do without most of its animals for longer than there has been agriculture.

The synthesis is not mine. The management framework belongs to Savory. The species framework belongs to Martin. What I am doing, what this book is doing, is putting the two together and asking what follows. What follows is the subject of the next chapter: from the principle to the plate. What does it actually look like to eat from landscapes managed under proxy assemblages the fossil record endorses? The question is not abstract. The animals are real, some of them are already here, and the landscapes are waiting.

CHAPTER EIGHT

Rewilding the Plate

♦ ♦ ♦

The previous chapter ended with a principle: the fossil record tells you which animal belongs in which landscape, and the management framework tells you how to manage the animal once it is there. The principle is clean. What I want to do now is show what the principle looks like when it touches actual ground: real landscapes, real animals, real food being produced and consumed by real people, today, not in some aspirational future that requires a revolution before breakfast.

The vision this chapter describes is not hypothetical. Parts of it are already operating. Other parts are within reach of deliberate effort. The honest assessment of scale, what rewilded food systems can and cannot do at present, comes later in the chapter, because the honest assessment matters and the book's credibility depends on not overselling. But the starting point is what exists, because what exists is more substantial than most people in the food-systems conversation realize.

◆ WHAT ALREADY EXISTS ◆

Start with the feral hog.

Texas is home to an estimated 2.6 million feral hogs (*Sus scrofa*), descendants of domestic pigs escaped or released over the centuries since Spanish colonization, now established across every one of the state's 254 counties.[1] The animals are ecological demolition crews. They root through native grassland, destroy ground-nesting bird habitat, contaminate waterways, damage agricultural crops to the tune of an estimated $52 million per year in direct agricultural losses in Texas alone, and compete with native wildlife for mast and forage.[2] They are, by any ecological meas-

ure, an invasive species causing measurable harm. They are also, by any nutritional measure, a substantial source of animal protein living on land that did not ask for them.

The dual value is underappreciated in the food-systems conversation. Texas Parks and Wildlife Department reports that hunters harvest approximately 750,000 feral hogs per year in the state, a figure that does not keep pace with the animals' reproductive rate. Feral sows can produce two litters per year, averaging five to six piglets per litter, and populations can double in four months under favorable conditions.[3] The harvest, even at three-quarters of a million animals annually, is not reducing the population. It is converting a fraction of the ecological damage into usable food. The meat is lean, free-range in the most literal sense of the term, and produced without any of the inputs that industrial pork requires: no commodity feed, no confinement infrastructure, no antibiotics, no manure lagoons, no federal subsidies for corn. The animal feeds itself on the landscape, damages the landscape in the process, and the harvest simultaneously produces food and reduces the damage.

I am not romanticizing feral hogs. They are genuinely destructive, and their management is a serious and expensive problem. What I am pointing out is that the food-system value of their harvest is almost never counted in the food-systems literature, because the animals do not fit any of the categories the literature recognizes. They are not livestock. They are not game in the managed sense. They are not rewilded proxy species. They are an accident of colonial history that happens to produce millions of pounds of protein per year as a byproduct of pest management. The protein is real. The ecological benefit of the harvest is real. The fact that neither appears in any sustainability assessment I have encountered tells us something about the narrowness of what the assessments are measuring.

Bison are a different case: not an accident but a deliberate restoration, and one already well under way.

The American bison, *Bison bison*, is the only surviving North American megaherbivore from the Pleistocene assemblage this book has described. It survived the terminal extinction pulse, persisted through the Holocene in herds that European observers described in the millions, was reduced by the commercial hide hunt of the 1870s and 1880s to fewer than a thousand individuals, and has since been rebuilt to a continental population of roughly 500,000, most of them in commercial herds

managed as livestock, roughly 20,000 in conservation herds managed primarily for genetic diversity and ecological function.[4]

The conservation herds are the ones that matter for this chapter's argument. American Prairie, a private land conservation initiative in northeastern Montana, is assembling the largest contiguous tract of rewilded grassland in the temperate United States: currently over 450,000 acres of deeded and leased land along the Missouri River breaks, with a long-term acquisition target that would create a continuous habitat block adjoining the Charles M. Russell National Wildlife Refuge.[5] Bison are central to the project. American Prairie's herd, managed for genetic diversity and free-ranging behavior rather than for commercial production metrics, is intended to restore the grazing dynamics the northern Great Plains evolved under: concentrated impact, long recovery, nutrient redistribution across a landscape large enough that the animals' movement patterns can express themselves without fencing constraints.

The food-system dimension is real and deliberate. American Prairie's bison produce meat, marketed under the Wild Sky Beef brand, that is sold commercially as a revenue stream supporting the conservation mission. The animals are not supplemented with grain. They eat what the prairie grows. They are harvested by skilled marksmen in the field rather than transported to a processing facility, which eliminates both the transport stress and the infrastructure dependency. The product that reaches the consumer is, in a meaningful sense, food from a rewilded landscape: an animal that belongs on that landscape by any baseline the fossil record supports, managed under conditions that approximate the ecological dynamics the landscape evolved under, producing food as a function of its ecological role rather than in spite of it.

Tribal nations across the northern Plains are doing related work on a different institutional and cultural foundation. The InterTribal Buffalo Council, a cooperative of more than seventy-six tribal nations, has facilitated the restoration of bison to tribal lands across nineteen states, with a collective herd now numbering in the tens of thousands.[6] The restoration is simultaneously cultural and ecological: bison are central to the spiritual and subsistence traditions of many Plains nations, and their return to tribal land is an act of cultural restoration as much as an ecological one. The food produced from these herds feeds tribal communities directly, in some cases through programs that distribute bison meat as a healthier alternative to the commodity foods that federal nutrition programs have historically provided to

reservation communities. The survivability profile of these operations is notable: the animals require no purchased feed, no confinement infrastructure, and no industrial inputs beyond the fencing and water infrastructure that any large-herbivore operation requires. The dependency profile is thin. The system functions because the animal and the landscape have a relationship that predates agriculture itself.

Wild-caught fish from intact fisheries represent another existing component that is already producing food from systems the industrial model has not organized. The Alaska salmon fisheries are the clearest case. Alaska's constitution mandates sustainable management of its fisheries, and the state's management regime, built on the stock-assessment methodology that Ray Hilborn and Carl Walters systematized in their foundational *Quantitative Fisheries Stock Assessment*, is among the most rigorous in the world.[7] The system works by monitoring escapement: the number of spawning adults that pass upstream past counting stations to reproduce. Harvest is permitted only when escapement targets are met. In years when returns are low, fisheries are restricted or closed entirely. The result is a fishery that has sustained commercial-scale harvest for decades without the stock collapses that have characterized poorly managed fisheries elsewhere.

Hilborn and colleagues, in a 2020 analysis in *PNAS*, demonstrated that this kind of science-based management works at a global scale: fisheries under intensive management have, on the whole, been improving in stock status, while unmanaged or poorly managed fisheries continue to decline.[8] The finding matters for the food-systems argument because it establishes that wild-harvest food production from intact ecosystems is not inherently extractive or unsustainable. It is extractive when it is unmanaged. It is sustainable when the management matches the biology. Alaska's salmon fisheries produce hundreds of millions of pounds of protein per year from a system that requires no feed inputs, no freshwater diversion, no antibiotic administration, and no arable land. The system's primary requirement is that the ecosystem remain intact and that the harvest be governed by the stock's reproductive capacity. The dependency profile is thin in the same way the bison operation's is thin: what the system requires is the ecosystem itself, and management disciplined enough to not take more than the ecosystem can give.

Shellfish restoration on both coasts extends the marine case. Oyster reef restoration in the Chesapeake Bay, driven by federal and state partnership programs, has rebuilt reef acreage that was functionally destroyed by a century of overharvest

and disease pressure. A 2011 synthesis by Michael Beck and colleagues in *BioScience* documented that approximately eighty-five percent of the world's oyster reef habitat had been lost, and called for restoration as both an ecological and an economic priority: oyster reefs provide water filtration, shoreline stabilization, and habitat for commercially important fish and invertebrate species, in addition to producing a harvestable food product.[9] The restored reefs in the Chesapeake, on the Gulf Coast, and in Puget Sound are early-stage examples of what marine rewilding looks like when it produces food: a system in which the harvested organism is simultaneously performing ecological work — filtering water, building reef structure, providing habitat — and producing a food product for human consumption. The dependency profile is, again, thin. The oyster needs the water column. It does not need feed, antibiotics, or industrial infrastructure.

Managed elk and deer populations across much of the American West round out the existing picture. State fish and wildlife agencies manage ungulate populations through regulated hunting seasons calibrated to population surveys, habitat carrying capacity, and predator-prey dynamics. The system is imperfect (chronic wasting disease is a serious and growing problem in cervid populations, and the politics of predator management complicate the ecological picture), but the food produced is substantial. The US Fish and Wildlife Service estimates that hunters harvest approximately six million white-tailed deer per year nationally, producing roughly 600 million pounds of venison.[10] Elk harvests in the western states add to the total. This is food from landscapes that have been managed, at least in part, for the ecological relationships the animals participate in. It is not food from a rewilded system in the full sense this book describes — most state wildlife management operates under the 1492 baseline the book has critiqued — but it is food from animals on landscapes they belong to, managed for population health rather than for maximum commodity throughput, and it represents an existing infrastructure of harvest, processing, and distribution that a more deliberately rewilded food system could build upon.

WHAT COULD EXIST

The distance between what already exists and what could exist with deliberate effort is shorter than most people assume. It is a distance of policy, of infrastructure, and of imagination, not of technology or of ecological feasibility.

Knepp Estate in West Sussex, England, is the most thoroughly documented case of what happens when a conventional farm is released into a rewilding process, and Isabella Tree's 2018 account of the project in *Wilding* is the single best available narrative of rewilding as a food-producing enterprise.[11]

The Knepp story is worth telling in some detail, because the details are what make it instructive. The estate had been farmed conventionally for generations: arable crops, dairy cattle, the full infrastructure of a productive English lowland farm. By the late 1990s, the operation was losing money. The heavy Weald clay soils were marginal for arable production, and the costs of maintaining the fertility and drainage that intensive cropping required on those soils exceeded what the crops returned. Charlie Burrell and Isabella Tree made a decision that was, at the time, seen by neighboring farmers as somewhere between eccentric and irresponsible: they stopped farming the land conventionally and allowed it to begin reverting.

What happened over the subsequent two decades is one of the most closely observed ecological transitions in the European literature. The key decision — and the decision that makes Knepp relevant to this book rather than merely to the conservation literature — was to stock the land with proxy herbivores chosen for their ecological role rather than for their production characteristics. Old English longhorn cattle, Exmoor ponies, Tamworth pigs, and fallow deer were introduced, not managed in paddocks or rotations, but allowed to range freely across the estate, creating their own grazing and browsing patterns in response to the vegetation as it developed. The ponies browsed scrub and maintained open rides. The cattle grazed the grassland areas and created the dung-based invertebrate habitat that supports insectivorous birds. The pigs rooted through the woodland floor, creating the soil disturbance that germination of many native plant species requires. The deer browsed the woodland edge. Each species was performing a different ecological function, and the combination of functions produced a landscape that neither species could have produced alone. The results, documented by ecological surveys conducted by Imperial College London and other research institutions, were

striking. Species that had been absent from the region for decades reappeared: turtle doves, nightingales, peregrine falcons, purple emperor butterflies, and all five British species of owl now breed on the estate.[12] Soil biological activity increased. Water retention improved. The landscape shifted from a simplified agricultural surface into a structurally complex mosaic of scrub, wood pasture, grassland, and wetland, a mosaic that ecologists recognized as closer to the natural vegetation structure of lowland Britain than either the arable farmland it had replaced or the dense closed-canopy woodland that "natural" often implies.

The food produced from Knepp is not incidental. The estate sells pasture-fed organic meat from its free-roaming herds — beef, venison, pork — and the revenue from the meat operation, combined with income from ecotourism and conservation stewardship payments, has made the estate financially viable in a way the conventional farming operation had not been for years. The animals are not supplemented with purchased feed. They eat what the rewilding landscape produces. The meat is marketed as a premium product, and the market absorbs it. The estate demonstrates, at a scale large enough to be taken seriously (roughly 3,500 acres), that a rewilded landscape can produce food, support ecological recovery, and generate revenue simultaneously.

Tree's documentation of the Knepp process is careful about what the project does and does not prove. It proves that a specific piece of marginal English farmland, released from conventional management and stocked with proxy herbivores, produced measurable ecological recovery and a financially viable food product over a twenty-year period. It does not prove that every piece of farmland would respond the same way, or that rewilding is universally more profitable than conventional agriculture, or that the Knepp model scales to feed a nation. What it proves is that the model works: that rewilding and food production are not in opposition, that the landscape can do both, and that the animals doing the ecological work are also the animals producing the food.

CASE

The Knepp Experiment

> Knepp Estate in West Sussex, England — 3,500 acres of former Weald-clay farmland released from conventional management beginning in 2001 — is the most thoroughly documented ecological rewilding experiment in Europe.

Charlie Burrell and Isabella Tree stocked the land with proxy herbivores chosen for ecological function rather than production characteristics: Old English longhorn cattle, Exmoor ponies, Tamworth pigs, fallow deer. No paddocks, no rotations, no purchased feed. Within two decades, turtle doves, nightingales, peregrine falcons, purple emperor butterflies, and all five British owl species were breeding on the estate.

The meat operation (beef, venison, pork), combined with ecotourism and conservation stewardship payments, made the estate financially viable where the preceding conventional operation had failed. Tree's 2018 *Wilding* is the narrative record, and the ecological monitoring literature around Knepp continues to grow.

American Prairie, which I described above in its current form, represents the most ambitious version of this model currently under development in the United States. The long-term vision — a continuous habitat block of several million acres along the Missouri River — would create a landscape large enough for bison herds to express genuinely free-ranging behavior, for predator-prey dynamics to operate without intensive human management, and for the prairie ecosystem to function at something approaching its pre-industrial ecological capacity. The food-system implications are direct: bison managed at that scale, on that landscape, under those conditions, would produce food as a function of their ecological role. The harvest would be governed by population dynamics rather than by production targets. The product would be genuinely wild-harvested meat from a genuinely rewilded landscape.

Smaller-scale rewilding efforts on private and public land across the United States represent a distributed version of the same trajectory. Ranchers in the southern Plains who have replaced cattle with bison on shortgrass prairie are running operations that the fossil record endorses and that produce food with dependency profiles substantially thinner than conventional beef. Landowners in the Southwest who are experimenting with managed burro and horse populations on degraded rangeland are testing, in practice, the proxy-selection hypothesis Chapter 7 described.

Coastal restoration programs that combine oyster reef rebuilding with regulated harvest are producing food from ecosystems they are simultaneously restoring.

The distributed model deserves attention on its own terms, because the distributed model is how most food-system transitions actually proceed. The shift from draft horses to tractors did not happen by decree. It happened farm by farm, across decades, as individual operators made individual decisions based on what worked for their land and their economics. A shift toward rewilded food production would likely follow the same pattern: individual ranchers choosing bison over cattle on land that responds better to bison, individual landowners leasing acreage for conservation grazing rather than for commodity production, individual coastal communities investing in reef restoration that produces both ecological benefit and harvestable shellfish. Each individual decision is small. The aggregate is a trajectory, and trajectories compound. A hundred small operations producing food from rewilded landscapes, spread across a dozen bioregions, constitute a proof-of-concept network that no single demonstration site can provide. They test the model across soils, climates, regulatory environments, and market conditions. They generate the data that policy needs to scale what works. And they produce food while they do it: food that people eat, food that enters local and regional markets, food that gradually makes the idea of eating from rewilded systems less exotic and more ordinary.

None of these operations is large enough, individually, to register in the national food-supply statistics. Collectively, they represent a direction the food system could move in, with deliberate support, at a pace that is honest about what is currently possible and ambitious about what could become possible.

◆ THE HONEST SCALING ASSESSMENT ◆

The trajectory matters, and stating it honestly matters more.

Rewilded food systems cannot, today, feed every person currently eating from industrial systems. The numbers do not support that claim and I am not making it. The roughly 500,000 bison in the United States, even if all of them were managed as conservation herds on rewilded grassland, would produce a small fraction of the roughly 27 billion pounds of beef the US market consumes annually.[13] The Alaska salmon fisheries, impressive as they are, produce roughly 700 million to 1 billion pounds of fish per year, a substantial total, but a fraction of the roughly 6.3 billion

pounds of seafood Americans consume annually.[14] Feral hog harvest, managed elk and deer harvest, restored shellfish, and the scattered rewilding operations on private land add real volume without changing the fundamental arithmetic. Rewilded food systems today occupy a margin. The margin is real and productive, but it is a margin.

The argument this book is making is not that the margin should replace the center overnight. The argument is about trajectory: about which direction the food system should be moving, about which investments in land, infrastructure, and policy would produce the most durable food systems over the next century, and about what the framework developed in this book reveals when it is applied honestly to the options available.

Three things are true simultaneously, and the honest position requires holding all three.

First, rewilded food systems score well across all four axes of the framework this book has developed. They score well on the economic axis where they have been given the chance to operate: Knepp is financially viable, American Prairie's bison operation generates revenue, Alaska's salmon fisheries support a multi-billion-dollar industry. They score well on the social axis: tribal bison restoration has direct community-health and cultural-restoration dimensions, managed hunting supports rural economies and wildlife management infrastructure, coastal shellfish restoration employs restoration workers and supports fishing communities. They score well on the environmental axis: by definition, a system that produces food as a function of ecological restoration is not trading ecological health against food production but producing both together. And they score well on the survivability axis, which is the axis that matters most for the argument this book has been building.

The survivability score is where the distinction sharpens. Recall the axis as Chapter 4 defined it: survivability asks what a system requires to keep functioning, and what happens when those requirements are not met. An industrial feedlot requires continuous commodity grain, continuous diesel, continuous antibiotics, continuous federal subsidies, and continuous stable climate. A regenerative pasture operation requires most of what the industrial system requires for its monogastric enterprises, as Chapter 6 documented. A bison herd on rewilded prairie requires grass, water, and management. The dependency profile is not zero — the management requires human skill, the fencing requires materials, the harvest requires

processing infrastructure — but the profile is structurally thinner than anything the industrial or regenerative alternatives can offer. A system whose primary input is an intact ecosystem is a system that does not collapse when the commodity supply chain is disrupted, because it was never on the commodity supply chain. It does not require diesel for feed delivery, because the feed is growing under the animals' feet. It does not require antibiotics for disease management, because the stocking densities and movement patterns that produce disease pressure in confined systems do not exist. It does not require federal subsidies to make the economics work, because the economics are not built on the gap between input costs and commodity prices that the subsidies exist to bridge.

This is not a theoretical advantage. It is the practical consequence of a food system organized around ecological relationships rather than around industrial inputs. Consider what this means concretely. When the 2022 disruption of Ukrainian grain exports drove US feed costs to levels that Chapter 4 described as existential for some pastured-poultry operations, American Prairie's bison herd was unaffected. The animals did not eat commodity grain. The price of corn was irrelevant to their diet, to their management, and to the economics of their harvest. When the Texas winter storm of 2021 knocked out grid power across the state and killed livestock in operations dependent on electrically pumped water and heated shelters, free-ranging elk and deer in the same state continued to drink from streams and shelter in draws, as they had always done. The resilience was not engineered. It was inherent in the relationship between the animal and the landscape. That relationship predates, and does not depend on, any of the industrial infrastructure that the conventional system treats as a permanent condition. The poultry operation in the Piedmont that opened Chapter 4, the one that scored well on three axes and collapsed when the corn supply was disrupted, could not have collapsed on the fourth axis if it had never been on the commodity corn supply chain in the first place. Rewilded food systems are not on that supply chain. Their vulnerability surfaces are different: they are vulnerable to ecosystem degradation, to climate shifts that alter the vegetation the animals depend on, to disease events in wild or semi-wild populations, to political decisions about land use and access. Those vulnerabilities are real and should not be minimized. They are also vulnerabilities of a fundamentally different kind from the ones the industrial system carries, because they are vulnerabilities of the ecosystem itself rather than vulnerabilities of the industrial substrate layered on top of the ecosystem.

Second, the margin is expandable. The amount of land currently managed for rewilded food production is a small fraction of the land that could be managed that way. American Prairie's eventual footprint, if the acquisition and leasing program reaches its long-term target, would be measured in millions of acres, a single project on a single section of the northern Great Plains. Equivalent projects on equivalent landscapes across the continent — the shortgrass prairies of the central Plains, the basin-and-range country of the Southwest, the Gulf Coast marshlands, the eastern hardwood silvopastoral opportunities — would multiply the acreage by an order of magnitude. The constraint is not ecological feasibility. It is capital, policy, and the cultural willingness to manage landscapes for ecological function and food production simultaneously rather than treating the two as competing uses.

Third, the question is one of trajectory, not of overnight replacement. No serious person proposes dismantling the industrial food system on Monday and replacing it with bison and salmon on Tuesday. The industrial system feeds people. It feeds people badly, in the sense that the system's externalities are substantial and its survivability is poor, but it feeds them, and any transition away from it has to account for what people eat during the transition. The trajectory this book proposes is the deliberate expansion of rewilded food systems alongside — not instead of — the existing infrastructure, with the understanding that every acre moved from industrial production to rewilded production is an acre that scores better on all four axes and is more likely to still be producing food in fifty years under conditions that may look nothing like today's.

The trajectory framing is not a concession. It is the honest version of the argument. A food system that scores well on all four axes, that produces food as a function of ecological restoration rather than in spite of ecological degradation, and that does not depend on the persistence of an industrial substrate whose vulnerabilities the last decade has made starkly visible — that system deserves expansion. It deserves investment. It deserves the policy support and the cultural attention that have historically been directed at the systems it would gradually, partially, honestly replace. The argument is not that rewilded systems are the only answer. The argument is that they are the most underexplored answer, and that the framework this book has developed reveals them as the direction most consistent with the conditions the next century is likely to impose.

FROM THE FRAMEWORK TO THE FORK

The systems described in this chapter are not abstractions. They produce food that people eat. The bison from American Prairie and from tribal herds becomes steaks and roasts and ground meat that appears in kitchens and restaurants. The salmon from Alaska's managed fisheries fills freezers and feeds families. The venison from managed elk and deer harvests is processed and consumed by millions of American hunters and their communities every year. The pork from feral hog harvest in Texas, still underutilized relative to the volume available, feeds the hunters who take the animals and, increasingly, supplies processors who sell it into regional markets. The beef and pork and venison from Knepp feeds people in West Sussex and beyond.

What eating from rewilded systems looks like, in practice, is not exotic. It looks like a bison burger. It looks like grilled salmon. It looks like venison stew. It looks like oysters from a restored reef. It looks like food: recognizable, preparable, delicious in the ways that animal protein from well-managed systems is always delicious, because the animals ate what they were supposed to eat and lived on land that supported them. The plate is not strange. The system behind the plate is different, and the difference is what the framework measures.

There is a deeper point here about the relationship between the food and the landscape it comes from, and the point is worth stating because it connects to the book's larger argument about what sustainability means. When a bison grazes the northern Plains, it is not merely producing meat. It is performing the ecological work the prairie requires — the trampling, the grazing, the nutrient deposition, the seed-coat scarification — and the meat is a consequence of that work. When an oyster filters water in a restored Chesapeake reef, it is not merely producing a shellfish product. It is cleaning the estuary, building reef structure, creating habitat for juvenile fish and crustaceans, and the harvestable oyster is a consequence of that function. When a managed deer herd is held at carrying capacity by regulated hunting, the hunting is performing the predator function that wolves and cougars once performed at broader scale, and the venison is a consequence of that ecological substitution. In each case, the food is not the purpose of the system: the ecological function is the purpose, and the food is what the function produces. This inverts the relationship that industrial agriculture assumes. Industrial agriculture organizes

landscapes around food production and treats ecological consequences as externalities to be managed. Rewilded food systems organize landscapes around ecological function and treat food production as an output of that function. The inversion is not merely philosophical. It produces systems that score differently on the framework this book has developed, because the framework asks what a system requires to keep functioning, and a system whose purpose is ecological function requires the ecosystem — not the industrial substrate — to keep functioning.

The previous chapter established the principle that the fossil record provides the species guide for which animals belong in which landscapes. This chapter has shown that the principle is not merely a principle. It is an operational reality in the places where it has been given the chance to operate. Bison on the northern Plains are the fossil record's recommendation in practice. Salmon in Alaska's rivers are an intact fishery managed for persistence rather than for maximum extraction. Elk across the western mountains are a managed population of animals that belong where they stand. Even the feral hog — the accidental case, the invasive species converted into food — demonstrates that the landscape produces animal protein whether we plan for it or not, and that the question is whether we harvest it intelligently.

I want to pause on that inversion, because it is the thing this chapter has been building toward. The food systems described in Chapters 5 and 6 (industrial production and regenerative agriculture, respectively) both organize the landscape around the production of food. They differ enormously in how they do it, and the regenerative version is substantially better by every measure this book has applied. But both start from the same premise: the landscape is a production surface, and the question is how to manage the production surface responsibly. Rewilded food systems start from a different premise: the landscape is an ecosystem, the ecosystem has a set of relationships it evolved to maintain, and the food is what those relationships produce when they are allowed to operate. The distinction is not semantic. It produces food systems with fundamentally different dependency profiles, fundamentally different vulnerability surfaces, and fundamentally different relationships to the conditions the next century is likely to impose.

The next chapter takes the serious critics of Pleistocene rewilding head-on. The vision this chapter has described is not uncontested. Biologists have raised legitimate concerns about proxy-species equivalence, about ecological risk, about the cultural and political feasibility of the proposals the rewilding literature has advanced. Those

concerns deserve the engagement Chapter 9 will give them. What this chapter has tried to establish, before the critics are heard, is that the vision is not speculative. Parts of it are already here. The animals are real. The landscapes are real. The food is real. The question is not whether rewilded food systems can work: they already do, at the scales where they have been tried. The question is whether the scales can grow, and what it would take to grow them honestly. The framework says the direction is right. The next chapter asks whether the objections are answerable.

CHAPTER NINE

Proxies and Critics

♦ ♦ ♦

In 2005, Josh Donlan and twelve co-authors published a one-page essay in *Nature* titled "Re-wilding North America."[1] The essay proposed, in compressed form, what the previous three chapters of this book have been working toward at length: that the ecological histories of the Americas make the case for restoring large vertebrates, including Old World proxies for extinct Pleistocene species, to landscapes that had co-evolved with them. Donlan and his co-signatories included Paul Martin, Michael Soulé, Dave Foreman, and several prominent conservation biologists. The essay was short, ambitious, and deliberately provocative. It ended with a call for an "optimistic agenda for twenty-first century conservation" centered on the phased introduction of large mammals to carefully selected North American sites.

The response was substantial and, in large part, skeptical.

Within a year, multiple peer-reviewed rebuttals had appeared. The critiques ranged from the procedural to the ecological to the political, made by serious scholars whose concerns were grounded in real knowledge of conservation biology, invasion ecology, and the practical realities of land management. The critiques deserve to be engaged, not summarized, not caricatured, not dismissed. This chapter's job is to present the strongest version of each major critique, engage it on its merits, and arrive at a position the reader can evaluate for themselves.

I want to say at the outset what this chapter does not do. It does not decry the critics. The scholars who questioned the Donlan proposal were doing what scholars are supposed to do: subjecting an ambitious claim to rigorous scrutiny and identifying where the claim outran its evidence. Some objections I accept

substantially. Some I think are answerable on empirical grounds. Some miss the larger point, for reasons I will try to articulate. In no case do I think the objections are trivial, and in no case do I think the person making them was wrong. The rewilding proposal asks for much. The scrutiny is proportionate to the ask.

THE PROXY-EQUIVALENCE PROBLEM

The most ecologically grounded critique of Pleistocene rewilding was articulated by Tim Caro in a 2007 commentary in *Trends in Ecology & Evolution* titled "The Pleistocene Re-wilding Gambit."[2] Caro is a behavioral ecologist whose career has been built on careful empirical work — mostly on African mammalian ecology, where the large-bodied fauna that the rewilding proposal seeks to restore or proxy is still intact. His critique cuts to the heart of the proxy-selection argument this book has been making.

Caro's central concern is what might be called the proxy-equivalence assumption: that a closely related living species will perform the same ecological function as its extinct relative when placed in the same landscape. The rewilding proposal depends on this assumption. Donlan and colleagues proposed that African or Asian elephants could serve as proxies for mammoths, African lions or cheetahs as proxies for the ex-
tinct American lion and cheetah, and Bactrian camels or dromedaries as proxies for *Camelops*. The assumption underlying each proposal is that phylogenetic proximity — the degree of relatedness between the proxy and the extinct species — predicts ecological equivalence. A close relative, the assumption holds, will occupy a similar niche, exert similar effects on vegetation and soil, and restore something like the ecological function the extinct species performed.

Caro argued the assumption is unsupported by available evidence. Ecological function is not a simple derivative of phylogenetic relatedness. Two closely related species can occupy very different niches if they have diverged in diet, behavior, habitat preference, or body size. An African elephant is not a mammoth. The lineages diverged millions of years ago and adapted to different climatic regimes, vegetation communities, and predator assemblages. Placing an African elephant on the Great Plains and expecting it to perform the ecological role of *Mammuthus columbi* is an extrapolation from phylogeny to ecology that available evidence does

not license. The same concern applies to every proxy pairing the Donlan proposal advanced. A modern dromedary is not *Camelops*. A modern African cheetah is not *Miracinonyx*. The degree to which proxy actual behavior in the actual landscape would replicate the extinct species' ecological function is an empirical question that cannot be answered from phylogeny alone.[2]

Caro went further. He pointed out that introductions of large-bodied species into novel ecosystems have, historically, produced ecological consequences that introducing parties did not anticipate and could not control. The history of species introductions is not reassuring. Cane toads in Australia, rabbits in Australia, Nile perch in Lake Victoria—each was introduced with some expectation of ecological benefit, and each produced cascading damage still being managed decades or centuries later. The proxy-species proposal, Caro argued, was asking conservationists to introduce large-bodied animals with enormous individual ecological footprints, capable of transforming vegetation structure and soil dynamics across large areas, into ecosystems that have not experienced those animals for twelve thousand years. The ecosystems have reorganized in the interim. The vegetation communities have shifted. The predator-prey dynamics have stabilized around a different cast of species. Introducing a proxy megaherbivore into a system operating without megaherbivores for a hundred and twenty centuries is not a restoration in any straightforward sense. It is an experiment with uncertain outcome, and the scale of potential consequences, both positive and negative, is proportional to the size of animals being introduced.

I find this critique serious and, in its strongest form, largely correct. The proxy-equivalence assumption *is* unsupported as a blanket proposition—phylogenetic proximity does not guarantee ecological equivalence. The history of species introductions warrants caution. Caro is right that the burden of evidence lies with proponents, not skeptics, and that evidence as of 2007 was not sufficient to justify continental-scale proxy introductions on phylogenetic reasoning alone.

Where the critique reaches its limit is in what follows from it. Caro's argument establishes that the proxy-equivalence question is empirical—that it cannot be assumed but must be tested. It does not establish that the question is unanswerable, or that the answer will necessarily be negative—it establishes only that the empirical work remains undone. The proposition that a dromedary camel would browse the thorny vegetation of the American Southwest in a manner ecologically similar to *Camelops* is not a proposition that can be confirmed or denied from phylogeny. It is a

proposition that can be tested by putting dromedaries on Southwest rangeland and measuring what happens to vegetation, soil, the animal, and the surrounding ecological community. The test would need to be designed carefully, monitored rigorously, and conducted at a scale small enough that negative outcomes could be managed without ecological catastrophe. It would need control sites and long time horizons, the kind of sustained empirical attention that conservation biology gives to reintroduction programs for endangered species. What it would not need is certainty in advance. It would need a hypothesis worth testing and a methodology capable of testing it. And people with the nerve to actually do it.

FIELD NOTE

Why Not Goats?

I have been asked, more than once, why I do not just propose goats.

The question is reasonable on its face. Goats are easy to permit. The regulatory pathway exists. The market for goat meat exists, modestly. They eat thorny browse that cattle will not touch. California already deploys them, by the thousands of acres, for fuel-load reduction in chaparral. Goats are right there. Why bring in something exotic?

My honest first answer to that question, given to a person whose work I respect over an exchange that escalated faster than I had planned, was a tit-for-tat: "Why not camels?" That is not a real answer. The ludicrous question deserved an equally ludicrous non-response, and that is what it got from me at the time. The better answer, the one this Field Note exists to make, is body size, and there is a literature behind it.

The 2023 Pringle review in *Current Biology* synthesizes a decade of large-herbivore impact studies and lands on a finding that has been clean across exclosure experiments and observational comparisons: small herbivores do not functionally compensate for large ones. Body size governs ecological impact in ways that scale across multiple variables, including biomass consumption, seed dispersal, depth of trampling, structure of browse pressure, and generation of ground disturbance. Above roughly one hundred kilograms, the regulation of the population shifts from top-down predation to bottom-up food limitation. Above roughly one thousand kilograms, the animal becomes essentially invulnerable to non-human predators and exerts ecological influence that no smaller animal can replicate.[10] The review's framing is direct: livestock are poor surrogates for wild large

herbivores precisely because the body-size profile of common livestock species is concentrated in the smaller end of the range that the wild species occupied.

The numbers locate the argument. Goats run thirty to fifty kilograms. Cattle run five to seven hundred. The dromedary is in the cattle range. The Bactrian goes higher, six hundred to a thousand. *Camelops*, the animal the proposal is actually about, sat above the cow and probably above the modern Bactrian. Goats and camels are not interchangeable proxies for the same lost role. They occupy different positions on the body-size scale, and the ecological impact of the larger animal is not something the smaller animal can produce by being more numerous.

It also deserves saying that cattle themselves are proxies for the wild aurochs the European landscape lost. The fact that nobody calls them out as substituted does not make them less so. Cattle have monopolized the regenerative-agriculture conversation, but they are not the only large herbivore, and they are not the largest available. Camels, particularly Bactrians, are larger. Their ecological footprint, on a per-animal basis, is correspondingly greater. The argument for proposing them is not that goats are bad. It is that goats cannot do what *Camelops*-sized animals did, and the landscape was calibrated to *Camelops*-sized animals for most of its evolutionary history.

The eye-rolling I have received on this is real. It does not, by itself, change what the fossil record says or what the body-size literature documents.

CASE

Pleistocene Park

The proxy hypothesis the previous section discussed has, for thirty years, been undergoing a sustained empirical test. Sergey Zimov, a Russian geophysicist, started Pleistocene Park in 1996 in Yakutia, in the eastern Siberian Arctic. The premise is the proxy proposal carried to its conclusion: reassemble, at landscape scale, the large-herbivore community that occupied the mammoth steppe before the terminal Pleistocene losses, and watch what the land does.

The experimental enclosure now covers roughly one hundred fifty square kilometers and holds Yakutian horses, plains bison transferred from Alaska, musk-oxen, reindeer, yak, and a small number of Bactrian camels. The mammoth itself is, of course, absent; the project's framing has always been that the mammoth's *function* in the system — heavy-bodied browsing, snow disruption, dung-driven nutrient cycling — could be reconstructed from a community of available large herbivores even without the original keystone species.

What has been measured. Soil temperatures under grazed plots run five to ten degrees Celsius colder than ungrazed adjacent ground, because the herbivores compact and disturb the winter snow that otherwise insulates the soil and accelerates permafrost thaw. Vegetation has shifted in grazed plots from moss-and-shrub tundra toward a grass-dominated cover that more closely resembles pollen records of the Pleistocene mammoth steppe. Carbon dynamics — the project's larger claim, that grazer-driven cooling could buffer permafrost methane release at climatically meaningful scales — remain under measurement and dispute.

What it is not. Pleistocene Park is small relative to the landscape its hypothesis is about; it is not a controlled experiment in the strict statistical sense; and it is operating in a political environment that has constrained both the science and the international collaboration the project relies on. Critics including Caro 2007 would note that a single uncontrolled site does not settle the proxy-equivalence question even after thirty years. They are right that it does not settle it. They are also wrong, in the form the critique typically takes, that nothing has been learned: thirty years of measurements at a real site running a real version of the hypothesis is more empirical attention than the proxy debate had received from any other quarter when the project began.

The point of naming Pleistocene Park here is not to argue that it has succeeded or failed. It is to point out that, when the previous section said the experiment had not been run, that statement was true for the American case and not true for the Eurasian one. Somebody has been running it for thirty years. The question is whether anyone will run an analogous experiment on the continent the book is actually about.

The proxy-equivalence question is a question about experimental design rather than about ecological principle. The principle that an ecosystem missing its keystone herbivores functions differently than one in which they are present is well established in the ecological literature and has been demonstrated empirically in multiple contexts. The question is whether a specific proxy species, in a specific landscape, restores a specific ecological function to a measurable degree. That question is answerable by experiment, not by argument. The fact that the answer is not yet known is an argument for conducting the experiment, not for declining to conduct it.

Donlan and colleagues, in their fuller 2006 treatment in *The American Naturalist*, anticipated this line of response.[3] They proposed a phased approach: beginning with

species that are already present in North America in managed or feral populations (horses, burros, camels in small numbers), moving to species with closer ecological parallels to extinct fauna (bolson tortoises, which have since been the subject of a small-scale reintroduction program), and proceeding to larger introductions only as evidence accumulated from earlier phases. The phased approach directly responds to the proxy-equivalence concern: rather than assuming equivalence and introducing at scale, it proposes testing equivalence at small scale and scaling only what the evidence supports.

The phased approach is the right framework. It also requires honesty about what it can and cannot deliver. A small-scale experiment with dromedaries on a single Arizona ranch would tell us something about dromedary browsing behavior on Sonoran vegetation. It would not tell us what a self-sustaining population of dromedaries would do to the landscape over decades. The experimental evidence accumulates incrementally, and the increments may not resolve the proxy-equivalence question as cleanly as either side would like. What the experiments would produce is data: specific, empirical, measurable data about the interaction between proxy species and the landscapes they are being considered for. The current state of the conversation, in which advocates assert equivalence and critics deny it on the basis of argument rather than evidence, is less useful than a conversation in which both sides work from data generated by carefully designed field experiments.

◆ THE POLITICAL AND CULTURAL OBSTACLE ◆

The second major critique came from Daniel Rubenstein and colleagues, most forcefully in a 2006 paper in *Biological Conservation* titled "Pleistocene Park: Does Re-wilding North America Represent Sound Conservation for the 21st Century?"[4] Rubenstein, a behavioral ecologist at Princeton whose work on equid ecology is among the most cited in the field, brought different concerns than Caro had raised. Where Caro focused on the ecological risks of proxy introductions, Rubenstein focused on the practical, political, and cultural barriers.

The Rubenstein critique is, in essence, a feasibility argument. Suppose the ecology works. Suppose the proxy-equivalence question, when tested, yields favorable results for at least some species-landscape pairings. The proposal still requires land,

regulatory approval, and the consent of communities living adjacent to the introduced animals, or at least their tolerance. It requires a political and cultural environment in which the introduction of large, potentially dangerous animals onto working landscapes is regarded as an acceptable use of public and private land. Rubenstein and colleagues argued that none of these conditions obtained in the United States of the early twenty-first century, and that the distance between the ecological vision and the political reality was large enough to make the proposal practically unworkable.

The argument was specific. Americans do not, on the whole, want large predators near their homes, livestock, or children. The political conflict over wolf reintroduction in the Northern Rockies, which had been ongoing for a decade at the time of the Rubenstein paper, illustrated the scale of resistance that even a single native predator species could generate. Extending the reintroduction concept to African lions, elephants, or cheetahs, the authors suggested, was to move from a politically difficult proposal to a politically impossible one. The same reasoning applied to megaherbivores. A free-ranging elephant on the Great Plains would be an extraordinary management challenge: property damage, liability exposure, risk to human safety, conflict with existing agricultural land use. None of these concerns were ecological. All were practical, and all were, in the judgment of the critics, sufficient to render the proposal dead on arrival in any real political context.

Rubenstein and Rubenstein returned to this line of argument a decade later, in a 2016 letter to *PNAS* prompted by Svenning and colleagues' synthesis on trophic rewilding.[5] The updated critique noted that the intervening decade had not produced the political or cultural conditions necessary for the kind of introductions the Donlan proposal envisioned, and that the conversation about rewilding had largely shifted from Pleistocene-proxy introductions toward trophic rewilding, a framework that emphasized the restoration of ecological processes (top-down regulation, herbivory, seed dispersal) rather than the reintroduction of specific Pleistocene analogs. The shift, they suggested, itself acknowledged that the original proposal's ambitions exceeded its political reach.

I accept this critique in significant part. The cultural and political obstacles to large-scale megafaunal proxy introductions in the United States are real, substantial, and not likely to be resolved by ecological argument alone. A proposal that requires Americans to accept free-ranging elephants on the Great Plains is not a proposal that

will be enacted in any political environment I can foresee. The obstacles are not irrational—they are the reasonable concerns of people who live on working landscapes and have legitimate interests in what lives on those landscapes alongside them. The rewilding conversation cannot proceed as though those interests do not exist or as though they will yield to the force of ecological reasoning.

Where the critique warrants a response, and here the response is more modest than the original Donlan proposal, is in the scope of what is being proposed. The political obstacles that Rubenstein identifies are real for the proposal as Donlan et al. stated it in 2005: a continental-scale reintroduction program involving African megafauna. The political obstacles are substantially different for a more modest version: small-scale proof-of-concept experiments, conducted on private or willing public land, using species that are already present in the Americas, at scales small enough that the management challenges are manageable and the risks are bounded.

Consider what a modest proof-of-concept program would actually look like. A rancher in southern Arizona, one of many whose land is degraded enough and whose relationship to the conventional cattle economy is marginal enough to be open to alternatives, places twenty dromedary camels on a thousand acres of Sonoran desert rangeland under a research protocol developed in partnership with a university. The camels are fenced. The vegetation response is monitored. The soil biology is sampled. The animal behavior is documented. The experiment runs for five years. At the end of five years, the data either support the hypothesis that dromedary browsing produces measurable ecological benefit on that landscape, or they do not. If they do, the experiment is expanded to additional sites, species, and scales. If they do not, the experiment is ended and the hypothesis is revised.

The political barriers to this version of the proposal are not zero. They include permitting, liability, neighbor relations, and the cultural unfamiliarity of camels on Arizona rangeland. But they are barriers of a different order than those to placing elephants on the Great Plains. They are navigable, and in fact are already being navigated in analogous contexts: Bolson tortoise reintroduction in New Mexico, experimental bison management on tribal and private land across the northern Plains, and the ongoing management of feral horse and burro populations on public land in the West. Each of these involves the placement of large-bodied animals on working landscapes under regulatory frameworks that already exist.

The political critique is strongest when aimed at the most ambitious version of the proposal. It is weakest when aimed at the most modest version. The distance between the two versions is not small. One asks a nation to accept a fundamentally different relationship to its landscapes. The other asks a handful of willing landowners and researchers to run a bounded experiment and publish the results. The book's ask is the modest version: not a continental program but a set of experiments, not certainty but data, not political transformation but the kind of incremental, evidence-based expansion that conservation biology already practices with endangered-species reintroduction programs.

◆ THE BASELINE-AS-FANTASY CRITIQUE ◆

The third critique cuts deeper because it challenges not a specific proposal but the frame within which all rewilding proposals are made.

The argument, advanced by various scholars with different emphases, is that the Pleistocene baseline is itself a kind of conservation fantasy, a longing for a world that existed under conditions so different from the present that its restoration is not merely difficult but conceptually incoherent. Conservation, under this view, should focus on protecting what currently exists: the species, ecosystems, and ecological processes that are present now, under current conditions, and that face measurable threats from current human activity. The late-Pleistocene assemblage is interesting as paleontology, not a useful reference point for conservation in the Anthropocene. The world has moved on. The ecosystems have reorganized. The question is not how to get back to a Pleistocene state that no living person has experienced but how to protect and improve the ecological communities we actually have.

This is not a fringe position. Versions of it appear in thoughtful conservation writing across the last two decades, and the version embedded in the trophic-rewilding framework that Svenning and colleagues articulated in their 2016 synthesis in *PNAS* is, in part, a response to the force of the objection.[6] Svenning and colleagues shifted emphasis from "restore the Pleistocene assemblage" to "restore the ecological processes that large-bodied animals perform," a shift that sidesteps the historical-baseline question by making the argument functional rather than historical. If a landscape needs top-down regulation by large predators, the relevant question is

whether large predators can provide that function now, not whether the specific predators being considered are phylogenetically correct proxies for Pleistocene species.

I respect the pragmatism of the trophic-rewilding reframe. It is a useful bridge between the Donlan proposal's ambitions and the political and ecological realities the proposal confronted. Where the baseline-as-fantasy critique ultimately falls short is in what it asks us to accept.

The argument that conservation should focus on protecting what currently exists takes the current ecological state as its reference point. But the current ecological state of North America, as Chapter 2 of this book argued at length, is not a stable equilibrium that happens to have arrived at its present form through natural processes—it is a depleted state, one that resulted from the loss of more than forty species of large mammals in near time, a loss that reshaped every terrestrial ecosystem on the continent. The vegetation communities are different. The soil dynamics are different. The predator-prey relationships are truncated. The seed dispersal networks are broken. The grasslands that evolved under proboscidean herbivory operate differently without it. The canyons that evolved under camelid browsing are being colonized by thorny vegetation the camelids used to suppress. The pronghorn still runs at speeds calibrated to a predator that no longer exists.

To argue that conservation should protect "what currently exists" is to argue that we should protect the depleted state, that we should defend the post-extinction baseline against further degradation without asking whether the baseline itself represents a condition worth defending on ecological grounds. It is the argument from Chapter 2 returning in a different register: the Columbian Curtain, extended. Except that here the curtain is not drawn at 1492. It is drawn at the present moment, and what it hides is the same thing the Columbian Curtain hides: the depth of the loss and the possibility that addressing the loss would produce ecological outcomes that the current baseline cannot.

Granted, the comparison has limits. The Pleistocene is gone in a way that 1491 is not. The ecosystems have had twelve thousand years to reorganize, and the reorganization is not nothing. New ecological relationships have formed, new equilibria have been reached, and disrupting those equilibria carries its own risks. Caro's concern about the consequences of introducing megafauna into systems that

have stabilized without them is a version of this point, and it is a legitimate one. The twelve-thousand-year gap is real. The ecosystems on the other side of it are not frozen in time, waiting for the return of the animals they lost. They have moved, and the moving is itself an ecological fact.

But the moving has not been unconstrained. The ecosystems have reorganized within the limits of what was available to them—and what was available was a diminished cast of species. A grassland that loses its proboscideans does not replace them with an ecological equivalent. It reorganizes around the absence: vegetation structure shifts, nutrient cycling changes, fire regime adjusts, soil biology adapts to a world with less large-mammal disturbance. The reorganization is real, but it is a reorganization around a loss, not a recovery from one. The new equilibrium is a depleted equilibrium. To take the depleted equilibrium as the baseline and to argue that conservation should protect it against disturbance is to naturalize the depletion, to treat the consequence of a mass extinction as the reference condition for all subsequent ecological management.

I do not think this is a defensible position when stated plainly. The fact that it is the de facto position of most conservation practice in the United States does not make it defensible—it makes it familiar.

The response is not to demand that every conservation agency immediately adopt a Pleistocene baseline—that would be as politically unrealistic as the most ambitious version of the Donlan proposal, and for similar reasons. The response is to recognize that the current baseline is contingent, that it results from a specific historical event (the late-Pleistocene megafaunal extinction), that it represents a depleted state rather than a natural one, and that proposals to address the depletion deserve evaluation on their ecological merits rather than dismissal on the grounds that the baseline is the baseline.

◆ WHAT THE WOLF TAUGHT US ◆

There is one case in which the rewilding conversation moved from argument to evidence, and the evidence is worth pausing on, because it is the closest thing the field has to a completed experiment.

In 1995 and 1996, gray wolves were reintroduced to Yellowstone National Park after an absence of approximately seventy years. The reintroduction was politically contentious: ranching communities adjacent to the park opposed it vigorously, and the legal and regulatory battles surrounding it lasted years before and after the animals arrived. The wolves were not Pleistocene proxies. They were a species reintroduced to a landscape it had occupied within living human memory, under a regulatory framework that already existed for endangered-species recovery. It matters here not as a Pleistocene-rewilding experiment but as something simpler: a large-bodied animal returned to a landscape that had reorganized in its absence, with the reorganization measured before, during, and after the return.

What happened was substantial. William Ripple and Robert Beschta, in a 2012 synthesis covering the first fifteen years of wolf presence in the park, documented a series of measurable ecological changes that cascaded from the reintroduction through the park's trophic structure.[7] Elk, which had been the park's dominant herbivore in the wolves' absence, altered their behavior and distribution in response to predation risk. Riparian areas that had been heavily browsed by elk—willows, aspens, and cottonwoods along streams and rivers—began to recover as elk browsing pressure declined. Luke Painter and colleagues, in a 2015 study in *Ecology*, documented that the proportion of young aspen browsed by elk in northern Yellowstone dropped from ninety percent in 1998 to thirty-seven to sixty-three percent by 2012, with corresponding increases in aspen recruitment and height growth in areas of high wolf activity.[8] The vegetation recovery had downstream effects: stream-bank stabilization improved as willow and cottonwood root systems reestablished, channel morphology began to shift, and beaver — absent from much of the park's northern range for decades — returned to streams where the recovering riparian vegetation provided both food and building material.

The trophic cascade was not simple. Creel and Christianson, in a 2009 study also in *Ecology*, found that individual elk willow consumption actually increased in the presence of wolves, suggesting that vegetation recovery was driven primarily by elk population reduction rather than by behavioral avoidance alone.[9] The mechanism was more complex than the initial "landscape of fear" narrative suggested. What was not in dispute was the outcome: the reintroduction of a top predator to a landscape that had reorganized in its absence produced measurable, cascading, and largely beneficial ecological effects within two decades.

The Yellowstone wolf case is not a perfect analogy for the proxy-species proposals this book has discussed. Wolves are a native species reintroduced to a landscape from which they had been absent for decades, not centuries or millennia. The ecological continuity between the pre-removal and post-reintroduction periods is much tighter than the continuity between the late Pleistocene and the present. The political and regulatory framework for wolf reintroduction, while contested, was grounded in the Endangered Species Act, a legal infrastructure that does not exist for Pleistocene-proxy introductions.

What the wolf case does demonstrate—and this is what I want us to pause on before moving to the chapter's close—is that the principle works. An ecosystem missing a keystone species reorganizes around the absence. When the species is returned, the ecosystem begins to reorganize around the presence. The reorganization is measurable, cascading, and ecologically significant. The fact that the wolf was absent for seventy years rather than twelve thousand does not change the principle; it changes the degree of uncertainty about the outcome. A twelve-thousand-year absence produces more reorganization, more ecological distance between the pre-removal and the present state, and more uncertainty about what reintroduction would produce. The uncertainty is real. It is not a reason to decline the experiment but a reason to design the experiment carefully.

◆ THE ASK ◆

The book has now made its case across eight chapters. It has argued that the dominant framework for evaluating food systems is incomplete. It has proposed a fourth axis — survivability — that captures what the existing framework misses. It has applied the framework to industrial systems, to regenerative agriculture, and to the rewilded food systems that score best across all four axes. It has taken the fossil record seriously as a guide to which animals belong in which landscapes. It has described what rewilded food systems look like where they already exist, and what they could look like with deliberate effort.

This chapter has taken the critics seriously. Caro's concern about proxy equivalence is valid: the assumption that phylogenetic proximity equals ecological equivalence is unsupported as a blanket claim and should be tested rather than assumed. Rubenstein's concern about political and cultural feasibility is valid: the

most ambitious version of the rewilding proposal exceeds what the current political environment can absorb, and the conversation has to meet that reality where it is. The baseline-as-fantasy critique has force: the Pleistocene is gone, the ecosystems have reorganized, and any intervention carries risks that must be weighed against the potential benefits.

What the book asks for, in light of all three critiques, is not certainty—not a continental program, not the introduction of elephants to the Great Plains or lions to the Rockies, not a demand that the conservation establishment abandon its existing frameworks or that ranching communities accept megafauna on their land without consent.

What the book asks for is proof of concept. Rewilding hypotheses can be tested at small scales: camels on a single ranch in Arizona, managed burro populations in the Grand Canyon watershed, bison on tribal and private land across the northern Plains (where this is already happening, with measurable results), wolves in specific Northern Rocky Mountain ecosystems (where this has already been done, with the trophic cascade data to show for it). Each of these is a bounded experiment that produces data and can be expanded if the data are favorable and ended if they are not.

The proof-of-concept approach does not require resolving the proxy-equivalence debate in advance; it generates the data the debate needs. It does not require overcoming the political obstacles to continental-scale megafaunal introduction; it works within the political space that already exists for small-scale research and land-management experimentation. It does not require accepting the Pleistocene baseline as the one true reference point for conservation; it requires only the recognition that the current baseline is contingent, that the depletion it represents is real, and that proposals to address the depletion deserve to be evaluated rather than dismissed.

The experiments are not hypothetical. Some of them have already been run. Wolves in Yellowstone. Bison on tribal and conservation land across the northern Plains. Bolson tortoises in New Mexico. Feral horse and burro management—contested and imperfect, but generating ecological data in real time about what happens when equids occupy landscapes their lineage shaped. Each of these is a data point that says something about the interaction between large-bodied animals and the landscapes they are placed on. The data are not yet sufficient to answer the proxy-equivalence question for every species-landscape pairing the rewilding literature has

proposed. They are sufficient to show that the question is answerable, that it yields to empirical investigation rather than resolving only to argument.

The critics have earned the book the obligation to state its ask carefully. The ask is this: try the experiments honestly, at small scales, with rigorous monitoring and genuine willingness to accept negative results. Try them on landscapes where the fossil record and the current ecological evidence both point to an animal that is not there. Measure what happens. Publish the results. Let the data accumulate. And then decide not from argument, not from longing, not from the fossil record alone, but from the evidence of what actually happens when the animals return to the land that shaped them.

The critics say the proposal is uncertain. They are right. The response is not to retreat from the uncertainty but to design experiments that reduce it. The response is not certainty.

The response is science.

CHAPTER TEN

Water and Indigenous Systems

♦ ♦ ♦

The argument this book has been making rests on a particular frame. The Pleistocene baseline, the proposal that restoration should be guided by what the fossil record tells us about the ecological assemblages that preceded the near-time extinctions, is the frame the previous nine chapters have developed, applied, and defended against its serious critics. The frame is, I believe, defensible. It is not the only one.

This chapter exists because intellectual honesty requires saying so. The Pleistocene baseline is one way of answering the question "what should this landscape look like?" It is not the only way the question has been answered, and it is not the only way the question should be answered. Indigenous peoples across the Americas managed their landscapes for millennia between the terminal Pleistocene extinctions and the arrival of Europeans, and the management traditions they developed constitute a parallel intellectual framework for thinking about restoration, one that the Pleistocene frame should be in dialogue with rather than in competition with, and one that the rewilding conversation has been slower to engage than it should have been.

The chapter also exists because the rewilding conversation, as conducted in the conservation biology literature and in this book through Chapter 9, has been almost entirely terrestrial. The oceans are missing. Kelp forests, oyster reefs, pelagic fisheries, the entire marine domain: the water column has been largely absent from popular rewilding discourse, and the absence is a gap the book needs to address before it moves into practical application. The marine systems are as depleted as the terrestrial ones, and the case for their restoration is, if anything, more empirically advanced.

Two expansions, then. The first is a framework expansion: from the Pleistocene baseline alone to the Pleistocene baseline in dialogue with Indigenous management traditions. The second is a domain expansion: from terrestrial rewilding alone to terrestrial rewilding alongside marine restoration. Both expansions strengthen the book's argument by making it less parochial. Neither contradicts what has come before.

A PARALLEL TRADITION

Robin Wall Kimmerer opens *Braiding Sweetgrass* with a creation story, the Skywoman story of the Potawatomi, and then spends the next three hundred pages demonstrating what it looks like when Indigenous ecological knowledge and Western scientific training are held in the same mind simultaneously.[1] The book is not an argument against Western science. It is an argument that Western science, when practiced in isolation from the traditions of people who have lived with and managed specific landscapes for thousands of years, is working with an unnecessarily thin dataset. The observation has force. It has force because Kimmerer is not making it from outside the scientific tradition. She is a plant ecologist at the SUNY College of Environmental Science and Forestry, a researcher who publishes in peer-reviewed journals, and she is making the argument from inside both traditions at once.

What Kimmerer describes, and what a substantial body of scholarship by Indigenous researchers has documented across multiple ecosystems, is a form of landscape management that does not fit cleanly into either the conservation or the agriculture categories that Western land-use thinking provides. The management is neither preservation (leaving the landscape alone) nor cultivation (reorganizing the landscape around a production target). It is something in between, and the in-between space is where the framework this book has been building meets a tradition it needs to reckon with.

M. Kat Anderson's *Tending the Wild*, a study of Native American management of California's ecosystems, provides the most thoroughly documented case for the scale and sophistication of this management.[2] Anderson spent years working with Indigenous elders and with the ethnobotanical and fire-ecology literatures to reconstruct what California's landscapes looked like under active Indigenous management before European contact. The picture that emerges is not wilderness in

the sense that the word is usually used. It is a managed landscape: managed through fire, through selective harvesting, through the cultivation of specific plant communities, through the deliberate maintenance of habitat for specific animal species. The management was not incidental. It was systematic, it was transmitted across generations, and it produced landscapes that early European observers described as paradisiacal — open, productive, teeming with game — without understanding that the condition they were admiring was the product of human management rather than the absence of it.

Anderson documented that California's Indigenous peoples used fire as a primary management tool across virtually every vegetation type in the state. Grasslands were burned to maintain open habitat and to promote the growth of seed-bearing grasses and bulb-producing plants. Oak woodlands were burned to reduce competition from conifers, to promote acorn production, and to clear the understory for ease of harvest. Chaparral was burned on rotation to produce a mosaic of successional stages, each of which provided different resources: early successional stages for basket materials and herbaceous food plants, later stages for browse and mast.[3] The burning was not random. It was timed, targeted, and calibrated to the specific ecology of each vegetation type. The resulting landscapes were not natural in the sense of being unmanaged, and they were not agricultural in the sense of being reorganized around a single crop. They were something the Western vocabulary does not have a clean word for: managed ecosystems that produced food, fiber, and habitat as functions of ecological management rather than as alternatives to it.

The scale of this management was not marginal. Anderson estimates that Indigenous burning in California shaped millions of acres annually, a figure that challenges the assumption, embedded in the word "wilderness," that the landscapes Europeans encountered were unmanaged. The ecological consequences of the management's cessation after European contact were rapid and, in many cases, still unfolding. The dense conifer forests that now dominate much of California's Sierra Nevada, forests that burn catastrophically in modern wildfires because they have accumulated fuel loads that historical fire regimes would have prevented, are not the forests that existed under Indigenous management. They are the forests that grew in the absence of Indigenous management, the same way the grasslands that Chapter 7 described are the grasslands that exist in the absence of their Pleistocene grazers. The parallel is structural. In both cases, the current state of the landscape is a consequence

of a relationship that has been interrupted, and the interruption is legible in the landscape's present condition.

The parallels to the rewilding argument are striking, and they are also incomplete. The rewilding framework this book has developed asks: what did this landscape look like before the near-time losses, and how do we restore the ecological relationships the losses disrupted? The Indigenous management framework asks a different question: how did the people who lived on this landscape between the Pleistocene and 1492 manage it, and what can that management tradition teach us about the landscape's productive capacity under human stewardship? Both questions are valid. Both produce different answers. The difference matters, and collapsing the two frameworks together would be a mistake that serves neither well.

◆ TWO BASELINES, NOT ONE ◆

The Pleistocene baseline and the Indigenous management baseline are not the same thing, and I want to be precise about why.

The Pleistocene baseline looks backward past human presence on the continent: past the near-time extinctions, past the arrival of the first peoples, to the ecological assemblages that existed before the megafaunal losses. The question it asks is ecological in the deep sense: what was the full complement of species and relationships this landscape evolved under, and what has been lost? The answer involves animals — mammoths, ground sloths, camelids, equids — that no living human on this continent has ever seen in the wild. The restoration it envisions is a restoration of ecological function that predates human management of any kind.

The Indigenous management baseline looks at a different window: the millennia between the Pleistocene losses and European colonization. During that window, Indigenous peoples across the Americas were not passive inhabitants of a depleted landscape. They were active managers of it. The management traditions they developed were responses to the landscape as it existed after the megafaunal extinctions, a landscape already missing its mammoths and ground sloths, already reorganized around a reduced fauna, and managed within those constraints by people whose knowledge of the land's ecology was deep, specific, and accumulated over hundreds of generations. The restoration this framework envisions is a restoration of human-landscape relationships that were disrupted not by the

Pleistocene extinctions but by European colonization, forced removal, and the suppression of Indigenous land-management practices.

The two baselines produce different prescriptions. A Pleistocene baseline applied to the American Southwest asks about camelids and equids, the animals the fossil record identifies as co-evolutionary partners of the landscape's vegetation, as Chapter 7 described. An Indigenous management baseline applied to the same landscape asks about fire regimes, about the cultivation of agave and mesquite, about the management of riparian corridors, about the specific practices of the O'odham, the Diné, the Apache, and other peoples whose management shaped the region's ecology for thousands of years before Spanish contact. Both sets of questions are worth asking. They are not the same questions, and they lead to different places.

Lyla June Johnston, a scholar of Diné and Tsétsêhéstâhese descent whose work on Indigenous regenerative land management has drawn attention across disciplinary boundaries, has articulated a version of this distinction with particular clarity. Johnston's research documents Indigenous land management practices across North America that produced what she describes as food forests: complex, multi-story, species-diverse managed ecosystems that provided food, medicine, fiber, and habitat simultaneously.[4] The management was not the management of wild landscapes by people who happened to live in them. It was the management of landscapes by people who understood themselves as participants in the ecological relationships the landscape maintained, people who saw their role not as external managers imposing order on nature but as keystone species whose management was itself a form of ecological function.

The keystone-species framing is the point at which the two traditions come closest to each other, and it is also the point at which the distinction between them is most important. The Pleistocene rewilding framework asks: which keystone species are missing from this landscape, and how do we restore them? The Indigenous management framework asks: what happens when humans function as keystone species, when human management is itself the ecological process the landscape depends on? The first question imagines restoration as the return of non-human animals to landscapes they have been absent from. The second imagines restoration as the return of human management practices to landscapes they have been absent from. Both involve restoring something the landscape has lost. What they propose to restore is different.

Notice that the two frameworks produce different models of what restoration looks like in practice. The Pleistocene frame, in its strongest version, imagines large tracts of land managed for ecological function with relatively minimal human intervention: proxy species reintroduced and then allowed to interact with the landscape in something approaching the self-organizing dynamics the Pleistocene assemblage would have produced. The Indigenous management frame imagines something closer to what Kimmerer calls the "honorable harvest": a landscape under continuous, attentive, knowledgeable human management, in which the management itself is the ecological process and the food is what the management yields. The two models are not opposites. They exist on a continuum, and any real landscape restoration would likely draw from both. But the emphases are different, and the difference in emphasis reflects a difference in what each tradition identifies as the primary loss to be restored.

I do not think one framework is correct and the other mistaken. I think they are complementary in a way that neither is sufficient to capture alone. A landscape like the American Southwest has lost both its Pleistocene megafauna and its Indigenous management traditions. A complete restoration framework for that landscape would need to engage both losses: the twelve-thousand-year absence of the camelids and equids, and the five-hundred-year suppression of the fire regimes, the managed harvesting, and the cultivation practices that Indigenous peoples maintained for the intervening millennia. The two losses are different in kind, different in timescale, and different in what their restoration would require. They are both real.

◆ WHAT THIS CHAPTER DOES NOT DO ◆

I want to be explicit about the limits of this engagement, because the limits matter.

This chapter does not speak for Indigenous peoples. It engages the published work of Indigenous scholars (Kimmerer, Johnston, Anderson) on the terms those scholars have set. Their work represents a parallel intellectual tradition with its own epistemology, its own standards of evidence, and its own relationship to the landscapes it describes. The Pleistocene framework this book has developed is not a superior framework into which Indigenous knowledge can be slotted as supporting evidence. It is a different framework, developed from a different disciplinary tradition, asking a different set of questions. The dialogue between the two is

valuable precisely because it is a dialogue between genuinely different perspectives, not because one subsumes the other.

This chapter does not romanticize Indigenous land management as a monolith. The practices Anderson documents in California are specific to California's ecosystems and to the peoples who managed them. The practices Johnston describes across the continent vary enormously by region, by nation, by ecosystem, and by the specific cultural and ecological knowledge of the specific peoples involved. There is no single "Indigenous land management practice" any more than there is a single "Western agricultural practice." The diversity is the point. The scholarship this chapter engages is valuable because it is specific: specific peoples, specific landscapes, specific practices, specific ecological outcomes.

This chapter does not instrumentalize Indigenous knowledge as support for the book's argument. The Pleistocene rewilding case stands or falls on its own evidence, as the previous chapters have presented it. What the Indigenous management literature adds is not evidence for the Pleistocene case but a parallel framework that the Pleistocene case should be in conversation with, a framework that reveals dimensions of the restoration question the Pleistocene frame alone cannot see. The conversation is additive. It is not appropriative.

◆ THE WATER COLUMN ◆

The Indigenous management framework just described establishes something the Pleistocene frame alone cannot: a model of restoration in which the restored relationship, not just the restored species, is the unit of ecological function. Fire regimes, managed harvests, food forests maintained across centuries: these are relationships between people and landscapes that produced measurable ecological outcomes and that were lost not in the Pleistocene but within the last five hundred years. The question the framework raises is whether restoration of this kind can be demonstrated at scales and on timescales that make the case legible to policymakers and skeptics, not just to ecologists and Indigenous scholars who already understand it.

The marine domain is where that demonstration is furthest along. The rewilding conversation, in both the popular and the peer-reviewed literature, has been overwhelmingly terrestrial. The animals that dominate the discussion — mammoths,

bison, wolves, camels, horses — are land animals. The landscapes that dominate the discussion — grasslands, forests, deserts, savannas — are land surfaces. The oceans have been present in the book only as a source of analogy (Daniel Pauly's shifting baseline syndrome in Chapter 2, the Alaska salmon fisheries in Chapter 8) rather than as a domain with its own restoration framework and its own set of losses to reckon with. That absence is a gap, and the gap matters for a specific reason: marine restoration operates on a faster clock than terrestrial restoration, and the results are correspondingly more concrete. If the argument this book has been making — that restoring ecological relationships produces food systems that score well on all four axes — needs a proof of concept, the water column is where the proof is most advanced.

The marine losses are enormous. They are also, unlike the Pleistocene megafaunal extinctions, primarily historical rather than prehistoric. They occurred within the window of human record-keeping, often within the last two centuries, and the documentation of what was lost is correspondingly detailed.

Kelp forests are the marine equivalent of old-growth terrestrial forests: structurally complex, biologically diverse, and foundational to the ecosystems they anchor. Kira Krumhansl and colleagues, in a 2016 analysis in *PNAS*, assembled the most comprehensive global dataset on kelp forest change to date, drawing on more than a hundred time series spanning multiple decades across the world's temperate and subpolar coastlines.[5] The synthesis found that kelp forests had declined in thirty-eight percent of the ecoregions studied, with particularly severe losses in parts of the North Atlantic, the southern coast of Australia, and several Pacific regions. The declines were driven by a combination of ocean warming, sea urchin population explosions following the removal of urchin predators — otters, large predatory fish — and coastal water-quality degradation from nutrient runoff and sedimentation. The pattern echoes the terrestrial trophic cascades this book has described: remove the predator, and the herbivore population expands, and the foundation species collapses, and the entire community reorganizes around the loss.

DEFINITION

Trophic Cascade

A trophic cascade is the amplification of ecological change through predator-prey relationships.

When a top predator is removed, herbivore populations expand without control, consuming the foundation species (kelp, grass, coral) those herbivores depend on. The herbivores eventually starve, but not before the foundation species has been destroyed. The system reorganizes around the loss: kelp forests become urchin barrens, grasslands become shrublands, coral reefs become algal deserts.

The cascade demonstrates that ecosystems are held in shape by relationships rather than by species individually. The predator does not merely eat the herbivore. It holds the entire structure in place by eating the herbivore. Remove the predator and the structure inverts. The pattern runs both directions, and the restoration record (sea otters, wolves, large sharks) shows the inversion can often be reversed if the predator returns in time.

The difference between the kelp story and the mammoth story is that the kelp story is reversible on timescales that matter for policy. Where sea otter populations have recovered, through legal protection under the Marine Mammal Protection Act and through active reintroduction programs along the Pacific coast, kelp forests have recovered with them. James Estes and colleagues documented this relationship in foundational work beginning in the 1970s, showing that the presence or absence of sea otters was the primary determinant of whether a given stretch of coastline supported a kelp forest or an urchin barren.[6] The mechanism is straightforward trophic regulation: otters eat urchins, urchins eat kelp, and the presence of the predator controls the herbivore, which allows the foundation species to persist. Remove the otters and the system collapses to an urchin barren. Return the otters and the system recovers to a kelp forest. The time required for recovery, once the predator is restored, is measured in years rather than in millennia. The marine trophic cascade operates on a faster clock than the terrestrial one, and the restoration results are correspondingly more visible within a human career.

Oyster reefs present a parallel case with an additional dimension: the harvested organism is simultaneously performing ecological work. Michael Beck and colleagues, in their 2011 synthesis in *BioScience*, documented that approximately eighty-five percent of the world's oyster reef habitat had been lost, primarily through

overharvest, disease, and habitat degradation.[7] The loss is not merely a fisheries loss. Oyster reefs are ecosystem engineers, the marine equivalent of the keystone herbivores this book has described on land. A healthy oyster reef filters water: a single adult oyster can filter approximately fifty gallons of water per day, removing suspended sediment, excess nitrogen, and algal biomass from the water column.[8] Oyster reefs provide structural habitat for hundreds of species of fish, crustaceans, and invertebrates. They stabilize shorelines against wave erosion. They sequester carbon and nitrogen in their shell matrix. The loss of eighty-five percent of the world's oyster reefs is not a loss of one species. It is the loss of an ecosystem function that maintained water quality, supported fisheries, and protected coastlines across every temperate and subtropical estuary where oysters once grew.

The restoration case is as strong for oyster reefs as for any terrestrial system the book has described. Federal and state partnership programs in the Chesapeake Bay, the Gulf of Mexico, and Puget Sound have rebuilt reef acreage on substrates of recycled shell and limestone, and the restored reefs have shown rapid colonization by native oyster populations, with measurable improvements in water clarity, fisheries productivity, and shoreline stability within years of construction.[9] The food-system dimension is direct: restored oyster reefs produce a harvestable food product while simultaneously performing the ecological work the estuary depends on. The dependency profile is thin in exactly the way the survivability framework measures: the oyster needs the water column, and the water column needs the oyster, and neither requires commodity feed, antibiotics, diesel, or federal subsidies to keep the relationship functioning.

The food-system dimension of kelp forest restoration is also worth noting. Kelp forests are not merely habitat — they are productive ecosystems that support fisheries directly. The rockfish, lingcod, sea urchin, and abalone populations that kelp forests sustain are commercially harvested species. A coastline that supports a kelp forest supports a fishery. A coastline that has collapsed to an urchin barren supports neither. The restoration calculus is not an abstraction: it is a calculation about whether a specific stretch of coast produces food or does not, and the variable that determines the answer is whether the trophic structure is intact.

The broader marine picture extends the argument. The distinction between farmed salmon and wild Alaskan salmon is not merely a culinary preference or a marketing category. It is a distinction between two fundamentally different food

systems with fundamentally different dependency profiles. Farmed Atlantic salmon — produced in net-pen aquaculture operations predominantly in Norway, Chile, Scotland, and British Columbia — requires continuous inputs of processed fish meal and fish oil derived from wild forage-fish harvests, continuous antibiotic and antiparasitic administration to manage the disease pressure inherent to high-density aquaculture, and continuous management of the waste stream the operations produce.[10] A major farmed-salmon operation is, in structural terms, the marine equivalent of a terrestrial CAFO: a confined animal feeding operation in the water column, with many of the same dependency vulnerabilities the terrestrial version carries. Wild Alaskan salmon, managed under the stock-assessment methodology Chapter 8 described, requires the river system to remain intact, the ocean ecosystem to remain productive, and the harvest to remain within the bounds the stock's reproductive capacity sets. The dependency profile is the ecosystem itself. The difference between the two systems, scored on the survivability axis, is the same difference the book has documented on land: a system organized around industrial inputs versus a system organized around ecological function.

The salmon case connects, moreover, to the Indigenous framework this chapter has been developing. Pacific Northwest salmon fisheries were managed by Indigenous peoples for thousands of years before European contact, under systems that sustained both the harvest and the ecological function the salmon perform: nutrient transport from ocean to riparian forest, bear and eagle feeding, marine-derived nitrogen deposition in inland soils. The industrial replacement of those management systems with commodity extraction is a marine parallel to the terrestrial replacement of Indigenous fire management with fire suppression. In both cases, a management tradition that sustained the ecosystem's productive capacity across millennia was replaced by an extractive regime that degraded it within decades. In both cases, the restoration question is not merely ecological but cultural: whose knowledge of the system is being restored alongside the system itself?

What marine ecosystems would look like restored to historical baselines is a question the fisheries-history literature has begun to answer in remarkable detail. Jeremy Jackson and colleagues, in a landmark 2001 paper in *Science*, assembled historical, archaeological, and paleontological evidence to reconstruct the ecological state of coastal marine ecosystems before the onset of industrial exploitation.[11] The reconstruction was sobering. Coastal ecosystems that are now managed as though

their current depleted state were normal had, within the last few centuries, supported populations of marine vertebrates — whales, sea turtles, sharks, large predatory fish, seabirds — at densities that no living marine scientist has observed in the wild. The Chesapeake Bay, before industrial oyster dredging, contained oyster reefs sufficient to filter the bay's entire water volume in approximately a week; the current remnant population takes approximately a year to filter the same volume.[12] Caribbean coral reefs, before the sequential removal of green turtles, manatees, and parrotfish, maintained algal communities in states that no modern reef ecologist has studied in the absence of the animals whose grazing controlled them. The marine shifting baseline, in Jackson's reconstruction, is at least as severe as the terrestrial one this book has documented.

The convergence of the marine and Indigenous frameworks is worth noting here, because it is not incidental. Many coastal Indigenous peoples managed marine systems with the same sophistication Anderson documented for terrestrial ones. The clam gardens of the Pacific Northwest, intertidal rock-wall terraces constructed by First Nations peoples to increase the productivity of clam beds, illustrate this pattern precisely. Research by Dana Lepofsky, Nicole Smith, and colleagues has documented that clam gardens increased clam productivity by factors of two to four compared with unmodified beaches, through deliberate modification of substrate, tidal exposure, and sediment dynamics.[13] The management was not fishing in the extractive sense. It was mariculture: the active management of a marine ecosystem to increase its food-producing capacity while maintaining the ecological relationships the system depended on. The clam gardens were, in structural terms, the marine equivalent of the managed fire regimes Anderson documented on land: human management that enhanced rather than degraded the ecosystem's productive capacity, transmitted across generations, and disrupted by colonization.

CASE

The Clam Gardens of the Pacific Northwest

Between what is now British Columbia and Southeast Alaska, First Nations peoples engineered the intertidal zone for millennia through the construction of rock-wall terraces that modified substrate angle, sediment dynamics, and tidal exposure.

These clam gardens increased clam biomass by factors of two to four compared to unmodified beaches, not through extraction but through deliberate ecological manipulation that enhanced the beach's productive capacity. Research by Dana Lepofsky, Nicole Smith, and colleagues has documented systematic construction, transmission across generations, and rapid recovery of productivity under restoration.

Clam gardens represent the marine equivalent of California's Indigenous fire management: human-engineered ecosystems that produced food while maintaining, and often increasing, the ecological health they depended on. The archaeological record of the practice extends back thousands of years.

The marine domain, in short, has its own version of every argument this book has made about land. It has its own missing keystone species. It has its own depleted baselines mistaken for natural states. It has its own trophic cascades demonstrating what happens when the missing species return. It has its own food systems — wild fisheries, restored shellfish, intact pelagic ecosystems — that produce food as functions of ecological health rather than as alternatives to it. And it has been almost entirely absent from the popular rewilding conversation, which has focused on charismatic terrestrial megafauna to the exclusion of the water column that covers seventy percent of the planet's surface.

◆ INTEGRATION ◆

The Indigenous management framework adds something the Pleistocene frame alone cannot provide: a model of restoration that includes people. The Pleistocene rewilding conversation, as it has developed in the conservation biology literature, has tended to imagine restoration as the return of non-human animals to landscapes from which humans are largely absent, or at least from which human management is minimized. Indigenous management traditions offer a different vision: restoration as the return of human-landscape relationships in which human management is itself an ecological function. The two visions are not incompatible. A landscape could support both the return of proxy megafauna and the restoration of Indigenous fire and harvesting practices. The proxy species and the management practices would interact (as they always have, throughout the history of human habitation of every

landscape on Earth), and the interaction would be the ecology of a managed, inhabited, restored system rather than the ecology of a wilderness emptied of people.

The marine framework adds something the terrestrial frame cannot: completed experiments. The terrestrial rewilding proposals this book has described operate on timescales of decades to centuries. The marine domain has already demonstrated what restoration looks like when it works: kelp forests recovering in years, oyster reefs becoming functional habitat within seasons, fish populations rebounding within a few reproductive generations under calibrated harvest. The clam gardens of the Pacific Northwest sit at the intersection of both expansions: Indigenous mariculture that enhanced marine productivity for centuries before colonization, disrupted by contact, and now being documented in ways that make its restoration empirically tractable.

Both expansions operate within the four-axis framework the book has developed. Indigenous management traditions, where they have been practiced and documented, produce food systems that score well on all four axes — economically viable within their scale, socially equitable by design, environmentally generative rather than extractive, and highly survivable because their dependency profiles are thin and their relationship to the landscape is constitutive rather than instrumental. Marine restoration programs produce food systems with the same profile: the ecosystem remains intact, the management remains disciplined, and the food is what the intact ecosystem produces when the management holds.

The book's argument is not weaker for admitting that the Pleistocene frame is one frame among several. It is stronger. A single-frame argument is brittle: it stands or falls on the validity of one analytical lens. An argument that holds the Pleistocene frame in dialogue with Indigenous management traditions and with marine restoration is an argument with multiple foundations. The frames support each other not by agreeing on everything but by covering each other's gaps. If the Pleistocene proxy-selection hypothesis proves more difficult to test than its advocates hope, the Indigenous management framework and the marine restoration framework remain. If the political obstacles to terrestrial rewilding prove as formidable as Rubenstein has warned, marine restoration offers a domain where the ecological case is already demonstrated and the political barriers are lower.

Part IV of this book, which this chapter opens, is about doing. The preceding chapters made the case. The remaining chapters ask what follows from it: how the framework adapts to where readers actually live, what practical guidance it offers, what must change in policy and culture for the trajectory to shift. The Pleistocene frame asks what the land was before the losses. The Indigenous frame asks what the land was under the management of the people who knew it best. The marine frame asks what the water was before the baselines shifted. Each question produces a different restoration target. Together, they produce a picture of what this continent could be — on land and at sea, in managed landscapes and in rewilded ones — that is richer, more honest, and more defensible than any single frame can draw alone.

CHAPTER ELEVEN

Bioregional and Urban

♦ ♦ ♦

The framework this book has developed does not produce a single answer. It produces different answers for different places, and the difference is not a weakness in the framework. It is the framework working correctly.

The previous chapters built the argument in general terms: a four-axis assessment, a proxy-selection principle guided by the fossil record, a set of restoration frameworks (Pleistocene, Indigenous, marine) that converge on the proposition that food systems organized around ecological function score better across every axis than food systems organized around industrial inputs. The argument, as stated, is continental in scope. It applies, in principle, to every landscape on the continent. What it does not do, stated at that level, is tell anyone what the argument means for the specific piece of ground they are standing on. A rancher in southern Arizona, a fisherman on the Gulf Coast, a family in the hardwood hills of New England, and a software engineer in a Chicago apartment all inhabit different ecological realities, and the framework, if it is worth anything, must be able to speak to each of them in terms that are specific rather than abstract.

This chapter does two things. First, it walks through specific bioregional examples to show what the four-axis framework looks like when it is applied to particular ground: specific landscapes, specific species assemblages, specific food-system possibilities. Second, it addresses the readers who do not have ground to apply it to: the urban readers, the apartment dwellers, the people whose relationship to the food system runs through a grocery store rather than through a pasture. Most readers of this book will be those people. The chapter takes them seriously.

THE ARID SOUTHWEST

Chapter 7 established the proxy-selection argument for the Southwest through the Savory food-truck conversation and the Grand Canyon fossil record. I will not repeat that material here. What I want to do is take the argument one step further: from the proxy-selection principle to a worked picture of what a food system guided by that principle would look like across the region.

The arid Southwest — the Sonoran and Chihuahuan basins, the Colorado Plateau, the sky-island ranges of southeastern Arizona and southwestern New Mexico — is a landscape whose vegetation is still calibrated to animals that have been absent for twelve thousand years. The thorny, browse-heavy plant communities that dominate the region — creosote, mesquite, catclaw acacia, palo verde, saltbush, yucca, prickly pear — evolved under browsing pressure from camelids and ground sloths whose foraging kept the woody vegetation in check and whose dung cycled nutrients through soils that receive less than twelve inches of rainfall per year.[1] The current management regime puts cattle on this landscape. The cattle graze what grass exists, avoid the browse, and leave the woody species to expand unchecked. The result, documented across decades of range-science literature, is mesquite encroachment: the steady conversion of open grassland to dense mesquite thicket, driven not by climate change but by the absence of the browsers that historically suppressed the mesquite.[2] The range is not degrading because it is being used. It is degrading because it is being used by the wrong animal.

A food system guided by the fossil record would look different. Dromedary camels browsing mesquite and palo verde on Sonoran rangeland; Bactrian camels in the cold-winter and saline basins to the north and west. Horses and burros (already present in the region by the hundreds of thousands, currently managed as nuisance populations on BLM land) reframed as proxy species occupying an ecological role their lineage held for millions of years before a twelve-thousand-year interruption.[3] The browse species that cattle refuse are the species the fossil record's animals consumed. The mesquite pods that accumulate uneaten under the current regime are a calorie-dense feed source for camelids whose digestive systems evolved to process exactly that kind of material. The prickly pear fruits that drop and rot are fruits the ground sloths once dispersed. The landscape is not short of food. It is short of the mouths the food was made for.

The products of such a system would be unfamiliar to the American consumer but not to the global one. Camel milk (higher in fat and protein than cow's milk, lower in lactose, and already a significant commercial product in the Middle East, East Africa, and parts of Central Asia) is produced by an animal that thrives on arid-adapted vegetation without irrigation or supplemental feed.[4] Horse and burro meat, while culturally unfamiliar in the United States, is consumed widely in Europe, Central Asia, and South America, and the protein is lean, low in intramuscular fat, and nutritionally comparable to venison. The cultural barriers are real, and I do not dismiss them. The ecological logic is also real. The framework does not insist that everyone eat camel. It observes that a food system organized around the animals the Southwest's landscape evolved under would score better on every axis (environmental integrity, survivability, and the ecological dimension of social equity) than the cattle-based system currently imposed on that landscape.

DEFINITION

Bactrian and Dromedary

The proxy literature names "camels" as the candidate replacement for *Camelops*, and the book has followed that convention. The proxy decision is, in fact, two decisions. There are two living Old World camel species, they occupy distinct ecological niches, they tolerate different climates, and the difference matters for which American landscape gets which animal.

The dromedary, *Camelus dromedarius*, is the single-humped camel of the Sahara, the Arabian Peninsula, and the Horn of Africa. It is heat-adapted: a sparse coat, body mass typically four to seven hundred kilograms, and a metabolism optimized for high-temperature, low-humidity environments. Dromedaries thrive on hot-desert browse, including acacia, mesquite, palo verde, and the thorny succulents, and tolerate water deprivation that would kill cattle or sheep within days. They are the species the Saudi-Gulf trade is built around, and the species the US Camel Corps experimented with in the 1850s.

The Bactrian, *Camelus bactrianus*, is the two-humped camel of the Central Asian steppe, the Gobi, and the Mongolian and northern Chinese deserts. It is the wide-temperature-tolerance specialist of the camelid lineage. A dense winter coat, body mass commonly six hundred to a thousand kilograms, and a tolerance range that runs from forty degrees below zero in steppe winters to over a hundred degrees Fahrenheit in summer rangeland use. Bactrians eat plants other camels treat as marginal: saxaul, salt-stressed *Atriplex*, and the halophytic shrubs of the

Amaranthaceae and Chenopodiaceae families that store sodium at concentrations toxic to most herbivores. They drink saline water that other livestock will not, with documented tolerance above two percent salinity. The Bactrian's foot pad, broader and heavier than the dromedary's, also performs a service the dromedary does less of: it fractures the biological soil crusts and physical caps that form on overgrazed and compacted desert ground, opening the surface to water infiltration and seedling establishment.

The halophyte-eating ability is the unsung edge. Halophytes — *Atriplex*, *Salicornia*, *Halocnemum*, *Salsola* — accumulate salt in their tissues at concentrations that exclude most herbivores from using them. The plants are dominant on salt-affected soils across the western United States: irrigated farmland where a century of input water has accumulated sodium and chloride beyond what conventional crops can tolerate, and natural saline playas where the geology has produced the same outcome. The Bactrian, alone among large livestock species, can convert halophyte biomass into meat, milk, and other commodities while removing salt from the system through the animal's tissues and excreted in urine. This is phytoremediation with a productive output, and the working land it could be applied to is not abstract: several million acres of the western US carry soil salinity sufficient to limit conventional agriculture, and most of that acreage is not currently producing food.

The utility extends well beyond restoration. Camel meat is lean, high in protein, and an established staple across much of the Old World. Camel milk is higher in protein and several minerals than cow's milk, lower in lactose, and increasingly used in the United States by the autism community for behavioral and gastrointestinal symptoms — documented in small randomized clinical trials by Al-Ayadhi (2013) and Bashir (2014), and rising in the marketplace. A 2026 systematic review by Alshebrami and colleagues further documents the nutritional and immunological case for camel milk as a candidate infant formula, particularly for infants with cow’s milk protein allergy or lactose intolerance. Lactoferrin, a milk protein with antimicrobial, antiviral, and immunomodulatory activity, occurs in camel milk at high concentration; recent work has identified that camel tears carry it as well, and a small extraction industry has begun to form around it. Bactrian wool, comparable in fineness to cashmere, supports the substantial commercial fiber industry of Mongolia and northern China. The animal is not a single-product animal. It is a multi-product animal in a regulatory environment that does not yet recognize multi-product animals as a category.

The bioregional decision follows. The hot southern deserts (Sonoran, Chihuahuan, lower Mojave) are dromedary country: the climate matches the

species. The cold-winter Great Basin, the Colorado Plateau, the Wyoming Red Desert, and the saline-affected playas across the basin-and-range are Bactrian country: the cold tolerance, the halophyte browse, the soil-crust breaking, and the salt-water tolerance all line up. *Camelops*, the extinct American camel the proposal is actually about, was probably closer to the Bactrian in body mass and in temperature tolerance, and the paleogenomic work of Heintzman and colleagues now places it as sister to the Old World *Camelus* lineage rather than to *Lama* and *Vicugna*. The right answer to "which camel" is therefore both, allocated by landscape.

CASE

A Million Camels Looking for a Continent

The supply question for the Southwest's camelid scenario has a counterintuitive answer: there are too many camels in the world already, and one continent is killing them rather than using them.

Australia carries the world's largest population of feral dromedaries — descended from camels imported in the nineteenth century to support exploration and freight across the interior, released into the bush after the rail and motor lines came in, and reproducing across a continent whose vegetation suits them well. By the late 2000s, federal and state surveys placed the population at approaching one million animals, expanding at roughly ten percent per year, and producing the kind of damage to fences, water infrastructure, and native vegetation that puts a species into the legal category of pest.

The federal response was the Australian Feral Camel Management Project, a four-year program (2009–2013) funded at roughly nineteen million Australian dollars, which culled more than one hundred forty thousand camels, primarily through aerial gunning from helicopters across the central deserts. The program reduced the population to an estimated three hundred thousand by program end. By 2020, with reproduction outpacing residual culling effort, the population had recovered enough that the Anangu Pitjantjatjara Yankunytjatjara Lands authorized an emergency cull during drought, killing approximately ten thousand additional camels over five days when the animals began damaging community water infrastructure searching for moisture. The current population is again estimated in the high hundreds of thousands and rising.

The Gulf states are running the inverse experiment. Saudi Arabia is the world's largest importer of camel meat and live camels, anchored in a regional camel-meat

economy that runs through the United Arab Emirates, Qatar, Oman, Kuwait, and Bahrain at multi-billion-dollar scale. The kingdom's interest is not nostalgic. Camel meat is a staple protein, camel milk a premium commodity, and camel racing a substantial industry. The trade has scaled to the point that the regional shipping infrastructure now includes dedicated multi-deck livestock-carrier vessels — purpose-built ships, not chartered aircraft — moving live animals across the Indian Ocean at volumes that air freight cannot economically support.

The contrast is the same contrast the feral-hog discussion will draw at smaller scale and inside one country. Australia treats the surplus animal as a damage vector and pays its taxpayers to kill it. Saudi Arabia treats the same animal as a food source and invests in infrastructure to import it. The United States, which has the landscape that wants the animal and a federal regulatory architecture that does not yet have a category for it, treats the question as not yet asked. Three continents, three outcomes, one species.

The animal-welfare reading writes itself. If the choice for a particular cohort of feral Australian camels is between aerial gunning and live transport across an ocean, the second outcome is the better outcome for the animal. If the choice is between aerial gunning and a transport pathway that ends with the camels released onto American Pleistocene-functional rangeland under a research protocol of the kind Chapter Thirteen describes, the gap between actual and possible widens further.

The supply problem the Southwest scenario faces is not, in the end, a supply problem. It is a regulatory and logistical problem: which animals are permitted to cross which oceans on which paperwork. The animals exist. The market for them exists. The transport infrastructure exists. What does not yet exist is the American regulatory category that would permit the third option.

The survivability advantage is particularly sharp. A camel on Sonoran rangeland requires no irrigation, no commodity feed, no antibiotics, and no subsidized grain. It drinks infrequently, tolerates heat that would kill cattle, and feeds itself on vegetation the landscape produces without human input. The dependency profile is as thin as anything the framework can measure. When the corn supply is disrupted, when the diesel price spikes, when the aquifer drops another foot, the camel is still eating mesquite. The landscape is still producing. The system is still functioning, because the system's primary input is the landscape itself, and the landscape has been producing browse for twelve thousand years without asking anyone for help.

THE NORTHERN GREAT PLAINS

The Great Plains are the bioregion where the framework's prescription comes closest to what is already being practiced. The shortgrass prairie evolved under bison, and bison are still here: roughly 500,000 of them, most in commercial herds, roughly 20,000 in conservation herds managed for genetic diversity and ecological function.[5] The proxy-selection question, for this landscape, is largely answered. The animal the prairie evolved under is the animal that should be on the prairie. The question is not whether bison belong on the Plains. It is what happens when you take the relationship seriously.

American Prairie, in northeastern Montana, is the most ambitious attempt to answer that question on the ground. The project, which began acquiring private ranchland and grazing leases in Phillips County in 2004, is assembling a contiguous landscape of more than three million acres — private land, BLM grazing allotments, and the Charles M. Russell National Wildlife Refuge — and managing it for a functioning prairie ecosystem with bison as the keystone grazer. The herd, as of 2025, numbers roughly 800 animals on approximately 90,000 acres of deeded and leased land, managed under a protocol that allows the bison to graze freely across the landscape without the rotational confinement that cattle operations impose. The results are visible from satellite imagery: the bison-grazed sections show a mosaic of close-cropped and tall-grass patches (the heterogeneous structure the prairie's plant diversity depends on) that the adjacent cattle-grazed sections, managed under standard rotational grazing, do not produce. The difference is not management philosophy. It is the behavior of the animal. Bison graze selectively, return to preferred patches repeatedly, wallow in depressions that become ephemeral wetlands used by breeding shorebirds and amphibians, and move across the landscape at scales that create the spatial variability the prairie's full community of grasses, forbs, insects, and ground-nesting birds requires. Cattle, even under the best rotational management, do not produce this mosaic because they are not bison and do not behave like bison on grass.

The food-system dimension of American Prairie is still developing, but the economics of bison production on native grassland are already legible from the commercial herds that have been operating across the northern Plains for decades. A bison cow on native prairie feeds herself year-round. She requires no commodity

grain, no supplemental hay in most winters if the range is adequate, no antibiotics (bison are notably resistant to the respiratory diseases that plague cattle in confined conditions), and no subsidized inputs of any kind. The animal's maintenance cost is the grass, and the grass is what the prairie produces without being asked. The calves grow on their mothers' milk and on the prairie's forage, and the harvested animals (processed through the small but growing network of bison-certified processing facilities in Montana, South Dakota, and Colorado) produce a lean, nutrient-dense meat that commands a retail premium of roughly two to three times the price of commodity beef. The premium reflects consumer demand for a product perceived as healthier and more environmentally responsible, and it is large enough to sustain the operations at their current scale.

The framework adds specificity to what this arrangement means in survivability terms. The cattle ranching economy of the northern Plains operates under the dependency structure the book has documented: commodity markets set the price, federal subsidies smooth the volatility, and the rancher's margin depends on conditions the rancher does not control. A bison operation on native prairie operates under a different structure. The animal feeds itself on the prairie's own production. The harvest is governed by the herd's population dynamics and the range's carrying capacity rather than by the commodity cycle. The ecological services the bison provide while alive — the grazing that maintains grass vigor, the wallowing, the selective foraging — are not incidental. They are the ecological process that keeps the prairie functioning as prairie, and the food the operation produces is a byproduct of that maintenance rather than an extraction from it. The survivability score is high not because bison are exotic or novel but because the relationship between bison and prairie is the oldest working partnership on the landscape, and it requires the least from outside itself to keep functioning.

The tribal dimension is equally important and should not be treated as secondary. The InterTribal Buffalo Council, a consortium of more than seventy tribal nations, manages bison herds across reservations from Montana to Oklahoma, and the herds serve cultural, ecological, and food-sovereignty purposes simultaneously. For the Blackfeet, the Fort Peck Assiniboine and Sioux, the Rosebud Sioux, and other northern Plains nations, the bison's return to tribal land is not merely an ecological restoration. It is the restoration of a relationship that was deliberately severed as an instrument of colonial policy. The nineteenth-century slaughter of the bison herds

was, among other things, a military strategy to deprive Plains nations of their primary food source and force dependence on federal commodity rations. The return of the bison to tribal management is, in this context, simultaneously a food-system intervention, an ecological restoration, and a reassertion of sovereignty over a relationship the federal government spent a century trying to destroy. The framework scores the tribal bison operations well because the operations score well. The historical weight behind the scores is something the framework does not capture but the reader should know.

The scaling question for the Plains is not ecological but institutional. The land exists. The northern Great Plains contain millions of acres of native grassland, much of it under cattle grazing leases that could, in principle, support bison instead. The animal exists. The ecological relationship is intact. What does not yet exist at scale is the market infrastructure, the processing capacity, and the policy framework that would allow bison production on native grassland to move from a conservation-niche market to a meaningful fraction of the region's protein supply. The USDA's inspection requirements, the geographic concentration of bison-certified processing facilities, and the absence of bison from the commodity-support programs that underwrite cattle production all create friction that is institutional rather than ecological. The constraint is not the landscape. The constraint is the institutional architecture built around cattle, and the question is whether that architecture can be expanded to accommodate the animal the landscape actually wants.

◆ TWO FURTHER BIOREGIONS, BRIEFLY ◆

The Gulf Coast and the Pacific Northwest each deserve extended treatment, and each illustrates a different dimension of the framework's application. I will sketch them here rather than develop them fully, because the principles the Southwest, Plains, and New England sections have established apply to these regions with modifications of detail rather than of structure.

The Gulf Coast (from the Texas barrier islands through the Louisiana marshes to the Florida panhandle) is where the marine and terrestrial frameworks converge most visibly. On land, the region's Pleistocene fauna included tapirs, capybaras, llamas, and ground sloths, a Neotropical assemblage whose proxy-selection question is more complex than for the arid Southwest or the grassland Plains.[6] On water, the Gulf is

where the oyster-reef restoration arguments from Chapter 10 come into sharpest focus: reef restoration programs across four states are rebuilding habitat that simultaneously filters water, stabilizes shoreline, and produces a harvestable food product whose dependency profile is the estuarine ecosystem itself.[7] The wild-caught Gulf shrimp fishery, battered by competition from farmed imports whose dependency profile runs through the industrial substrate, is a food system the framework can distinguish from its competitor on the fourth axis alone.[8]

The Pacific Northwest (western Oregon and Washington, coastal British Columbia) is the bioregion where all three frameworks converge on a single watershed. The salmon is the organizing species: Pacific salmon connect the ocean to the interior through their spawning migrations, transporting marine-derived nitrogen upstream into riparian forests where it is measurable in the growth rings of old-growth trees miles from the coast.[12] The terrestrial side runs through the region's extraordinary soil productivity and the Indigenous management traditions Chapter 10 described: the fire-maintained prairies of the Willamette Valley, the camas cultivation, the oak savannas that Europeans mistook for natural meadows.[13] The shellfish dimension (Puget Sound's geoduck, Dungeness crab, and Pacific oyster industries) adds marine food systems whose dependency profiles run through ecosystem health rather than the industrial supply chain. The Northwest's food-system potential, viewed through the framework, is a network of interlocking marine and terrestrial systems whose collective survivability depends on the health of the watershed that connects them. Restore the salmon, and the riparian forests receive their nitrogen. Restore the otters, and the kelp forests recover, and the nearshore fisheries follow. The systems are coupled, and restoring one restores the conditions the others depend on.

◆ THE NEW ENGLAND HARDWOODS ◆

New England is a landscape that has been farmed, logged, abandoned, and reforested within the span of written American history. The stone walls that run through second-growth forest across Connecticut, Massachusetts, Vermont, and New Hampshire mark the boundaries of fields that were cleared in the eighteenth and nineteenth centuries and released back to forest as New England agriculture moved west or collapsed under competition from the Midwest's deeper soils.[9] The forest

that has regrown is young by ecological standards (most of it is less than a hundred and fifty years old), and it is missing most of the large-bodied fauna that the region's Pleistocene and early-Holocene forests supported.

The mastodon was the eastern hardwoods' dominant large browser. Its range extended from the Great Lakes to the Gulf Coast, and its feeding behavior (browsing the understory, stripping bark, creating canopy gaps) shaped the structure of the deciduous forest in ways the current forest does not reflect.[10] There is no living proxy for a mastodon. The framework does not pretend otherwise. What the framework does suggest is that the New England food system should lean into the ecological character of the landscape it sits on — a forest landscape, not a grassland — and that the food-system possibilities of managed forest are more extensive than the region's current agricultural economy recognizes.

The food-system potential of the New England forest is not hypothetical. It is already being demonstrated at small scale across the region, though the formal economy barely registers it. Consider what the forest produces without being asked: mast crops — acorns, beechnuts, hickory nuts, black walnuts — that fall by the ton across millions of acres every autumn and are consumed primarily by deer, turkeys, and squirrels because no commercial infrastructure exists to harvest them at scale for human use. Maple syrup, already a $150-million-per-year industry in the Northeast, produced from trees that require no planting, no irrigation, no fertilizer, and no annual disturbance: a food product whose dependency profile is the forest itself and a late-winter freeze-thaw cycle. Wild mushrooms — chanterelles, hen-of-the-woods, black trumpets, morels — that fruit reliably in the same forests, sought by restaurant buyers who pay $15 to $30 per pound for a product that grows on dead wood without human intervention. Ramps, fiddleheads, sumac, elderberry, and the broader category of wild edibles that the foraging economy has begun to rediscover but that the formal food system still treats as marginal curiosities rather than as products of an ecosystem's inherent productivity.

What distinguishes these products from niche delicacies is that the ecosystem produces them as a function of its own ecological processes. The maple sap flows because the tree is cycling nutrients between root and crown in late winter. The mushrooms fruit because the mycelial network is decomposing wood and cycling carbon. The mast falls because the oaks and hickories are reproducing. None of these require the forest to be anything other than a forest. The food is what the forest does.

The economic question is not whether the forest produces value (it does, continuously) but whether the institutional and market infrastructure exists to capture that value as food rather than leaving it on the ground.

Silvopastoral systems in the New England hardwoods take the argument one step further: not merely harvesting what the forest produces passively but managing the forest as a food-producing ecosystem through deliberate integration of animals under the canopy. Pigs on forest floor, processing mast and rooting through the understory in a pattern that mimics the soil disturbance the extinct peccaries once produced. The heritage breeds suited to this work — Berkshires, Large Blacks, breeds whose foraging behavior and fat-deposition patterns were selected over centuries for woodland production rather than confinement — produce a pork whose flavor and fat profile reflect the mast-based diet in ways that commodity pork, raised on corn and soy in climate-controlled buildings, does not. A pig finished on acorns and beechnuts under a New England hardwood canopy is producing Ibérico-grade pork (the same mast-finishing principle that makes Spanish *jamón ibérico* among the most valued cured meats in the world) from an ecosystem that is reproducing the mast without human input. The economics of mast-finished pork are not commodity economics. The product commands a premium because it is a different product, and the premium is supportable because the input costs are low: the feed is what falls from the trees.

Sheep and goats browsing the understory in managed rotation add a second layer. The Janzen-Martin anachronism fruits — pawpaws, honey locust, Kentucky coffee tree — are eastern species whose oversized fruits, thick pods, and large seeds are calibrated to dispersal by animals that have been absent for twelve thousand years.[11] Cultivating these species in silvopastoral systems would begin to restore a plant-animal relationship that has been one-sided since the mastodons disappeared. The pawpaw, in particular, is a fruit whose commercial potential has been recognized only recently: a custard-textured, tropical-flavored native fruit that grows in the understory of eastern hardwood forests and requires almost no management once established. The tree is shade-tolerant, pest-resistant, and productive for decades. It is also almost entirely absent from the commercial food system because the fruit is soft, perishable, and does not ship well in the cold-chain infrastructure built for apples and oranges. The market limitation is logistical, not ecological. The fruit exists. The

demand exists. What does not exist is the supply-chain infrastructure calibrated to a forest product rather than an orchard product.

The New England case illustrates the framework's sensitivity to history in a way the other bioregions do not. The region's agricultural landscape was created by clearing forest. Its ecological trajectory, in the absence of active management, is reforestation. A food-system framework that asks New England to maintain open pasture at scale is asking the landscape to maintain a condition it is actively trying to leave. A framework that asks New England to produce food from its forest (the ecological state the landscape is returning to on its own) is working with the trajectory rather than against it. The survivability implications are direct: a food system aligned with the landscape's own ecological direction requires less energy to maintain than one that fights that direction.

The stone walls in the woods are the evidence. They mark the boundaries of a food system that required constant labor to hold the forest back (clearing, plowing, fencing, draining) on soils that were thinner and rockier than the Midwest bottomland that ultimately outcompeted them. The forest won that contest. It reclaimed the fields within decades of abandonment, and the walls now run through maple and oak and birch as monuments to a wager the landscape refused to honor. The next food system for this landscape should not make the same wager against the same opponent. It should ask, instead, what the forest is willing to produce. And the answer, if you look at what the forest is already producing without being asked, is more than the formal economy has begun to measure.

CASE

Two Future-Food Categories Through the Four-Axis Test

The framework has to hold up against food systems we are still building, not just food systems we have inherited. Two of the most-discussed categories in 2026 are cultivated meat (sometimes called cellular agriculture or lab-grown meat) and closed-loop aquaponics. Below is how each scores when the framework is applied at the level of present-day implementation, with the caveat that both categories are evolving and each score moves with the technology.

Cultivated meat. Economic viability is the axis that has, as of this writing, been the hardest. Production costs have fallen by orders of magnitude since the 2013 Mosa Meat demonstration but remain well above grocery-shelf parity for

unsubsidized retail. Social fit is mixed: regulatory approval has come in Singapore, the United States, and Israel, but consumer acceptance varies regionally and the supply chain (bioreactors, growth media, regulatory infrastructure) does not employ the agricultural workforce the displaced production did. Environmental fit depends entirely on where the energy comes from; with low-carbon electricity, life-cycle greenhouse-gas emissions can be substantially below conventional beef, but with grid-average electricity, the advantage shrinks. Survivability is where the framework gets least friendly: the system depends on continuous high-purity inputs (growth media, sterile bioreactor environments, reliable cold chain) that are themselves outputs of the industrial system. A cultivated-meat plant cannot operate during a sustained electricity or supply-chain disruption. The system is, on the survivability axis, more brittle than conventional pasture and roughly comparable to industrial broilers.

Closed-loop aquaponics. Economic viability depends heavily on scale and on energy cost; small operations carry high per-unit cost but high product premium. Social fit is high in urban and peri-urban contexts where the production is visible and integrates with restaurant or institutional buyers. There is a closed-loop aquaponics system running underneath a restaurant in downtown Kansas City as I write this; the produce reaches the kitchen of the restaurant above without leaving the building. Environmental fit is generally favorable: water reuse approaches ninety percent and inputs do not cycle through external soils. Survivability is the interesting axis. The system carries some of the same input-fragility as cultivated meat — pumps, filters, water heaters, climate control all depend on continuous power — but the system's smaller scale and shorter supply chain mean a more graceful failure mode. A neighborhood aquaponics system loses its fish before its bok choy. A cultivated-meat plant loses everything when the bioreactor's temperature drifts.

The four-axis test does not declare either system a failure or a success. What it does is force the conversation to happen on more than one axis. Cultivated meat may pass the environmental axis spectacularly while failing the survivability axis just as spectacularly. Aquaponics may pass survivability at small scale while losing economic viability at the scales it would need to feed a city. Both observations are true. Neither is a takedown. The framework's job is to surface the trade-offs, not to deliver a verdict.

THE URBAN READER

I have been writing, so far in this chapter, as though the reader has land. Most readers of this book do not. Most readers live in cities, in apartments or houses without acreage, and their relationship to the food system runs through a grocery store, a farmers' market, a restaurant menu, or a delivery app. If the book's argument ends at the pasture gate, it loses its audience. The argument does not end there.

The framework this book has developed is, at its core, an evaluative tool. It tells you how to assess a food system (any food system, at any scale) by asking four questions: is it economically viable, is it socially equitable, is it environmentally sound, and will it survive when conditions change? Those questions are as useful to a consumer in a grocery aisle as they are to a rancher on a thousand acres. The consumer cannot raise bison or manage a kelp forest. The consumer can ask, of the food they are buying, what system it comes from, what that system depends on, and whether what it depends on is likely to persist. The framework does not require land. It requires attention.

The urban reader has three forms of agency that the book's argument speaks to directly, and none of them require moving to the country.

The first is purchasing. Every food purchase is a vote for the system that produced it. This is not a novel observation, but the framework gives it specificity that the familiar formulation lacks. The question is not merely "is this organic" or "is this local" (labels whose limitations Chapter 1 documented) but "what does the system behind this product depend on, and how does it score on the fourth axis?" Wild-caught salmon from a managed fishery and farmed salmon from a net-pen operation sit side by side in the same case. The framework distinguishes them. Bison from a conservation herd on native prairie and beef from a feedlot finished on commodity corn both produce a burger. The framework distinguishes them. The distinction is not a matter of taste or preference. It is a structural difference in the systems' relationships to the conditions the next century will impose, and the consumer who understands the distinction is making a different kind of choice than the consumer who does not.

I do not want to overstate this. Individual purchasing decisions, even aggregated across millions of consumers, will not by themselves restructure the food system. The structural constraints — subsidy architecture, commodity markets, processing

infrastructure, regulatory frameworks — are larger than any individual's grocery cart. The consumer who buys bison instead of feedlot beef has made a choice that the framework supports. That choice, multiplied, shifts demand. Shifted demand creates market signals. Market signals, over time, influence what gets produced and how. The mechanism is real but slow, and it operates within a larger system that individual choices alone cannot redirect. The purchasing power of urban consumers is a lever. It is not the only lever.

The second form of agency is political. The policy architecture that sustains the current food system — the farm bill, the commodity support programs, the regulatory definitions that Chapter 1 described, the land-management frameworks that determine what happens on public land — is built and rebuilt by political decisions that urban voters participate in. The farm bill is reauthorized roughly every five years. Its provisions determine which crops are subsidized, which conservation programs are funded, which land-management practices are incentivized. An urban voter who understands the survivability framework understands that the subsidy architecture is propping up systems that score poorly on the fourth axis while systems that score well receive little or no support. That voter can advocate for policy that redirects support toward the systems the framework identifies as more durable.

Conservation policy is equally relevant. The management of public lands — BLM rangeland, national forest, national wildlife refuge — is governed by plans that are subject to public comment and political influence. The question of whether feral horses on BLM land are managed as nuisance animals or reframed as proxy species occupying an ecological role is a policy question, not an ecological one. The ecology is what it is regardless of the policy. The policy determines whether the ecology is managed in a way that produces food, restores function, and scores well on the framework, or whether it continues to be managed under assumptions the book has spent ten chapters questioning. Urban voters participate in these decisions. Their participation matters, not least because urban voters constitute the majority of the electorate and the majority of the political pressure on the representatives who write the farm bill and confirm the heads of the agencies that manage public land.

The third form of agency is advocacy. The experiments this book has called for (proof-of-concept trials with proxy species on willing private land, expanded bison restoration, marine-restoration programs that produce food alongside ecological recovery) require funding, institutional support, and public attention. Urban readers

are the primary audience for the public-attention component. They are the readers of the journalism, the supporters of the nonprofits, the donors to the research programs, the consumers whose demand signals tell producers what is worth producing. The book's argument reaches the landscape through the people who care about the landscape, and most of those people live in cities.

I want to be direct about this because the alternative is a book that talks past its own readers. A food-systems book that speaks only to people with acreage is a book written for a small and shrinking fraction of the American population. The framework this book has developed is relevant to everyone who eats, which is everyone. The urban reader is not a bystander to the food-system conversation. The urban reader is the majority of the food-system conversation, and the conversation will go nowhere without them.

◆ SPECIFICITY AS FEATURE ◆

The chapter has walked through five bioregions, and the five have produced five different pictures. The Southwest wants camelids and equids on browse. The northern Plains want bison on grass. The Gulf Coast wants silvopastoral systems and restored marine ecosystems. New England wants food from its forests. The Pacific Northwest wants its salmon and its managed lowland agriculture integrated across a single watershed.

The variation is the point. The four-axis framework does not produce a single prescription for every landscape. It produces a method for arriving at the prescription the landscape calls for, and the prescriptions differ because the landscapes differ. The arid basins of Arizona and the rain-soaked valleys of Oregon have different evolutionary histories, different vegetation communities, different fossil records, different Indigenous management traditions, and different food-system possibilities. A framework that produced the same answer for both would be ignoring everything that makes each landscape what it is. The framework's value lies precisely in its sensitivity to the specific. It asks: what did this landscape evolve under? What does this landscape need? What does this landscape produce when its ecological relationships are allowed to function? The answers are local. They have to be.

This specificity is also, I think, what distinguishes the framework from the approach it is meant to improve upon. The industrial food system treats the continent as a single production surface: corn from Iowa to feed cattle in Texas to produce beef for consumers in New York. The commodity system is designed to erase local difference, to make a bushel of corn from Nebraska interchangeable with a bushel from Indiana, to make a pound of beef from a Kansas feedlot indistinguishable from a pound from a Colorado one. The erasure of local difference is the system's efficiency, and the efficiency is real. It is also the mechanism by which the system's survivability deficit is produced, because a system that treats every landscape the same cannot respond to the ways in which landscapes are not the same.

The framework proposed in this book asks the opposite question. It asks what each landscape is, specifically, and what each landscape can do, specifically, when its ecological relationships are allowed to operate. The answers are diverse. The diversity is the feature. A food system that produces camel milk in Arizona, bison on the Montana plains, oysters in the Chesapeake, forest-raised pork in Vermont, and salmon in the Columbia River watershed is a food system that is not vulnerable to a single point of failure, because it is not organized around a single commodity, a single animal, a single supply chain, or a single set of assumptions about what the future will look like. It is a food system that survives because it is many food systems, each adapted to its own place, each drawing on its own ecological relationships, each capable of continuing to function when the conditions that any one of the others depends on change.

The four-axis framework does not produce a universal answer. It produces the right question — what does this place need, what does this place produce, what will this place still be doing in a hundred years? — and the right question, asked of every landscape on the continent, produces a set of answers that are collectively more durable, more diverse, and more honestly sustainable than any single answer could be.

The next chapter asks what these answers look like at the scale of an individual meal, an individual household, an individual set of economic constraints. The framework has been applied to landscapes. It is time to apply it to the plate.

CHAPTER TWELVE

Eating Toward Vision

♦ ♦ ♦

The framework is built. Over eleven chapters, this book has assembled a way of seeing food systems that most sustainability conversations leave incomplete — four axes rather than three, survivability alongside economic viability, social equity, and environmental integrity. The question now is what to do with it.

This chapter is not a meal plan. It does not tell anyone what to eat for breakfast. It offers no prescriptions, no ranked lists of approved foods, no weekly menus calibrated to a particular body weight or health outcome. What it offers instead is a set of thinking tools: ways to evaluate what is in front of you at the point of purchase, ways to ask better questions of the people who produce your food, and ways to think about the real cost of what you are buying without either ignoring that cost or pretending it does not matter.

The distinction matters. A meal plan is a finished answer. A framework is a way of generating answers that fit circumstances the author cannot predict: your bioregion, your household budget, your access to producers, your cooking capacity, your values. The framework respects the distance between the person who wrote this book and the person reading it. That distance is real, and any guidance that pretends it does not exist is guidance that will fail the moment circumstances diverge from the author's assumptions.

This is also, it should be said, the chapter of this book most likely to slip into a register the rest of the book has worked to avoid. Practical guidance chapters in food-systems writing have a well-established failure mode: they begin by acknowledging that food choices are complex and structurally constrained, and then they proceed to tell the reader what to buy as if those constraints do not exist. The chapter you are

reading was written with that failure mode in mind. Where it succeeds in avoiding it, the framework is doing its job. Where it does not — and the author invites the reader to notice — the draft needs revision.

◆ SEEING THROUGH LABELS ◆

The reader who has stayed with this book since Chapter 1 entered it at a meat counter in Overland Park, Kansas, standing in front of grass-fed beef that turned out to be grain-finished. The labels said one thing; the practice was another. The observation that opened the book was that this gap between label and practice is not a failure of the labeling system — it is what the labeling system is designed to do. Labels communicate less, not more, because communicating more would make visible the practices that the current food system depends on remaining invisible.

Eleven chapters later, the reader has the framework to see through this. The four-axis assessment does what labels cannot: it asks not just whether an animal was raised on pasture or whether a crop was grown without synthetic pesticides, but whether the system that produced the food can survive the removal of its key inputs. A label tells you what a product *is*, within a narrow and sometimes misleading set of categories. The framework asks what a product *requires* — and what happens when those requirements are not met.

I have written certification standards for sixteen years. The irony is not lost on me. The framework is still better than the label.

Consider a carton of eggs labeled "cage-free." The term certifies a specific thing: the hens were not kept in battery cages.[1] It does not certify that they went outdoors. It does not certify what they were fed. It does not certify the stocking density of the barn they were housed in. It does not address survivability at all. The operation may depend entirely on commodity soy feed shipped from a thousand miles away, and the label will not tell you this.[2] The consumer who reads "cage-free" and imagines hens on grass is not being stupid. They are being managed. The label is doing exactly what it was designed to do: communicating enough to justify a price premium while leaving the structural dependencies of the operation invisible.

The framework changes this. When you stand in front of that carton and ask the four-axis questions — what does this system require economically, what are its social

conditions, what is its environmental footprint, and what happens to it under stress — the label becomes what it actually is: a single data point in a much larger assessment. It is not worthless. Cage-free is better than battery-caged, for the hens and for the person eating the eggs. But it is radically incomplete, and treating it as a sufficient guide to a food system's sustainability is the mistake the labeling architecture is designed to encourage.

The same analysis applies across the label landscape. "Organic" certifies the absence of specific synthetic inputs but does not certify biodiversity, does not address survivability, and does not distinguish between a diversified small farm and a ten-thousand-acre monoculture that has simply substituted approved inputs for prohibited ones.[3] "Grass-fed" has been diluted by the USDA's 2016 withdrawal of the federal standard, leaving the term's meaning dependent on which certifying body (if any) stands behind it.[4] "Pasture-raised" means something different depending on whether it appears on a chicken label (where meaningful outdoor access standards exist through some third-party certifiers) or a beef label (where the term has no federal definition at all).[5] "Humane" is perhaps the most variable of all: the gap between the American Humane Certified standard and the Animal Welfare Approved standard is enormous, and the consumer who sees "humane" on both products has no way to know this from the label alone.[6]

The pattern extends beyond animal products. "Natural" on a processed-food label has no federal definition beyond the prohibition of artificial flavors, added colors, and synthetic substances — a bar so low that high-fructose corn syrup from an industrial monoculture system qualifies.[17] "Sustainable" appears on seafood labels backed by certifiers whose standards range from rigorous stock-assessment science to industry self-reporting.[18] "Local" has no regulated definition at all; a product grown three hundred miles away in a system identical to the industrial commodity system it purports to oppose may be labeled "local" in any jurisdiction where the seller chooses to do so.

Each of these terms occupies the same structural position. Each certifies something real but narrow. Each communicates less than the consumer assumes. And each leaves the survivability axis, the question the framework adds to the standard three, entirely unaddressed. No label currently in widespread use tells the consumer what happens to the system that produced this food when its key inputs are disrupted.

None of this means labels are useless. It means they are insufficient. The framework does not replace labels; it contextualizes them. A label is a claim about one dimension of a food system. The framework asks about all four dimensions, and it asks about the dependencies that no label is designed to reveal.

◆ WHAT TO ASK A PRODUCER ◆

Labels mediate between producers and consumers at scale. They exist because most people buying food will never meet the person who produced it. But where direct purchasing is possible (at a farmers' market, through a CSA, from a ranch that sells direct, from a fishing cooperative), the framework generates a different set of questions than the ones most consumers have been taught to ask.

The conventional questions are about outcomes: Is this grass-fed? Is this organic? Were these hens on pasture? These are reasonable questions, and the answers matter. But they are questions about the product. The framework generates questions about the system.

What does your operation require to function? This is the survivability question, translated into a conversation. A producer who raises cattle on pasture and finishes them on grain purchased from a regional feed mill has a different dependency profile than a producer who raises cattle entirely on managed grassland with no purchased inputs. Both may produce excellent beef. Both may be doing good work. But their operations have different vulnerabilities, and the consumer who understands this is making a more informed choice than the consumer who knows only that both products are "grass-fed." The question is not accusatory. It is genuinely curious, in the way that Chapter 4 framed the survivability axis, not as a judgment but as a description of what a system needs to persist.

How many species does your operation depend on? This is a question most producers have not been asked before, and the answer is often revealing. A cattle ranch running a single breed on monoculture pasture has a different ecological profile than a diversified operation running cattle, sheep, and poultry across managed multi-species grassland. Ecological diversity on the production side correlates with resilience, a principle the book has traced from Pleistocene ecosystems through to the present. The producer whose operation depends on a single species, a single breed, a single feed source, and a single market channel has concentrated risk in the same way

a monoculture concentrates risk in crop agriculture. The producer who has diversified across species, markets, and input sources has distributed it.

Where does your feed come from? For any operation that purchases feed (and as Chapter 6 established, even the best pasture-based poultry operations purchase substantial quantities), this question reveals the structural tether to the commodity system. The answer is not disqualifying. A producer who buys feed from a regional mill sourcing from regional farms has a shorter, more resilient supply chain than a producer whose feed originates in the industrial commodity belt. Neither is fully independent of the larger system, but the degree of dependence differs, and that difference matters on the survivability axis.[7]

What happens to your operation if diesel doubles in price? If your primary feed supplier goes under? If the county changes its zoning? These are stress-test questions, and they sound unusual in the context of buying a dozen eggs. But they are the questions the framework generates, because survivability is about what happens under stress, not what happens under current conditions. A producer who has thought about these questions — who has contingency plans, who has diversified their inputs, who has reduced their dependence on any single external system — is running an operation that scores differently on the fourth axis than a producer who has not.

What does your land look like in twenty years if you keep doing what you are doing? This is the environmental-integrity question extended through time. Soil health, water quality, biodiversity, carbon sequestration: these are trajectory questions, not snapshot questions. A producer who is building soil organic matter year over year is on a different trajectory than a producer who is mining it, even if both operations look similar today.[8]

These questions are not tests. They are conversations. A producer who answers them openly is telling you something valuable about their operation. A producer who is uncomfortable with them is also telling you something, though the discomfort may simply reflect the fact that most customers do not ask questions like these and the producer has not had occasion to articulate what they know.

The questions work differently in different purchasing contexts. At a farmers' market, they are face-to-face conversations. Through a CSA, they might be questions posed in a newsletter or at a farm visit. For direct-purchase meat or seafood, they

might be part of the research a buyer does before committing to a quarter-beef or a fishing share. The form adapts; the substance is the same. The framework generates questions about systems, not just about products.

A note on the limits of this approach: direct purchasing is not available to everyone. The farmers' market with the knowledgeable rancher selling quarter-beefs is not equally distributed across geography, income levels, or time availability. The questions above are tools for contexts where they apply. In contexts where they do not (where the grocery store is the only option, where the nearest farmers' market is forty miles away, where the household's schedule does not permit a Saturday morning at a farm stand), the framework still operates, but it operates on the information available rather than on the information ideal. A reader who applies the four-axis assessment to the options actually in front of them at a conventional grocery store is using the framework correctly. The framework does not require access to a particular kind of retail environment. It requires a way of thinking about whatever retail environment the reader has.

◆ THINKING BIOREGIONALLY ◆

Chapter 10 introduced the principle that restoration looks different in different landscapes: that the Pleistocene baseline generates different answers in the arid Southwest than in the Pacific Northwest, and that Indigenous management traditions provide a parallel framework adapted to specific places over millennia. The same principle applies to food choices. What scores well on the four axes in one bioregion may score differently in another, because the ecological and economic conditions that determine survivability are local.

In the Pacific Northwest, wild-caught salmon from well-managed fisheries represents one of the highest-scoring protein sources available on all four axes. The fishery is ecologically adapted, the management science is among the most rigorous in the world, the economic structure supports coastal communities, and the system's survivability depends on intact river ecosystems rather than industrial inputs.[19] In the landlocked interior of the country, that same salmon arrives by truck or plane, its survivability profile degraded by the transportation infrastructure it now depends on. The fish is the same; the system that delivers it is not.

In the Southern Plains, the feral hog population represents an ecological problem (invasive, destructive, and growing) that is simultaneously a food-system opportunity. Wild-harvested feral hog scores at the top of the survivability axis: the protein source requires no feed, no housing, no veterinary inputs, and no supply chain beyond the harvest itself. It also addresses an ecological damage vector. The framework scores this differently than a Pacific Northwest reader's access to managed salmon, but both represent the same principle: the highest-scoring food choices are often the ones most adapted to the specific ecological conditions of the place where the reader lives.

CASE

The Feral Hog Paradox

Feral hogs descended from nineteenth-century escaped domestic stock and twentieth-century imported Eurasian wild boar now number roughly 2.6 million in Texas alone, with rapidly expanding populations across the Southeast and lower Midwest. USDA estimates the national herd at six to nine million animals and growing.

They cause roughly $1.5 billion in annual agricultural damage in the United States, destroy ground-nesting bird habitat, and accelerate soil erosion on the landscapes they colonize. They are, by any standard ecological accounting, an invasive species problem.

They are also a high-scoring protein source: wild-harvested feral hog requires zero feed, zero housing, zero veterinary care, and no supply chain beyond harvest. The harvest addresses the ecological damage while producing food. The contradiction is not a contradiction. It is what it looks like when a damage vector is also a food system.

CASE

Broken Arrow Ranch

One operation worth naming in this chapter is Broken Arrow Ranch, a Texas game-meat processor that handles wild-harvested feral hog, axis deer, blackbuck, and other free-ranging species across the Texas Hill Country and surrounding

regions. The model is straightforward: trained mobile-harvest crews work directly on the ranches where the animals are, field-dress the animals there, and bring them back to the Broken Arrow facility for processing.

The animals were never in a feedlot. They were never in a confined-feeding pipeline. They never required transportation to a centralized slaughterhouse, never received veterinary inputs, never ate purchased grain. They lived on the land and were harvested from it, and the operation that processes them is one of the few that has built a regulatory pathway through the existing USDA architecture for wild-harvested protein at scale.

Paleo Approved certified Broken Arrow under its animal standards, and the operation more than met the standards as written. The standards, in some sense, were written with an operation like this in mind — an operation working with, rather than against, the animals already on the land.

The contrast is worth holding alongside the survivability scoring of Chapter Five. The federal government, through the USDA and various state agencies, currently funds aerial-gunning programs in which contracted shooters in helicopters cull feral hogs across Texas rangeland, treating the animals as an invasive nuisance to be reduced as cheaply as possible. The same animals, harvested rather than gunned and routed through an operation like Broken Arrow's, become protein for human food, revenue for the landowner, and processing work for the local economy. Both are management responses to the same population. One costs the taxpayer money and produces nothing. The other costs the taxpayer nothing and produces protein. The difference is structural rather than ecological — a function of which regulatory pathway exists and which does not.

Broken Arrow does not solve the broader feral-hog problem, and it is not at the scale to feed a region. What it demonstrates is that a regulatory pathway for wild-harvested protein at any scale can exist. If it can exist for feral hog and axis deer in Texas, it can in principle exist for whatever the next animal is.

This is not an argument for localism as ideology. It is an argument for localism as ecological logic. The framework does not score "local" as inherently superior. A greenhouse tomato grown a mile away using natural gas heating and hydroponic nutrient solution may score worse on survivability than a field tomato shipped from a region where tomatoes grow without climate-controlled infrastructure. What the framework scores is the fit between a food system and the ecological, economic, and social conditions of the place where it operates. That fit is often local, because

ecological conditions are local. But "local" is a geographic descriptor, not a sustainability credential, and the framework treats it as such.

The bioregional principle also connects to Chapter 10's discussion of Indigenous management traditions. Where those traditions are accessible and active — through tribal food sovereignty projects, through Indigenous-managed fisheries and wild-harvest cooperatives, through the growing body of work that Robin Wall Kimmerer and M. Kat Anderson have documented — they represent a knowledge base calibrated to specific places over centuries. A reader in the Great Lakes region who has access to manoomin (wild rice) harvested under Anishinaabe management traditions is encountering a food system that has been adapted to its specific ecological context for longer than the agricultural systems surrounding it have existed. The framework scores this favorably not because it is Indigenous — identity is not a sustainability metric — but because the system's ecological fit, survivability profile, and intergenerational knowledge base are measurable strengths on the axes the framework evaluates.

◆ THINKING ABOUT COST ◆

This is the section of this chapter most likely to go wrong, so it is worth being direct about what it will and will not do.

It will not pretend that food from well-managed, ecologically sound, survivable systems costs the same as food from industrial systems. It does not. A dozen eggs from a pasture-based operation with diversified feed sources will cost more at the point of purchase than a dozen eggs from an industrial cage-free barn. A pound of grass-finished beef from a ranch that purchases no commodity feed will cost more than a pound of conventional ground beef. Wild-caught Alaskan salmon will cost more than farmed Atlantic salmon. This is true, and pretending otherwise would insult the reader's intelligence and their lived experience.

It will not lecture anyone about priorities. The financial constraints people operate under are real, they are various, and they are not the business of this book to judge. A household spending thirty percent of its income on housing in a city where wages have not kept pace with costs is not making a lifestyle choice when it buys the cheaper eggs. It is navigating a set of structural conditions that no individual purchasing decision can resolve. The framework is a tool for thinking, and it is only

useful if it can be applied honestly within the actual circumstances of the person using it.

What the cost discussion can do honestly is make visible what the price difference represents.

Industrial food is cheap at the point of purchase because a substantial share of its actual costs are externalized: borne by someone other than the buyer.[9] The environmental remediation costs of CAFO waste lagoons are borne by municipalities and by the communities living near them. The healthcare costs associated with antibiotic resistance, which the FDA's own reporting links to agricultural antibiotic use, are borne by the healthcare system and by the individuals who develop resistant infections.[10] The soil-depletion costs of continuous monoculture are borne by future farmers who will inherit degraded land. The subsidy costs that keep commodity corn and soy artificially cheap are borne by taxpayers. A 2021 analysis by the Rockefeller Foundation estimated that the total externalized cost of the U.S. food system (including environmental, health, and social impacts) exceeds $2.1 trillion annually, roughly three times the direct consumer expenditure on food.[11]

DEFINITION

Externalized Costs

Externalized costs are expenses borne by parties who are not party to the transaction that produced them.

When an industrial hog operation exports waste to groundwater, the cost of remediation falls to municipalities and affected residents rather than to the operator. When agricultural antibiotic use produces resistant bacteria, the cost appears as healthcare spending rather than as feed expense. When federal commodity subsidies hold corn below production cost, the cost is transferred from producer to taxpayer. When monoculture depletes topsoil, the cost is transferred to future farmers.

The price at checkout reflects only the portion of total cost the producer has internalized. The 2019 Rockefeller Foundation analysis this chapter cites estimated the externalized cost of the U.S. food system at roughly $2.1 trillion annually, approximately three times the direct consumer expenditure on food.

This does not mean industrial food is secretly more expensive than pastured food, if you do the math correctly. It means the cost comparison at the register is structurally misleading. The cheaper product is cheaper in part because some of its costs have been removed from the price and distributed elsewhere in the economy. The more expensive product is more expensive in part because it has internalized costs that the cheaper product has externalized. When a consumer pays $8 for a dozen pastured eggs instead of $3 for a dozen industrial eggs, some portion of that $5 difference represents costs the industrial operation has offloaded onto the public.[12]

Understanding this does not change anyone's grocery budget. It changes the frame within which the budget operates. The question shifts from "why is the good food so expensive?" to "why is the cheap food so cheap, and who is paying the difference?" This is not a rhetorical question. It has answers, and the answers are specific: taxpayers, through commodity subsidies; rural communities, through environmental degradation and depressed wages; future generations, through topsoil depletion and aquifer drawdown; the healthcare system, through the downstream costs of antibiotic resistance and diet-related disease.

Within that frame, the practical question becomes: given my actual budget and my actual constraints, where do the highest-return choices lie? The framework suggests a way of thinking about this that respects financial reality.

Not all four-axis improvements cost more. Some of the highest-survivability food choices are among the cheapest. Dried beans and lentils — which require minimal processing, store indefinitely, grow in diverse climates, fix their own nitrogen, and depend on almost no external inputs — score remarkably well across all four axes and cost less per serving than almost any animal protein.[13] Whole grains from regional mills, where available, offer better survivability profiles than processed grain products at comparable or lower cost. Frozen vegetables, which retain most of their nutritional value and eliminate the waste associated with fresh produce that spoils before it is eaten, are an economically efficient choice that the fresh-is-always-better orthodoxy tends to obscure.[14]

Where animal products are concerned, the cost gradient correlates with the degree to which the operation has internalized its costs. The choices that represent the largest per-dollar improvement over industrial systems are often not the most expensive options on the shelf. Choosing conventional eggs from a regional producer

over conventional eggs shipped from a distant industrial operation may represent a meaningful survivability improvement at minimal additional cost. Choosing whole chickens over boneless breasts (and learning to use the entire animal) reduces the per-serving cost of higher-quality poultry substantially. Buying less meat of better provenance, rather than the same quantity at a higher price point, is a strategy that the framework supports and that household budgets can often accommodate.[15]

The wild-harvest and restoration-based protein sources that Chapter 8 described — feral hog in Texas, managed deer and elk across much of the West, wild-caught fish from well-managed fisheries — represent a different cost structure entirely. For households with access to hunting or fishing, these are among the most affordable protein sources available, and they score at the top of the survivability axis because they depend on intact ecosystems rather than industrial inputs. The barriers here are access and knowledge rather than price, and they are barriers that communities, not individuals, are best positioned to address.[16]

There is also the question of time, which functions as a cost even when it does not appear on a receipt. Cooking from whole ingredients — whole chickens rather than boneless breasts, dried beans rather than canned, whole grains rather than processed — costs less money and more time. For households where time is the binding constraint, this trade-off is not trivially navigable, and the framework does not pretend otherwise. What the framework can do is make visible that the processed-food economy has externalized time costs onto the consumer in the same way the industrial food system has externalized environmental costs onto the public: by creating a default in which the time-saving option is also the option most deeply embedded in the industrial supply chain. Understanding this structure does not add hours to anyone's day. It does clarify why the choices available at the intersection of limited time and limited budget tend to cluster at the low end of the survivability axis — and why changing this requires structural intervention, not individual willpower.[20]

The honest position on cost is this: the framework cannot make sustainably produced food cheaper than it is. What it can do is help a reader identify which choices, within their actual budget, represent the highest improvement across the four axes. Some of those choices cost more. Some cost the same. Some cost less. The framework makes the differences visible without pretending the reader's constraints are not real.

WHAT THE FRAMEWORK CANNOT DO

The framework is a tool for individual decisions. It helps a reader evaluate what is in front of them, ask better questions, and make choices that are more fully informed than the ones the labeling system is designed to support. This is worth doing. Individual choices aggregate, and the direction they aggregate in matters.

But individual choices, even fully informed ones, cannot produce the systemic changes the framework reveals as necessary. The reader who buys pastured eggs instead of industrial eggs is making a better choice on all four axes. They are not, by that choice, reforming the subsidy structure that makes industrial eggs artificially cheap. They are not changing the zoning laws that make small-scale pastured poultry operations difficult to establish near population centers. They are not funding the research that would test proxy-species reintroductions at meaningful scale. They are not restructuring the commodity feed system that tethers even the best regenerative operations to industrial inputs.

Consider the structural ceiling that Chapter 6 described. The best pasture-based poultry operation in the country still purchases tens of thousands of pounds of commodity feed annually, most of it produced under industrial monoculture conditions. No amount of consumer preference for pastured chicken, expressed at any price point, changes this structural dependency. The dependency exists because the alternative (a poultry operation that grows its own feed on-site, at the scale needed to support commercial production) requires land, labor, and capital configurations that the current agricultural economy does not reward and current zoning law often prohibits. The consumer's choice is constrained by a structure the consumer's choice cannot modify.

The gap between what individual choices can accomplish and what systemic change requires is the subject of the next chapter. It is worth naming here because the alternative (the suggestion that conscious consumerism is sufficient to transform a food system) is a form of dishonesty the framework does not permit. The framework reveals structural dependencies. Structural dependencies require structural responses.

This is not a reason for despair, and it is not a reason to conclude that individual choices are meaningless. It is a reason to hold both things at once: the choices matter, and they are not enough. The reader who has followed this book through eleven

chapters of framework, evidence, and vision is equipped to make better choices at the point of purchase. They are also equipped to understand why better choices at the point of purchase, alone, will not produce the food systems this book has described as possible.

What those systems require (at the level of policy, culture, economics, and institutional will) is the question Chapter 13 takes up. The framework has done what a framework can do. It has made visible what was hidden, connected what was fragmented, and given the reader a way of seeing that the labeling architecture was designed to prevent. What remains is whether the seeing produces not just better shopping but better systems.

The reader who stood at the meat counter in Chapter 1, confused by labels that communicated less than they appeared to, now has something the labels were never going to provide: a way of asking what a food system actually requires, and what happens when those requirements are not met. That is not a meal plan. It is something more durable — a way of thinking that survives the next relabeling, the next marketing cycle, the next round of standards dilution. The framework is portable. It travels with the reader who has internalized it, into whatever purchasing context they encounter next.

The framework, applied honestly, produces this double awareness. It equips the reader to make better individual choices. It simultaneously reveals why individual choices are insufficient. Both of these are real. The temptation to collapse one in favor of the other, to conclude either that individual action is everything or that individual action is nothing, is a temptation the framework resists, because the framework is descriptive. It describes what is actually happening in the systems that produce food. What is actually happening is that individual choices matter and structural conditions matter more.

That is what this chapter can offer. Not a prescription, but a practice of seeing.

◆ NOTES ◆

1. USDA Agricultural Marketing Service. "Shell Eggs from Farm to Table." The USDA defines cage-free as hens housed in a building or enclosed area with

unlimited access to food and water during the laying cycle, but does not require outdoor access. Updated 2023.

2. Lusk, J. L. (2019). Consumer preferences for cage-free eggs and impacts of retailer pledges. *Agribusiness*, 35(2), 129–148. Lusk's work documents the gap between consumer perception of cage-free claims and the actual production conditions the label certifies.
3. Seufert, V., Ramankutty, N., & Foley, J. A. (2012). Comparing the yields of organic and conventional agriculture. *Nature*, 485(7397), 229–232. The study notes that organic certification addresses input restrictions but does not directly measure biodiversity outcomes or system resilience.
4. The USDA withdrew its grass-fed marketing claim standard in 2016, leaving the term's definition to individual certifying bodies. The American Grassfed Association, the Food Alliance, and A Greener World each maintain their own standards with different requirements for pasture access, supplemental feeding, and finishing practices. See: Cheung, R., & McMahon, P. (2017). "Back to Grass: The Market Potential for U.S. Grassfed Beef." Stone Barns Center for Food & Agriculture.
5. USDA Food Safety and Inspection Service labeling guidelines permit "pasture-raised" claims on poultry when supported by producer documentation, but the agency does not define the term with specific stocking-density or access requirements. Third-party certifiers such as Humane Farm Animal Care (Certified Humane) set specific standards — 108 square feet per bird for poultry — that the federal label does not require. No comparable federal or widely adopted third-party standard governs "pasture-raised" claims on beef.
6. The American Humane Certified program permits indoor-only production systems for poultry and does not require outdoor access. The Animal Welfare Approved program (now A Greener World) requires continuous outdoor pasture access and prohibits beak trimming. Both programs certify products as "humane." See: Consumer Reports. (2017). "Food Labels Survey." Consumers Union.
7. The National Sustainable Agriculture Coalition's 2022 report on regional feed systems documents the supply-chain resilience advantages of locally and regionally sourced animal feed, including shorter transportation distances, reduced exposure to commodity-market price volatility, and stronger

producer relationships that permit quality verification. NSAC. (2022). "Building Regional Feed Supply Chains."

8. LaCanne, C. E., & Lundgren, J. G. (2018). Regenerative agriculture: merging farming and natural resource conservation profitably. *PeerJ*, 6, e4428. The study documents measurable differences in soil organic matter, water infiltration, and insect biodiversity between regenerative and conventional operations in the Northern Plains over a multi-year assessment period.
9. Hendrickson, M. K. (2015). Resilience in a concentrated and consolidated food system. *Journal of Environmental Studies and Sciences*, 5(3), 418–431. Hendrickson's analysis documents how concentration in the food system distributes environmental and social costs away from the point of sale and onto communities, ecosystems, and public budgets.
10. Centers for Disease Control and Prevention. (2019). *Antibiotic Resistance Threats in the United States, 2019.* The report estimates more than 2.8 million antibiotic-resistant infections and 35,000 deaths annually in the U.S. and identifies agricultural antibiotic use as a contributing factor in the development and spread of resistant organisms. FDA. (2022). *Summary Report on Antimicrobials Sold or Distributed for Use in Food-Producing Animals.* Documents that approximately 6.0 million kilograms of medically important antimicrobials were sold for food-animal use in 2021.
11. Rockefeller Foundation. (2021). *True Cost of Food: Measuring What Matters to Transform the U.S. Food System.* The report estimates that the total cost of the U.S. food system — including health impacts ($1.1 trillion), environmental impacts ($900 billion), and biodiversity and livelihood impacts — exceeds $3.2 trillion annually, against approximately $1.1 trillion in direct consumer spending. The externalized portion thus represents roughly two additional dollars of cost for every dollar spent at the register.
12. This is a simplified illustration. The precise allocation of externalized costs to specific products is methodologically complex and contested. What is not contested is that significant costs of industrial food production are borne by parties other than the buyer. For a rigorous treatment of externality accounting in food systems, see: Pretty, J. N., et al. (2005). Farm costs and food miles: An assessment of the full cost of the UK weekly food basket. *Food Policy*, 30(1), 1–19.

13. Havemeier, S., et al. (2017). Dietary guidance for pulses: the challenge and opportunity to be part of both the vegetable and protein food groups. *Annals of the New York Academy of Sciences*, 1392(1), 58–66. Pulses — dried beans, lentils, chickpeas — require minimal water relative to animal protein, fix atmospheric nitrogen, store without refrigeration, and produce protein at a fraction of the land and energy cost of animal sources.
14. Bouzari, A., Holstege, D., & Barrett, D. M. (2015). Mineral, fiber, and total phenolic retention in eight fruits and vegetables: a comparison of refrigerated and frozen storage. *Journal of Agricultural and Food Chemistry*, 63(3), 951–956. The study found no consistent nutritional advantage of fresh over frozen for the eight produce items examined, and frozen storage eliminated spoilage-related waste entirely.
15. Heller, M. C., & Keoleian, G. A. (2015). Greenhouse gas emission estimates of U.S. dietary choices and food loss. *Journal of Industrial Ecology*, 19(3), 391–401. The study documents that reducing food waste and shifting toward less-processed, more whole-animal consumption patterns can reduce the environmental footprint of animal-protein consumption without requiring wholesale dietary change.
16. Nugent, R. (2011). Bringing agriculture to the table: How agriculture and food can play a role in preventing chronic disease. Chicago Council on Global Affairs. For the specific economics of wild-harvest protein, see also: Tidball, K. G., & Stedman, R. C. (2013). Positive dependency and virtuous cycles: from resource dependence to resilience in urban social-ecological systems. *Ecological Economics*, 86, 292–299.
17. FDA. "Use of the Term Natural on Food Labeling." The FDA has not established a formal definition for the term "natural" beyond its longstanding policy that the term means "nothing artificial or synthetic (including all color additives regardless of source) has been included in, or has been added to, a food that would not normally be expected to be in that food." This does not address production methods, environmental practices, or system dependencies.
18. Jacquet, J., et al. (2010). Seafood stewardship in crisis. *Nature*, 467(7311), 28–29. The study documents significant variation in the rigor and enforcement of seafood sustainability certifications, noting that some

programs rely on industry self-assessment while others require independent stock assessment and ecosystem impact evaluation.

19. Hilborn, R., et al. (2020). Effective fisheries management instrumental in improving fish stock status. *PNAS*, 117(4), 2218–2224. Alaska's salmon fishery management, based on in-season stock assessment and escapement-goal enforcement, represents one of the most rigorous fisheries management systems in the world and has maintained sustainable yields for decades.
20. Darmon, N., & Drewnowski, A. (2015). Contribution of food prices and diet cost to socioeconomic disparities in diet quality and health: a systematic review and analysis. *Nutrition Reviews*, 73(10), 643–660. The analysis documents that time cost and monetary cost interact to constrain diet quality, and that the most nutrient-dense and ecologically sound food choices often require both more money and more preparation time than processed alternatives.

CHAPTER THIRTEEN

What Must Change

♦ ♦ ♦

Sixteen years from the meat counter, the question I asked there has not gotten smaller. It has gotten more concrete. The framework Chapter 4 named was the answer to the question that started in front of a label that did not say what I thought it said, and the chapters since have been the test of whether the answer holds up against every food system the framework can be pointed at. Most of what follows is what is left when the test is done.

This book has been, for the most part, an argument directed at readers, at people who buy food, think about food, and have some capacity to choose what food they buy. The framework it has proposed, the four-axis assessment of food systems, is a tool for individual evaluation: a way for a person standing in a grocery store or a farmers' market or a restaurant to think more clearly about what the food in front of them depends on, what it costs in ways the price does not capture, and whether the system that produced it could survive conditions different from the ones it was designed for. The practical guidance the previous chapter offered was guidance for individuals making decisions within the food system as it currently exists. That guidance is real, and I stand behind it.

It is also not sufficient.

The limits of consumer choice as an engine of systemic change are well documented in the food-systems literature and are worth stating directly, because the temptation to end a book like this with an appeal to individual action is strong and the appeal is, on its own, inadequate to the scale of the problem the book has described. A 2012 analysis by Daniel Welch in the *Journal of Consumer Culture* examined the structural assumptions behind the "vote with your fork" framing that has

dominated the sustainable-food conversation since Michael Pollan's *The Omnivore's Dilemma* helped popularize it, and concluded that consumer choice operates within a system whose structure is determined by forces the consumer does not control: commodity policy, land-use regulation, subsidy allocation, infrastructure investment, and the path dependencies of an industrial food system built over seventy years of policy and capital accumulation.[1] The consumer can choose among the options the system offers. The consumer cannot, by choosing, change what options the system offers.

This is not a criticism of the readers who have followed this book's argument. It is a structural observation about the relationship between individual action and systemic architecture. A reader who adopts the four-axis framework, shifts purchasing toward rewilded and sustainably managed food sources, supports local producers operating outside the industrial commodity system, and advocates for the experiments the book has called for is doing everything the individual can do. If a hundred thousand readers did the same, the effect on the food system would be measurable but bounded. If a million readers did the same, the effect would be larger but still bounded by the same structural constraints. The industrial commodity system, the subsidy architecture, the land-use regulatory framework, the conservation baselines, and the institutional infrastructure that governs what kind of food production is permitted, incentivized, and scaled in the United States are not determined by consumer purchasing decisions. They are determined by policy, by regulation, by the allocation of public resources, and by the institutional inertia of a system that has been building in its current direction since the middle of the twentieth century.

The book's argument, if it is correct, has implications that extend past the individual reader. If the four-axis framework reveals structural vulnerabilities in the dominant food system that three-axis assessment cannot see, then the response to those vulnerabilities cannot be purely individual. If the fossil record provides a guide to which animals belong in which landscapes, and if the proxy-species experiments the book has called for are worth conducting, then the institutional and regulatory conditions under which those experiments could occur need to exist. If the rewilding vision the book has described, in which food is produced from landscapes managed under ecologically appropriate species assemblages, represents a direction the food

system should move toward, then the policy environment needs to make that direction possible rather than impossible.

This chapter walks through what would have to change at the structural level for the book's vision to move from argument to practice. I want to be direct about the difficulty. None of what follows is easy. None of it is politically assured. I do not have a theory of how these changes get enacted, because I do not think such a theory exists in honest form. What I have is a description of the structural conditions that would need to be different, offered in the hope that naming them clearly is a precondition for working toward them, even when the path from naming to enactment is not visible.

SUBSIDY REFORM

The federal subsidy structure is the floor on which the current food system stands. The commodity support system, administered through successive iterations of the farm bill since the Nixon-era restructuring of agricultural policy in the early 1970s, has shaped what the United States grows, how it grows it, where the economic incentives in agriculture point, and what kinds of food production are financially viable at scale. The structural effects are not subtle. Between 1995 and 2021, the federal government disbursed approximately $478 billion in agricultural subsidies, with the majority flowing to commodity crop producers — corn, soybeans, wheat, cotton, and rice — and to crop insurance premium subsidies that further underwrite commodity production risk.[2] The distribution is concentrated: Environmental Working Group analyses of USDA subsidy data have consistently shown that the top ten percent of subsidy recipients receive the majority of payments, that the payments flow disproportionately to the largest operations, and that the crops subsidized are overwhelmingly the commodity inputs to the industrial food system the previous chapters have scored.[3]

The consequence of this structure for the food system the book has described is direct. When commodity corn is subsidized below the cost of production, the feedlot system that depends on cheap corn is effectively subsidized as well. The Tufts analysis documented roughly $35 billion in implicit savings to the industrial livestock sector from below-cost feed between 1997 and 2005, and the structural dynamic has not fundamentally changed since.[4] When crop insurance subsidies reduce the

financial risk of planting commodity crops on marginal land, the incentive to convert that land to other uses (perennial pasture, agroforestry, rewilding) is suppressed. When conservation payments through the Conservation Reserve Program and its successors pay landowners to take land out of production but do not create pathways for that land to be managed under the kinds of ecological restoration the book has described, the policy creates a binary — produce commodities or produce nothing — that forecloses the middle option of producing food from ecologically managed landscapes.

DEFINITION

Subsidy Incidence

Subsidy incidence is the economic principle that the party receiving a subsidy check is not necessarily the party capturing its economic benefit.

A federal payment supporting corn production below cost is technically paid to the corn farmer, but the real beneficiary is whoever purchases the subsidized corn downstream. Between 1997 and 2005, the industrial livestock sector captured an estimated $35 billion in implicit savings through below-cost feed enabled by commodity subsidies, the Tufts figure this chapter cites. The incidence of the subsidy determined what kind of livestock production became economically viable at scale.

Restructuring agricultural support toward ecosystem-service payments rather than commodity price supports would shift that incidence, and with it the kinds of land use the subsidy system incentivizes. Where the money ends up determines what gets built on the land.

Restructuring the subsidy system toward ecological outcomes would change the economics of food production in the United States in ways that the current system's beneficiaries have understandable reasons to resist. The restructuring would mean, concretely, that federal agricultural spending would be redirected, partially or substantially, from commodity price supports and crop insurance premium subsidies toward payments for ecosystem services: soil carbon accumulation, biodiversity outcomes, water quality improvements, and the maintenance of ecological function on working landscapes. The concept is not novel. The Conservation Stewardship

Program already makes payments for conservation practices on working land, and the 2018 farm bill expanded some of these provisions. What does not yet exist is a subsidy architecture in which the ecological outcomes the book's framework measures, including survivability and the restoration of species-landscape relationships, are weighted as heavily in the allocation of federal agricultural spending as commodity production volumes currently are.

Julie Witcover and colleagues, in a 2019 analysis for the UC Davis Policy Institute for Energy, Environment, and the Economy, examined the potential effects of redirecting a fraction of federal commodity subsidies toward ecosystem-service payments and found that even modest redirections could produce measurable shifts in land-use patterns, particularly on marginal cropland where the return to commodity production is lowest and the ecological return to alternative management is highest.[5] The finding is intuitive: if the subsidy system pays you to grow corn on land that is marginal for corn, you grow corn; if the subsidy system pays you to manage that land for ecological outcomes, you manage for ecological outcomes. The incentive structure determines the land use. Changing the incentive structure changes the land use. The policy question is not whether this mechanism works. It is whether the political conditions exist to redirect the incentive.

I am not confident they do. Farm bill politics in the United States are among the most entrenched in the federal legislative process. The commodity support programs have institutional constituencies in the agricultural committees of Congress, in the commodity organizations, in the agribusiness firms that supply inputs to commodity producers, and in the rural communities whose economies have been organized around commodity production for generations. Those constituencies have successfully defended the basic architecture of the subsidy system through more than a dozen farm bill reauthorizations since 1973. Structural reform of the subsidy system is not impossible. It has happened before: the 1996 farm bill's elimination of deficiency payments and the subsequent reinstatement of support mechanisms under different names in the 2002 and subsequent bills illustrate both that the system can be restructured and that restructured systems tend to reconverge on the patterns they replaced.[6] I do not think the political difficulty is a reason to stop talking about the structural need. I do think it is a reason to be honest about the distance between the need and the enactment.

◆ CONSERVATION BASELINES ◆

Chapter 2 of this book argued that the Columbian Curtain, the conservation paradigm that treats 1492 as the ecological baseline, shapes what counts as native, what counts as damage, and what counts as restoration. The argument was analytical. The policy implications are concrete.

Conservation policy in the United States operates, in the main, under a post-Columbian baseline. The Endangered Species Act protects species that are threatened or endangered relative to their post-Columbian range and abundance. The National Park Service manages park ecosystems toward reference conditions derived from the early historical record. The Fish and Wildlife Service's management of wildlife refuges is oriented toward the maintenance of habitat for species present in the post-Columbian fauna. The regulatory architecture, in short, is built on the baseline the book has argued should be reexamined.

Shifting the conservation baseline to include near-time frames would not require abandoning the work that has been done under the post-Columbian paradigm. The bald eagle recovery, the alligator recovery, the wolf reintroduction in the Northern Rockies: all of these are genuine conservation achievements accomplished under existing baselines, and none of them would be undermined by an expanded baseline. What the shift would do is create regulatory space for a category of restoration that the current baseline forecloses: the reintroduction of species, or proxies for species, that were present in the near-time fauna but absent from the post-Columbian baseline. Under the current regulatory architecture, a proposal to introduce camels to the American Southwest as proxies for *Camelops* is not merely unusual. It is categorically outside the frame of what conservation policy is designed to evaluate. The camel is not listed under the Endangered Species Act, because it is not a native species under the Act's baseline. It is not a candidate for reintroduction under any existing wildlife management program, because no existing program operates under a baseline that recognizes its lineage's presence on the continent. The regulatory system does not say "no" to the proposal because the proposal has been evaluated and found wanting. The regulatory system does not have a category in which the proposal can be evaluated at all.

Creating that category, a regulatory pathway for the evaluation of near-time proxy reintroductions, would be a structural change of considerable significance. It would

not require the approval of any specific reintroduction. It would require only that the regulatory system acknowledge the existence of the question and create a process for answering it: a framework for evaluating proposals, assessing ecological risks, setting monitoring requirements, and establishing the conditions under which a proof-of-concept experiment could proceed. The framework would need to be rigorous. The ecological risk assessments would need to meet or exceed the standards applied to endangered-species reintroductions. The monitoring requirements would need to be sustained and independently evaluated. The bar should be high, because the stakes of getting it wrong (introducing an animal that causes unforeseen ecological damage) are real, as Caro and Rubenstein have argued and as this book has acknowledged.

The institutional model closest to what would be needed is the experimental-population designation under Section 10(j) of the Endangered Species Act, which allows the Fish and Wildlife Service to designate reintroduced populations of listed species as "experimental" and manage them under more flexible regulatory provisions than fully protected populations receive.[7] The gray wolf reintroductions in Yellowstone and central Idaho were conducted under 10(j) designations, which allowed the managing agencies to address the legitimate concerns of livestock operators and adjacent landowners through management flexibility that full ESA protection would not have permitted. The mechanism is not directly applicable to proxy-species introductions, since a camel is not a listed species, but the conceptual architecture is relevant: a regulatory pathway that acknowledges the experimental nature of the introduction, provides for rigorous monitoring, establishes conditions for expansion or termination based on empirical results, and creates a legal framework within which the experiment can proceed without being foreclosed by categorical exclusion.

Jens-Christian Svenning and colleagues, in their 2016 synthesis on trophic rewilding in *PNAS*, argued that the scientific case for expanding conservation baselines to include Pleistocene-era ecological assemblages has strengthened substantially over the last decade, and that the institutional frameworks governing conservation practice have not kept pace with the science.[8] The gap between the scientific literature and the regulatory architecture is not unique to this issue. Regulatory frameworks routinely lag behind the science they are meant to implement. But the gap is particularly consequential here, because it forecloses an entire category of restoration work that the science has identified as worth pursuing.

FURTHER STRUCTURAL CHANGES

Three additional areas of structural change bear noting, each briefer than the subsidy and baseline questions but each a real barrier to the vision the book has described.

The first is land-use regulation. Current agricultural zoning and conservation-easement structures assume a binary: commodity production or conservation set-aside. Ecologically managed food production — the Knepp model, the bison operations, the proxy-browsing experiments — is a third thing, and the regulatory apparatus does not have a well-developed framework for it. The Conservation Reserve Program pays landowners to take land out of production and maintain vegetative cover, but CRP contracts generally restrict the commercial use of enrolled acreage.[9] The result is a policy binary that forecloses the option of producing food from an ecologically managed landscape. What would need to change is the creation of a third land-use classification with its own ecological-outcome standards, its own monitoring requirements, and its own eligibility for public support. The classification would need to be rigorous enough to prevent greenwashing and flexible enough to accommodate the diversity of landscapes the book has described. State-level habitat exchange programs for greater sage-grouse offer institutional models that could be adapted.[10]

The second is regulatory space for proxy-species experiments. The proof-of-concept work the book has called for faces a permitting landscape that was not built with it in mind. A dromedary camel on Arizona rangeland under a university research protocol might need permits from the state department of agriculture, the state game and fish department, and the county — each operating under a different statutory framework, none designed for ecologically motivated species introductions. The friction is not deliberate. It is a side effect of regulatory architecture built for different purposes. What is needed is a streamlined pathway that integrates federal, state, and local authority into a single review process with clear criteria and defined timelines, modeled on the ESA's Section 10(j) experimental-population designation, which enabled the Yellowstone wolf reintroductions and the California condor recovery.[7] The institutional capacity to manage large-bodied animal introductions on working landscapes exists. It needs to be adapted to a new category of question.

FIELD NOTE

The Standard Already Exists

I should say what is true about my own relationship to the regulatory architecture this chapter has been describing.

I wrote a camel standard. By 2013, or 2014 at the latest, under the Paleo Approved program, more than a decade before the book in your hands began to argue that the American Southwest could use the animal. The standard ran the categories a third-party certification standard runs: grass-fed and grass-finished, humanely treated, free range, antibiotic-free, hormone-free. It carried a banned-feedstuffs list — corn, soy, cereal grains except oats, milk replacers containing antibiotics or growth hormones, non-protein nitrogen, antibiotics, hormones. It carried a traceability requirement of one hundred percent from birth to harvest, with forty-eight-hour trace-back. It carried a welfare-indicator inventory the auditor uses to flag environmental issues: calluses, swelling, injuries, bruising, infections, laminitis, hematomas, abscesses, broken bones and tails, scars. The definitions section ran more than thirty terms, from Baleage to Vitamin, because a standard that cannot be audited is not a standard.

I used it. We certified Desert Farms, the US camel-milk operation, by visiting their producer outside Springfield, Missouri, walking the operation, and applying the standard against what we saw. We saw Walid Abdul-Wahab, the founder, again at the Ancestral Health Symposium in 2014. I could not have certified without writing the standard. The standard existed because I had written it.

I am telling you this in Chapter Thirteen because the chapter has been describing a regulatory architecture that does not yet have a federal category for keystone-proxy species. That description is correct as far as it goes. What it is not is complete. The federal pathway does not exist. The third-party certification pathway, for the specific animal this book is most directly about, does. It existed by 2014. The Paleo Approved program no longer runs, and the standard is no longer certifying anyone. But the document was written, the categories were defined, and the framework for what an audited American camel operation would have to demonstrate was laid out, by an author who is now asking the country to consider operations like the ones the standard was designed to certify.

If a federal pathway is built, it will not be built from scratch. There is already a draft. There is already an example of using it to certify a real producer in a real state. The harder work, what a federal architecture would have to add to a third-party standard to make it generally applicable, is the work the rest of this chapter has been naming. The pathway has been waiting on someone with federal-

rulemaking authority to take the categories seriously. The categories have been there.

The third is fisheries management. Here the institutional infrastructure is, in some respects, further along than the terrestrial equivalent. The Magnuson-Stevens Act provides the statutory framework, the regional fishery management councils provide governance, and Hilborn's stock-assessment methodology provides the scientific tools. The Alaska salmon fisheries are a working example of sustainable wild-harvest food production at commercial scale.[11] What would need to change is the extension of this model to marine restoration — kelp forests, oyster reefs, coastal wetlands — under regulatory frameworks that treat restored marine ecosystems as productive systems rather than merely as conservation amenities, and that treat the food-production value of restoration as part of the justification for public investment.[12]

◆ WHAT WOULD HAVE TO BE TRUE ◆

I have described five areas of structural change: subsidy reform, conservation-baseline expansion, land-use regulatory modernization, regulatory pathways for proxy-species experiments, and the extension and strengthening of science-based fisheries management. Each is individually difficult. Together they constitute a reorientation of the policy framework governing food production and land management in the United States, a reorientation that is, by any honest assessment, politically formidable.

I want to name what would have to be true for these changes to become possible, because naming the preconditions honestly is more useful than pretending the changes are around the corner.

There would have to be a political constituency for ecological food-system reform that is large enough and organized enough to influence farm bill deliberations. Farm bill politics are currently dominated by the commodity agriculture constituency, the nutrition-assistance constituency (SNAP and related programs account for the majority of farm bill spending), and the conservation constituency, which has achieved meaningful but incremental gains over successive reauthorizations. A constituency for the kind of structural reform this chapter describes — one that

includes producers operating outside the commodity system, conservation organizations working on near-time baselines, Indigenous nations managing land under traditional and restored practices, and consumers who have adopted some version of the framework this book proposes — does not currently exist as a coordinated political force. It exists in fragments. The fragments have not yet found each other.

There would have to be a sustained research program generating the empirical evidence that policy change requires. The proof-of-concept experiments the book has called for are not just interesting ecological questions. They are the evidentiary foundation without which no policy advocate can credibly argue for regulatory change. A policymaker being asked to create a permitting pathway for proxy-species experiments will ask what the evidence shows. If the evidence does not yet exist because the experiments have not been conducted, the argument stalls. The research comes first. The policy follows the research.

There would have to be an honest reckoning with the distributional consequences of subsidy reform. Redirecting federal agricultural spending from commodity support to ecosystem-service payments would produce winners and losers, and the losers would include producers and communities whose livelihoods are organized around the current commodity system. A reform that does not account for the transition costs — that does not provide pathways for commodity producers to shift toward ecological management, that does not invest in the rural infrastructure and retraining that the transition would require — is a reform that imposes its costs on the people least able to absorb them. The people most affected by subsidy reform are not the agribusiness corporations that capture the largest share of the payments. They are the family-scale commodity producers and the rural communities whose economies have been organized around commodity production for generations. A just transition would need to address their interests, and addressing their interests would need to be part of the reform's design rather than an afterthought.

There would have to be, finally, a cultural shift in how Americans think about the land they live on: a shift from treating the post-Columbian landscape as the default ecological condition to recognizing it as one state in a longer ecological history. This is the shift the Columbian Curtain argument was designed to support. It is not a shift that a book or a policy proposal can produce on its own. It is a shift that happens, if it happens, through the accumulation of evidence, education, and

experience: through people visiting rewilded landscapes and seeing what they look like, through students learning near-time ecology alongside contemporary ecology, through the slow accretion of a different ecological imagination. I do not know how long this takes. I know it has not happened yet.

CASE

When the Test Stops Being a Thought Experiment

This chapter has been arguing that the survivability axis matters because the conditions under which our food systems were optimized may not hold. The argument is not abstract.

The most-discussed grain-export chokepoint in the world as I am writing this is the Odesa region of the Black Sea, through which Ukraine has historically shipped roughly four-fifths of its grain and oilseed output. The export corridor has been under episodic military pressure since 2022, has operated at reduced capacity for stretches of every year since, and remains among the most concentrated routes by which carbohydrate calories reach the food-insecure populations of Africa and the Middle East. When that route falters, the World Food Programme reports an immediate caloric cascade thousands of miles away. The chokepoint is not theoretical.

The most-discussed nitrogen-fertilizer chokepoint is the Strait of Hormuz, through which a substantial fraction of the world's seaborne natural-gas trade transits, and which therefore sets the marginal global price of the natural gas that feeds the Haber-Bosch process. Roughly half of the nitrogen currently in living human bodies traces back to that industrial synthesis. Vaclav Smil has been making the point in print for thirty years: four billion people are alive who would not be alive without it. The dependency is not a hypothetical future risk. The dependency is the current architecture, and the pricing of its key input cycles through a strait that has been militarily contested repeatedly in this century.

These are not the only chokepoints. Phosphate runs through North Africa. Potash runs through Belarus and Russia. Seed stock concentrates in a handful of multinational firms. The cold chain runs on refrigerants whose supply is its own chokepoint. Each of these is a single point of failure for a different essential input to industrial food production — and each of them compounds with the climate-driven failure modes (drought, heatwave, flood) that act on the same production base from a different angle. The list is not new and it is not partisan. It is the

structural reality of a global food system that has been optimized for cheap energy and stable trade lanes for the better part of a century.

The four-axis framework was built to make exactly this kind of dependency visible. An industrial food system that scores well on economic viability, on social fit, and even on present-day environmental footprint can score catastrophically on survivability if its essential inputs cycle through a few politically contested geographies. The reader does not have to take any particular position on the geopolitics to take the survivability score seriously. The dependency exists regardless of who holds office anywhere.

This is no longer a question only the Pentagon planners are asking. Singapore's *30 by 30* program is an explicit state-level commitment to producing thirty percent of the country's nutritional needs domestically by the decade's end — built around exactly the survivability axis this chapter has been describing. Israel's national food-security planning treats the dependency profile as an input on equal footing with conventional defense planning. China's strategic grain reserves, held at a scale that has been publicly estimated at roughly half of the world's wheat stockpiles, are the same calculation in a different vocabulary. The Pentagon, the war colleges, and the food-systems-resilience desks of the major think tanks are running the same math. The countries doing this work are not all on the same political team, and the math does not care which team they are on.

The United States enters this conversation carrying a structural disadvantage of its own, and a specific opportunity. The disadvantage is that the continent it operates on is, by the ecological measure that matters most for the fourth axis, already depleted — running its food systems on land that lost most of its large-bodied animals twelve and a half thousand years ago and has been operating below ecological capacity ever since. The Americas paid that survivability cost more acutely and on a more recent timescale than most other inhabited continents, and the cost is still on the books. The opportunity is that the disadvantage is one we can get ahead of. The framework this book has built points at what getting ahead of it would actually look like. The chapter that ends this book makes the case.

The Walking Dead question that opened this work was a thought experiment when I asked it fifteen years ago. It is no longer.

FIELD NOTE

The Minimum Viable Flip

The structural changes this chapter has named are individually difficult and collectively formidable. None of them is a precondition for running the smallest version of the experiment. The MVP, to borrow a frame from outside ecology, is what Eric Reis, writing about startups rather than ecosystems, called a minimum viable product: not the cheapest version of the project, but the smallest version that can be falsified.

A hundred camels in an osage orange orchard is the long-tail outcome, not the MVP. The MVP is something more like: forty acres of distressed basin-and-range ground, six to ten camels (or feral burros, or whatever combination of available proxies for what the fossil record says belongs there can actually be permitted), a small planting of a co-evolved tree species, and a monitoring protocol that captures soil carbon, water infiltration, plant diversity, and animal body condition at fixed intervals over three to five years. Bounded. Instrumented. Designed to fail informatively if it fails. The point is not to prove the orchard hypothesis — the MVP cannot do that. The point is to generate the first real signal where signal currently does not exist, at a cost low enough that the permitting pathway and the funding architecture can be tested against a concrete proposal rather than a theoretical one.

The case for running the MVP rests on a separate argument this book has been making in different vocabulary throughout. The current food system is fragile in the sense Chapter Four named: optimized for stable conditions and brittle when those conditions change. The dominant agricultural modality is engineered toward an equilibrium of stable rainfall, stable energy prices, stable trade lanes, stable phosphate, stable nitrogen, stable insect ecology, stable microbial ecology. The engineering succeeds only as long as the equilibrium holds. The Dust Bowl was a fragility cascade. The Irish Potato Famine was a fragility cascade. Panama disease in bananas, late blight, foot-and-mouth, BSE, the broiler flock culls of the last decade — all fragility cascades. Each one was solved by reaching for more of the input the system already depended on, which deepened the specialization, which set up the next cascade.

A keystone-driven, ecologically diverse system would not be immune to disturbance. It would be the opposite of fragile to it. Fire, drought, herd movement, mass mortality — these are not anomalies in a Pleistocene-functional landscape. They are the conditions the landscape is calibrated to. Take the disturbance away and the system degrades. Put it back and the system reorganizes.

Intellectual honesty asks the policy framework to plan for both worlds. The one in which the dominant agricultural modality keeps refining itself, throws ever-better solutions at its dependencies, and reaches the better version of what it

already is. That world is worth planning for. So is the other one. If the input chains tighten, if the climate signal shifts past what the cultivars can tolerate, if a microbial event takes out a single-genome crop the way late blight took out the lumper potato, the contingency is not "find a cleverer monoculture." The contingency is "have the alternative system already running, somewhere, at any scale, with enough data that it can be expanded."

That is the case for funding the MVP even if the dominant system is fine. Diversity of crops is what survivability looks like at the field level. Diversity of approaches is what survivability looks like at the system level. We did not learn nothing from the Dust Bowl. We learned, at minimum, that running every farm by the same playbook is what makes a Dust Bowl possible. The same logic applies one level up. The MVP is a way of buying optionality before the option needs to be exercised.

A note on the limits of the MVP frame, because the metaphor will do work I do not want it to do unless I name the limit. In a software MVP, *fail* means the product did not find market fit and *discard* means you stop running it. The cost of the failure is the cost of the time and capital spent. There are no other consequences. You shut the server down, you pay the people, you move on.

In an ecological MVP, *fail* and *discard* do not work the same way. If a hundred camels are introduced into an osage orange orchard in the Southwest and the experiment fails, the camels are still there. They are large mammals with reproductive capacity and a shared continent. Discarding them means a euthanasia program, a re-export program, or a containment program — every one of which is a moral, financial, and political event. The Australian feral camel population, now around a million animals, exists because the original introduction's failure mode was not to remove the camels. Failure was a direction, not a stop button.

I am not retreating from the MVP frame. I think it remains the right metaphor for how to design the experiment — small scale, defined success criteria, willingness to revise. What I am doing is naming what the metaphor's edges are, because the edges are where most importing of one field's logic into another field goes badly.

The honest version of the camel-hypothesis MVP is: small initial population, on enclosed land, with explicit conditions under which the experiment is judged a failure, and an explicit, pre-committed plan for what happens to the animals if the

failure conditions are met. The plan has to exist before the camels arrive. Without that plan, the framework I have built does not authorize the experiment. With it, the experiment is a real test of the survivability axis at a real scale. The camels are not a metaphor.

◆ CLOSING ◆

This book has made its argument. It has proposed a framework, applied it, followed it into territory the food-systems conversation has mostly not visited, and arrived at a set of conclusions I believe the evidence supports. The framework identifies structural vulnerabilities in the dominant food system. The fossil record identifies a direction for restoration that most of the sustainability literature has not considered. The existing proof-of-concept work — the bison operations, the feral hog harvests, the intact fisheries, the Knepp restoration — demonstrates that rewilded food production is not an abstraction. The serious critics have been engaged, and the responses to their critiques are, I think, defensible.

What the book cannot do is enact what it proposes. The distance between an argument and its realization is not a distance the argument itself can cross. The structural changes this chapter has described are preconditions for the book's vision to scale beyond the readers who adopt the framework individually. Those changes are politically difficult, institutionally complex, and dependent on conditions — a political constituency, an evidentiary base, a cultural shift — that do not yet fully exist.

I am not ending with certainty about outcomes, because certainty about political outcomes is not something the evidence supports, and the book's commitment to stating what it can defend requires applying that standard here as well. I am ending with a description of the door. The door is the set of structural conditions under which the experiments could be tried, the baselines could be expanded, the subsidy system could be restructured, and the food system could begin to move in the direction the framework points. The door exists. It is not open. Whether it opens depends on work that this book can contribute to but cannot complete.

The epilogue returns to the orchard. What remains after the argument, after the framework, after the policy, is the question the prologue already asked: is the experiment worth trying?

The book has laid out what it would take. The answer to whether or not the experiment is worth trying is not the book's to give.

It is the reader's.

EPILOGUE

One Hundred Camels

♦ ♦ ♦

The orchard does not exist yet. I want to return to it anyway.

I opened this book with a camel: *Camelops hesternus*, yesterday's camel, native to North America for most of the history of its genus, gone for thirteen thousand years, remembered mostly in bones and the occasional museum mount. I opened it with an osage orange tree, producing its softball-sized fruit for a mouth that is no longer there. I opened it with a wish: to see one hundred camels in an osage orange orchard, just to see if it could work. The wish was not a program. It was the thing the program would be for.

The book between the prologue and this page has been the program, or at least the argument for one. Thirteen chapters of framework, evidence, application, and honest reckoning with what the framework cannot do. The reader who has come this far knows what the four-axis assessment measures and what it reveals about the dominant food system. The reader knows what the Columbian Curtain hides and what the near-time window makes visible. The reader has walked through the Polyface math, the Savory food-truck conversation, the fossil record of the Grand Canyon, the critics' strongest arguments, the marine restorations, the Indigenous management traditions that operate under a parallel frame. The reader has sat with Chapter 13's description of the structural changes that would need to happen for any of this to move from argument to practice, and the reader has noticed — I hope — that the chapter was honest about the distance between the argument and the enactment.

I do not want to summarize any of that. The reader has just read it. What I want to do, here at the end, is return to the image that started it, because the image has changed, not in its content but in what it carries.

When I described the orchard in the prologue, it carried one thing: a longing. A specific want, plainly stated, for a specific arrangement of animals and trees on a specific kind of ground. The longing was personal. It was mine. I did not try to dress it as policy or as science, because it was not those things. It was the reason I had spent years thinking about the questions the book asks, and it seemed honest to say so at the beginning, before the analytical apparatus had been set up and before the reader had any reason to take the analysis seriously.

The longing has not gone away. I still want to see those camels. I still want to know what the fruit looks like when it has a mouth. But the image is no longer sitting in the space it sat in at the prologue's close — a private wish, carried by a person the reader had just met, before the argument had been made. It is sitting, now, on the other side of the argument. The orchard has accumulated meaning it did not have sixty thousand words ago.

The camels in the orchard are not just camels anymore. They are the proxy-selection principle from Chapter 7 — the fossil record consulted, the co-evolutionary relationship identified, the right animal placed in the right landscape for reasons the paleontology supports. They are the survivability argument from Chapter 4 — animals that feed themselves on what the landscape produces, with a dependency profile thinner than anything the industrial system can offer. They are the proof-of-concept experiment from Chapter 9 — bounded, monitored, designed to generate data rather than to prove a point. They are the structural change from Chapter 13 — the permitting pathway that does not yet exist, the regulatory category that has not been created, the political constituency that has not yet found itself. The hundred camels carry all of that now. They carried none of it before.

The osage orange carries something different than it did at the beginning, too. In the prologue, it was an anachronism: a tree stranded in a present that had forgotten what it was for, producing fruit for a disperser that had been gone for a hundred and twenty centuries. That much has not changed. What has changed is the frame around it. The tree is no longer an isolated curiosity, a bit of natural-history trivia about the deep past. It is an instance of a pattern the book has traced across the entire

continent: the grasslands still calibrated to grazers that are gone, the thorny shrubs still armed against browsers that are absent, the pronghorn still running from nothing, the kelp forests and oyster reefs stripped to fractions of their historical extent, the Indigenous fire regimes suppressed, the soil communities waiting for trampling patterns that have not been produced in twelve thousand years. The osage orange, in the prologue, was one tree with one absent partner. In the epilogue, it is a representative of a continent whose ecological vocabulary has been spoken at half-volume for longer than there has been agriculture, and whose restoration (if it is attempted, if it is permitted, if the experiments are funded and the baselines are expanded and the political will materializes) would be the most ambitious act of ecological repair in the history of the species attempting it.

That is a large thing to rest on one tree and a hundred hypothetical camels. I am aware of the disproportion. The book has been aware of it from the beginning. The analytical argument is not proportional to the longing. The longing is outsized, personal, and not fully defensible on the evidence the book has assembled. The evidence supports a framework. The framework supports a set of experiments. The experiments might support a direction. The direction might, over decades, produce a food system organized around ecological function rather than industrial inputs. None of that is the same as a hundred camels in an orchard. The orchard is where the argument started, and it is further from realization than any of the intermediate steps the book has described.

I think this disproportion is honest rather than embarrassing. The book began with a wish because the wish is what generated the work. The work is what the book contains. The work does not fulfill the wish, and it would be dishonest to pretend otherwise. What the work does is take the wish seriously enough to ask what it would actually require (what framework, what evidence, what engagement with critics, what policy changes, what institutional capacity, what cultural shift) and to lay that out as clearly as I can, knowing that I do not control most of the conditions the answer depends on.

The book has not solved the problem it describes. It has not demonstrated that Pleistocene rewilding works at scale, because the experiments have not been conducted. It has not demonstrated that the four-axis framework produces better policy outcomes than the three-axis model, because the framework has not been adopted by any policymaking body. It has not demonstrated that Americans will eat

camel, or that the permitting pathway will be created, or that the farm bill will be restructured, or that the cultural imagination will expand to include a continent whose memory is older than its paperwork. The book has made an argument. The argument is partial, contested, and dependent on conditions I do not control. I have tried to make it carefully and to be honest about where the evidence is strong, where it is provisional, and where I am asking for something the evidence does not yet support.

What I am asking for, in the end, is not certainty. It is not even, really, that the experiment be mine. I read Stephen Hawking when I was young, the way you do, with a flashlight, understanding half of it, and ever since I have wanted to see a black hole and throw popcorn at its event horizon. The popcorn was mine. The want was mine. The physics were going to go on doing what they do without my participation, and the participation was never the point. The point was to watch the popcorn go in and know, finally, what happens at the edge. Or a refrigerator. Just to watch. I will most likely die before I get to throw popcorn at an event horizon, or a refrigerator. What will not die is that I really, truly, just want to see. The wanting does not go away. It hasn't now for thirty years.

FIELD NOTE

The Popcorn

The popcorn thing is real. I was twelve, maybe thirteen, and I had read Stephen Hawking the way you read Stephen Hawking when you are that age, grasping the shape of the thing without grasping the math.

It has stirred me deeply, ever since. And like camels, it has taken up residence in my mind. The point of the popcorn, I understood only much later, was not the popcorn. It was that some wants are worth carrying even when they exceed what the carrier can do about them. Throwing popcorn at the event horizon, what does that achieve, meaningfully? Seeing whether a hundred camels can reverse desertification, seeing whether we can learn something useful from running an experiment like that — that is a wish worth having.

I am not embarrassed to end a book of analytical argument talking about popcorn and refrigerators and event horizons and camels in an osage orange orchard. Some wishes are the reason for the analysis. At the very least, I have full faith that at least one of these wishes could plausibly be granted within my lifetime.

> And if I had to choose one — and that would be a difficult choice — I would choose camels. There is no utility in finding out what happens to popcorn, or refrigerators, tossed into an event horizon. There is utility in finding out what camels can do for land-flipping and reversing desertification, while producing meat, milk that the autism community has been turning to in growing numbers, camel baby formula powder, hump fat, leather, and "camel cashmere," to name a few of the commodities I am aware of. And this does not even begin to include the money from potential safaris and tourism drawn by the attraction of large megafauna in a space where they seem foreign and exotic.

What I am describing at the end of this book is the same kind of wanting. I want to know whether a parcel of desertified ground in the basin and range country of California or Nevada — real ground, not hypothetical, ground that is sick and losing what little it has left to lose — can be flipped. Not flipped the way a house gets flipped, exactly, though that is the register I mean. Bought distressed. Returned to function by putting the right animals back on it and the right trees in it, running as a closed-loop system for a decade with honest monitoring. Sold, or kept, or left to the commons, whichever way the experiment wants to go once the experiment has been run. What I want to know is whether the ground can be healed. I see desertification as a disease. I do not see it as a settled condition. I do not see it as something the ground has chosen, or earned, or is stuck with. I see it as something nobody has yet been permitted to treat.

I have said as much at Allan Savory conferences and been eye-rolled by people in that audience whom I love and respect and whose work I admire. I do not care. I am not trying to win the argument at the conference. I am trying to find out whether the answer exists. Those are not the same project, and the second one is the one I am interested in. The experiment is the point.

What I am asking for, when I say the experiment is the point, is the smallest version of the experiment that can give a real answer — what Chapter Thirteen named the minimum viable flip. Forty acres of distressed ground. Six to ten camels. A monitoring protocol that can detect signal in three to five years. That is the test the question deserves.

I think it is going to work. I cannot see how it does not. The animals are calibrated to the landscape. The landscape is calibrated to the animals. Take the disturbance

away and the system degrades. Put it back and the system reorganizes. The hypothesis is the most plausible thing I have ever proposed in print, and the evidence the book has assembled supports it.

I am also aware that I have been wrong before. The Revision sidebars are in the book for a reason. The first manuscript I turned in was answering a smaller question than the one Martin was asking, and I did not know it until I read him. I have changed my mind on monocultures, on the overkill hypothesis, on the right tone for an argument like this one. The change-of-mind track record is part of why I am unwilling to oversell this. The same uneasiness that has me unable to oversell it is also why I think we should run it. The way you find out whether an idea works is to test it. Dumber experiments have been done.

If it works — if the ground responds, if the camels feed themselves on what the landscape produces, if the soil carbon and the water infiltration and the plant diversity move in the directions the framework predicts — then the hundred camels in an osage orange orchard is not a wish. It is the next acreage.

The experiment does not need to be mine. It needs to be somebody's. The ground is there. The camels are there: feral in parts of the American West, stabled on hobby ranches, eating mesquite because that is what camels eat when you put them on ground that grows it. The osage orange is there, falling and rotting and asking for a mouth nobody has yet put in front of it. The regulatory pathway does not exist, but that is what Chapter 13 was for. The patron, the university, the protocol, the acreage: none of these exist yet in the configuration the experiment requires, and I do not control whether they come together. What I can do is name what I am asking for, and hope the person who can do it reads this far.

I do not know what the data would show. I think the hypothesis is worth testing, or I would not have written the book. I think the ecological logic is sound: that a continent missing its keystone herbivores for twelve thousand years is functioning below its capacity, that the fossil record provides a guide to restoration that the current conservation baseline does not see, and that food systems organized around ecological function are more survivable than food systems organized around industrial inputs. I think these things because the evidence the book has assembled supports them. I do not think them with the certainty I once brought to simpler versions of this argument, because the revisions the book has documented have

taught me something about the relationship between confidence and evidence that the earlier version of me had not yet learned.

The orchard is still hypothetical. The camels are not standing in it. The osage oranges are still falling, still rotting in place, still producing for no one. Somewhere in the arid basin country of the Southwest, a dromedary on a hobby ranch is eating mesquite, because mesquite is what camels eat when you put them on ground that grows it, and neither the camel nor the mesquite requires an argument to know what they are for.

I would still like to see those camels. I would still like to know what happens when the fruit has a mouth. I would like, more than either of those things, to know whether the ground can be flipped — whether a parcel that has been desertifying for a century can, under the right animals and the right trees and enough time, come back. I will probably not be the person who finds out. I want to be alive when somebody does.

AFTERWORD

The Rabbit Holes Adjacent

♦ ♦ ♦

The book you have just read began with a question at a meat counter in 2009. The question, as the Field Note in Chapter One described, was about the gap between what a label said and what the package contained. The book that came out of that question is the one in your hands. It is about agricultural systems, about the survivability axis, about an experiment with one hundred camels in an osage orange orchard.

What the book does not say — and what I want to say briefly here, because it is part of the same story even though it did not fit inside the four corners of the argument — is that the question, having opened, did not close. Other rabbit holes opened alongside the agricultural one. They are still opening.

The first thread to pull free from the agricultural one was the human gut microbiome. The two questions braided in my head almost from the start. What was being fed to the animals; what was being fed to the soil; what those feedings did to the microbial communities the soil and the gut both depend on. As the work has continued, the braid has tightened: the dysbiotic gut microbiomes that show up across so many modern conditions look, on close inspection, like downstream effects of heavy-metal exposure and mismetallation rather than independent etiologies. The work I have done on that braid is now organized under two reference resources I built for my own continuing study: **Wikibiome** (wikibiome.com), focused on the connection between microbes, disease, and heavy metals; and **Microbiome Medicine** (microbiomemedicine.com), focused on microbiome and metallomic signatures

and on how to reverse disease through interventions that reduce heavy-metal exposure, including dietary protocols like the low-nickel diet.

The next thread, by then no longer separable from the first, was heavy metals as a system-level contamination question. The same agricultural systems that had simplified the soil microbial communities had also been concentrating heavy metals in those soils for decades — cadmium in phosphate-fertilized cropland, arsenic legacies from organoarsenical poultry litter, atmospheric lead, nickel from manure and fertilizer applications. Once I started looking, the metals were everywhere in the food supply, almost always invisible at the label and almost always unmonitored at the producer. The standards work I had been doing through the Paleo Foundation began turning toward metals testing. That work is now organized under a label called **HMTc — Heavy Metal Tested & Certified** (heavymetaltested.com), and around a public reference called the **Heavy Metal Index** (heavymetalindex.com), which catalogs heavy-metal contamination across food categories and supply chains alongside published remediation strategies.

I name these here for two reasons. The first is honest disclosure: these are my projects, and the book you are reading was written alongside them. The second is more practical: if anything in this book made you want to look further, the four sites name the threads I have already pulled. Whatever you find on them is more current than what is in the book, because the work is still moving.

Two specific lines of that work are documented in preprints already in circulation. *Heavy Metals in Fertilizers: A Historical Analysis of Contamination Trends (1960–2025)* traces the rising heavy-metal load in agricultural fertilizers across six and a half decades, reconstructing the four-phase historical arc from the unregulated industrial expansion of the 1960s through the 1990s peak in cadmium and nickel concentrations to the partial regulatory recovery of the present (Zenodo, https://doi.org/10.5281/zenodo.18439158). *Heavy Metals, Microbial Metallomics, and the US Obesity Epidemic: A Mechanistic Examination of a Population-Level Metabolic Disruption* advances what I call the *heavy metal dysbiosis hypothesis* — the proposition that the dysbiotic microbiome signatures observed across the modern US population, and the metabolic conditions associated with them, are mechanistically downstream of chronic dietary heavy-metal exposure and microbial mismetallation rather than independent etiologies. The paper draws on US population data, including a NHANES analysis in which insulin resistance was found to mediate

roughly seventy-four percent of the association between urinary nickel and metabolic dysfunction-associated steatotic liver disease (Zenodo, https://doi.org/10.5281/zenodo.18434951). The two papers together carry the analytical case the book gestures at: that the agricultural thread, the microbiome thread, and the heavy-metal thread are one thread.

All four began with the same question I was asking at the meat counter in 2009. None of them are finished. Sixteen years in, I am still standing where the camel and the chicken and the man behind the counter put me: convinced that the systems I had trusted were not telling the truth, and committed to figuring out what was actually true, by whatever route the question opens next.

The book is the agricultural thread. The other threads are at:

- WIKIBIOME.COM — microbes, disease, and the heavy-metals connection
- MICROBIOMEMEDICINE.COM — microbiome and metallomic signatures, dietary interventions
- HEAVYMETALTESTED.COM — HMTc certification (Heavy Metal Tested & Certified)
- HEAVYMETALINDEX.COM — heavy-metal contamination in food, plus remediation strategies

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Notes

PROLOGUE · ONE HUNDRED CAMELS

1. Martin, P. S. (2005). *Twilight of the Mammoths: Ice Age Extinctions and the Rewilding of America.* University of California Press. Martin synthesizes the Late Pleistocene loss of more than forty North American large-mammal species, including *Camelops hesternus* and *Hemiauchenia*, dated to approximately 13,000 years before present in the North American fossil record.

2. Stuart, A. J. (2014). "Late Quaternary megafaunal extinctions on the continents: a short review." *Geological Journal,* 50(3), 338–363, https://doi.org/10.1002/gj.2633. Stuart's synthesis documents that of the large-bodied terrestrial mammals (genera with body mass typically ≥ 44 kg) present at the Last Glacial Maximum, North America lost approximately 72 percent of its megafaunal genera during the Late Pleistocene–early Holocene window, with comparable losses in South America and somewhat smaller losses in Eurasia. See also MacPhee, R. D. E. (2018). *End of the Megafauna: The Fate of the World's Hugest, Fiercest, and Strangest Animals.* W. W. Norton, for the broader synthesis of cause-of-extinction debates and the comparative continental record.

3. Martin (2005). The "Columbian Curtain" is Martin's own framing for the conservation paradigm that uses 1492 as an ecological baseline, treating European contact as the effective starting line for questions of native and non-native status.

4. Janzen, D. H., & Martin, P. S. (1982). Neotropical anachronisms: the fruits the gomphotheres ate. *Science*, 215(4528), 19–27. The foundational paper on megafaunal fruit syndrome and ecological anachronism.

CHAPTER ONE · THE MEAT COUNTER

1. United States Department of Agriculture, Agricultural Marketing Service. (2007). "United States Standards for Livestock and Meat Marketing Claims, Grass (Forage) Fed Claim for Ruminant Livestock and the Meat Products Derived from Such Livestock." *Federal Register,* 72(199), 58631–58637. The standard defined grass-fed as meaning the animal's diet was derived solely from forage after weaning; grain and grain byproducts were excluded from the permitted diet.

2. The voluntary nature of the USDA grass-fed marketing claim standard meant that producers who did not apply for certification could still use the term "grass-fed" on their packaging, subject only to the general USDA-FSIS labeling approval process for meat products. See USDA Food Safety and Inspection Service, "Labeling Policies" guidance documents, particularly the label approval process for special claims under 9 CFR 317.4.

3. United States Department of Agriculture, Agricultural Marketing Service. (2016). "Withdrawal of United States Standards for Livestock and Meat Marketing Claims, Grass (Forage) Fed Claim for Ruminant Livestock and the Meat Products Derived from Such Livestock." *Federal Register,* 81(12), 3073–3074. Effective January 12, 2016.

4. United States Department of Agriculture, Food Safety and Inspection Service. "Meat and Poultry Labeling Terms." The FSIS defines "free-range" or "free-roaming" for poultry as meaning "producers must demonstrate to the Agency that the poultry has been allowed access to the outside." No specifications for duration, frequency, or size of outdoor area are included in the operative definition. See FSIS Directive 7120.1, "Safe and Suitable Ingredients Used in the Production of Meat, Poultry, and Egg Products," Appendix A, and the FSIS labeling guidance for animal-raising claims.

5. Organic Foods Production Act of 1990, 7 U.S.C. §§ 6501–6523; implementing regulations at 7 CFR Part 205. The National Organic Program standards specify allowed and prohibited substances, require annual inspection by accredited certifying agents, and define organic production as "a production system that is managed in accordance with the Act and regulations in this part to respond to site-specific conditions by integrating cultural, biological, and mechanical practices that foster cycling of resources, promote ecological balance, and conserve biodiversity."

6. The term "natural" as applied to meat and poultry is defined by USDA-FSIS as meaning "a product containing no artificial ingredient or added color and is only minimally processed." The definition addresses processing, not production practices; it says nothing about how the animal was raised, what it was fed, or what inputs were used in its production. See 9 CFR 317.16 and FSIS Policy Memo 055B. For non-meat food products regulated by the FDA, "natural" has no formal regulatory definition and has been the subject of repeated citizen petitions and comment periods without resolution. See FDA, "Use of the Term 'Natural' on Food Labeling," Docket FDA-2014-N-1207.

CHAPTER TWO · THE COLUMBIAN CURTAIN

1. MacFadden, B. J. (1992). *Fossil Horses: Systematics, Paleobiology, and Evolution of the Family Equidae.* Cambridge University Press, pp. 1–14 and chapter 5. MacFadden's synthesis of Equidae evolution is the standard reference for the North American origin of the genus.

2. Scott, E., & Cox, S. M. (2008). "Late Pleistocene distribution of *Equus* (Mammalia: Equidae) in western North America." In *Geology and Vertebrate Paleontology of Western and Southern*

North America, Natural History Museum of Los Angeles County Science Series 41, pp. 189–208. For the specific North American late-Pleistocene equid taxa and their chronology.

3. Orlando, L., et al. (2013). "Recalibrating *Equus* evolution using the genome sequence of an early Middle Pleistocene horse." *Nature,* 499(7456), 74–78. The Thistle Creek genome.

4. Weinstock, J., et al. (2005). "Evolution, systematics, and phylogeography of Pleistocene horses in the New World: a molecular perspective." *PLoS Biology,* 3(8), e241.

5. Martin, P. S. (2005). *Twilight of the Mammoths: Ice Age Extinctions and the Rewilding of America.* University of California Press, chapter 3 and chapter 9. Martin's synthesis of the North American near-time losses and the date range of the terminal Pleistocene pulse.

6. Bureau of Land Management. "Wild Horse and Burro Program: Program Data." Current public population estimates are published annually by the BLM's National Wild Horse and Burro Program office; the AML (appropriate management level) framework and the discrepancy between AML and current population numbers are detailed in the Program Data documentation. See also the 2020 National Academies of Sciences, Engineering, and Medicine review of BLM wild horse and burro management, which describes the same discrepancy.

7. MacPhee, R. D. E., ed. (1999). *Extinctions in Near Time: Causes, Contexts, and Consequences.* Kluwer Academic / Plenum. The introductory chapter and the volume's synthesis chapters are the primary references for the near-time frame as used here.

8. Koch, P. L., & Barnosky, A. D. (2006). "Late Quaternary extinctions: state of the debate." *Annual Review of Ecology, Evolution, and Systematics,* 37, 215–250. Koch and Barnosky's review assembles the continent-by-continent loss figures used here.

9. Smith, F. A., et al. (2016). "Body size downgrading of mammals over the late Quaternary." *Science,* 351(6280), 310–313. Smith and colleagues quantify the loss of large-mammal body mass across the late Quaternary, with regional breakdowns that include the North American figures referenced here.

10. Graham, R. W., et al. (2016). "Timing and causes of mid-Holocene mammoth extinction on St. Paul Island, Alaska." *PNAS,* 113(33), 9310–9314. For the latest securely dated North American mammoth populations, on a Bering Sea refugium, dating to roughly 5,600 years before present. Mainland North American mammoths are generally gone by roughly 10,500–11,000 BP; the refugial populations extend the range somewhat.

11. Martin, P. S., Sabels, B. E., & Shutler, D., Jr. (1961). "Rampart Cave coprolite and ecology of the Shasta ground sloth." *American Journal of Science,* 259, 102–127. For the classic description of preserved Shasta ground sloth dung and the reconstruction of late-Pleistocene browse based on it.

12. See the Executive Order 13112 definition of "invasive species" as operationalized by the US National Invasive Species Council: a non-native species whose introduction does or is likely to cause economic or environmental harm, or harm to human health. The "non-native"

qualifier in the operative federal and most state-level definitions is assessed against the pre-Columbian biota. See also Richardson, D. M., et al. (2000). "Naturalization and invasion of alien plants: concepts and definitions." *Diversity and Distributions,* 6(2), 93–107, for the parallel academic formulation in invasion biology.

13. Pauly, D. (1995). "Anecdotes and the shifting baseline syndrome of fisheries." *Trends in Ecology & Evolution,* 10(10), 430. A short note that has since anchored a substantial literature; see also Jackson, J. B. C., et al. (2001). "Historical overfishing and the recent collapse of coastal ecosystems." *Science,* 293(5530), 629–638, for a major application of the shifting-baseline concept to marine historical ecology.

14. Hansen, R. M. (1978). "Shasta ground sloth food habits, Rampart Cave, Arizona." *Paleobiology,* 4(3), 302–319. For the systematic reconstruction of *Nothrotheriops shastensis* diet from preserved coprolites, which remains one of the most detailed diet reconstructions for any extinct megaherbivore.

15. Stock, C., & Harris, J. M. (1992). *Rancho La Brea: A Record of Pleistocene Life in California* (7th ed.). Natural History Museum of Los Angeles County Science Series 37. For the systematic description of the La Brea faunal record, including *Panthera atrox*, *Smilodon fatalis*, and the dire wolf *Aenocyon dirus*, with chronological framework.

16. Heintzman, P. D., Zazula, G. D., Cahill, J. A., Reyes, A. V., MacPhee, R. D. E., & Shapiro, B. (2015). "Genomic data from extinct North American *Camelops* revise camel evolutionary history." *Molecular Biology and Evolution,* 32(9), 2433–2440. https://doi.org/10.1093/molbev/msv128. The paleogenomic analysis of three Hunker Creek (Yukon) *Camelops* specimens recovered both mitochondrial and partial nuclear genomes, placing the genus as sister to the Old World *Camelus* clade and dating the Camelops–Camelus divergence to the Late Miocene.

CHAPTER THREE · NEAR TIME

1. Libby, W. F. (1952). *Radiocarbon Dating.* University of Chicago Press. Libby's foundational monograph on the method for which he received the Nobel Prize in Chemistry in 1960.

2. Martin, P. S. (2005). *Twilight of the Mammoths: Ice Age Extinctions and the Rewilding of America.* University of California Press, chapter 1 and chapter 9, and table 1 (pp. 10–11). Martin's canonical summary of the North American generic losses concentrated in the terminal Pleistocene pulse around 13,000 calendar years before present.

3. Martin (2005), chapter 1, table 1 and the accompanying discussion of species-level losses. The forty-five-kilogram threshold is the standard used in the near-time literature for "large" mammals and is consistent with MacPhee, R. D. E., ed. (1999), *Extinctions in Near Time: Causes, Contexts, and Consequences,* Kluwer Academic / Plenum.

4. Scott, E., & Cox, S. M. (2008). "Late Pleistocene distribution of *Equus* (Mammalia: Equidae) in western North America." In *Geology and Vertebrate Paleontology of Western and Southern*

North America, Natural History Museum of Los Angeles County Science Series 41, pp. 189–208. For the terminal Pleistocene chronology of the North American equids.

5. Matheus, P. E. (1995). "Diet and co-ecology of Pleistocene short-faced bears and brown bears in eastern Beringia." *Quaternary Research,* 44(3), 447–453. For isotopic reconstruction of *Arctodus* diet and ecological role; see also Figueirido, B., Pérez-Claros, J. A., Torregrosa, V., Martín-Serra, A., & Palmqvist, P. (2010). "Demythologizing *Arctodus simus,* the 'short-faced' long-legged and predaceous bear that never was." *Journal of Vertebrate Paleontology,* 30(1), 262–275, for a contrary view of the animal's ecology; the loss of *Arctodus* is uncontested, whatever its exact trophic role.

6. Martin (2005), chapter 1 and chapter 2, on the residual post-extinction fauna; and Koch, P. L., & Barnosky, A. D. (2006). "Late Quaternary extinctions: state of the debate." *Annual Review of Ecology, Evolution, and Systematics,* 37, 215–250, for the continent-by-continent loss figures.

7. Barnosky, A. D., Koch, P. L., Feranec, R. S., Wing, S. L., & Shabel, A. B. (2004). "Assessing the causes of Late Pleistocene extinctions on the continents." *Science,* 306(5693), 70–75, for the intercontinental pattern of selective loss of large terrestrial mammals and its implications.

8. Mosimann, J. E., & Martin, P. S. (1975). "Simulating overkill by Paleoindians." *American Scientist,* 63(3), 304–313, for the simulation work demonstrating that a small human population advancing through naive megafaunal prey could plausibly produce the observed extinction pattern in a few centuries. Also Martin, P. S., & Klein, R. G., eds. (1984). *Quaternary Extinctions: A Prehistoric Revolution.* University of Arizona Press, for the field's mid-1980s synthesis; Martin, P. S., & Wright, H. E., eds. (1967). *Pleistocene Extinctions: The Search for a Cause.* Yale University Press, for the earlier collected volume that established the debate.

9. Martin (2005), chapter 9, "Models in Collision: Climatic Change versus Overkill," p. 168 (quotation regarding Australia/Americas/New Zealand non-concordance).

10. Alley, R. B. (2000). "The Younger Dryas cold interval as viewed from central Greenland." *Quaternary Science Reviews,* 19(1–5), 213–226, for the Greenland ice-core record and the place of the Younger Dryas within the broader pattern of late-Quaternary climate variability; also Dansgaard, W., et al. (1993). "Evidence for general instability of past climate from a 250-kyr ice-core record." *Nature,* 364(6434), 218–220, for the earlier comparable oscillations.

11. Martin (2005), chapter 9, p. 165, quoting E. O. Wilson's 1992 characterization as epigraph and offering the subsequent "I will lay aside impartiality" declaration.

12. For a representative version of this refined climate-centric account, see Grayson, D. K., & Meltzer, D. J. (2003). "A requiem for North American overkill." *Journal of Archaeological Science,* 30(5), 585–593; and Grayson, D. K. (2007). "Deciphering North American Pleistocene extinctions." *Journal of Anthropological Research,* 63(2), 185–213. Grayson and

Meltzer are the most persistent recent critics of the strong overkill formulation; their counter-model is best read as an example of what a refined climate-centric account looks like.

13. Broecker, W. S. (2006). "Was the Younger Dryas triggered by a flood?" *Science,* 312(5777), 1146–1148, for the canonical statement of the Lake Agassiz meltwater mechanism; and Carlson, A. E., Clark, P. U., Haley, B. A., Klinkhammer, G. P., Simmons, K., Brook, E. J., & Meissner, K. J. (2007). "Geochemical proxies of North American freshwater routing during the Younger Dryas cold event." *PNAS,* 104(16), 6556–6561, for geochemical evidence constraining the routing. Tarasov, L., & Peltier, W. R. (2005). "Arctic freshwater forcing of the Younger Dryas cold reversal." *Nature,* 435(7042), 662–665, offers an alternative Arctic-routing reconstruction. The mechanism remains actively argued; the onset of the cooling is not.

14. Vartanyan, S. L., Garutt, V. E., & Sher, A. V. (1993). "Holocene dwarf mammoths from Wrangel Island in the Siberian Arctic." *Nature,* 362(6418), 337–340, for the Wrangel Island chronology; and Graham, R. W., et al. (2016). "Timing and causes of mid-Holocene mammoth extinction on St. Paul Island, Alaska." *PNAS,* 113(33), 9310–9314, for the St. Paul Island chronology and the case for non-anthropogenic mechanisms in the refugial populations' terminal phase.

15. Firestone, R. B., et al. (2007). "Evidence for an extraterrestrial impact 12,900 years ago that contributed to the megafaunal extinctions and the Younger Dryas cooling." *Proceedings of the National Academy of Sciences,* 104(41), 16016–16021.

16. Pinter, N., Scott, A. C., Daulton, T. L., Podoll, A., Koeberl, C., Anderson, R. S., & Ishman, S. E. (2011). "The Younger Dryas impact hypothesis: a requiem." *Earth-Science Reviews,* 106(3–4), 247–264; and Holliday, V. T., Surovell, T., Meltzer, D. J., Grayson, D. K., & Boslough, M. (2014). "The Younger Dryas impact hypothesis: a cosmic catastrophe." *Journal of Quaternary Science,* 29(6), 515–530. Both papers marshal the principal lines of critique; the first is particularly forceful on the markers, the second on the chronological and archaeological record.

17. Wittke, J. H., et al. (2013). "Evidence for deposition of 10 million tonnes of impact spherules across four continents 12,800 y ago." *PNAS,* 110(23), E2088–E2097.

18. Kennett, J. P., et al. (2015). "Bayesian chronological analyses consistent with synchronous age of 12,835–12,735 Cal B.P. for Younger Dryas boundary on four continents." *PNAS,* 112(32), E4344–E4353.

19. Sandom, C., Faurby, S., Sandel, B., & Svenning, J.-C. (2014). "Global late Quaternary megafauna extinctions linked to humans, not climate change." *Proceedings of the Royal Society B,* 281(1787), 20133254. The paper's Table 1 and its climate-versus-human-arrival regression analysis are the key results.

20. Stuart, A. J. (2015). "Late Quaternary megafaunal extinctions on the continents: a short review." *Geological Journal,* 50(3), 338–363. Stuart's synthesis is particularly useful for readers

who want the continent-by-continent interaction of climate and human pressure laid out compactly.

CHAPTER FOUR · THE FOURTH AXIS

1. United States Department of Agriculture, Economic Research Service. (2023). *Feed Outlook.* USDA-ERS, multiple issues, 2012 and 2022. Corn futures exceeded $8.00/bushel during the 2012 drought, the highest nominal prices in the history of the US corn market at that time. The 2022 spike, driven by the combination of drought stress, post-pandemic supply-chain disruption, and the loss of Ukrainian grain exports, pushed feed costs to levels not seen since the 2012 event in inflation-adjusted terms. For pastured-poultry-sector impacts specifically, see trade-press accounts in *Acres U.S.A.*, *Growing for Market*, and *Stockman Grass Farmer* covering the 2012 and 2022 seasons.

2. The three-pillar model of sustainability — economic, social, environmental — evolved through several institutional formulations in the late twentieth century. For the institutional genealogy, see Purvis, B., Mao, Y., & Robinson, D. (2019). "Three pillars of sustainability: in search of conceptual origins." *Sustainability Science,* 14, 681–695, which traces the model's emergence through the Brundtland Report, the 1992 Rio Declaration, and the subsequent corporate and institutional sustainability literature.

3. World Commission on Environment and Development. (1987). *Our Common Future.* Oxford University Press. The Brundtland definition of sustainable development appears in Chapter 2 of the report.

4. For the ISO 14000 family of environmental management standards as applied to food-system assessment, see International Organization for Standardization. (2015). *ISO 14001:2015 — Environmental management systems — Requirements with guidance for use.* For life-cycle assessment methodology applied to food systems specifically, see Notarnicola, B., Sala, S., Anton, A., McLaren, S. J., Saouter, E., & Sonesson, U. (2017). "The role of life cycle assessment in supporting sustainable agri-food systems: A review of the challenges." *Journal of Cleaner Production,* 140, 399–409.

5. Global Reporting Initiative. (2021). *GRI Standards: Consolidated Set of GRI Sustainability Reporting Standards.* Global Sustainability Standards Board. The GRI Standards are the most widely adopted sustainability reporting framework globally, organizing disclosure around economic, social, and environmental performance categories, and have been applied to food and agriculture cooperatives through the GRI 13 sector standard for Agriculture, Aquaculture and Fishing Sectors (2022).

6. McBride, W. D., & Greene, C. (2009). *Characteristics, Costs, and Issues for Organic Dairy Farming*. Economic Research Report No. 82, United States Department of Agriculture, Economic Research Service. See also Su, Y., Cook, M. L., & Park, T. (2019). "The impact of the farmgate milk price on the survival of U.S. organic dairy farms." *Renewable Agriculture*

and Food Systems, 34(6), 558–568, documenting the role of the organic price premium in organic dairy farm survival, and the sector's repeated contraction episodes when premium compression has reduced the gap between organic and conventional milk prices.

7. Waltenburg, M. A., Victoroff, T., Rose, C. E., et al. (2020). "Update: COVID-19 among workers in meat and poultry processing facilities — United States, April–May 2020." *Morbidity and Mortality Weekly Report,* 69(27), 887–892. Centers for Disease Control and Prevention. The report documents infection clusters at 115 meat and poultry processing facilities across 19 states in the first two months of the pandemic, resulting in facility closures and throughput reductions that produced simultaneous live-animal surpluses on farms and retail meat shortages.

8. Henson, C. B., Luedke, A., Allured, D. J., & Hoerling, M. P. (2022). "The February 2021 cold event in Texas and the south-central US: Historical context and attribution." *Journal of Climate,* 35(18), 5927–5943. For the agricultural impacts specifically, see Texas A&M AgriLife Extension Service. (2021). *Preliminary Texas Agricultural Losses from Winter Storm Uri.* The estimated crop and livestock losses across the Texas agricultural sector exceeded $600 million, concentrated in Rio Grande Valley citrus, livestock mortality in operations without backup power, and cold-chain disruption for produce exports.

9. Glauber, J., & Laborde, D. (2022). "How will Russia's invasion of Ukraine affect global food security?" *International Food Policy Research Institute Blog,* February 24, 2022. See also Ben Hassen, T., & El Bilali, H. (2022). "Impacts of the Russia-Ukraine war on global food security: towards more sustainable and resilient food systems?" *Foods,* 11(15), 2301, for a systematic assessment of the trade disruption's effects on global grain markets and food security in import-dependent regions.

10. Boyer, J. S., Byrne, P., Cassman, K. G., Cooper, M., Delmer, D., et al. (2013). "The U.S. drought of 2012 in perspective: a call to action." *Global Food Security,* 2(3), 139–143. The 2012 drought affected approximately eighty percent of US agricultural land and reduced the national corn yield by roughly twenty-seven percent relative to trend, driving commodity corn prices above $8/bushel.

11. Holling, C. S. (1973). "Resilience and stability of ecological systems." *Annual Review of Ecology and Systematics,* 4, 1–23. Holling's distinction between stability (the speed of return to equilibrium after disturbance) and resilience (the magnitude of disturbance a system can absorb before shifting to a qualitatively different state) remains foundational in ecological theory and has been widely imported into sustainability science.

12. Tendall, D. M., Joerin, J., Kopainsky, B., Edwards, P., Shreck, A., Le, Q. B., Kruetli, P., Grant, M., & Six, J. (2015). "Food system resilience: Defining the concept." *Global Food Security,* 6, 17–23. The paper reviews the disparate uses of "resilience" in the food-systems literature and proposes a consolidated definition, noting that the concept remains underspecified in most applied assessment frameworks.

13. Taleb, N. N. (2012). *Antifragile: Things That Gain from Disorder*. Random House. Taleb's three-part classification — fragile (harmed by volatility), robust (unaffected by volatility), antifragile (benefiting from volatility) — applies to food systems with particular force. The argument that optimization for efficiency under stable conditions produces fragility under unstable conditions is developed throughout the book and is directly relevant to the industrial food-system case.

CHAPTER FIVE · SCORING THE INDUSTRIAL SYSTEM

1. USDA National Agricultural Statistics Service. (2019). *2017 Census of Agriculture.* United States Department of Agriculture. The census documents that approximately 99 percent of US meat animals are finished in confined operations, reflecting the consolidation of animal finishing into the CAFO model over the last half century.

2. USDA Economic Research Service. (2023). *Meat Price Spreads.* ERS tracks retail, wholesale, and farm-level prices for beef, pork, and chicken in nominal and inflation-adjusted terms. The long-term trend in inflation-adjusted poultry prices is a decline of roughly forty percent between 1960 and 2020, driven by the efficiency gains of industrial confined production.

3. Pew Commission on Industrial Farm Animal Production. (2008). *Putting Meat on the Table: Industrial Farm Animal Production in America.* Johns Hopkins Bloomberg School of Public Health / Pew Charitable Trusts. The commission's assessment documented the externalized costs of industrial animal production across environmental, public health, and rural economic dimensions, concluding that the retail price of industrial meat does not reflect its full cost of production.

4. Centers for Disease Control and Prevention. (2019). *Antibiotic Resistance Threats in the United States, 2019.* US Department of Health and Human Services. The report documented more than 2.8 million antibiotic-resistant infections and more than 35,000 deaths annually in the United States, with associated economic costs estimated in the tens of billions of dollars.

5. US Food and Drug Administration. (2022). *2021 Summary Report on Antimicrobials Sold or Distributed for Use in Food-Producing Animals.* FDA Center for Veterinary Medicine. The report documented that approximately 66 percent of medically important antimicrobials sold in the United States in 2021 were intended for use in food-producing animals.

6. World Health Organization. (2001). *WHO Global Strategy for Containment of Antimicrobial Resistance.* WHO. The strategy identified sub-therapeutic antibiotic use in livestock as a significant driver of antimicrobial resistance affecting human health. For the mechanism of resistance selection and transfer, see also van den Bogaard, A. E., & Stobberingh, E. E. (2000). "Epidemiology of resistance to antibiotics: Links between animals and humans." *International Journal of Antimicrobial Agents,* 14(4), 327–335.

7. US Environmental Protection Agency. (2003). *National Pollutant Discharge Elimination System Permit Regulation and Effluent Limitation Guidelines and Standards for Concentrated Animal Feeding Operations: Final Rule.* 68 Fed. Reg. 7176. The EPA's regulatory preamble documented water quality impairment attributable to animal feeding operations.

8. Palmquist, R. B., Roka, F. M., & Vukina, T. (1997). "Hog operations, environmental effects, and residential property values." *Land Economics,* 73(1), 114–124. See also Herriges, J. A., Secchi, S., & Babcock, B. A. (2005). "Living with hogs in Iowa: The impact of livestock facilities on rural residential property values." *Land Economics,* 81(4), 530–545. Both studies document measurable declines in residential property values near large-scale confined animal operations.

9. Starmer, E., Witteman, A., & Wise, T. A. (2006). *Feeding the Factory Farm: Implicit Subsidies to the Broiler Chicken Industry.* Global Development and Environment Institute, Tufts University. The analysis calculated the savings to the industrial livestock sector from purchasing feed grain at prices below the full cost of production, attributable to federal commodity support payments.

10. US Government Accountability Office. (2005). *Workplace Safety and Health: Safety in the Meat and Poultry Industry, While Improving, Could Be Further Strengthened.* GAO-05-96. The report documented injury and illness rates in meat and poultry processing substantially above national manufacturing averages.

11. Waltenburg, M. A., Victoroff, T., Rose, C. E., et al. (2020). "Update: COVID-19 among workers in meat and poultry processing facilities — United States, April–May 2020." *Morbidity and Mortality Weekly Report,* 69(27), 887–892.

12. Wing, S., & Wolf, S. (2000). "Intensive livestock operations, health, and quality of life among eastern North Carolina residents." *Environmental Health Perspectives,* 108(3), 233–238. See also Donham, K. J., Wing, S., Osterberg, D., Flora, J. L., Hodne, C., Thu, K. M., & Thorne, P. S. (2007). "Community health and socioeconomic issues surrounding concentrated animal feeding operations." *Environmental Health Perspectives,* 115(2), 317–320.

13. Wilson, S. M., Howell, F., Wing, S., & Sobsey, M. (2002). "Environmental injustice and the Mississippi hog industry." *Environmental Health Perspectives,* 110(Suppl 2), 195–201. The study documented the disproportionate siting of industrial hog operations near low-income and minority communities in Mississippi.

14. MacDonald, J. M., & McBride, W. D. (2009). *The Transformation of U.S. Livestock Agriculture: Scale, Efficiency, and Risks.* USDA Economic Research Service, Economic Information Bulletin No. 43. The report traces the consolidation of US livestock production from the 1980s through the 2000s.

15. USDA Natural Resources Conservation Service. (2009). *Animal Manure Management.* NRCS Conservation Practice Standard. The 500-million-ton annual estimate is widely cited across USDA and EPA regulatory literature.

16. National Oceanic and Atmospheric Administration. (2024). *Gulf of Mexico Dead Zone.* NOAA National Centers for Coastal Ocean Science. The annual survey measures the extent of the hypoxic zone, which has averaged roughly 5,400 square miles over the last five years.

17. Gerber, P. J., Steinfeld, H., Henderson, B., Mottet, A., Opio, C., Dijkman, J., Falcucci, A., & Tempio, G. (2013). *Tackling Climate Change Through Livestock: A Global Assessment of Emissions and Mitigation Opportunities.* Food and Agriculture Organization of the United Nations. The FAO assessment estimated global livestock supply chain emissions at 14.5 percent of anthropogenic greenhouse gas emissions.

18. Pimentel, D., & Pimentel, M. (2008). *Food, Energy, and Society* (3rd ed.). CRC Press. Pimentel's analysis of energy and grain inputs in feedlot beef production provides the feed conversion ratios cited here.

19. USDA National Agricultural Statistics Service. (2023). *Acreage Report.* NASS. Corn, soybeans, and wheat together account for the majority of planted acreage in the United States, with corn alone covering approximately 90 million acres in a typical year.

20. Orden, D., Paarlberg, R., & Roe, T. (1999). *Policy Reform in American Agriculture: Analysis and Prognosis.* University of Chicago Press. For the structural history of commodity support programs and their effects on production incentives, see also Sumner, D. A. (2014). "American farms keep growing: Size, productivity, and policy." *Journal of Economic Perspectives,* 28(1), 147–166.

21. USDA Economic Research Service. (2023). *Farm Structure and Contracting.* ERS. The long-term trend in farm numbers and average farm size is tracked in the Census of Agriculture and in ERS structural analyses.

22. For diet-related chronic disease costs, see Ward, Z. J., Bleich, S. N., Long, M. W., & Gortmaker, S. L. (2021). "Association of body mass index with health care expenditures in the United States by age and sex." *PLoS ONE,* 16(3), e0247307. The broader economic burden of diet-related disease, including cardiovascular disease, type 2 diabetes, and certain cancers, has been estimated at several hundred billion dollars annually across the US healthcare system.

23. Pimentel, D. (2006). "Soil erosion: A food and environmental threat." *Environment, Development and Sustainability,* 8, 119–137. Pimentel estimated average US cropland erosion at roughly ten times the natural soil-formation rate, with total annual soil loss of approximately 1.8 billion tons.

24. USDA Natural Resources Conservation Service. (2015). *Summary Report: 2012 National Resources Inventory.* NRCS. The NRI uses the Revised Universal Soil Loss Equation to estimate erosion rates across US cropland.

25. Tilman, D., Cassman, K. G., Matson, P. A., Naylor, R., & Polasky, S. (2002). "Agricultural sustainability and intensive production practices." *Nature,* 418(6898), 671–677. The paper documented that agricultural intensification was the largest driver of terrestrial biodiversity loss globally.

26. Burow, K. R., Nolan, B. T., Rupert, M. G., & Dubrovsky, N. M. (2010). "Nitrate in groundwater of the United States, 1991–2003." *Environmental Science & Technology,* 44(13), 4988–4997. The USGS assessment documented the geographic extent and agricultural correlation of nitrate contamination in US groundwater.

27. Rillig, M. C. (2004). "Arbuscular mycorrhizae and terrestrial ecosystem processes." *Ecology Letters,* 7(8), 740–754. For the specific effects of fungicide application on mycorrhizal networks, see also Zaller, J. G., Heigl, F., Ruess, L., & Grabmaier, A. (2014). "Glyphosate herbicide affects belowground interactions between earthworms and symbiotic mycorrhizal fungi in a model ecosystem." *Scientific Reports,* 4, 5634.

28. Foley, J. A., Ramankutty, N., Brauman, K. A., Cassidy, E. S., Gerber, J. S., Johnston, M., Mueller, N. D., O'Connell, C., Ray, D. K., West, P. C., Balzer, C., Bennett, E. M., Carpenter, S. R., Hill, J., Monfreda, C., Polasky, S., Rockström, J., Sheehan, J., Siebert, S., Tilman, D., & Zaks, D. P. M. (2011). "Solutions for a cultivated planet." *Nature,* 478(7369), 337–342.

29. Doccioli, C., Sera, F., Francavilla, A., Cupisti, A., & Biggeri, A. (2024). "Association of cadmium environmental exposure with chronic kidney disease: A systematic review and meta-analysis." *Science of the Total Environment,* 906, 167165 (https://doi.org/10.1016/j.scitotenv.2023.167165), for the direct meta-analytic evidence linking environmental cadmium exposure to declining estimated glomerular filtration rate (eGFR) in adult populations. Malik, I. S., Khan, Z. I., Ahmad, K., Nadeem, M., Akhtar, S., Ashfaq, A., Faisal, M., Alatar, A. A., Munir, M., & Ullah, S. (2025). "Food chain risk assessment of cadmium accumulation and the relationships among diversely irrigated pasture-reared ruminants." *Scientific Reports* (https://doi.org/10.1038/s41598-025-13628-6), for primary measurements of cadmium accumulation along soil–water–forage–ruminant–milk pathways with health-hazard indices (BCF, EF, HRI, THQ) exceeding unity. See also Alengebawy, A., Abdelkhalek, S. T., Qureshi, S. R., & Wang, M.-Q. (2021). "Heavy Metals and Pesticides Toxicity in Agricultural Soil and Plants: Ecological Risks and Human Health Implications." *Toxics,* 9(3), 42 (https://doi.org/10.3390/toxics9030042), for the broader review of heavy-metal pathways in agricultural soils, and Madjar, R. M., & Vasile Scăeţeanu, G. (2025). "An Overview of Heavy Metal Contamination in Water from Agriculture: Origins, Monitoring, Risks, and Control Measures." *Sustainability,* 17(16), 7368 (https://doi.org/10.3390/su17167368), for the broader survey of heavy-metal pathways from agricultural inputs into water and food systems. On the arsenic-from-poultry-litter pathway specifically, see Nachman, K. E., Baron, P. A., Raber, G., Francesconi, K. A., Navas-Acien, A., & Love, D. C. (2013). "Roxarsone, inorganic arsenic, and other arsenic species in chicken: A U.S.-based market basket sample." *Environmental Health Perspectives,* 121(7), 818–824 (https://

doi.org/10.1289/ehp.1206245), which documents arsenic species in conventional and organic chicken meat following decades of organoarsenical drug use and contextualizes the FDA's subsequent regulatory withdrawal of roxarsone, carbarsone, and arsanilic acid approvals. For the broadest authoritative synthesis of metal contamination, food-chain pathways, and remediation strategies, see Hou, D., O'Connor, D., Igalavithana, A. D., Alessi, D. S., Luo, J., Tsang, D. C. W., Sparks, D. L., Yamauchi, Y., Rinklebe, J., & Ok, Y. S. (2020). "Metal contamination and bioremediation of agricultural soils for food safety and sustainability." *Nature Reviews Earth & Environment,* 1, 366–381 (https://doi.org/10.1038/s43017-020-0061-y). On the persistence of heavy-metal loads even in inputs marketed as clean, see Su, K., Zhang, Q., Chen, A., Wang, X., Zhan, L., Rao, Q., Wang, J., & Yang, H. (2024). "Heavy metals concentrations in commercial organic fertilizers and the potential risk of fertilization into soils." *Scientific Reports* (https://doi.org/10.1038/s41598-024-79681-9), in which inductively coupled plasma mass spectrometry of seventy-four representative commercial organic fertilizers found 32 percent exceeding the EU standard for copper, 76 percent for zinc, and 85 percent for nickel. For the regional lead legacy specifically, see Filippelli, G. M., Risch, M., Laidlaw, M. A. S., Nichols, D. E., & Crewe, J. (2015). "Geochemical legacies and the future health of cities: A tale of two neurotoxins in urban soils." *Elementa: Science of the Anthropocene,* 3, 000059 (https://doi.org/10.12952/journal.elementa.000059). The Filippelli group's work documents the persistent atmospheric-deposition lead burden in upper soil horizons across the central United States. For the comprehensive US-wide soil-geochemistry baseline against which agricultural contamination is measured, see Smith, D. B., Cannon, W. F., Woodruff, L. G., Solano, F., Kilburn, J. E., & Fey, D. L. (2014). *Geochemical and Mineralogical Maps for Soils of the Conterminous United States.* US Geological Survey Open-File Report 2014–1082 (https://doi.org/10.3133/ofr20141082). For primary measurements of cadmium concentrations in phosphate fertilizers across European countries, see Nziguheba, G., & Smolders, E. (2008). "Inputs of trace elements in agricultural soils via phosphate fertilizers in European countries." *Science of the Total Environment,* 390(1), 53–57 (https://doi.org/10.1016/j.scitotenv.2007.09.031), reporting cadmium concentrations in phosphate-fertilizer products from sixteen European countries with population-weighted means consistent with the 12–14 mg/kg peak figure cited in the body. See also Roberts, T. L. (2014). "Cadmium and Phosphorous Fertilizers: The Issues and The Science." *Procedia Engineering,* 83, 52–59 (https://doi.org/10.1016/j.proeng.2014.09.012), for the global review of cadmium concentrations in commercial phosphate fertilizers and the regulatory trajectory across major producing regions. For the historical synthesis of fertilizer contamination trends from 1960 through 2025 that integrates these primary measurements with the longer time-series, see Pendergrass, K. (2026). "Heavy Metals in Fertilizers: A Historical Analysis of Contamination Trends (1960–2025)." *Food Metallomics* (Zenodo preprint, https://doi.org/10.5281/zenodo.18439158). On heavy-metal pathways into endocrine-receptor signaling, including the cadmium-as-metalloestrogen hypothesis and its implications for breast cancer risk, see Aquino, N. B., Sevigny, M. B., Sabangan, J., & Louie, M. C. (2012). "Role of Cadmium and Nickel in

Estrogen Receptor Signaling and Breast Cancer: Metalloestrogens or Not?" *Journal of Environmental Science and Health, Part C,* 30(3), 189–224 (https://doi.org/10.1080/10590501.2012.705159).

30. Maier, R. J., & Benoit, S. L. (2019). "Role of Nickel in Microbial Pathogenesis." *Inorganics,* 7(7), 80 (https://doi.org/10.3390/inorganics7070080). The Maier and Benoit review (University of Georgia Center for Metalloenzyme Studies) catalogs nickel-dependent virulence factors across pathogens, including the urease and hydrogenase enzymes that allow *Helicobacter pylori* to colonize the gastric environment, and the nickel-dependent enzymes used by virulent *E. coli, Klebsiella, Streptococcus, Staphylococcus,* and *Mycobacterium tuberculosis.* The asymmetry of essential-for-pathogens / nonessential-for-hosts is the underlying biochemical claim. For the population-level extension — that dietary nickel and cadmium loads, mediated by microbial mismetallation, are mechanistically implicated in the US obesity epidemic and related metabolic conditions — see Pendergrass, K. (2026). "Heavy Metals, Microbial Metallomics, and the US Obesity Epidemic: A Mechanistic Examination of a Population-Level Metabolic Disruption." *Microbial Metallomics* (Zenodo preprint, https://doi.org/10.5281/zenodo.18434951).

31. El-Naggar, A., Ahmed, N., Mosa, A., Niazi, N. K., Yousaf, B., Sharma, A., Sarkar, B., Cai, Y., & Chang, S. X. (2021). "Nickel in soil and water: Sources, biogeochemistry, and remediation using biochar." *Journal of Hazardous Materials,* 419, 126421 (https://doi.org/10.1016/j.jhazmat.2021.126421). The review covers nickel sources in agricultural soils (phosphate fertilizers, manure applications, atmospheric deposition), the biogeochemistry of metal speciation and bioavailability, and the role of soil microbial communities in mediating those processes — the same communities suppressed by the fungicide regime modern grain production depends on. Alengebawy et al. (2021) cited above also covers the soil-microbial dimension of metal toxicity. For direct experimental evidence of the metal–mycorrhiza interaction in a cropping system, see Zalewska, M. "Effect of Soil Contamination by Lead, Nickel and Cadmium and VA-Mycorrhizal Fungi on Yield and Heavy Metal Concentration in Roots and Aboveground Biomass of Oat," a pot-experiment study (Justus-Liebig-Universität Giessen) documenting that soil contamination at 50 mg Ni, 10 mg Cd, and 100 mg Pb per kg significantly decreased oat root and shoot biomass, with metals accumulating primarily in roots, and arbuscular mycorrhizal infection modulating uptake.

32. Pimentel, D., & Pimentel, M. (2008). *Food, Energy, and Society* (3rd ed.). CRC Press. The energy-input analysis for corn production documents that nitrogen fertilizer production and application accounts for roughly forty percent of total energy inputs.

33. Scanlon, B. R., Faunt, C. C., Longuevergne, L., Reedy, R. C., Alley, W. M., McGuire, V. L., & McMahon, P. B. (2012). "Groundwater depletion and sustainability of irrigation in the US High Plains and Central Valley." *Proceedings of the National Academy of Sciences,* 109(24), 9320–9325.

34. Boyer, J. S., Byrne, P., Cassman, K. G., Cooper, M., Delmer, D., et al. (2013). "The U.S. drought of 2012 in perspective: a call to action." *Global Food Security,* 2(3), 139–143.

35. Schlenker, W., & Roberts, M. J. (2009). "Nonlinear temperature effects indicate severe damages to U.S. crop yields under climate change." *Proceedings of the National Academy of Sciences,* 106(37), 15594–15598. The study projected that under continued warming, the frequency of growing-season temperatures exceeding crop damage thresholds in the US Corn Belt would increase substantially.

36. USDA National Agricultural Statistics Service. (2024). *2022 Census of Agriculture: Cover Crop Data.* The census documented an increase in US cover crop acreage from approximately 10.3 million acres in 2012 to approximately 18.3 million acres in 2022.

37. Brown, Gabe. (2018). *Dirt to Soil: One Family's Journey into Regenerative Agriculture.* White River Junction, VT: Chelsea Green Publishing. Brown's book is the primary published accounting for the specific soil organic matter increases, elimination of synthetic fertilizer and fungicide purchases, reduction of herbicide use, and water infiltration improvements on the 5,000-acre operation near Bismarck, North Dakota. The NRCS and SARE have independently profiled the ranch's soil conservation outcomes in technical site documentation, which corroborates the order of magnitude of the organic matter and infiltration changes Brown reports.

38. LaCanne, C. E., & Lundgren, J. G. (2018). "Regenerative agriculture: merging farming and natural resource conservation profitably." *PeerJ,* 6, e4428. The peer-reviewed study compared matched pairs of regenerative and conventional corn operations in the Northern Plains and found that the regenerative operations were more profitable than their conventional neighbors despite lower per-acre yields, because reduced input costs more than compensated for yield differences. This is the same structural mechanism Brown's operation exemplifies — profitability driven by cost reduction rather than yield maximization — and it provides independent, peer-reviewed verification that the pattern is not unique to Brown's case.

CHAPTER SIX · PASTURE, HONESTLY

1. Salatin, J. (1998). *You Can Farm: The Entrepreneur's Guide to Start and Succeed in a Farming Enterprise.* Polyface Inc., chapters on soil and pasture management. Salatin's published accounting of the soil organic matter trajectory on Polyface from the early 1960s baseline to the late-1990s figure. The continental cropland average is drawn from USDA NRCS summaries, which place typical cultivated mineral soil organic matter in the one-to-three-percent range depending on region and management history. For the NRCS synthesis see Natural Resources Conservation Service (2012). *Soil Quality Indicator Sheets: Soil Organic Matter.* USDA-NRCS.

2. Natural Resources Conservation Service. (2018). *Summary Report: 2017 National Resources Inventory.* USDA-NRCS, Washington, D.C. The NRI is the authoritative US federal source

for cultivated-cropland erosion rates under the RUSLE2 methodology; current average sheet-and-rill plus wind erosion rates sit in the four-to-five-ton-per-acre-per-year range on conventionally managed cropland. See also Thaler, E. A., Larsen, I. J., & Yu, Q. (2021). "The extent of soil loss across the US Corn Belt." *PNAS,* 118(8), e1922375118, for the continental-scale reconstruction of cumulative topsoil loss across the major US cropping regions since the onset of European-style agriculture, which supports the ninety-centimeter-plus figure in the body text.

3. Machmuller, M. B., Kramer, M. G., Cyle, T. K., Hill, N., Hancock, D., & Thompson, A. (2015). "Emerging land use practices rapidly increase soil organic matter." *Nature Communications,* 6, 6995, for the direct long-term measurement of soil organic carbon accumulation under managed-intensive pasture rotation in the US Southeast. See also Teague, W. R., et al. (2011). "Grazing management impacts on vegetation, soil biota and soil chemical, physical and hydrological properties in tall grass prairie." *Agriculture, Ecosystems & Environment,* 141(3–4), 310–322, for the comparison of adaptive multi-paddock, heavy continuous, and light continuous grazing on tall-grass prairie.

4. The National Chicken Council's *Animal Welfare Guidelines and Audit Checklist for Broilers* (2017 revision, subsequent updates) is the industry's own reference standard for stocking density and housing conditions in US commercial broiler production. US federal statute does not impose a specific on-farm stocking density for broiler chickens; the Poultry Products Inspection Act regulates processing rather than husbandry. For the regulatory gap see Matheny, G., & Leahy, C. (2007). "Farm-animal welfare, legislation, and trade." *Law and Contemporary Problems,* 70(1), 325–358.

5. Leonard, C. (2014). *The Meat Racket: The Secret Takeover of America's Food Business.* Simon & Schuster, for the contemporary account of the vertical-integration structure of US broiler production and the contract-grower financial model. See also MacDonald, J. M. (2008). "The economic organization of U.S. broiler production." *USDA Economic Research Service Economic Information Bulletin* No. 38, for the same material from the USDA side.

6. Carpenter, S. R., Mooney, H. A., Agard, J., et al. (2009). "Science for managing ecosystem services: Beyond the Millennium Ecosystem Assessment." *PNAS,* 106(5), 1305–1312, for the broader framework. For the specific rural-community-indicator work tied to livestock consolidation, see Lobao, L., & Stofferahn, C. W. (2008). "The community effects of industrialized farming: social science research and challenges to corporate farming laws." *Agriculture and Human Values,* 25, 219–240, which synthesizes fifty years of sociological studies on the relationship between agricultural structure and rural community outcomes.

7. Teague, W. R., Apfelbaum, S., Lal, R., Kreuter, U. P., Rowntree, J., Davies, C. A., Conser, R., Rasmussen, M., Hatfield, J., Wang, T., Wang, F., & Byck, P. (2016). "The role of ruminants in reducing agriculture's carbon footprint in North America." *Journal of Soil and Water Conservation,* 71(2), 156–164.

8. McLauchlan, K. K. (2006). "Effects of soil disturbance on ecosystem properties: a global meta-analysis." *Ecological Applications,* 16(6), 2245–2252, for the cultivation-versus-perennial-cover comparison on soil biota. For avian biodiversity differences between row-crop and pasture landscapes, see Rodenhouse, N. L., Best, L. B., O'Connor, R. J., & Bollinger, E. K. (1995). "Effects of agricultural practices and farmland structures." In *Ecology and Management of Neotropical Migratory Birds* (T. E. Martin & D. M. Finch, eds.), Oxford University Press, pp. 269–293.

9. Bharati, L., Lee, K.-H., Isenhart, T. M., & Schultz, R. C. (2002). "Soil-water infiltration under crops, pasture, and established riparian buffer in Midwestern USA." *Agroforestry Systems,* 56(3), 249–257, for the direct infiltration-rate comparison among cropping, pasture, and riparian-buffer treatments in a Midwest setting.

10. Pollan, M. (2006). *The Omnivore's Dilemma: A Natural History of Four Meals.* Penguin Press, Part III (the Polyface section), for the description of the farm's scale and the approximate annual broiler count.

11. Salatin, J. (1999). *Pastured Poultry Profit$: Net $25,000 in 6 Months on 20 Acres.* Polyface Inc., for the feed-conversion, finished-weight, and ration specifications used in the calculation in the body text.

12. Salatin (1999), ration specifications; the corn and soy proportions have been independently confirmed in subsequent profiles of the farm, including Ostrander, M. (2012). "How to feed the world: A conversation with Joel Salatin." *Yes! Magazine,* Spring 2012 issue. The exact proportions have varied slightly across editions of Polyface's documentation and across seasons of operation; the working figures used here are 49.7% corn and 30.87% soy, which yield the 71,568-pound and 44,452-pound annual totals.

13. Salatin, J. (2007). *Everything I Want to Do Is Illegal: War Stories from the Local Food Front.* Chelsea Green; and Salatin, J. (2011). *Folks, This Ain't Normal: A Farmer's Advice for Happier Hens, Healthier People, and a Better World.* Center Street. For discussion of feed-sourcing philosophy and the preference for local non-GMO sources over anonymous commodity channels.

14. McWilliams, J. E. (2012, April 12). "The myth of sustainable meat." *The New York Times,* opinion section.

15. Salatin, J. (2012). "The sheer ignorance of James McWilliams and the New York Times." Polyface Farm Facebook post and blog, April 2012, with expanded versions appearing in subsequent essays collected in Salatin, J. (2016). *The Marvelous Pigness of Pigs: Respecting and Caring for God's Creation.* FaithWords, and in interviews in the *Stockman Grass Farmer* and *Acres U.S.A.* in the months following the *Times* op-ed.

16. Pollan, M. (2006), chapter 2, for the history of the 1973 Farm Bill's transition from supply-management to direct-payment commodity support and its consequences for corn and soy pricing. See also Winders, B. (2009). *The Politics of Food Supply: U.S. Agricultural Policy in*

the World Economy. Yale University Press, for the structural account of how commodity-support architecture has shaped US grain markets since the 1970s.

17. Garnett, T., Godde, C., Muller, A., Röös, E., Smith, P., de Boer, I. J. M., zu Ermgassen, E., Herrero, M., van Middelaar, C., Schader, C., & van Zanten, H. (2017). *Grazed and Confused? Ruminating on Cattle, Grazing Systems, Methane, Nitrous Oxide, the Soil Carbon Sequestration Question — And What It All Means for Greenhouse Gas Emissions.* Food Climate Research Network, University of Oxford.

18. Teague, W. R., & Kreuter, U. P. (2020). "Managing grazing to restore soil health, ecosystem function, and ecosystem services." *Frontiers in Sustainable Food Systems,* 4, 534187, for the Teague-side response to the *Grazed and Confused?* aggregation approach and the case for separate analysis of adaptive multi-paddock grazing as a management class.

19. Crews, T. E., Carton, W., & Olsson, L. (2018). "Is the future of agriculture perennial? Imperatives and opportunities to reinvent agriculture by shifting from annual monocultures to perennial polycultures." *Global Sustainability,* 1, e11; and DeHaan, L. R., & Van Tassel, D. L. (2014). "Useful insights from evolutionary biology for developing perennial grain crops." *American Journal of Botany,* 101(10), 1801–1819, for the Land Institute's and allied perennial-grain breeding programs and their state of play. The Kernza intermediate wheatgrass program is the most commercially advanced current example.

CHAPTER SEVEN · THE WRONG PROXY IN THE RIGHT PLACE

1. Martin, P. S. (2005). *Twilight of the Mammoths: Ice Age Extinctions and the Rewilding of America.* University of California Press, chapter 5 ("Grand Canyon Suite"). The Rampart Cave deposits were first systematically excavated in the 1930s and have yielded one of the most complete records of Shasta ground sloth (*Nothrotheriops shastensis*) ecology in the American Southwest. The stratified dung deposits span approximately 40,000 to 11,000 radiocarbon years before present.

2. Hansen, R. M. (1978). "Shasta ground sloth food habits, Rampart Cave, Arizona." *Paleobiology,* 4(3), 302–319. Hansen's composition analysis of ground sloth dung from Rampart Cave is the primary dietary reconstruction for the species; the roughly fifty-percent globe mallow (*Sphaeralcea laxa*) figure, eighteen-percent Mormon tea (*Ephedra*) pollen, and the near-total absence of grasses are drawn from this study.

3. Martin (2005), chapter 10 ("Restoration"), pp. 185–186. Martin's argument that the thorny, armed vegetation of the arid Southwest evolved under heavy browsing pressure from now-extinct large herbivores, and that the anti-herbivore defenses of these plants are calibrated to animals that have been absent for approximately 13,000 years.

4. Martin (2005), p. 185. The full passage reads: "From fossils we know that the grasses, forbs, shrubs, and trees of the Americas coevolved with a much greater variety of large herbivores than exists today."

5. Savory, A., & Butterfield, J. (1999). *Holistic Management: A New Framework for Decision Making*. Island Press. The foundational text of holistic management theory, laying out the relationship between large-herbivore impact, grassland health, and desertification reversal.

6. Martin (2005), chapter 5. The Stanton's Cave faunal assemblage, discussed alongside the Rampart Cave deposits, includes predominantly Harrington's mountain goat (*Oreamnos harringtoni*) bones, with smaller numbers of bighorn sheep, two small species of *Equus*, and some bison. Martin notes that Harrington's goat bones and horn sheaths accounted for the majority of the approximately 120 ungulate bones identified from the cave.

7. Donlan, J., Greene, H. W., Berger, J., Bock, C. E., Bock, J. H., Burney, D. A., Estes, J. A., Foreman, D., Martin, P. S., Roemer, G. W., Smith, F. A., & Soulé, M. E. (2005). "Re-wilding North America." *Nature,* 436(7053), 913–914. The initial public statement of the Pleistocene rewilding proposal, arguing that ecological history justifies the managed introduction of large vertebrates — including Old World proxies for extinct New World species — to North American landscapes.

8. Donlan, C. J., Berger, J., Bock, C. E., Bock, J. H., Burney, D. A., Estes, J. A., Foreman, D., Martin, P. S., Roemer, G. W., Smith, F. A., Soulé, M. E., & Greene, H. W. (2006). "Pleistocene rewilding: an optimistic agenda for twenty-first century conservation." *The American Naturalist,* 168(5), 660–681. The extended treatment of the rewilding proposal, with detailed discussion of proxy-species selection, ecological risk assessment, and phased implementation strategies for multiple North American biomes.

9. Janzen, D. H., & Martin, P. S. (1982). "Neotropical anachronisms: the fruits the gomphotheres ate." *Science,* 215(4528), 19–27. The foundational paper establishing the concept of megafaunal fruit syndrome and identifying specific New World fruit species — including osage orange, honey locust, and pawpaw — as ecological anachronisms whose dispersal biology is calibrated to large-bodied herbivores that went extinct in the late Pleistocene.

10. Martin (2005), chapters 9 and 10. Martin discusses the late-Quaternary disappearance of tapirs (*Tapirus*) from Florida, California, Kansas, and Arizona, including from elevations above 6,000 feet on the Colorado Plateau, and argues that modeling their range contraction as a retreat to tropical refugia under predation pressure is more parsimonious than a climatic explanation, given that packrat midden records from the relevant regions do not show the kind of tropical-to-temperate climate shift that would be required to force a thermally driven range collapse.

11. Zimov, S. A. (2005). "Pleistocene Park: return of the mammoth's ecosystem." *Science,* 308(5723), 796–798. Sergey Zimov's initial description of the Pleistocene Park experiment in northeastern Siberia, which reintroduced horses, bison, muskoxen, and other large

herbivores to test the hypothesis that megafaunal grazing can restore the mammoth-steppe grassland ecosystem and reduce permafrost thaw.

12. Rubenstein, D. R., Rubenstein, D. I., Sherman, P. W., & Gavin, T. A. (2006). "Pleistocene Park: does re-wilding North America represent sound conservation for the 21st century?" *Biological Conservation,* 132(2), 232–238. The most careful of the early critiques of the Donlan et al. rewilding proposal, raising concerns about ecological risk, invasive-species dynamics, and the assumption that proxy species will fill the ecological roles of their extinct counterparts.

CHAPTER EIGHT · REWILDING THE PLATE

1. Mayer, J. J., & Brisbin, I. L., Jr. (2009). *Wild Pigs: Biology, Damage, Control Techniques and Management.* Savannah River National Laboratory, SRNL-RP-2009-00869. The Texas population estimate of approximately 2.6 million feral hogs is drawn from Texas Parks and Wildlife Department surveys; see also Timmons, J. B., Mellish, J. M., Higginbotham, B., Griffin, J. L., Lopez, R. R., & Sumners, J. A. (2012). "Feral hog population growth, density, and harvest in Texas." Texas A&M AgriLife Extension Service, SP-472.

2. Anderson, A., Slootmaker, C., Harper, E., Holderieath, J., & Shwiff, S. A. (2016). "Economic estimates of feral swine damage and control in 11 US states." *Crop Protection,* 89, 89–94. The $52 million annual direct agricultural loss figure for Texas is conservative and accounts only for crop damage; total economic losses including infrastructure, ecological, and disease-vector costs are substantially higher.

3. Texas Parks and Wildlife Department. (2023). "Feral Hogs — Texas Distribution and Biology." TPWD Wildlife Division fact sheet. The annual harvest estimate and the reproductive rate data — two litters per year, five to six piglets per litter, population doubling time of four months under favorable conditions — are drawn from TPWD's published management summaries.

4. Hedrick, P. W. (2009). "Conservation genetics and North American bison (*Bison bison*)." *Journal of Heredity,* 100(4), 411–420. For the current continental population structure and the distinction between commercial and conservation herds. The roughly 500,000 total and 20,000 conservation-herd figures are consistent with current estimates from the IUCN Bison Specialist Group and the Wildlife Conservation Society's bison program.

5. American Prairie. (2024). "About American Prairie." americanprairie.org. The 450,000-acre current footprint and the long-term acquisition target are drawn from the organization's public documentation. See also Freese, C. H., Aune, K. E., Boyd, D. P., Derr, J. N., Forrest, S. C., Gates, C. C., Gogan, P. J. P., Grassel, S. M., Halbert, N. D., Kunkel, K., & Redford, K. H. (2007). "Second chance for the plains bison." *Biological Conservation,* 136(2), 175–184, for the ecological rationale underlying large-scale bison restoration in the northern Great Plains.

6. InterTribal Buffalo Council. (2024). "About ITBC." itbcbuffalonation.org. The council's membership of seventy-six-plus tribal nations and the collective herd numbering in the tens of thousands are drawn from ITBC's published program data. See also Zontek, K. (2007). *Buffalo Nation: American Indian Efforts to Restore the Bison.* University of Nebraska Press, for the cultural and historical context of tribal bison restoration.

7. Hilborn, R., & Walters, C. J. (2013). *Quantitative Fisheries Stock Assessment: Choice, Dynamics and Uncertainty.* Springer. The foundational reference for the stock-assessment methodology that underlies Alaska's fisheries management regime. Alaska's constitutional mandate for sustained-yield fisheries management is codified in Article VIII, Section 4 of the Alaska State Constitution.

8. Hilborn, R., Amoroso, R. O., Anderson, C. M., Baum, J. K., Branch, T. A., Costello, C., de Moor, C. L., Faraj, A., Hively, D., Jensen, O. P., Kurota, H., Little, L. R., Mace, P., McClanahan, T., Melnychuk, M. C., Minto, C., Osio, G. C., Parma, A. M., Pons, M., ... Ye, Y. (2020). "Effective fisheries management instrumental in improving fish stock status." *Proceedings of the National Academy of Sciences,* 117(4), 2218–2224.

9. Beck, M. W., Brumbaugh, R. D., Airoldi, L., Carranza, A., Coen, L. D., Crawford, C., Defeo, O., Edgar, G. J., Hancock, B., Kay, M. C., Lenihan, H. S., Luckenbach, M. W., Toropova, C. L., Zhang, G., & Guo, X. (2011). "Oyster reefs at risk and recommendations for conservation, restoration, and management." *BioScience,* 61(2), 107–116.

10. US Fish and Wildlife Service. (2022). *2022 National Survey of Fishing, Hunting, and Wildlife-Associated Recreation.* US Department of the Interior. The six-million annual white-tailed deer harvest figure and the estimated venison poundage are derived from the Survey's hunting statistics and standard dressed-weight calculations for white-tailed deer.

11. Tree, I. (2018). *Wilding: The Return of Nature to a British Farm.* Picador. Tree's account of the Knepp rewilding project is the primary reference for the ecological and economic outcomes described in this chapter.

12. Greenaway, F., & Hill, D. (2004–present). Ecological monitoring reports for the Knepp Wildland Project, conducted by Imperial College London and associated research institutions. Species recolonization data — turtle doves, nightingales, peregrine falcons, purple emperor butterflies, all five British owl species — are documented in successive monitoring reports and summarized in Tree (2018), chapters 14–18.

13. USDA Economic Research Service. (2024). *Livestock and Meat Domestic Data.* The approximately 27-billion-pound annual US beef consumption figure is drawn from USDA-ERS per-capita disappearance data extrapolated to the current US population.

14. National Marine Fisheries Service. (2023). *Fisheries of the United States, 2022.* NOAA. Alaska salmon harvest tonnage and US total seafood consumption figures are drawn from NMFS annual fisheries statistics.

CHAPTER NINE · PROXIES AND CRITICS

1. Donlan, J., Greene, H. W., Berger, J., Bock, C. E., Bock, J. H., Burney, D. A., Estes, J. A., Foreman, D., Martin, P. S., Roemer, G. W., Smith, F. A., & Soulé, M. E. (2005). "Re-wilding North America." *Nature,* 436(7053), 913–914. The initial public statement of the Pleistocene rewilding proposal.

2. Caro, T. (2007). "The Pleistocene re-wilding gambit." *Trends in Ecology & Evolution,* 22(6), 281–283. Caro's critique focuses on the proxy-equivalence assumption and the risks of introducing large-bodied species into ecosystems that have reorganized in their absence.

3. Donlan, C. J., Berger, J., Bock, C. E., Bock, J. H., Burney, D. A., Estes, J. A., Foreman, D., Martin, P. S., Roemer, G. W., Smith, F. A., Soulé, M. E., & Greene, H. W. (2006). "Pleistocene rewilding: an optimistic agenda for twenty-first century conservation." *The American Naturalist,* 168(5), 660–681. The fuller treatment of the rewilding proposal, with discussion of phased implementation and ecological risk assessment.

4. Rubenstein, D. R., Rubenstein, D. I., Sherman, P. W., & Gavin, T. A. (2006). "Pleistocene Park: does re-wilding North America represent sound conservation for the 21st century?" *Biological Conservation,* 132(2), 232–238. The most careful of the early critiques, focusing on political, cultural, and practical feasibility rather than on ecological principles alone.

5. Rubenstein, D. R., & Rubenstein, D. I. (2016). "From Pleistocene to trophic rewilding: a wolf in sheep's clothing?" *Proceedings of the National Academy of Sciences,* 113(1), E1. A response to Svenning et al.'s trophic rewilding synthesis, arguing that the shift from Pleistocene-proxy to trophic-process framing does not resolve the underlying feasibility and risk concerns.

6. Svenning, J.-C., Pedersen, P. B. M., Donlan, C. J., Ejrnæs, R., Faurby, S., Galetti, M., Hansen, D. M., Sandel, B., Sandom, C. J., Terborgh, J. W., & Vera, F. W. M. (2016). "Science for a wilder Anthropocene: synthesis and future directions for trophic rewilding research." *Proceedings of the National Academy of Sciences,* 113(4), 898–906. The synthesis that reframed the rewilding conversation around ecological process rather than Pleistocene species proxies.

7. Ripple, W. J., & Beschta, R. L. (2012). "Trophic cascades in Yellowstone: the first 15 years after wolf reintroduction." *Biological Conservation,* 145(1), 205–213. The most comprehensive early synthesis of the ecological effects of wolf reintroduction in Yellowstone, documenting cascading effects from predator reintroduction through elk behavior and distribution to riparian vegetation recovery.

8. Painter, L. E., Beschta, R. L., Larsen, E. J., & Ripple, W. J. (2015). "Recovering aspen follow changing elk dynamics in Yellowstone: evidence of a trophic cascade?" *Ecology,* 96(1), 252–263. Documents the decline in elk browsing pressure on young aspen in northern Yellowstone — from ninety percent browsed in 1998 to thirty-seven to sixty-three percent by 2012 — with corresponding increases in aspen recruitment in areas of high wolf activity.

9. Creel, S., & Christianson, D. (2009). "Wolf presence and increased willow consumption by Yellowstone elk: implications for trophic cascades." *Ecology,* 90(9), 2454–2466. A study that complicates the simple "landscape of fear" narrative by showing that individual elk willow consumption increased in the presence of wolves, suggesting that the primary mechanism of vegetation recovery was elk population reduction rather than behavioral avoidance alone.

10. Pringle, R. M., Abraham, J. O., Anderson, T. M., Coverdale, T. C., Davies, A. B., Dutton, C. L., Gaylard, A., Goheen, J. R., Holdo, R. M., Hutchinson, M. C., Kimuyu, D. M., Long, R. A., Subalusky, A. L., & Veldhuis, M. P. (2023). "Impacts of large herbivores on terrestrial ecosystems." *Current Biology,* 33(11), R584–R610. https://doi.org/10.1016/j.cub.2023.04.024. The review synthesizes a decade of large-herbivore impact studies and identifies body size as the master variable governing ecosystem effects, with documented thresholds at roughly 100 kg (top-down to bottom-up regulation) and 1000 kg (effective invulnerability to non-human predation). The review's key conclusion for proxy-rewilding is that small herbivores do not functionally compensate for large ones, which is why livestock are poor surrogates for the wild large-herbivore species they have displaced.

CHAPTER TEN · WATER AND INDIGENOUS SYSTEMS

1. Kimmerer, R. W. (2013). *Braiding Sweetgrass: Indigenous Wisdom, Scientific Knowledge, and the Teachings of Plants.* Milkweed Editions. Kimmerer is a member of the Citizen Potawatomi Nation and a plant ecologist whose work bridges Traditional Ecological Knowledge and Western botanical science.

2. Anderson, M. K. (2005). *Tending the Wild: Native American Knowledge and the Management of California's Natural Resources.* University of California Press. Anderson's study draws on decades of ethnobotanical fieldwork and collaboration with Indigenous elders across California.

3. Anderson (2005), chapters 6–10. Anderson documents fire management practices across grassland, oak woodland, chaparral, and mixed conifer ecosystems, with specific fire-return intervals and management objectives for each vegetation type, reconstructed from both ethnographic and fire-scar evidence. See also Lightfoot, K. G., & Parrish, O. (2009). *California Indians and Their Environment: An Introduction.* University of California Press, for additional synthesis of pre-contact landscape management.

4. Johnston, L. J. (2022). "Kincentric ecology and the restoration of a 4000-year-old food forest." Paper presented at the Ecological Society of America Annual Meeting, Montréal, Canada. Johnston's research on Indigenous food forests across North America draws on both archaeological evidence and oral tradition to reconstruct the scale and sophistication of pre-contact landscape management. See also Kimmerer, R. W. (2002). "Weaving traditional ecological knowledge into biological education: A call to action." *BioScience,* 52(5), 432–438, for a peer-reviewed articulation of the relationship between Indigenous ecological knowledge and Western scientific frameworks.

5. Krumhansl, K. A., et al. (2016). "Global patterns of kelp forest change over the past half-century." *PNAS,* 113(48), 13785–13790. The study compiled 862 kelp time series from 34 ecoregions, covering a temporal span of several decades, and found net declines in 38% of the ecoregions studied.

6. Estes, J. A., & Palmisano, J. F. (1974). "Sea otters: their role in structuring nearshore communities." *Science,* 185(4156), 1058–1060. The foundational paper documenting the trophic cascade between sea otters, sea urchins, and kelp forests. See also Estes, J. A., et al. (2011). "Trophic downgrading of planet Earth." *Science,* 333(6040), 301–306, for the broader synthesis of top-predator removal effects across marine and terrestrial ecosystems.

7. Beck, M. W., et al. (2011). "Oyster reefs at risk and recommendations for conservation, restoration, and management." *BioScience,* 61(2), 107–116. The synthesis documented that 85% of oyster reef habitat had been lost globally, making oyster reefs among the most threatened marine habitats on Earth.

8. Newell, R. I. E. (2004). "Ecosystem influences of natural and cultivated populations of suspension-feeding bivalve molluscs: a review." *Journal of Shellfish Research,* 23(1), 51–61. Newell's review provides the filtration-rate estimates for individual oysters and models the water-quality consequences of historical versus current oyster populations in the Chesapeake Bay.

9. Zu Ermgassen, P. S. E., et al. (2012). "Historical ecology with real numbers: past and present extent and biomass of an imperiled estuarine habitat." *Proceedings of the Royal Society B,* 279(1742), 3393–3400. For the Chesapeake Bay restoration monitoring data specifically, see the NOAA Chesapeake Bay Office restoration program reports and Schulte, D. M., Burke, R. P., & Lipcius, R. N. (2009). "Unprecedented restoration of a native oyster metapopulation." *Science,* 325(5944), 1124–1128.

10. Naylor, R. L., et al. (2000). "Effect of aquaculture on world fish supplies." *Nature,* 405(6790), 1017–1024. Naylor and colleagues documented that farmed salmon production requires approximately 2.5 to 5 kg of wild-caught fish (processed into fish meal and oil) per kilogram of salmon produced, creating a net reduction in the total fish available for human consumption. For antibiotic and antiparasitic use in salmon aquaculture, see Burridge, L., Weis, J. S., Cabello, F., Pizarro, J., & Bostick, K. (2010). "Chemical use in salmon aquaculture: a review of current practices and possible environmental effects." *Aquaculture,* 306(1–4), 7–23.

11. Jackson, J. B. C., et al. (2001). "Historical overfishing and the recent collapse of coastal ecosystems." *Science,* 293(5530), 629–638. Jackson and colleagues assembled historical, archaeological, and paleoecological data from multiple coastal ecosystems to reconstruct pre-exploitation baselines and document the sequential removal of megafauna, large predators, and small herbivores that produced the modern depleted state.

12. Newell, R. I. E. (1988). "Ecological changes in Chesapeake Bay: Are they the result of overharvesting the American oyster, *Crassostrea virginica*?" In *Understanding the Estuary:*

Advances in Chesapeake Bay Research, Chesapeake Research Consortium Publication 129, pp. 536–546. Newell's estimate that pre-exploitation oyster populations could filter the bay's volume in approximately one week, compared with approximately one year for the current remnant population, has been widely cited and was subsequently refined in Newell (2004).

13. Lepofsky, D., Smith, N. F., Cardinal, N., Harper, J., Morris, M., Gitla (Elroy White), Bouchard, R., Kennedy, D. I. D., Salomon, A. K., Puckett, M., & Rowell, K. (2015). "Ancient shellfish mariculture on the Northwest Coast of North America." *American Antiquity,* 80(2), 236–259. See also Groesbeck, A. S., Rowell, K., Shellfish, D., Giber, A. K., & Salomon, A. K. (2014). "Ancient clam gardens increased shellfish production: adaptive strategies from the past can inform food security today." *PLoS ONE,* 9(3), e91235, for the quantitative comparison of clam productivity between garden and non-garden sites.

CHAPTER ELEVEN · BIOREGIONAL AND URBAN

1. Martin, P. S. (2005). *Twilight of the Mammoths: Ice Age Extinctions and the Rewilding of America.* University of California Press, chapter 10. Martin's discussion of the co-evolutionary relationship between arid-Southwest browse vegetation and the Pleistocene browsing fauna — camelids, ground sloths, and equids — whose foraging behavior shaped the plant community's defensive architecture.

2. Archer, S. (1995). "Tree-grass dynamics in a *Prosopis*-thornscrub savanna parkland: reconstructing the past and predicting the future." *Ecoscience,* 2(1), 83–99. Archer's foundational analysis of mesquite encroachment dynamics in the Sonoran and Chihuahuan regions, documenting the shift from open grassland to dense shrubland in the absence of large-bodied browsers. See also Van Auken, O. W. (2009). "Causes and consequences of woody plant encroachment into western North American grasslands." *Journal of Environmental Management,* 90(10), 2931–2942, for the broader pattern across western rangelands.

3. Bureau of Land Management. "Wild Horse and Burro Program: Program Data." The BLM estimates the current free-roaming horse and burro population across ten western states at roughly 83,000 animals on public land, with additional populations on tribal and private land. See also the 2013 National Academies of Sciences, Engineering, and Medicine review, *Using Science to Improve the BLM Wild Horse and Burro Program: A Way Forward,* for the management challenges and the gap between current population levels and the agency's appropriate-management-level estimates.

4. Farah, Z., Mollet, M., Younan, M., & Dahir, R. (2007). "Camel dairy in Somalia: Limiting factors and development potential." *Livestock Science,* 110(1–2), 187–191. For the nutritional profile of camel milk — higher fat and protein content, lower lactose, higher vitamin C than bovine milk — see also Al Haj, O. A., & Al Kanhal, H. A. (2010). "Compositional, technological and nutritional aspects of dromedary camel milk." *International Dairy Journal,* 20(12), 811–821.

5. Hedrick, P. W. (2009). "Conservation genetics and North American bison (*Bison bison*)." *Journal of Heredity,* 100(4), 411–420. For current population structure and the distinction between commercial and conservation herds.

6. Martin (2005), chapters 9 and 10. Martin discusses the late-Quaternary presence of tapirs, capybaras, and ground sloths across the Gulf Coast and subtropical Southeast, with range extensions into Florida and the coastal plain. See also Webb, S. D. (2006). "The Great American Biotic Interchange: patterns and processes." *Annals of the Missouri Botanical Garden,* 93(2), 245–257, for the Neotropical character of the Gulf Coast's Pleistocene fauna.

7. Beck, M. W., et al. (2011). "Oyster reefs at risk and recommendations for conservation, restoration, and management." *BioScience,* 61(2), 107–116. The roughly eighty-five-percent loss of global oyster reef habitat documented in this synthesis includes severe losses in the Gulf of Mexico.

8. For the environmental and dependency profile of farmed shrimp production in Southeast Asia, see Naylor, R. L., Goldburg, R. J., Primavera, J. H., Kautsky, N., Beveridge, M. C. M., Clay, J., Folke, C., Lubchenco, J., Mooney, H., & Troell, M. (2000). "Effect of aquaculture on world fish supplies." *Nature,* 405(6790), 1017–1024. The contrast between wild-caught Gulf shrimp and imported farmed shrimp is a direct application of the survivability axis to a consumer-facing comparison.

9. Foster, D. R. (1992). "Land-use history (1730–1990) and vegetation dynamics in central New England, USA." *Journal of Ecology,* 80(4), 753–771. Foster's reconstruction of the clearing, farming, abandonment, and reforestation cycle in New England is the standard reference for the region's ecological history within the period of European settlement.

10. Fisher, D. C. (2009). "Paleobiology and extinction of proboscideans in the Great Lakes region of North America." In *American Megafaunal Extinctions at the End of the Pleistocene* (G. Haynes, ed.), Springer, pp. 55–75. For the mastodon's range, diet, and ecological role in the eastern deciduous forests.

11. Janzen, D. H., & Martin, P. S. (1982). "Neotropical anachronisms: the fruits the gomphotheres ate." *Science,* 215(4528), 19–27. The ecological-anachronism concept, as applied to eastern North American fruit species whose dispersal biology is calibrated to large-bodied herbivores absent since the late Pleistocene.

12. Helfield, J. M., & Naiman, R. J. (2001). "Effects of salmon-derived nitrogen on riparian forest growth and implications for stream productivity." *Ecology,* 82(9), 2403–2409. The study documented measurable marine-derived nitrogen in the foliage and growth rings of riparian trees near salmon spawning streams, establishing the quantitative link between salmon migration and terrestrial nutrient cycling.

13. Boyd, R., ed. (1999). *Indians, Fire, and the Land in the Pacific Northwest.* Oregon State University Press. The collected essays document the extent and sophistication of Indigenous fire

management in the Pacific Northwest, including the maintenance of Willamette Valley prairies and Puget Sound oak savannas through deliberate burning.

14. Al-Ayadhi, L. Y., & Elamin, N. E. (2013). "Camel milk as a potential therapy as an antioxidant in autism spectrum disorder (ASD)." *Evidence-Based Complementary and Alternative Medicine,* 2013, 602834. https://doi.org/10.1155/2013/602834. The randomized clinical trial in 60 children with autism spectrum disorder, using 500 ml/day of camel milk over two weeks, documented improvements in Childhood Autism Rating Scale scores and elevated antioxidant markers including superoxide dismutase, glutathione peroxidase, and myeloperoxidase relative to control milk.

15. Bashir, S., & Al-Ayadhi, L. Y. (2014). "Effect of camel milk on thymus and activation-regulated chemokine in autistic children: double-blind study." *Pediatric Research,* 76(2), 166–170. https://doi.org/10.1038/pr.2014.79. The double-blind trial of camel milk in autistic children documented reduced thymus-and-activation-regulated chemokine (TARC) levels and improved Childhood Autism Rating Scale scores, supporting the immunomodulatory mechanism Al-Ayadhi 2013 had identified.

16. Alshebrami, R. M., Ahmed, W. E., & Almujaydil, M. S. (2026). "Camel milk as a promising infant's formula (IF) to improve the health and nutritional status of infants and young children: a systematic review." *International Journal of Chemical and Biochemical Sciences,* 29(23). https://doi.org/10.62877/14-IJCBS-26-29-23-14. The systematic review documents the nutritional, immunological, and antioxidant case for camel milk as an alternative infant formula, particularly for infants with cow's-milk-protein allergy, lactose intolerance, or compromised gut function.

CHAPTER THIRTEEN · WHAT MUST CHANGE

1. Welch, D. (2012). "What can we do about sustainable consumption? Reframing consumer policy for shifted priorities." *Journal of Consumer Culture,* 12(3), 379–399. Welch's analysis examines the structural limitations of consumer-choice-based approaches to food system sustainability, arguing that the options available to consumers are shaped by policy, infrastructure, and industry structure that consumer purchasing decisions cannot, by themselves, alter.

2. Environmental Working Group. (2022). *Farm Subsidy Database.* EWG. The database compiles USDA subsidy disbursement records from 1995 through 2021; the cumulative figure of approximately $478 billion includes commodity payments, crop insurance premium subsidies, conservation payments, and disaster payments across all farm bill programs.

3. Environmental Working Group. (2022). *Farm Subsidy Database: Concentration Analysis.* EWG. The EWG analyses have consistently shown that the top ten percent of subsidy recipients receive approximately seventy-five percent of total payments, and that the

payments are concentrated in commodity crop production — corn, soybeans, wheat, cotton, and rice — rather than in diversified, specialty, or ecologically managed production systems.

4. Starmer, E., Witteman, A., & Wise, T. A. (2006). *Feeding the Factory Farm: Implicit Subsidies to the Broiler Chicken Industry.* Global Development and Environment Institute, Tufts University. The analysis calculated $35 billion in below-cost feed savings to the industrial livestock sector between 1997 and 2005, attributable to federal commodity support payments that held corn and soybean prices below the cost of production.

5. Witcover, J., Murphy, S., & Kaffka, S. (2019). *Policy Options for Alternative Uses of Agricultural Land.* UC Davis Policy Institute for Energy, Environment, and the Economy. The analysis modeled land-use responses to alternative subsidy structures on marginal cropland in the US Corn Belt and found measurable shifts toward non-commodity land uses under ecosystem-service payment scenarios.

6. Orden, D., Paarlberg, R., & Roe, T. (1999). *Policy Reform in American Agriculture: Analysis and Prognosis.* University of Chicago Press, for the history of farm bill restructuring from the 1996 Federal Agriculture Improvement and Reform Act through its subsequent modifications. See also Winders, B. (2009). *The Politics of Food Supply: U.S. Agricultural Policy in the World Economy.* Yale University Press, for the structural account of how commodity-support architecture reconverges after attempted reforms.

7. Endangered Species Act of 1973, Section 10(j), 16 U.S.C. § 1539(j). The experimental-population provision allows the Fish and Wildlife Service to designate reintroduced populations as "experimental" and manage them under regulations that may be less restrictive than those applying to naturally occurring populations. The gray wolf reintroductions in the Greater Yellowstone Ecosystem and central Idaho (1995–1996) were conducted under 10(j) designations. See also Bangs, E. E., & Fritts, S. H. (1996). "Reintroducing the gray wolf to central Idaho and Yellowstone National Park." *Wildlife Society Bulletin,* 24(3), 402–413.

8. Svenning, J.-C., Pedersen, P. B. M., Donlan, C. J., Ejrnæs, R., Faurby, S., Galetti, M., Hansen, D. M., Sandel, B., Sandom, C. J., Terborgh, J. W., & Vera, F. W. M. (2016). "Science for a wilder Anthropocene: Synthesis and future directions for trophic rewilding research." *PNAS,* 113(4), 898–906. The synthesis argues that the scientific literature on trophic rewilding has advanced substantially and that institutional frameworks for conservation have not kept pace with the evidence base.

9. USDA Farm Service Agency. (2023). *Conservation Reserve Program: Program Fact Sheet.* FSA. CRP contracts restrict commercial use of enrolled acreage; limited haying and grazing are permitted under emergency provisions and managed-harvest provisions, but the program's primary orientation is toward maintaining vegetative cover rather than toward active ecological management that includes food production.

10. For habitat exchange programs as institutional models for integrating ecological outcomes with working-land management, see Wilcove, D. S., & Lee, J. (2004). "Using economic and

regulatory incentives to restore endangered species: Lessons learned from three new programs." *Conservation Biology,* 18(3), 639–645. The sage-grouse habitat exchange programs operating in Wyoming, Montana, Nevada, and other western states provide a state-level model for crediting ecological outcomes on working landscapes.

11. Hilborn, R., Amoroso, R. O., Anderson, C. M., Baum, J. K., Branch, T. A., Costello, C., de Moor, C. L., Faraj, A., Hively, D., Jensen, O. P., Kurota, H., Little, L. R., Mace, P., McClanahan, T., Melnychuk, M. C., Minto, C., Osio, G. C., Parma, A. M., Pons, M., Segurado, S., Szuwalski, C. S., Wilson, J. R., & Ye, Y. (2020). "Effective fisheries management instrumental in improving fish stock status." *PNAS,* 117(4), 2218–2224. The analysis demonstrated that fisheries under intensive science-based management have been improving in stock status, while unmanaged fisheries continue to decline.

12. Hilborn et al. (2020). The analysis documents significant disparities in management quality across the world's fisheries, with well-managed jurisdictions (US, EU, Australia, New Zealand) showing improving stock status and poorly managed jurisdictions showing continued decline.

13. On the Haber-Bosch dependency cited in the "When the Test Stops Being a Thought Experiment" Case sidebar, see Smil, V. (2022). *How the World Really Works: The Science Behind How We Got Here and Where We're Going*. Viking, for the trade-readable statement of the dependency; and Erisman, J. W., Sutton, M. A., Galloway, J., Klimont, Z., & Winiwarter, W. (2008). "How a century of ammonia synthesis changed the world." *Nature Geoscience,* 1(10), 636–639 (https://doi.org/10.1038/ngeo325), for the peer-reviewed estimate that roughly half of the nitrogen in current human bodies traces to industrial fixation.

14. On the Ukraine grain-corridor chokepoint cited in the same sidebar, see World Food Programme situation reports for the Black Sea grain corridor (ongoing series) and the Food and Agriculture Organization's grain-export tracking. On the broader chokepoint architecture — phosphate, potash, seed concentration — see US Geological Survey, *Mineral Commodity Summaries* (annual; covers phosphate-rock and potash production-and-reserve concentration); and Bonny, S. (2017). "Corporate Concentration and Technological Change in the Global Seed Industry." *Sustainability,* 9(9), 1632 (https://doi.org/10.3390/su9091632), for the documentation of seed-stock concentration in a handful of multinational firms.

15. On the institutional convergence (Pentagon, war colleges, national-security planners running food-resilience math), see the Center for Strategic and International Studies Global Food Security Project (csis.org/programs/global-food-security-program), which collects the public-facing strategic-planning work and provides the standard institutional entry point for the framing.

Technical Appendix: The Heavy-Metal Load

This appendix carries the full pathway-by-metal exposition that Chapter 5 names but does not develop. It is here for the reader who wants to follow the chemistry into the soil microbiology, the regulatory record, and the primary literature; the body of the book scores the axis on its strength without requiring the reader to pass through any of this material. Readers who came for the framework can skip the appendix without losing the argument. Readers who want the audit trail will find it more usefully consolidated here than scattered across the chapter notes.

◆ CADMIUM ◆

Phosphate rock, the source of the phosphorus fertilizer industrial grain monoculture relies on, contains cadmium as a co-occurring trace element. Concentrations in commercial phosphate fertilizers peaked at twelve to fourteen milligrams per kilogram during the 1990s and have declined to roughly nine milligrams per kilogram in the 2010s under tightened EU and national standards (Nziguheba and Smolders 2008; Roberts 2014; Pendergrass 2026 preprint). The soils that received the high-load fertilizer carry the legacy: five to twenty-five times background cadmium concentrations in regions of intensive phosphate-fertilizer application. Cadmium concentrates in cropland soils under repeated phosphate application and moves into the food chain through plant uptake, with measurable accumulation in staple grains; leafy vegetables routinely exceed WHO permissible limits by five- to ten-fold (Su et al. 2024 inductively coupled plasma mass spectrometry of seventy-four representative commercial organic fertilizers found 32 percent exceeding the EU standard for copper, 76 percent for zinc, and 85 percent for nickel). Corresponding accumulation in human kidney tissue is documented across populations on phosphate-fertilized

food systems. The Doccioli et al. 2024 systematic review and meta-analysis links environmental cadmium exposure to declining estimated glomerular filtration rate (eGFR) in adult populations; Malik et al. 2025 documents cadmium accumulation along the soil–water–forage–ruminant–milk pathway with health-hazard indices (BCF, EF, HRI, THQ) exceeding unity.

◆ ARSENIC ◆

Arsenic entered US cropland through approved organoarsenical poultry-feed additives — roxarsone, carbarsone, and arsanilic acid — and the litter from medicated flocks that was spread on fields. The FDA's withdrawal of those drug approvals after 2011 tightened the input but did not remediate decades of accumulated soil load. Nachman et al. 2013 documented arsenic species in conventional and organic chicken meat following decades of organoarsenical drug use, contextualizing the FDA's subsequent regulatory action. Soil-borne arsenic continues to move through plant uptake and to appear in rice, leafy greens, and root crops grown on legacy-contaminated land.

◆ LEAD ◆

Atmospheric lead deposition is a smaller modern contributor than it was during the leaded-gasoline era, but the legacy load remains in the upper soil horizons of much of the US Corn Belt. Filippelli et al. 2015 documents the persistent atmospheric-deposition lead burden in upper soil horizons across the central United States; the Smith et al. 2014 USGS soil-geochemistry baseline provides the comprehensive US-wide reference against which agricultural contamination is measured.

◆ NICKEL ◆

Nickel cycles into agricultural soils through phosphate fertilizers, manure applications, and atmospheric deposition. El-Naggar et al. 2021 reviews the biogeochemistry of nickel speciation and bioavailability in agricultural systems, including the role of soil microbial communities in mediating those processes. The Maier and Benoit 2019 review (University of Georgia Center for Metalloenzyme

Studies) catalogs nickel-dependent virulence factors across pathogens, including the urease and hydrogenase enzymes that allow *Helicobacter pylori* to colonize the gastric environment and the nickel-dependent enzymes used by virulent *E. coli, Klebsiella, Streptococcus, Staphylococcus,* and *Mycobacterium tuberculosis.* Nickel itself has no established essential role in human nutrition. The asymmetry — essential for pathogens, nonessential for hosts — is what makes agricultural nickel accumulation a microbial-system question rather than a contamination question alone. The dysbiotic gut-microbiome signatures associated with an increasing list of modern conditions, including the metabolic disruptions linked to the US obesity epidemic, are now being examined as downstream effects of this asymmetry, mediated by mismetallation in microbial communities exposed to chronic dietary loads of nickel and cadmium (Pendergrass 2026, "Heavy Metals, Microbial Metallomics, and the US Obesity Epidemic").

◆ THE MYCORRHIZAL CONNECTION ◆

The soil microbial communities that mediate metal speciation, sequestration, and bioavailability are not incidental to the heavy-metal pathway. They determine whether a given soil cadmium load reaches a wheat kernel or stays bound. They are also the communities the chemical-input regime suppresses. Hou et al. 2020 (Nature Reviews Earth & Environment) provides the synthesis of metal contamination, food-chain pathways, and remediation strategies; Zalewska's pot-experiment study at Justus-Liebig-Universität Giessen documented that soil contamination at 50 mg Ni, 10 mg Cd, and 100 mg Pb per kg significantly decreased oat root and shoot biomass, with metals accumulating primarily in roots and arbuscular mycorrhizal infection modulating uptake. The metal-cycling pathway is what the three-axis sustainability framework is not measuring; the survivability axis is what makes the cost legible.

◆ A NOTE ON THE AUTHOR'S PREPRINTS ◆

Two of the citations in this appendix are to preprints I have placed on Zenodo: *Heavy Metals in Fertilizers: A Historical Analysis of Contamination Trends (1960–2025)* (https://doi.org/10.5281/zenodo.18439158) and *Heavy Metals, Microbial*

Metallomics, and the US Obesity Epidemic (https://doi.org/10.5281/zenodo.18434951). The disclosure note in the front matter addresses the conflict of interest these preprints implicate. They are cited here because they synthesize the primary literature in ways no other single source currently does, and the synthesis is what the body chapter rests on. The reader is welcome to read the primary sources directly — they are listed in the chapter notes — and to score the synthesis against them.

Author's Afterword: How This Book Got Here

I finished the first version of this book in 2012. It was a Paleo manuscript for Victory Belt, I turned it in, and a week or two later I picked up Paul Martin's *Twilight of the Mammoths* for the first time.

I read it in two days. I knew before I was halfway through that I had not written the book I wanted to write. The book I had turned in was answering a smaller question than the one Martin was asking. It was built on a baseline I had not realized was a choice: 1492, or something like it.

A year after I finished the manuscript, I had already begun to disagree with several of its premises. Two years later I had reversed position on others. The Paleo book did not publish. This one did, fourteen years later.

I am not embarrassed by the first version. The person who wrote it was doing her honest best with the evidence she had read. But one of the best things that ever happened to me was that the book did not go to press in 2012. I would have regretted it for the rest of my life.

It took fourteen years and five restarts. According to the files on my hard drive, I picked the manuscript up again in 2015, and set it down. Picked it up in 2016, set it down. 2018, the same. 2022, again. Now, 2026, here I am yet again, grappling with an idea that will not die. I think I have moved on. The niggling thought persists. Like throwing popcorn — or a refrigerator — into an event horizon, the idea follows me everywhere I go. Revision is not something that happened to this book. It is what the book is.

In April of 2025, Justin Bieber gave a Coachella performance that was, by the standards of polished pop, a mess. He sat at a desk. He played YouTube clips of his

own past, the embarrassing ones included. He took the audience through his actual history, not the curated one, and let them watch him put it together in real time. Some people balked. I understood the performance the moment I saw it. He was looking back. He was bringing the audience along.

This book is, in its register, a Coachella set. I am not handing the reader a polished argument from a polished position. I am taking them on the adventure that produced the argument, and the Field Notes scattered through these chapters are the looking-back: the meat counter in 2009 and what I was actually like that day, the standards I have spent sixteen years writing, the times I lost the argument by losing my temper, the times I changed my mind and regretted not having changed it sooner. The book without those moments would be a tidier book. It would also be a less honest one. Looking back means something.

I have tried to show my work.

IN GRATITUDE

Acknowledgments

♦ ♦ ♦

What follows is both acknowledgment and apology. Many of the people thanked here put up with a younger version of me whose insufferability is documented elsewhere in this section, and the gratitude carries the apology — sometimes openly, sometimes not. The people involved know.

Paul S. Martin did not know me. He died in 2010. His book, *Twilight of the Mammoths*, and his ideas became the spine of this one — I have never been able to get his ideas out of my mind. Whatever this book gets right about the deep history of this continent, the credit is his. Whatever it gets wrong is mine.

To my mother, Kimberly, who jumped without the parachute. You have been by my side, my biggest believer, since day one. No matter how crazy it sounded, thank you for believing in the camels too. I could not have done a single piece of this work without you, and the book is yours as much as it is mine.

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To Erich Krauss at Victory Belt: the manuscript I turned in in 2012 was, by my own honest reading of it now, badly written, and the writer who turned it in was difficult to work with. I have had fourteen years of practice since, and looking back at what I sent you then is jarring. Yikes is the word. Thank you for the patience you extended to a young writer who had not yet figured out what she was doing or why. The book I wish I had been able to write then is the book I have finally written now, and the apology to you is part of why I wrote it.

To Allan Savory: The conversation that runs through Chapter Seven came out of that meeting and the meetings that followed it. We do not agree on everything. We agreed on enough. I think the world of you.

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Finally, to the reader who has come this far in a personal proof copy: thank you. The book is asking for something. It is asking for an experiment to be tried, a parcel of ground to be flipped, a missing mouth to be put back in front of a fruit that has been falling for twelve thousand years. Whether or not you can act on any of that, the fact that you read it through means the argument has reached at least one set of hands. That is more than most arguments get.

Colophon

The Sustainability Diet

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